NOTES ON COLLATERAL DISCOUNTING

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ABSTRACT. In these personal notes, we generalise the HJM framework[1] to multiple currencies and stochastic collateral. Then, we define the Linear Gaussian Collateral Model by specialising this framework to separable and deterministic volatility and constant funding spread thus generalising the Black-Scholes-Hull-White model to account for stochastic collateral. In this model, we derive the dynamic of all market observables (ZCBs, FX-forwards, FRAs, swap rates, currency basis spreads etc. . .) as well as their collateral convexity adjustments. Finally, we describe a procedure to calibrate this model accurately to market quotes and/or historical data. The convexity adjustments cover Libor-type and compounded overnight (RFR) coupons alike. The calibration builds the cross-currency discount curve jointly with the collateral basis volatility, with the curves either bootstrapped or fitted globally (nested least squares on piecewise flat and smooth curves, with rate futures). We also give Gaussian approximations of the terminal correlations of FX and swap rates, and apply the framework to repo convexity. Every result is checked numerically against an exact benchmark or Monte Carlo in a Python companion.

TODO LIST. All items below were addressed on 2026-10-09 and checked numerically (see the verification remarks in each section).

(1) *Done.* Gaussian approximation for swap rates and currency basis spreads: Proposition 41, Proposition 42, proofs in Appendix D.
(2) *Done.* Reconstruction formulas in the Quasi-Gaussian model: Theorem 4 and Remark 33.
(3) *Done.* Algorithm for the computation of covariance integrals: Algorithm 1.
(4) *Done.* Drifts and covariance matrices in the LGCM under risk neutral, T-forward and jumping numéraire measures: Proposition 46, Proposition 47.
(5) *Done.* Consistency of interpolation with model dynamic: subsection 10.7.

Remaining open points found on the way: the Gaussian approximations are for a single collateral basis block ($f \to d$); options on currency basis spreads with smile are not covered.

Open points as of 2026-10-13 (review of the notes):

(1) Deterministic volatility models: the dynamic of the currency basis spread (still "$dB = ??$"), the collateral convexity of resetting forward Libors (marked "???") and of currency basis spreads (empty subsection), the investigation of the funding spread assumption in the Libor collateral convexity, and the question whether the convexity adjustments of $Z^{f|d}$ and $Z^{d|f}$ offset or add up (the calibration notebook computes both).
(2) Calibration: caplets are priced (Proposition 43) but not calibrated, and mean reversions, correlations and factor weights are marked; the futures convexity adjustment is marked, not computed from the LGCM; correlations are for native swap rates only, not $Y^{f|d}$.
(3) Multi-currency model: odd and implied correlations, the triangle relation under stochastic collateral, off-market curves such as $Z^{\mathrm{EUR}|\mathrm{JPY}}$.
(4) Explicit parametrisations still marked FIXME in the deterministic volatility and LGCM reconstruction sections; comparison with the hypotheses of Kjaer.
(5) Appendices: the review of the Linear Gaussian Model (partly superseded by subsection 9.1 and subsection 9.4, with empty subsections), the second order mixtures of the normal approximations, the estimation of historical volatilities and correlations (headings only), and two sections marked "[TODO: Remove this]".

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0. INTRODUCTION

0.1. **Motivation.** Discount and index curves are built from market quotes for the prices of liquid rates instruments. They serve as interpolation scheme for these prices, allowing a market participant to value non-standard instruments. These liquid instruments typically come with a standardised CSA which we call the *native* collateral so the corresponding curves are only valid for the valuation of cashflows under the same CSA. In the OTC markets, non-native CSAs are extremely common but since hedging instruments, the standardised liquid instruments, have different CSAs, one has to rely on a model to value those instead.

Since the 2008 financial crisis, market participants recognised that currency basis and funding[2] basis spreads could not be neglected. This lead to a new paradigm where the USD Libor discounting practice was replaced by a multi-curve framework with a different discount curve for each payment currency and CSA. The current market practice for building these curves relies on the assumption that funding spreads are deterministic. The consensus around this practice comes from the following very appealing property: it allows to account for funding cost with very minimal changes to exotic pricing models: namely replacing the "risk-free" discount curve by the CSA-specific ones.

However, at the same time, currency basis and funding spreads also became more volatile which generated demand for models with stochastic spreads to quantify and hedge the associated risks. But, being based on deterministic spread assumptions, the CSA discount curves are fundamentally inconsistent with such models and can no longer be used as inputs. So stochastic spreads models require a new change of paradigm. Until now, derivatives modelling was always assumed to be linear and incremental when it turns out to be a circle! The recipe used to be: first build the curves for linear products, then the vol surfaces for vanilla options and finally dynamic models for exotics. And the CSA-discounting market practice allowed to maintain this hierarchy. But, under non-native collateral, even the simplest linear products are actually exotic streams of cash-flows whose valuation depend on the covariance between the underlying coupons and the collateral. So non-native discount and index curves need to be built at the same time as we calibrate the stochastic spread model.

0.2. **Examples.** Let us give a few examples where stochastic currency basis risk is material.

Example 1 (Currency basis options)**.** Just like the time value of a swaption comes from the volatility of the swap rate:

$$\mathrm{MTM}_t^{\mathrm{Swaption}} = A_t^{d|c}\mathbb{E}_t^{A^{d|c},T;M}\left[\left(S_{T;M}^{d|c}-K\right)_+\right] \tag{1}$$

the time value of an option to enter a currency basis swap depends on the volatility of the currency basis spread

$$\mathrm{MTM}_t^{\mathrm{CBSwaption}} = A_t^{f|c}\mathbb{E}_t^{A^{f|c},T;M}\left[\left(B_{T;M}^{f/d|c}-K\right)_+\right]. \tag{2}$$

Example 2 (CVA)**.** The CVA can be seen as an option on a netting set of trades. As a result, any netting set involving several currencies has currency basis volatility risk.

Example 3 (FVA)**.** FVA is defined as the cost of hedging a non-collateralised trade with collateralised instruments. By hedging with collateralised trades, the trader is subject to margin calls that he needs to fund by borrowing from the treasury. The FVA is thus defined as the difference between the price under CSA discounting and under treasury rate discounting. If either of those is correlated to the underlying, the FVA is exposed to collateral convexity.

0.3. **Structure of these notes.**

(1) *Valuation of collateralised trades*: reminder on risk-neutral pricing to set notations, and the standard instruments and market observables.
(2) *Market Models*: tentative formal definition of the class of models we are considering.
(3) *Deterministic Collateral Basis Model*: short description of the current standard market practice for the valuation of fixed and floating rates cash-flows (swaps, fx-forwards, cross-currency swaps, etc...).

[2]Libor-OIS spread or Euribor-EONIA spread for example.

(4) *Arbitrage free models with stochastic collateral*: extension of the HJM framework to fixed cash-flows under stochastic collateral, quanto adjustments and the dynamic of market observables (including compounded overnight coupons and currency basis spreads).
(5) *Collateral convexity adjustments and P&L*: change of collateral and of payment currency, the relation to McCloud's definition, and the P&L equation.
(6) *Deterministic volatility models*: specialisation to deterministic volatility to get explicit formulas[3] for convexity adjustments.
(7) *Quasi-Gaussian Collateral Model*: specialisation to separable volatility to get finite dimensional Markovian dynamic (Theorem 4).
(8) *Specification of the LGCM*. By combining the two previous sections we generalise the IR Linear Gaussian model and FX Black-Scholes model to account for stochastic collateral. The main twist is the closed form approximation for the volatility of the currency basis spread. The section ends with the Gaussian approximations of swap rates and currency basis spreads and the terminal correlations of FX and swap rates (subsubsection 8.3.4).
(9) *Implementation of the LGCM*: covariance integrals in closed form, convexity adjustments and exact Monte Carlo simulation under the risk-neutral, T-forward and jumping numéraire measures.
(10) *Calibration of the LGCM*. The main point is that the EUR|USD discount curve and the EUR collateral basis spot volatility should be *jointly* bootstrapped. The section covers the calibration instruments (including rate futures), the discount curves by bootstrap or by a nested global least squares fit (subsubsection 10.2.1), the volatilities, interpolation and consistency with the model dynamic, alternative calibrations and the multi-currency model.
(11) *Bonds, repo and repo convexity* (section 11): the repo rate as a collateral convexity adjustment, after McCloud, in the LGCM and in Hull–White.
(12) *Remarks*: model usage, P&L leaks, materiality, assessment and generalisations.

Appendices contain details of mathematical proofs, a review of the Linear Gaussian Model, normal approximations of currency basis distributions and a draft on the estimation of historical volatilities and correlations.

The numerical companion (`python/convexity`) implements the LGCM, its calibration and the benchmarks; the verification remarks quote its measurements, and the figures and tables are generated by `python -m convexity.figures`.

Floating coupons are either Libor-type (fixed at the start of the accrual period) or compounded overnight (RFR) coupons, which replaced Libor in GBP, JPY, CHF and USD. Both are treated alongside throughout (coupon factors (20)). Under native collateral every coupon type is model-free (Lemma 1). Under non-native collateral the collateral convexity of a compounded coupon has an extra in-arrears term (Proposition 26 and Proposition 27); in the examples of these notes it changes $\ln K$ by 10^{-8} to 10^{-7} per quarterly period and the calibrated curves by about 0.001bp (Remark 77).

[3]This could easily be extended to encompass displaced lognormal diffusions thus capturing the value of the forward, the atm vol and the atm skew of the calibration targets.

1. VALUATION OF COLLATERALIZED TRADES

Notations. In these notes, we will use the following notations:

- u, reference currency (USD)
- e, generic currency
- d, f, generic currency pair
- c, generic collateral
- i, j, generic collateral currencies
- t, valuation date
- T, expiry date
- θ, tenor of an index rate (typically 3M for a Libor)
- M, tenor of a swap
- ℓ, frequency of a swap (typically $\ell = 4$ if θ is 3M).

1.1. Risk-neutral pricing of collateralized trades.

We summarise the main results about the valuation of derivatives under perfect collateral in a perfect Brownian market (see Piterbarg [8] and McCloud [6]).

We will always assume that the following hypothesis holds.

Hypothesis (Frictionless market)**.** We will say that the market is *frictionless* if the following properties are satisfied:

(1) Absence of arbitrage.
(2) Assets are infinitely divisible
(3) There are no restriction on transactions
(4) Transactions are instantaneous
(5) There are no jumps

[TODO: Compare to hypothesis in Kjaer's Consistent XVA metrics]

We will always assume that the following hypothesis holds.

Hypothesis (Brownian market)**.** A Brownian market is a

(1) Brownian assumption: under the historical measure $\mathbb{P}$, asset prices are Ito processes driven by a finite dimensional Brownian motion $W^{\mathbb{P}}$.
(2) Absence of arbitrage.

Under these hypothesis, one can derive the whole risk-neutral pricing framework for perfectly collateralised trades as follows:

(1) For each currency e, there exist a unique measure $\mathbb{Q}^{e,\mathrm{RN}}$ equivalent to the historical measure $\mathbb{P}$ such that, the value $V^{e|c}$ of a self-financed portfolio in currency e under domestic collateral rate r^c satisfies the martingale property:

$$\mathbb{E}_t^{e,\mathrm{RN}}\left[e^{-\int_t^T r_u^c du} V_T^{e|c}\right] = V_t^{e|c} \tag{3}$$

i.e. $V^{e|c}/M^c$ is a $\mathbb{Q}^{e,\mathrm{RN}}$-martingale where $M_t^c = e^{\int_0^t r_u^c du}$. This measure[4] is called the *risk-neutral measure* for currency e.

(2) For non-native collateral i, the *effective discount rate* $r_t^{e|i}$ is *defined* as the rate such that the formula

$$\mathbb{E}_t^{e,\mathrm{RN}}\left[e^{-\int_t^T r_u^{e|i} du} V_T^{e|i}\right] = V_t^{e|i} \tag{4}$$

is still valid[5].

[4]This is the measure associated with the numeraire M^c. And more generally, for each numeraire N satisfying an integrability condition, there is an associated measure $\mathbb{Q}^N$.

[5]We could extend this to non-cash collateral. If a security S_t is posted as collateral, and it is fully rehypothecable, then the party holding the collateral can repo it, earning the corresponding repo rate and the exact same reasoning applies. As suggested by the usual FX-EQ analogy, posting foreign cash as collateral is essentially equivalent to posting a domestic security with repo rate $r_t^{f|d} = r_t^{d|d} - \mu_t^{df}$.

(3) The drift of the FX spot rate under the domestic risk-neutral measure is defined as

$$\mu_t^{df} := \mathbb{E}_t^{d,\mathrm{RN}}\left[\frac{dx_t^{df}}{x_t^{df}}\right] \tag{5}$$

It satisfies the relations[6]

$$\mu_t^{df} + \mu_t^{fd} = 0 \tag{6}$$

$$\mu_t^{de} + \mu_t^{ef} + \mu_t^{fd} = 0. \tag{7}$$

(4) The change of payment currency formula for effective discount rates is:

$$r_t^{d|i} = \mu_t^{df} + r_t^{f|i} \tag{8}$$

(5) The short collateral basis is defined as

$$q_t^{i\to j} = r_t^{e|j} - r_t^{e|i} \tag{9}$$

It does not depend on the payment currency e. So, for any currency e, we have the change of collateral formula:

$$r_t^{e|j} = r_t^{e|i} + q_t^{i\to j} \tag{10}$$

Moreover, we have the cocycle relations:

$$q_t^{i\to j} + q_t^{j\to i} = 0 \tag{11}$$

$$q_t^{i\to j} + q_t^{j\to k} + q_t^{k\to i} = 0 \tag{12}$$

(6) As a result, the dynamic of the spot FX rate is

$$\frac{dx_t^{df}}{x_t^{df}} = \left(r_t^{d|d} - r_t^{f|f} - q_t^{f\to d}\right)dt + \sigma_t^{df}\cdot dW_t^{d,\mathrm{RN}} \tag{13}$$

(7) The change of risk neutral measure is the usual one[7]:

$$\mathbb{E}_t^{d,\mathrm{RN}}\left[\frac{d\mathbb{Q}^{f,\mathrm{RN}}}{d\mathbb{Q}^{d,\mathrm{RN}}}\right] = e^{-\int_0^t \mu_u^{df}\,du}\frac{x_t^{df}}{x_0^{df}} = \mathcal{E}\left(\int_0^{\cdot}\sigma_v^{df}\cdot dW_v^{d,\mathrm{RN}}\right)_t \tag{14}$$

1.2. **Standard instruments and market observables.** We now list standard instruments of interest to us as well as the corresponding market quotes and set our notations:

- Zero coupon bonds: the value at t in currency e of the bond paying one unit of currency e at T under collateral i is denoted by:

$$Z_{t,T}^{e|i} = \mathbb{E}_t^{e,\mathrm{RN}}\left[e^{-\int_t^T r_u^{e|i}\,du}\right] \tag{15}$$

The measure associated with this numéraire is denoted by $\mathbb{Q}^{e|i,T}$. The instantaneous forward rate is defined by

$$r_{t,T}^{e|i} = -\partial_T \log Z_{t,T}^{e|i} \tag{16}$$

- Forward zero coupon bonds:

$$Z_{t,S,T}^{e|i} = \frac{Z_{t,T}^{e|i}}{Z_{t,S}^{e|i}} = e^{-\int_S^T r_{t,u}^{e|i}\,du} \tag{17}$$

[6]Very wonky remark!!! Mathematicians will recognise these relations as *cocycle* conditions. Let us write K^n for the vector space of $\mathfrak{X}^n$ valued processes $c = (c_t^x)_{x\in\mathfrak{X}^n}$. Define $d : K^n \to K^{n+1}$ by $dc_t^x = \sum_i (-1)^i c_t^{x^{(i)}}$ where $x^{(i)} = (x^1, \ldots, \widehat{x^i}, \ldots, x^n)$ is obtained by dropping the i-th component. This defines a co-homological complex. Moreover this complex has trivial cohomology. In other words, the cocycle relation implies that there exists $(r_t^x)_{x\in\mathfrak{X}}$ such that $\mu^{df} = r_t^d - r_t^f$ for all d, f. However, the choice of the processes $(r_t^x)_{x\in\mathfrak{X}}$ is not unique since adding a process (a collateral basis) c_t to all rates cancels out when taking the difference. In summary, one can interpret the FX-drifts $(\mu^{df})_{df\in\mathfrak{X}^2}$ as a 2-cocycle while the collateral basis $q^{i\to j}$ correspond to a coboundary.

[7]Given a semi-martingale L_t, we write $\mathcal{E}(L)_t$ for the corresponding Doleans-Date exponential martingale: $d\mathcal{E}(L)_t = \mathcal{E}(L)_{t_}\,dL_t$. Since we are only considering continuous semi-martingales, $\mathcal{E}(L)_t = e^{L_t - L_0 - \frac{1}{2}\langle L\rangle_t}$.

- Linear discount rates: $R^{e|i}_{t,T;\theta}$ is defined by

$$1+\theta R^{e|i}_{t,T;\theta} \;=\; \frac{Z^{e|i}_{t,T}}{Z^{e|i}_{t,T+\theta}} \;=\; e^{\int_T^{T+\theta} r^{e|i}_{t,u}\,du} \tag{18}$$

- Forward Libors. The par strike of an FRA in currency e under collateral i is

$$L^{e|i}_{t,T;\theta} \;=\; \mathbb{E}^{e|i,T+\theta}_t\left[L^e_{T;\theta}\right] \tag{19}$$

where $L^e_{T;\theta}$ is the value of the Libor index with tenor θ, fixing as T and paid at $T+\theta$.

- Coupon factors. Since the Libor transition, GBP, JPY, CHF and USD swaps and cross-currency swaps pay compounded overnight (RFR) coupons (SONIA, TONA, SARON, SOFR, ESTR) instead of Libor-type coupons. A floating coupon of currency e on the accrual period $[a,b]$, $\theta=b-a$, paid at b, is $C^e_{a,b}-1$ (plus a spread θs, added simply), where the *coupon factor* $C^e_{a,b}$ is one of:

$$\begin{aligned}
C^{e,\mathrm{L}}_{a,b} &= 1+\theta L^e_{a;\theta} && \text{(Libor, fixed at } a\text{)},\\
C^{e,\delta}_{a,b} &= \prod_{k=0}^{n-1}\left(1+\delta_k R^{e|e}_{t_k,t_k;\delta_k}\right) = \prod_{k=0}^{n-1}\frac{1}{Z^{e|e}_{t_k,t_{k+1}}} && \text{(daily compounding, exact)},\\
C^{e,0}_{a,b} &= \exp\left(\int_a^b r^{e|e}_u\,du\right) && \text{(continuous compounding)}.
\end{aligned} \tag{20}$$

Here $a=t_0<t_1<\ldots<t_n=b$ is the overnight grid, with steps $\delta_k=t_{k+1}-t_k=\delta$ except the last one, which is shortened to land on b ($\delta=1/365$ by default; day counts and calendars are out of scope), and the overnight fixing at t_k is the one-step simple rate of the native curve, $1+\delta_k R^{e|e}_{t_k,t_k;\delta_k}=1/Z^{e|e}_{t_k,t_{k+1}}$. The Libor factor is the one-step case $n=1$ times the funding spread: $C^{e,\mathrm{L}}_{a,b}=\kappa^e/Z^{e|e}_{a,b}$ under the constant funding spread hypothesis (29) ($\kappa^e=1-\theta S^{e|e,\div}$). The continuous factor is the limit of the daily one as $\delta\to 0$; it is an approximation of the traded coupon, whose error is quantified in Proposition 27. Payment lags, lookbacks, lockouts and observation shifts are out of scope.

We keep the notation of forward Libors for the forward of any coupon factor: under collateral i,

$$1+\theta L^{e|i}_{t,a;\theta} \;:=\; \mathbb{E}^{e|i,b}_t\left[C^e_{a,b}\right], \tag{21}$$

which is (19) for a Libor coupon. So the formulas below for swaps and cross-currency swaps hold for both coupon types, with the coupon convexity of the coupon type (Proposition 26).

- Futures. A future contract is economically equivalent to a FRA with domestic collateral earning a collateral rate of zero:

$$\mathrm{Fut}^e_{t,T;\theta} \;=\; \mathbb{E}^{e,\mathrm{RN}}_t\left[L^e_{T;\theta}\right]. \tag{22}$$

- Funding spreads. The additive funding spread[8] in currency e with collateral i is[9]:

$$S^{e|i,-}_{t,T;\theta} \;=\; R^{e|i}_{t,T;\theta}-L^{e|i}_{t,T;\theta} \tag{23}$$

[8]The Libor index corresponds to unsecured lending which means that its value encompasses credit risk. On the other hand, the discount rates correspond to perfectly collateralised lending so the funding spread quantifies the credit risk.

[9]This funding spread is negative to be consistent with conventions used at JPMorgan.

For mathematical reasons, we will mostly deal with the multiplicative funding spread[10] instead as defined by:

$$1-\theta S^{e|i,\div}_{t,T;\theta} = \frac{1+\theta L^{e|i}_{t,T;\theta}}{1+\theta R^{e|i}_{t,T;\theta}} \tag{24}$$

- Swap rates. The forward swap rate in currency e with schedule $T_0,\dots,T_{\ell M}$ under collateral i is

$$Y^{e|i}_{t,T;M} = \frac{\sum_{j=1}^{\ell M} Z^{e|i}_{t,T_0,T_j}\theta_j L^{e|i}_{t,T_{j-1},T_j}}{\sum_{k=1}^{m}\theta_k Z^{e|i}_{t,T_0,T_k}} \tag{25}$$

- FX forwards. The par strike of an FX forward under collateral i is

$$x^{df|i}_{t,T} = \mathbb{E}^{d|i,T}_t\left[x^{df}_T\right] = x^{df}_t\frac{Z^{f|i}_{t,T}}{Z^{d|i}_{t,T}}. \tag{26}$$

- Resetting forward Libor: $L^{f/d|i}_{t,T;\theta}$ is the par strike K of a contract with collateral i exchanging
  - at T, receive $N_d = x^{df}_T N_f$ in d, pay N_f in f
  - at $T+\theta$, pay $N_d(1+\theta L^d_{T;\theta})$ in d, receive $N_f(1+\theta K)$ in f.

  So, $L^{f/d|i}_{t,T;\theta}$ is characterised by

$$1+\theta L^{f/d|i}_{t,T;\theta} = \mathbb{E}^{f|i,T+\theta}_t\left[\frac{x^{fd}_{T+\theta}}{x^{fd}_T}(1+\theta L^d_{T;\theta})\right]$$

- Currency basis spread. A resetting currency basis swap has
  - foreign leg: pays in f, Libor plus spread $L^f_k + K$ and exchange of notionals N^f_k.
  - domestic leg: pays in d, Libor L^d_k and exchange of notionals resetting at the beginning of each period to match the corresponding foreign notional $N^d_k = x^{df}_{T_{k-1}}N^f_k$.

  It follows that, under collateral i, the par-spread of each swaplet is

$$B^{f/d|i,r}_{t,T_k;\theta} = L^{f/d|i,r}_{t,T_k;\theta} - L^{f|i}_{t,T_k;\theta} \tag{27}$$

[10] This definition and notation can seem a bit strange but they are meant to be consistent with the negative funding spread convention. One could flip the signs: $S^{e|i,+}_{t,T;\theta} = -S^{e|i,-}_{t,T;\theta}$, $S^{e|i,\times}_{t,T;\theta} = -S^{e|i,\div}_{t,T;\theta}$ and all the results would remain valid. However, they would break if we flipped the ratio in this definition. This is because, while both Discount and Libor forwards rates are martingales under the payment forward measure, there do not have symmetric economic roles. Discount forwards are determined by ratios of ZCBs which are prices of tradeable assets but there is no analog for Libor forwards. While we can always define pseudo-discount factors corresponding Libor forwards, these are not observable and not constrained by AOA. Specifying their dynamic requires additional model assumptions. For example, one could assume that the ratios $Z^{e|e,\text{spread}}_{t,T} := \frac{Z^{e|e}_{t,T}}{Z^{e|e,\text{proj}}_{t,T}}$ of pseudo-discount factors or the instantaneous forward spreads $s^{e|e}_{t,T} = r^{e|e,\text{proj}}_{t,T} - r^{e|e}_{t,T}$ are constant but it is not clear that this is economically or mathematically justified or that this leads to significant mathematical simplifications.

while for the whole swap[11] it is

$$B^{f/d|i,r}_{t,T;M} = \frac{\sum_{k=1}^{\ell M} Z^{d|i}_{t,T_k}\theta B^{f/d|i,r}_{t,T_{k-1};\theta}}{\sum_{k=1}^{\ell M} Z^{f|i}_{t,T_k}\theta} \tag{28}$$

[TODO: Add options, i.e. Swaptions, Caps, FX options and currency basis swaptions].

[11] For a generic non-resetting cross-currency swap, we can introduce $x^{df}_{t_{\mathrm{ref},i}} = N^d_i/N^f_i$. The previous formula generalises to

$$B^{f/d|i,*}_{t,T;M} = \frac{\sum_{k=1}^{\ell M}\left(Z^{d|i}_{t,T_k}\mathbb{E}^{d|i,T_k}_t\left[x^{df}_{\mathrm{ref},k}(1+\theta L^{d|i}_{T_{k-1};\theta})\right] - Z^{d|i}_{t,T_{k-1}}\mathbb{E}^{d|i,T_{k-1}}_t\left[x^{df}_{\mathrm{ref},k}\right]\right) - \sum_{k=1}^{\ell M}\left(Z^{f|i}_{t,T_k}\left(1+\theta L^{f|i}_{t,T_{k-1},T_k}\right) - Z^{f|i}_{t,T_{k-1}}\right)}{\sum_{k=1}^{\ell M}\theta Z^{f|i}_{t,T_k}}$$

In the case of constant notionals, i.e. deterministic $x^{df}_{t_{\mathrm{ref},i}}$, the par-spread is:

$$B^{f/d|i,nr}_{t,T;M} = \frac{\sum_{k=1}^{\ell M}\frac{x^{df}_{\mathrm{ref},k}}{x^{df}_t}\left(Z^{d|i}_{t,T_k}(1+\theta L^{d|i}_{T_{k-1};\theta}) - Z^{d|i}_{t,T_{k-1}}\right) - \sum_{k=1}^{\ell M}\left(Z^{f|i}_{t,T_k}\left(1+\theta L^{f|i}_{t,T_{k-1},T_k}\right) - Z^{f|i}_{t,T_{k-1}}\right)}{\sum_{k=1}^{\ell M}\theta Z^{f|i}_{t,T_k}}$$

2. MARKET MODELS

2.1. **Definition.**

Definition 1. A *market model* consists of the following data:

(1) A set of assets S^i
(2) A set of parameters $\theta \in \Theta$
(3) A stochastic process $(X_t^\theta)_{t\in[0,T],\theta\in\Theta}$
(4) Reconstruction formulas for asset prices, i.e. deterministic functions s.t. $S_t^i = S^i(\theta, t, X_t^\theta)$.

The model provides a dynamic for the value of any portfolios $\Pi_t = \sum_i w_{i,t} S_t^i$.

Calibration. The process of calibrating of the model corresponds to finding $\theta^* \in \Theta$ such that the model delivered prices of benchmark instruments match the market observed ones.

2.2. **Fixed cash-flows.**

Our focus is on fully collateralised assets. The most basic ones are fixed cash-flows with cash collateral in a single currency.

A market model for collateralised fixed cash-flows consists of the following data:

(1) a set of currencies $\mathfrak{X}$
(2) the following set of stochastic processes[12]:
   - for each currency $c \in \mathfrak{X}$, a reference native collateral rate $r_t^{c|c}$.
   - for each pair of currencies (d, f), an FX spot rate x_t^{df}.
   - for each pair of currencies x, c and each maturity T, a price process $Z_{t,T}^{x|c}$ for the value in currency x of the contract paying one unit of x at T under collateral c.

We will require that the model satisfies the following properties:

(1) Absence of arbitrage
(2) Calibration, i.e. the model reproduces prices of liquid market instruments at time $t = 0$.
(3) Regularity: discount curves $T \mapsto Z_{t,T}^{x|c}$ are derivable[13] to allow for risk management through computation of sensitivities.

2.3. **Floating rates instruments.**

Liquid long dated instruments are vanilla swaps, cross-currency swaps, tenor basis swaps etc…They all involve floating cash-flows associated with IR indices. So, in order to calibrate our market model, we will have to extend its scope to include these indices. There are two options for that:

(1) Assume that funding and tenor basis spreads are constant[14].
(2) Develop a model with stochastic funding and tenor basis spreads.

The second option adds a lot of complexity. Obviously the increased number of stochastic factors is one but more fundamentally, one can prove that the drift of instantaneous forward projection rates are not uniquely determined by AOA in a model with stochastic funding basis and there is no natural choice. Models of LMM type, on the other hand, do not have this problem because the dynamic of the market observable projection rates is modelled directly. This suggests that the right models would be of LMM type for the projection rate and funding spread but with a short discount rate[15] which makes them much more complex and high-dimensional.

We will focus on the first option instead.

Hypothesis (Constant funding spreads)**.** For every currency e and tenor θ, the native multiplicative funding spread

$$1 - \theta S_{t,T;\theta}^{e|e,\div} = \frac{1 + \theta L_{t,T;\theta}^{e|e}}{1 + \theta R_{t,T;\theta}^{e|e}} \tag{29}$$

is constant.

[12] Defined on a filtered probability space $(\Omega, \mathcal{A}, \mathcal{F})$

[13] They need to be at least absolutely continuous for instantaneous forward rates to exist.

[14] Or deterministic or at least independent of all other processes.

[15] It is not surprising that discount rates and index forwards require different modelling since they are fundamentally different.

From now on, we will assume the constant funding spreads hypothesis holds.

Remark 1 (Compounded overnight coupons need no hypothesis)**.** The hypothesis is about Libor-type indices. A compounded overnight coupon (20) is a product of one-step discount rates of the native curve, so it has no funding spread: under native collateral its forward is $Z^{e|e}_{t,a}/Z^{e|e}_{t,b}$ in any arbitrage free model (Lemma 1), i.e. $\kappa^e = 1$ exactly.

3. Market practice: The Deterministic Collateral Basis Model

3.0. **Introduction.** In this section, we detail the assumptions currently used by most market participants to value linear products. We then explain how they allow them to build discount curves and projection curves to value any linear product under any single currency cash collateral.

3.1. FX forward invariance.

Hypothesis (Constant collateral basis)**.** All collateral basis spreads $q_{t,T}^{e|i\to j}$ are constant:

$$\frac{Z_{t,T}^{e|i}}{Z_{t,T}^{e|j}} = \frac{Z_{0,T}^{e|i}}{Z_{0,T}^{e|j}} \tag{30}$$

This is equivalent to the following.

Hypothesis (FX forward invariance)**.** FX forward rates do not depend on the collateral:

$$x_{t,T}^{df|i} = x_{t,T}^{df|j} \tag{31}$$

Remark 2. In conjunction with the constant funding spread assumption, this implies that all spreads constant. Note however that these assumption only involve forward rates. We do not assume anything about the initial discount curves $Z_{0,T}^{f|d}$. In fact, we assume that the impact of the change of collateral on FX forwards an FRAs is entirely accounted for by the change of discount factor.

One could weakened the assumption a bit: our results would still hold if we assumed that all the basis spreads are independent of all the Libor and FX rates instead.

3.2. Valuation of resetting cross-currency swaps.

3.2.1. *Discount curves under reference collateral.* The first problem to define a market model is that for each new cross-currency swap expiry, we have:

(1) One new constraint: the master cross-currency equation (28).
(2) Three new unknowns[16]: $Z_{t,T_k}^{f|d}$ and $L_{t,T_k;\theta}^{f|d}$ and $L_{t,T_k;\theta}^{f/d|d,r}$.

This system of equations is clearly under-determined so we need to impose some constraints and/or include additional parameters.

3.2.2. *Deterministic collateral basis model.* The standard market practice is to assume that all collateral basis spreads are constant. This ensures that all collateral convexity adjustments in the valuation of cross-currency swaps can be neglected:

(1) The convexity adjustment due to non-standard collateral for foreign Libors: $L_{t,T_{i-1},T_i}^{f|d,\mathrm{DCB}} = L_{t,T_{i-1},T_i}^{f|f}$.
(2) The convexity adjustment due to the resetting notional $N_k^d = x_{t,T_{k-1}}^{df|d} N_k^f$ captured by $L_{t,T_{k-1};\theta}^{f/d|d}$.

3.2.3. *Main theorem.*

Theorem 1 (DCB)**.** Under the above assumptions,

$$R_{t,T;\theta}^{f|d} = L_{t,T;\theta}^{f|f} + B_{t,T;\theta}^{f/d|d,r} + S_{t,T;\theta}^{d|d,-}. \tag{32}$$

Remark 3. This formula is quite pleasing as it has an intuitive hedging interpretation. To hedge fixed cash-flows in foreign currency with domestic collateral, one should enter a foreign vanilla swap to swap the fixed foreign cash-flows into floating ones, a currency basis swap and a domestic funding basis swap[17]

[16] This is assuming that we can observe the currency basis spread $B_{t,T_k;\theta}^{f/d|d,r}$ for each swaplet which is not case. In practice, we only observe spot starting currency basis spreads for a given set of expiries. So some interpolation procedure is always required at the time of the bootstrapping.

[17] Can we make the replication argument more precise? If I enter,

- a foreign vanilla swap to swap the fixed foreign cash-flows into floating ones
- a currency basis swap to swap the foreign Ibor cash-flows into domestic ones
- a domestic funding basis swap to swap the domestic Ibor cash-flows into domestic O/N ones.

How should I set the notionals? What is the residual risk? Am I really left with a foreign fixed cash flow vs domestic OIS? Is this true at first order?

3.2.4. *Cross-currency curve stripping.* Under these simplifying assumptions, the master equation for the cross-currency discount curve is

$$\begin{aligned}
&\sum_{k=1}^{\ell M} \left(Z_{t,T_k}^{f|d} \left(1 + \theta (L_{t,T_{k-1},T_k}^{f|d} + B_{t,T;M}^{f/d|d,r}) \right) - Z_{t,T_{k-1}}^{f|d} \right) \\
&= \sum_{k=1}^{\ell M} x_{t,T_{k-1}}^{df|d} \left(Z_{t,T_k}^{d|d} (1 + \theta L_{T_{k-1};\theta}^{d|d}) - Z_{t,T_{k-1}}^{d|d} \right) \qquad (33)
\end{aligned}$$

In order to bootstrap the cross-currency discount curve, one writes this equation for two consecutive tenors M and N, and take the difference.

The problem is still under-determined since we only have one equation but each of the $Z_{t,T_k}^{f|d}$ for $T_k \in [T_{\ell M}, T_{\ell N}[$ is unknown.

A solution is to specify an interpolation scheme for the discount curve on the relevant segment. Typical choices are:

- Flat instantaneous forward rates: $r_{0,T}^{f|d}$ is constant for $T \in [T_{\ell M}, T_{\ell N}[$.
- Parabolic instantaneous forward rates: $r_{0,T}^{f|d}$ is a second order polynomial on $T \in [T_{\ell M}, T_{\ell N}]$ and it is derivable at $T = \ell M$.

The main point is that such a parametrisation reduces the knowledge of the discount factors in the relevant period to a single unknown:

$$\overline{r}_{0,[T_{\ell M},T_{\ell N}]}^{f|d} = \frac{1}{T_{\ell M} - T_{\ell N}} \int_{T_{\ell M}}^{T_{\ell N}} r_{0,u}^{f|d} \, du \qquad (34)$$

which is the average level of the instantaneous forward rate on this period.

Remark 4. The curve bootstrapping has some limitations. For example, it assumes that the schedules of the instruments are aligned. An alternative solution is to consider a global calibration. This allows the user to include more diversity in the calibration targets, and more granularity in the pillars. It is however more complex and it does not guarantee that risk is local: an instrument with maturity T can have exposure to benchmark instruments with arbitrary maturity.

3.3. **Off-market discount curves.**

3.3.1. *Discount curves under general collateral.* The second problem is that cross-currency swaps are only liquid against USD. So, there are no quotes to build a discount curve with non-native non-USD collateral such as $Z^{\mathrm{USD}|\mathrm{EUR}}$ or $Z^{\mathrm{EUR}|\mathrm{JPY}}$.

3.3.2. *Off-market USD discount curve.*

Proposition 1 (Off-market discount curve)**.** Under the FX-forward invariance assumption, the off-market discount curves $Z_{t,T}^{u|i}$ is characterised by:

$$Z_{t,T}^{u|i} = Z_{t,T}^{u|u} \frac{Z_{t,T}^{i|i}}{Z_{t,T}^{i|u}} \qquad (35)$$

This provides a way to build the curve.

3.3.3. *Off-market non-USD discount curves.*

Proposition 2 (Off-market discount curve)**.** Under the FX-forward invariance assumption, the off-market discount curves $Z_{t,T}^{e|i}$ is characterised by:

$$Z_{t,T}^{e|i} = Z_{t,T}^{e|u} \frac{Z_{t,T}^{i|i}}{Z_{t,T}^{i|u}} \qquad (36)$$

4. ARBITRAGE FREE MODELS WITH STOCHASTIC COLLATERAL

4.0. **Introduction.** The current market practice described in the previous section shows how to build a market model with *deterministic basis*. But the construction relies on assumptions that are inconsistent with any stochastic spreads between discount rates. As a result, the corresponding curves cannot be used as inputs to models with stochastic spreads without introducing arbitrage.

Our first objective in this section is to identify the arbitrage constraints on a stochastic multi-currency market model for collateralized fixed cashflows.

We assume that tenor basis (aka funding) spreads are deterministic (cf. Equation 30) so the stochasticity of collateral is essentially equivalent to the stochasticity of the currency basis.

4.1. Model specification.

Theorem 2. Let $\{Z^{x|c}_{t,T}\}_{t\leq T,x,c\in\mathfrak{X}}$ be an arbitrage-free market model for fixed collateralized cash-flows such that $T\mapsto Z^{x|c}_{t,T}$ are absolutely continuous. Then, given a reference currency $d\in\mathfrak{X}$, the model $\{Z^{x|c}_{t,T}\}_{t\leq T,x,c\in\mathfrak{X}}$ is uniquely specified by the following data: for each currency $f\in\mathfrak{X}$,

(1) Initial data for
- native discount curves $\{Z^{f|f}_{t_0,T}\}_{f\in\mathfrak{X}}$.
- collateral basis curves $\{\Xi Z^{f|f\to d}_{t_0,T}\}_{f\in\mathfrak{X}}$ or equivalently cross-currency discount curves $\{Z^{f|d}_{t_0,T}\}_{f\in\mathfrak{X}}$.
- FX spot rates $x^{df}_{t_0}$.

(2) Spot volatility processes for
- instantaneous forward discount rates denoted by $\{\sigma^{f|f}_{\cdot,T}\}$
- instantaneous forward collateral basis denoted by$\{\sigma^{f\to d}_{\cdot,T}\}$ (and equal to 0 if $f=d$)
- FX spot rates denoted by $\sigma^{df}_{\cdot}$ (and equal to 0 if $f=d$)

Remark 5. The theorem essentially says that arbitrage conditions imply two things:

(1) (Triangle constraints) the dynamic can be decomposed[18] with respect to a reference currency d.
(2) (Drift constraints) the risk-neutral drifts of the assets are entirely specified by their spot volatilities[19].

More explicitly, let us write,

$$\Sigma^{\alpha}_{t,T} = \int_t^T \sigma^{\alpha}_{t,u}\,du \tag{37}$$

$$\Sigma^{\alpha}_{t,T;\theta} = \int_T^{T+\theta} \sigma^{\alpha}_{t,u}\,du \tag{38}$$

Then, we have

$$dr^{e|e}_{t,T} = \left(\sigma^{e|e}_{t,T}\cdot\Sigma^{e|e}_{t,T}\right)dt+\sigma^{e|e}_{t,T}\cdot\left(dW^{d,\mathrm{RN}}_t-\sigma^{de}_t\,dt\right) \tag{39}$$

$$dq^{f|f\to d}_{t,T} = \left(\sigma^{f\to d}_{t,T}\cdot\Sigma^{f\to d}_{t,T}+\sigma^{f\to d}_{t,T}\cdot\Sigma^{f|f}_{t,T}+\sigma^{f|f}_{t,T}\cdot\Sigma^{f\to d}_{t,T}\right)dt+\sigma^{f\to d}_{t,T}\cdot\left(dW^{d,\mathrm{RN}}_t-\sigma^{df}_t\,dt\right) \tag{40}$$

We define the short rate and short collateral basis in the obvious way[20]:

$$r^{e|e}_t = r^{e|e}_{t,t} \tag{41}$$

$$q^{e\to d}_t = q^{e|e\to d}_{t,t} \tag{42}$$

[18] In mathematical terms, the market is a graph whose nodes correspond to currencies. Moreover this graph is complete. Once we choose a reference currency, there is an obvious spanning tree whose root is that reference currency. This is a very pedantic way to state things but it is interesting to remember when we consider the implementation.

[19] This is a generalisation of the standard HJM result.

[20] Note that instantaneous forward collateral basis does depend on the payment currency even if the short collateral basis does not! This is because the forward accounts for covariance between payment currency and collateral basis across time. This covariance tends to zero as the time interval goes to zero.

Then, the full dynamic is specified by

$$\frac{dZ_{t,T}^{d|i}}{Z_{t,T}^{d|i}} = \left(r_t^{d|d} + q_t^{d\to i}\right) dt - \left(\Sigma_{t,T}^{d|d} + \Sigma_{t,T}^{d\to i}\right) dW_t^{d,\mathrm{RN}} \tag{43}$$

$$\frac{dx_t^{df}}{x_t^{df}} = \left(r_t^{d|d} - r_t^{f|f} + q_t^{d\to f}\right) dt + \sigma_t^{df} \cdot dW_t^{d,\mathrm{RN}} \tag{44}$$

for domestic observables and

$$\frac{dZ_{t,T}^{e|i}}{Z_{t,T}^{e|i}} = \left(r_t^{e|e} + q_t^{d\to i} - q_t^{d\to e}\right) dt - \left(\Sigma_{t,T}^{e|e} + \Sigma_{t,T}^{d\to i} - \Sigma_{t,T}^{d\to e}\right) \cdot \left(dW_t^{d,\mathrm{RN}} - \sigma_t^{de} dt\right) \tag{45}$$

$$\frac{dx_t^{ef}}{x_t^{ef}} = \left(r_t^{e|e} - r_t^{f|f} + q_t^{d\to f} - q_t^{d\to e}\right) dt + \left(\sigma_t^{df} - \sigma_t^{de}\right) \cdot \left(dW_t^{d,\mathrm{RN}} - \sigma_t^{de}\, dt\right) \tag{46}$$

for foreign observables.

Remark 6. For short, we will write

$$\sigma_{t,T}^{d|i} = \sigma_{t,T}^{d|d} + \sigma_{t,T}^{d\to i} \tag{47}$$

$$\Sigma_{t,T}^{d|i} = \Sigma_{t,T}^{d|d} + \Sigma_{t,T}^{d\to i} \tag{48}$$

and use the same convention for the integrated vols. We will also write

$$\Sigma_{t,T}^{j\to i} = -\Sigma_{t,T}^{i\to j} \tag{49}$$

$$\Sigma_{t,T;\theta}^{j\to i} = -\Sigma_{t,T;\theta}^{i\to j} \tag{50}$$

so that

$$\Sigma_{t,T;\theta}^{d|j} = \Sigma_{t,T;\theta}^{d|i} + \Sigma_{t,T;\theta}^{i\to j} \tag{51}$$

4.2. **Deterministic collateral basis model.** We want to show that if the collateral basis volatilities $\sigma_{t,T}^{d\to i}$ are zero, we recover the DCB model.

This is straightforward since discount curve become deterministic.

4.3. **Quanto adjustments.** Quanto adjustments are convexity adjustments associated with the payment of an index in a non-native currency.

4.3.1. *Change of currency formula.*

Proposition 3. The Radon-Nikodym derivative of the change of T-forward measure is characterised by the exponential martingale

$$\Xi_{t,T}^{df|i} := \mathbb{E}_t^{d|i,T}\left[\frac{d\mathbb{Q}^{f|i,T}}{d\mathbb{Q}^{d|i,T}}\right] = \mathcal{E}\left(-\int_0^{\cdot} \Sigma_{u,T}^{df} \cdot dW_u^{d|i,T}\right)_t \tag{52}$$

where

$$\Sigma_{u,T}^{df} = \Sigma_{u,T}^{d|d} - \Sigma_{u,T}^{f|f} - \Sigma_{u,T}^{f\to d} + \sigma_u^{df} \tag{53}$$

is the volatility of the FX-forward.

4.3.2. *Quanto adjustments.*

Proposition 4.

4.4. **Dynamic of market observables.** We now derive the dynamic of market observables in order to calibrate the model.

4.4.1. *Discount factors.*

Proposition 5. The ZCBs have dynamic:

$$\frac{dZ^{e|i}_{t,T}}{Z^{e|i}_{t,T}} = \left(r^{e|e}_t + q^{e\to i}_t\right) dt - \Sigma^{e|i}_{t,T} \cdot \left(dW^{d,\mathrm{RN}}_t - \sigma^{de}_t\, dt\right) \tag{54}$$

4.4.2. *Forward discount rates.*

Proposition 6. The forward discount rate $R^{e|i}_{t,T;\theta}$ is a shifted exponential martingale under is a $\mathbb{Q}^{e|i,T+\theta}$:

$$\frac{d(1+\theta R^{e|i}_{t,T;\theta})}{1+\theta R^{e|i}_{t,T;\theta}} = \Sigma^{e|i}_{t,T;\theta} \cdot dW^{e|i,T+\theta}_t. \tag{55}$$

4.4.3. *FX forwards.*

Proposition 7. The FX forward $x^{df|i}_{t,T}$ is an exponential martingale under $\mathbb{Q}^{d|i,T}$:

$$\frac{dx^{df|i}_{t,T}}{x^{df|i}_{t,T}} = \Sigma^{df}_{t,T} \cdot dW^{d|i,T}_t \tag{56}$$

where

$$\Sigma^{df}_{t,T} = \Sigma^{d|d}_{t,T} - \Sigma^{f|f}_{t,T} + \Sigma^{d\to f}_{t,T} + \sigma^{df}_t. \tag{57}$$

Remark 7. Note that, a priori, the collateral does not explicitly appear in the volatility process of the FX-forward. The collateral only appears in the drift or equivalently in the initial value of the FX-forward when the model is calibrated to market quotes.

This is a pleasant surprise. Indeed, a priori, we should have expected one implied volatility surface for each FX pair and each collateral. And it is indeed the case if we consider the volatility smile as a function of absolute strikes. However, if we consider our volatility surface as a function $(\tau, y) \mapsto \sigma^{df}_{\mathrm{BS},t}(\tau, y)$ of time to expiry $\tau = T - t$ and log-moneyness forward $y = \ln(x^{df}_t / x^{df|i}_{t,t+\tau})$, then the model suggests that $\sigma^{df}_{\mathrm{BS},t}$ should be independent of the collateral i unless the volatility of the FX-forward is stochastic and correlated with the collateral.

One can reformulate this as saying that collateral convexity impacts the intrinsic value of an FX option but not its time value, at zeroth order in vol of vol[21].

4.4.4. *Forward Libors.*

Proposition 8. The forward Libor $L^{e|i}_{t,T;\theta}$ is a shifted exponential martingale under $\mathbb{Q}^{f|i,T+\theta}$:

$$\frac{d(1+\theta L^{e|i}_{t,T;\theta})}{1+\theta L^{e|i}_{t,T;\theta}} = \Sigma^{e|e}_{t,T;\theta} \cdot dW^{e|i,T+\theta}_t \tag{58}$$

Remark 8. Similarly to FX-forwards, this suggests that the Libor volatility is independent of the collateral, we can ignore collateral convexity on implied volatilities.

[21] Q: Can we get some estimate in the LV model where instantaneous correlation of the FX vol with the collateral basis is equal to that of the FX spot?

4.4.5. *Compounded overnight coupons.*

Lemma 1 (Native forward of a coupon factor)**.** Let $C^e_{a,b}$ be a daily compounded coupon factor (20) on the grid $a = t_0 < \ldots < t_n = b$ and $t \le b$. Write $t_* = \min\{t_k : t_k > t\}$ (with $t_* = b$ if there is none) and $A_t = \prod_{k:\, t_k \le t,\, k<n} 1/Z^{e|e}_{t_k,t_{k+1}}$ for the part of the coupon fixed at t (so $A_t = 1$ and $t_* = a$ if $t < a$). Then, in any arbitrage free model,

$$\mathbb{E}^{e|e,b}_t\left[C^{e,\delta}_{a,b}\right] \;=\; A_t\,\frac{Z^{e|e}_{t,t_*}}{Z^{e|e}_{t,b}}, \qquad \text{in particular} \qquad \mathbb{E}^{e|e,b}_t\left[C^{e,\delta}_{a,b}\right] \;=\; \frac{Z^{e|e}_{t,a}}{Z^{e|e}_{t,b}} \quad (t \le a). \tag{59}$$

The same holds for the continuous factor, $\mathbb{E}^{e|e,b}_t[C^{e,0}_{a,b}] = e^{\int_a^t r^{e|e}_u du}/Z^{e|e}_{t,b}$ for $a \le t \le b$ and $Z^{e|e}_{t,a}/Z^{e|e}_{t,b}$ for $t \le a$, and, under the constant funding spread hypothesis, for the Libor factor with the extra factor κ^e. So the native value of a coupon is $Z^{e|e}_{t,a}\kappa^e - Z^{e|e}_{t,b}$ ($\kappa^e = 1$ for compounded coupons) for every coupon type, and native swaps are model-free whatever the coupon type.

Proof. The value at s of a payoff X paid at b under native collateral is $V_s = \mathbb{E}^{e,\mathrm{RN}}_s[e^{-\int_s^b r^{e|e}_u du} X] = Z^{e|e}_{s,b}\mathbb{E}^{e|e,b}_s[X]$. Write $C = C_{<k}\,C_{\ge k}$ with $C_{<k} = \prod_{l<k} 1/Z^{e|e}_{t_l,t_{l+1}}$ (known at t_{k-1}) and $C_{\ge k} = \prod_{l\ge k} 1/Z^{e|e}_{t_l,t_{l+1}}$. We show by backward induction on k that the value at t_k of $C_{\ge k}$ paid at b is 1. For $k = n-1$ it is $Z^{e|e}_{t_{n-1},b}/Z^{e|e}_{t_{n-1},b} = 1$. If it holds for $k+1$, the value at t_k of $C_{\ge k}$ paid at b is the value at t_k of $1/Z^{e|e}_{t_k,t_{k+1}}$ units received at t_{k+1} (tower property at t_{k+1}, the factor being known at t_k), i.e. 1. Hence the value at t of C paid at b is A_t times the value at t of one unit at t_*, i.e. $A_t Z^{e|e}_{t,t_*}$, and we divide by $Z^{e|e}_{t,b}$. For the continuous factor and $t \le a$, $e^{-\int_t^b r^{e|e}_u du} C^{e,0}_{a,b} = e^{-\int_t^a r^{e|e}_u du}$, whose $\mathbb{Q}^{e,\mathrm{RN}}$-expectation is $Z^{e|e}_{t,a}$; for $a < t \le b$ the factor $e^{\int_a^t r^{e|e}_u du}$ is known at t and the rest is $e^{-\int_t^b r^{e|e}_u du} e^{\int_t^b r^{e|e}_u du} = 1$. For the Libor factor, this is (29) at $t = T = a$. □

Remark 9 (Native collateral: every coupon is model-free)**.** Under native collateral the coupons telescope: the value of a floating leg is $\sum_j (\kappa^e Z^{e|e}_{t,T_{j-1}} - Z^{e|e}_{t,T_j})$ and does not depend on the volatilities, the grid or the compounding (daily or continuous). This is why native OIS swaps and swaptions are unchanged when Libor coupons are replaced by compounded ones: the floating leg of an OIS swap is $Z^{e|e}_{t,T_0} - Z^{e|e}_{t,T_n}$ whatever the payment frequency. Collateral convexity only appears when the coupon is paid under a non-native collateral (Proposition 26). The lemma was checked numerically against the exact Gaussian benchmark of Remark 57: under native collateral it returns $Z^{f|f}_{t,a}/Z^{f|f}_{t,b}$ to $2.2\cdot 10^{-16}$ (relative), also inside a period and for a non-zero state z_t.

4.4.6. *Funding basis spreads.*

Proposition 9. The multiplicative funding spread $S^{e|i,\div}_{t,T;\theta}$ is a shifted exponential martingale[22] under $\mathbb{Q}^{e|i,T}$:

$$\frac{d(1-\theta S^{e|i,\div}_{t,T;\theta})}{1-\theta S^{e|i,\div}_{t,T;\theta}} \;=\; -\Sigma^{e\to i}_{t,T;\theta}\cdot dW^{e|i,T}_t. \tag{60}$$

Proof. By definition,

$$1-\theta S^{e|i,\div}_{t,T;\theta} \;=\; \frac{1+\theta L^{e|i,\div}_{t,T;\theta}}{1+\theta R^{e|i,\div}_{t,T;\theta}} \;=\; \frac{Z^{e|i}_{t,T+\theta}(1+\theta L^{e|i,\div}_{t,T;\theta})}{Z^{e|i}_{t,T}} \;=\; \frac{\mathrm{FRA}^{e|i}_{t,T;\theta}}{Z^{e|i}_{t,T}}. \tag{61}$$

[22]Note that this is the T-forward measure and not the he $(T+\theta)$-forward measure like $1+\theta L^{e|i,\div}_{t,T;\theta}$ are $1+\theta R^{e|i,\div}_{t,T;\theta}$. Also note that no such result holds for the additive funding spread (although they should be very close in practice).

This is the ratio of the PV of a tradeable asset by the numéraire $Z^{e|i}_{t,T}$ so it is a martingale under the corresponding measure $\mathbb{Q}^{e|i,T}$. It remains to compute its volatility. Ignoring finite variation terms, we have

$$\begin{aligned}\frac{d(1-\theta S^{e|i,\div}_{t,T;\theta})}{1-\theta S^{e|i,\div}_{t,T;\theta}} &= \frac{d(1+\theta L^{e|i}_{t,T;\theta})}{1+\theta L^{e|i}_{t,T;\theta}} - \frac{d(1+\theta R^{e|i}_{t,T;\theta})}{1+\theta R^{e|i}_{t,T;\theta}} + O(dt)\\ &= \Sigma^{e|e}_{t;T;\theta}\cdot dW^{e|i,T+\theta}_t - \Sigma^{e|i}_{t;T;\theta}\cdot dW^{e|i,T+\theta}_t + O(dt)\\ &= -\Sigma^{e\to i}_{t;T;\theta}\cdot dW^{e|i,T+\theta}_t + O(dt)\\ &= -\Sigma^{e\to i}_{t;T;\theta}\cdot dW^{e|i,T}_t\end{aligned}$$

□

Remark 10. Note that even if $1-\theta S^{d|d,\div}_{t,T;\theta}$ and $1-\theta S^{f|f,\div}_{t,T;\theta}$ are constant, $1-\theta S^{f|d,\div}_{t,T;\theta}$ is not, unless the collateral basis vol is zero.

4.4.7. *Resetting forward Libors.*

Proposition 10. The resetting forward Libor $L^{f/d|i}_{t,T;\theta}$ is a $\mathbb{Q}^{f|i,T+\theta}$-martingale:

$$1+\theta L^{f/d|i}_{t,T;\theta} = \mathbb{E}^{f|i,T+\theta}_t\left[\frac{x^{fd}_{T+\theta}}{x^{fd}_T}(1+\theta L^d_{T;\theta})\right] = (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T;\theta}\right] \tag{62}$$

In particular, if native funding spreads are constant,

$$1+\theta L^{f/d|d}_{t,T;\theta} = (1+\theta R^{f|d}_{t,T;\theta})(1-\theta S^{d|d,\div}_{t,T;\theta}) \tag{63}$$

and

$$\frac{d(1+\theta L^{f/d|d}_{t,T;\theta})}{1+\theta L^{f/d|d}_{t,T;\theta}} = \Sigma^{f|d}_{t,T;\theta}\cdot dW^{f|d,T+\theta} \tag{64}$$

Proof. The martingale property is clear from the definition. The more interesting part is the relation to the discount rate. We can rewrite the definition in terms in terms of domestic T-martingales:

$$\begin{aligned}1+\theta L^{f/d|i}_{t,T;\theta} &= \mathbb{E}^{f|i,T+\theta}_t\left[\frac{x^{fd}_{T+\theta}}{x^{fd}_T}(1+\theta L^d_{T;\theta})\right]\\ &= x^{fd|i}_{t,T+\theta}\mathbb{E}^{d|i,T+\theta}_t\left[x^{df}_T(1+\theta L^d_{T;\theta})\right]\\ &= x^{fd|i}_{t,T+\theta}\mathbb{E}^{d|i,T}_t\left[\frac{Z^{d|i}_{T,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}x^{df}_T(1+\theta L^d_{T;\theta})\right]\\ &= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]\\ &= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]\\ &= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}(1-\theta S^{d|i,\div}_{t,T;\theta})\frac{\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]}{x^{df|i}_{t,T}(1-\theta S^{d|i,\div}_{t,T;\theta})}\\ &= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T;\theta}\right]\end{aligned}$$

If $i = d$, $\mathbb{E}_t^{f|d,T}\left[1 - \theta S_{T;\theta}^{d|d,\div}\right] = 1 - \theta S_{0,T;\theta}^{d|d,\div}$ since funding spreads are constant and it follows that

$$\frac{d(1+\theta L_{t,T;\theta}^{f/d|d})}{1+\theta L_{t,T;\theta}^{f/d|d}} = \frac{d(1+\theta R_{t,T;\theta}^{f|d})}{1+\theta R_{t,T;\theta}^{f|d}} = \Sigma_{t,T;\theta}^{f|d} \cdot dW^{f|d,T+\theta}.$$

□

Remark 11. Under domestic collateral, the result can be easily extended to stochastic funding spreads by introducing the convexity adjustment due to the covariance of the funding basis and FX forward:

$$1+\theta L_{t,T;\theta}^{f/d|d} = (1+\theta R_{t,T;\theta}^{f|d})(1-\theta S_{t,T;\theta}^{d|d,\div}) e^{\int_t^T \Sigma_{u,T;\theta}^{d,\text{spread}} \cdot \Sigma_{u,T}^{df} du} \tag{65}$$

4.4.8. *Currency basis spread.*

Proposition 11. The currency basis spread $B_{t,T;M}^{f/d|i,*}$ (resetting or non-resetting) is a martingale under the measure $\mathbb{Q}^{f|i,T;M}$ associated with the annuity numéraire $A_{t,T;M}^{f|i,*}$.

Proof. Let us denote by $\text{CCS}_{t,T;M}^{f/d|i,*}$ the MTM at t in currency f of a cross-currency swap with zero spread. By definition, $B_{t,T;M}^{f/d|i,*} = -\frac{\text{CCS}_{t,T;M}^{f/d|i,*}}{A_{t,T;M}^{f|i,*}}$ so it is a martingale under the measure $\mathbb{Q}^{f|i,T;M}$ associated with the annuity numéraire $A_{t,T;M}^{f|i}$. □

In order to relate the dynamic of resetting currency basis spreads to that of discount factors and market observables (FX and Libor rates), we first consider the case of a swaplet. We restrict ourselves to the case of domestic collateral for now.

Theorem 3. The currency basis spread of a resetting cross-currency swaplet is:

$$B_{t,T;\theta}^{f/d|i,r} = \frac{1}{\theta}\left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(\mathbb{E}_t^{f|i,T}\left[\frac{1+\theta L_{T;\theta}^{d|i}}{1+\theta R_{T;\theta}^{d|i}}\right] - \frac{1+\theta L_{t,T;\theta}^{f|i}}{1+\theta R_{t;T;\theta}^{f|i}}\right) \tag{66}$$

In particular,

$$B_{t,T;\theta}^{f/d|d,r} = \left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(S_{t,T;\theta}^{f|d,\div} - S_{t_0,T;\theta}^{d|d,\div}\right) \tag{67}$$

Equivalently, with $Z_{t,T,T+\theta}^{f|d} = 1/(1+\theta R_{t,T;\theta}^{f|d})$,

$$Z_{t,T,T+\theta}^{f|d} B_{t,T;\theta}^{f/d|d,r} = S_{t,T;\theta}^{f|d,\div} - S_{t_0,T;\theta}^{d|d,\div} \tag{68}$$

Corrolary 1. The currency basis spread of a resetting cross-currency swap is:

$$\begin{aligned} B_{t,T;M}^{f/d|d,r} &= \frac{\sum_{j=1}^M Z_{t,T,T_j}^{f|d} B_{t,T_{j-1},\theta}^{f/d|d,r}}{\sum_{j=1}^M Z_{t,T_0,T_j}^{f|d}} \\ &= \frac{\sum_{j=1}^M Z_{t,T_0,T_k}^{f|d} (L_{t,T_{j-1},\theta}^{f/d|d,r} - L_{t,T_{j-1},\theta}^{f|d,r})}{\sum_{j=1}^M Z_{t,T_0,T_j}^{f|d}} \\ &= \frac{\sum_{j=1}^M Z_{t,T_0,T_{j-1}}^{f|d} (S_{t,T_{j-1},\theta}^{f|d,\div} - S_{t,T_{j-1},\theta}^{f/d|d,\div})}{\sum_{j=1}^M Z_{t,T_0,T_j}^{f|d}} \end{aligned} \tag{69}$$

Remark 12. The above expressions correspond to different representations $B_{t,T;M}^{f/d|d,r} = f(t, X_t)$ of the currency basis spread as function of a stochastic process. We use those to get approximations for the dynamic of $B_{t,T;M}^{f/d|d,r}$ by using Ito-Taylor expansions. The quality of the approximation is controlled by how close the function f is to a polynomial of low degree. For degree ≤ 1, the dynamic ends up being Gaussian if X is. This technique is

often known as weight freezing because the approximation amounts to replacing the stochastic weights in the exact expansion by their initial value.

The two representations as weighted sums above provide two opportunities to apply this technique to obtain Gaussian approximations. It would be interesting to compare those.

Remark 13 (Triangle relation for currency basis spreads)**.** Let d, e, f be three currencies. Then, non-resetting currency basis spreads satisfy:

$$B^{d/f|i,nr}_{t,T;M} \;=\; B^{d/e|i,nr}_{t,T;M} - \frac{A^{f|i}_{t,T;M}}{A^{d|i}_{t,T;M}} B^{f/e|i,nr}_{t,T;M}. \tag{70}$$

This follows from the fact that an f/e non-resetting cross-currency swap can be synthesised from d/f and d/e swaps. There is no analog for the resetting currency basis spreads.

Proposition 12 (Resetting swaplets with Libor or compounded coupons)**.** Consider a resetting cross-currency swaplet on $[a, b]$, $\theta = b - a$, under domestic collateral d, whose foreign leg pays the coupon factor $C^f_{a,b}$ plus the spread and whose domestic leg pays $C^d_{a,b}$ on the notional $N^d = x^{df}_a N^f$ reset at a, each coupon factor being Libor or compounded overnight (20). Then, for $t \le a$, in any arbitrage free model with constant funding spreads,

$$1 + \theta L^{f/d|d}_{t,a;\theta} \;:=\; \mathbb{E}^{f|d,b}_t\left[\frac{x^{fd}_b}{x^{fd}_a} C^d_{a,b}\right] \;=\; \kappa^d \frac{Z^{f|d}_{t,a}}{Z^{f|d}_{t,b}}, \tag{71}$$

$$\theta B^{f/d|d,r}_{t,a;\theta} \;=\; \left(1 + \theta L^{f/d|d}_{t,a;\theta}\right) - \mathbb{E}^{f|d,b}_t\left[C^f_{a,b}\right] \;=\; \kappa^d \frac{Z^{f|d}_{t,a}}{Z^{f|d}_{t,b}} - \kappa^f \frac{Z^{f|f}_{t,a}}{Z^{f|f}_{t,b}} K^C_t(a,b), \tag{72}$$

where $\kappa^e = 1$ for a compounded coupon and

$$K^C_t(a,b) \;:=\; \frac{\mathbb{E}^{f|d,b}_t\left[C^f_{a,b}\right]}{\mathbb{E}^{f|f,b}_t\left[C^f_{a,b}\right]} \;=\; \Xi^{f|f\to d}_{t,b}\left[C^f_{a,b}\right] \tag{73}$$

is the collateral convexity of the foreign coupon factor, computed in Proposition 26. In particular, the domestic resetting leg is model-free for both coupon types, and all the model dependence of the swaplet spread is in $K^C_t(a,b)$. The swap spread is (28) with these swaplets.

Proof. By the change of currency formula (subsection 1.2 and the proposition on FX forwards in the section on collateral convexity), $\mathbb{E}^{f|d,b}_t[\frac{x^{fd}_b}{x^{fd}_a} C^d_{a,b}] = x^{fd|d}_{t,b}\, \mathbb{E}^{d|d,b}_t[x^{df}_a C^d_{a,b}]$. By the tower property at a and Lemma 1 at $t = a$ (domestic coupon under native collateral), $\mathbb{E}^{d|d,b}_a[C^d_{a,b}] = \kappa^d / Z^{d|d}_{a,b}$, so

$$\mathbb{E}^{d|d,b}_t\left[x^{df}_a C^d_{a,b}\right] \;=\; \kappa^d\, \mathbb{E}^{d|d,b}_t\left[\frac{x^{df}_a}{Z^{d|d}_{a,b}}\right] \;=\; \kappa^d \frac{Z^{d|d}_{t,a}}{Z^{d|d}_{t,b}}\, \mathbb{E}^{d|d,a}_t\left[x^{df}_a\right] \;=\; \kappa^d \frac{Z^{d|d}_{t,a}}{Z^{d|d}_{t,b}}\, x^{df|d}_{t,a},$$

where the second equality is the change of numéraire from $Z^{d|d}_{\cdot,b}$ to $Z^{d|d}_{\cdot,a}$ for an $\mathcal{F}_a$-measurable payoff. With $x^{df|d}_{t,T} = x^{df}_t\, Z^{f|d}_{t,T}/Z^{d|d}_{t,T}$ and $x^{fd|d}_{t,b} = 1/x^{df|d}_{t,b}$ we get (71). The foreign leg is $\mathbb{E}^{f|d,b}_t[C^f_{a,b}] = K^C_t(a,b)\, \mathbb{E}^{f|f,b}_t[C^f_{a,b}]$, and Lemma 1 gives $\mathbb{E}^{f|f,b}_t[C^f_{a,b}] = \kappa^f\, Z^{f|f}_{t,a}/Z^{f|f}_{t,b}$. □

Remark 14. For Libor coupons (72) is (67) rewritten with $K^C = K^{\mathrm{L}}$, the Libor collateral convexity $(1 + \theta L^{f|d}_{t,a;\theta})/(1 + \theta L^{f|f}_{t,a;\theta})$. Switching the foreign coupon from Libor to compounded overnight changes only κ^f (to 1) and K^C; switching the domestic coupon changes only κ^d. For t inside the accrual period the foreign leg is Lemma 1 with the fixed part A_t, and the convexity only involves the fixings after t (Proposition 26).

5. Collateral convexity adjustments and P&L

In this section, we describe a basic framework to decompose the convexity adjustments associated with a change of collateral.

While our presentation is different, the results are equivalent to that of McCloud in [6].

5.1. Change of collateral formula.

Proposition 13. The Radon-Nikodym derivative of the change of T-forward measure is characterised by the exponential martingale

$$\xi_{t,T}^{d|i\to j} := \mathbb{E}_t^{d|i,T}\left[\frac{d\mathbb{Q}^{d|j,T}}{d\mathbb{Q}^{d|i,T}}\right] = \mathcal{E}\left(-\int_0^{\cdot}\Sigma_{u,T}^{i\to j}\cdot dW_u^{d|i,T}\right)_t \tag{74}$$

5.2. Collateral convexity adjustments.

Discount factors play a specific role in quantitative finance. They correspond to the price of tradeable assets (fixed cashflow of one unit of currency) but they are not directly observables: zero coupon bonds are not liquid assets. Nevertheless, they are ubiquitous because they provide a way to "factor out" inter-temporality: we can decompose the *present value* of a cash-flow into the product of the *discount factor* and the *forward value* of the cash-flow.

One can apply the same logic to collateral convexity adjustments, decomposing them into:

(1) Convexity adjustment to discount factors
(2) Convexity adjustment to forward rates.

Remark 15. The computation of convexity adjustment typically goes as follows:

(1) Consider a market quote
(2) Express it in terms of PV of tradeable assets (ideally a ratio).
(3) Derive its drift under a well-chosen measure (in case of a ratio, drift is zero under the measure associated with the denominator).
(4) Compute the collateral convexity adjustment as a covariance with the change of measure (if the drift is deterministic then the convexity adjustment only depends on the volatility).

5.3. Convexity adjustment to domestic discount factors.

We first consider fixed cash-flows/discount factors.

Definition 2. For any payment currency e, collateral i, j and maturity T, we define the corresponding *discount factor collateral convexity adjustment* as

$$\Xi Z_{t,T}^{e|i\to j} := \frac{Z_{t,T}^{e|j}}{Z_{t,T}^{e|i}} \tag{75}$$

Remark 16. By definition of the instantaneous collateral basis, it can be written as

$$\Xi Z_{t,T}^{e|i\to j} = \mathbb{E}_t^{e|i,T}\left[e^{-\int_t^T q_u^{i\to j}\,du}\right]. \tag{76}$$

5.4. Convexity adjustment to forward rates.

Definition 3. For any positive cashflow v_T^e paid at T in currency e, we define the corresponding *forward collateral convexity adjustment* as:

$$\Xi_{t,T}^{e|i\to j}[v_T^e] = \frac{\mathbb{E}_t^{e|j,T}[v_T^e]}{\mathbb{E}_t^{e|i,T}\left[v_T^e\right]} \tag{77}$$

Remark 17. By definition of the instantaneous collateral basis, it can be computed as

$$\Xi_{t,T}^{e|i\to j}[v_T^e] := \frac{\mathbb{E}_t^{e|i,T}\left[\xi_{T,T}^{e|i\to j}v_T^e\right]}{\mathbb{E}_t^{e|i,T}\left[\xi_{T,T}^{e|i\to j}\right]}. \tag{78}$$

Remark 18. Note that the symbol $\Xi_{t,T}^{e|i\to j}$ is neither linear nor a multiplicative operator[23]. However, it provides a nice way to express convexity adjustments in the case of lognormal variables. This property makes it especially relevant to the extension of the BS and LGM models where the standard changes of measures are given by lognormal processes.

Remark 19 (Coupon factors)**.** For a foreign coupon factor $C_{a,b}^{f}$ (20) paid at b under domestic collateral, the forward collateral convexity adjustment is the coupon collateral convexity of (73), $K_t^C(a,b) = \Xi_{t,b}^{f|f\to d}[C_{a,b}^{f}]$. The native forward in its denominator is model-free (Lemma 1); the numerator is not, because the coupon fixes at the dates t_k while the collateral basis accrues until b. For a Libor coupon this is $K_t^{\mathrm{L}}(a,b) = (1+\theta L_{t,a;\theta}^{f|d})/(1+\theta L_{t,a;\theta}^{f|f})$. The general HJM expression and the closed forms under deterministic volatilities are in Proposition 26.

5.5. **Change of collateral.**

5.5.1. *Multiplicative property.* The first (and trivial) property of collateral convexity adjustments is their multiplicative property with respect to collateral.

Proposition 14 (Change of collateral)**.** We have

$$\Xi \mathrm{Z}_{t,T}^{e|i\to j}\Xi \mathrm{Z}_{t,T}^{e|j\to k} = \Xi \mathrm{Z}_{t,T}^{e|i\to k} \tag{79}$$

$$\Xi_{t,T}^{e|i\to j}[v_T^e]\Xi_{t,T}^{e|j\to k}[v_T^e] = \Xi_{t,T}^{e|i\to k}[v_T^e] \tag{80}$$

Remark 20. This implies that it is enough to know the change of collateral from the payment currency. The others can be deduced according to:

$$\Xi \mathrm{Z}_{t,T}^{e|i\to j} = \frac{\Xi \mathrm{Z}_{t,T}^{e|e\to j}}{\Xi \mathrm{Z}_{t,T}^{e|e\to i}} \tag{81}$$

$$\Xi_{t,T}^{e|i\to j}[v_T^e] = \frac{\Xi_{t,T}^{e|e\to j}[v_T^e]}{\Xi_{t,T}^{e|e\to i}[v_T^e]} \tag{82}$$

5.5.2. *FX forwards.*

Proposition 15. Absence of arbitrage implies

$$x_{t,T}^{df|c} = x_t^{df}\frac{Z_{t,T}^{f|c}}{Z_{t,T}^{d|c}} \tag{83}$$

And, more generally,

$$\mathbb{E}_t^{d|c,T}\left[x_T^{df}v_T^f\right] = x_{t,T}^{df|c}\mathbb{E}_t^{f|c,T}\left[v_T^f\right]. \tag{84}$$

Proposition 16 (Triangular relation for FX forwards)**.** For any triple of currencies d, e, f and any collateral i, we have

$$x_{t,T}^{df|i} = x_{t,T}^{de|i}x_{t,T}^{ef|i} \tag{85}$$

Proof. Indeed,

$$x_{t,T}^{de|i}x_{t,T}^{ef|i} = x_t^{de}\frac{Z_{t,T}^{e|i}}{Z_{t,T}^{d|i}}x_t^{ef}\frac{Z_{t,T}^{f|i}}{Z_{t,T}^{e|i}} = x_t^{de}x_t^{ef}\frac{Z_{t,T}^{f|i}}{Z_{t,T}^{d|i}} = x_{t,T}^{df|i}. \tag{86}$$

□

[23]For mathematicians, it can be seen as the differential/coboundary operator in cohomology.

The FX triangle relation implies that the collateral convexity of the collateral convexity of FX forwards[24][25] is entirely determined by that of discount factors.

Proposition 17. For any currencies d, f and any collaterals i, j, we have

$$\Xi^{d|i\to j}_{t,T}[x^{df}_T] \;=\; \frac{x^{df|j}_{t,T}}{x^{df|i}_{t,T}} \;=\; \frac{\Xi \mathrm{Z}^{f|i\to j}_{t,T}}{\Xi \mathrm{Z}^{d|i\to j}_{t,T}} \tag{87}$$

Proof. We have

$$\Xi^{d|i\to j}_{t,T}[x^{df}_T] = \frac{\mathbb{E}^{d|j,T}_t\left[x^{df}_T\right]}{\mathbb{E}^{d|i,T}_t\left[x^{df}_T\right]} = \frac{x^{df|j}_{t,T}}{x^{df|i}_{t,T}} = \frac{x^{df}_t Z^{f|j}_{t,T}/Z^{d|j}_{t,T}}{x^{df}_t Z^{f|i}_{t,T}/Z^{d|i}_{t,T}} = \frac{Z^{f|j}_{t,T}}{Z^{f|i}_{t,T}} \Big/ \frac{Z^{d|j}_{t,T}}{Z^{d|i}_{t,T}} = \frac{\Xi \mathrm{Z}^{f|i\to j}_{t,T}}{\Xi \mathrm{Z}^{d|i\to j}_{t,T}} \tag{88}$$

□

5.6. **Change of payment currency.** The property of FX forwards generalises as follows.

Proposition 18 (Change of currency)**.** For any positive cashflow v^f_T paid at T in currency f,

$$\Xi^{d|i\to j}_{t,T}[x^{df}_T v^f_T] \;=\; \Xi^{d|i\to j}_{t,T}[x^{df}_T]\Xi^{f|i\to j}_{t,T}[x^{df}_T v^f_T] \;=\; \frac{\Xi \mathrm{Z}^{f|i\to j}_{t,T}}{\Xi \mathrm{Z}^{d|i\to j}_{t,T}}\Xi^{f|i\to j}_{t,T}[v^f_T]. \tag{89}$$

Equivalently

$$\Xi \mathrm{Z}^{d|i\to j}_{t,T}\Xi^{d|i\to j}_{t,T}[x^{df}_T v^f_T] \;=\; \Xi \mathrm{Z}^{f|i\to j}_{t,T}\Xi^{f|i\to j}_{t,T}[v^f_T]. \tag{90}$$

Proof.

$$\Xi^{d|i\to j}_{t,T}[x^{df}_T v^f_T] = \frac{\mathbb{E}^{d|j,T}_t\left[x^{df}_T v^f_T\right]}{\mathbb{E}^{d|i,T}_t\left[x^{df}_T v^f_T\right]} = \frac{x^{df|j}_{t,T}\mathbb{E}^{f|j,T}_t\left[v^f_T\right]}{x^{df|i}_{t,T}\mathbb{E}^{f|i,T}_t\left[v^f_T\right]} = \Xi^{d|i\to j}_{t,T}[x^{df}_T]\Xi^{f|i\to j}_{t,T}[v^f_T] = \frac{\Xi \mathrm{Z}^{f|i\to j}_{t,T}}{\Xi \mathrm{Z}^{d|i\to j}_{t,T}}\Xi^{f|i\to j}_{t,T}[v^f_T]$$

□

5.7. **Relation to McCloud's definition.**

Remark 21. In [6], McCloud defines convexity adjustment curves as[26]

$$\widehat{Q}^{e|i\to j}_{t,T}[v^e_T] \;:=\; \frac{\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}x^{ie}_T v^e_T\right]}{\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}\right]\mathbb{E}^{i|i,T}_t\left[x^{ie}_T v^e_T\right]} \tag{91}$$

[24]To continue with the cohomological point of view, FX forwards are trivial cocycles so they decompose into images of discount factors under the differential operators.

[25]Note: This does not extend to the case of a quanto FX forwards as the triangle relationship on the FX vols becomes relevant rather than just the forwards.

[26]The original notation is $Q^{ji|e}_{t,T}[v^e_T]$.

How does this relate to our definition?

$$\begin{aligned}
\widehat{Q}^{e|i\to j}_{t,T}[v^e_T] &= \frac{\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}x^{ie}_T v^e_T\right]}{\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}\right]\mathbb{E}^{i|i,T}_t\left[x^{ie}_T v^e_T\right]}\\
&= \frac{Z^{i|i}_{t,T}\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}x^{ie}_T v^e_T\right]}{Z^{i|i}_{t,T}\mathbb{E}^{i|i,T}_t\left[e^{-\int_t^T q^{i\to j}_u\,du}\right]\mathbb{E}^{i|i,T}_t\left[x^{ie}_T v^e_T\right]}\\
&= \frac{\mathbb{E}^{i,\mathrm{RN}}_t\left[e^{-\int_t^T r^{i|i}_u+q^{i\to j}_u\,du}x^{ie}_T v^e_T\right]}{\mathbb{E}^{i,\mathrm{RN}}_t\left[e^{-\int_t^T r^{i|i}_u+q^{i\to j}_u\,du}\right]\mathbb{E}^{i|i,T}_t\left[x^{ie}_T v^e_T\right]}\\
&= \frac{Z^{i|j}_{t,T}\mathbb{E}^{i|j,T}_t\left[x^{ie}_T v^e_T\right]}{Z^{i|j}_{t,T}\mathbb{E}^{i|i,T}_t\left[x^{ie}_T v^e_T\right]}\\
&= \Xi^{i|i\to j}_{t,T}[x^{ie}_T v^e_T]\\
&= \frac{\Xi\mathrm{Z}^{e|i\to j}}{\Xi\mathrm{Z}^{i|i\to j}}\Xi^{e|i\to j}_{t,T}[v^e_T]
\end{aligned}$$

So McCloud's convexity adjustments can be expressed in terms of ours according to:

$$\widehat{Q}^{e|i\to j}_{t,T}[v^e_T] = \frac{\Xi\mathrm{Z}^{e|i\to j}}{\Xi\mathrm{Z}^{i|i\to j}}\Xi^{e|i\to j}_{t,T}[v^e_T] \tag{92}$$

and, conversely, our convexity adjustments can be expressed in terms of McCloud's according to:

$$\frac{\widehat{Q}^{e|i\to j}_{t,T}[v^e_T]}{\widehat{Q}^{e|i\to j}_{t,T}[1^e_T]} = \Xi^{e|i\to j}_{t,T}[v^e_T] \tag{93}$$

Our definition splits McCloud's definition into two parts to emphasise the different roles played by discount factors and forward rates.

His formulation is interesting when one choses i to be the domestic currency of the model as it decomposes the convexity adjustment into expectations under the native domestic T-forward measure.

5.8. **P&L impact.**

5.8.1. *The P&L equation.*

$$V^{e|i}_t = \mathbb{E}^{e,\mathrm{RN}}_t\left[e^{-\int_t^T r^{e|i}_u\,du}V_T\right] \tag{94}$$

$$= Z^{e|i}_{t,T}\mathbb{E}^{e|i,T}_t[V^{e|i}_T] \tag{95}$$

(96)

Let's say the value of the portfolio under standard collateral is X, a function of the market observables S. Let Q be the collateral basis so that

$$\mathrm{MTM}_t = \mathbb{E}_t\left[QX\right] \tag{97}$$

$$= \mathbb{E}_t\left[Q\right]\mathbb{E}_t\left[X\right] + \mathrm{Cov}_t\left[Q,X\right] \tag{98}$$

$$\mathbb{E}^{\mathbb{P}}_t[d\mathrm{P\&L}_t] = \frac{1}{2}(\partial^2_S V)\left(\sigma^{S,2}_{t,\mathrm{real}} - \sigma^{S,2}_{t,\mathrm{mdl}}\right) \tag{99}$$

$$+\frac{1}{2}(\partial^2_Q V)\left(\sigma^{Q,2}_{t,\mathrm{real}} - \sigma^{Q,2}_{t,\mathrm{mdl}}\right) \tag{100}$$

$$+(\partial^2_{S,Q} V)\left(\rho^{S,Q}_{t,\mathrm{real}}\sigma^{S}_{t,\mathrm{real}}\sigma^{Q}_{t,\mathrm{real}} - \rho^{S,Q}_{t,\mathrm{mdl}}\sigma^{S}_{t,\mathrm{mdl}}\sigma^{Q}_{t,\mathrm{mdl}}\right). \tag{101}$$

In a model with deterministic basis Q, we have $\sigma^Q_{t,\mathrm{mdl}} = 0$ so

$$\mathbb{E}^{\mathbb{P}}_t[d\mathrm{P\&L}_t] = \frac{1}{2}(\partial^2_S V)\left(\sigma^{S,2}_{t,\mathrm{real}} - \sigma^{S,2}_{t,\mathrm{mdl}}\right) \tag{102}$$

$$+\frac{1}{2}(\partial^2_Q V)\sigma^{Q,2}_{t,\mathrm{real}} + (\partial^2_{S,Q} V)\rho^{S,Q}_{t,\mathrm{real}}\sigma^{S}_{t,\mathrm{real}}\sigma^{Q}_{t,\mathrm{real}}. \tag{103}$$

The basis Gamma and cross-Gamma ($\Gamma \times \Gamma$) terms are unhedged which leads to a drift in the P&L. From the quant point of vue they are more often called "P&L leaks" or "P&L bleed". Traders call these terms "deliveries": I hedge both my deltas, but when market moves, the difference between my predict and the realised P&L is not zero on average. Instead, I get "delivered" some additional P&L.

If the Gamma and cross-Gamma are negative, these deliveries are systematically negative and the desk hedging with this model will bleed P&L.

So, one should know either

(1) have a full model with stochastic Q factor
(2) know the sign of the $\Gamma \times \Gamma$ term to ensure the position is conservative.

6. Deterministic volatility models

6.1. **Model specification.** We now consider the case where the covariance of instantaneous forward rates is deterministic. This implies that the main market observables are deterministic functions of (displaced) lognormal processes so that we have explicit formulas for their collateral convexity adjustments.

Unless specified otherwise. in this section we assume that:

(1) Funding spreads are constant.
(2) Volatility matrices are deterministic.

6.2. **Dynamic of market observables.** We now derive the dynamic of market observables.

6.2.1. *FX forwards.*

Proposition 19. The FX forward $x_{t,T}^{df|i}$ is a lognormal martingale under $\mathbb{Q}^{d|i,T}$:

$$\frac{dx_{t,T}^{df|i}}{x_{t,T}^{df|i}} = \Sigma_{t,T}^{df} \cdot dW_t^{d|i,T} \tag{104}$$

where

$$\Sigma_{t,T}^{df} = \Sigma_{t,T}^{d|d} - \Sigma_{t,T}^{f|f} + \Sigma_{t,T}^{d\to f} + \sigma_t^{df}. \tag{105}$$

Remark 22. In the case of deterministic volatilities, FX forwards are lognormal processes with deterministic variance independent of the collateral. However, there is a convexity adjustment associated with the change of collateral. In other words, FX implied volatilities remains the same but there is a shift in the moneyness of the options corresponding to the covariance between the collateral basis and the log of the FX forward[27].

Proposition 20. FX options prices are given by the BS formula[28]:

$$\mathrm{Call}_t^{df|i}(T,K) = Z_{t,T}^{d|i} x_{t,T}^{df|i} \mathrm{BS}\left(y = \ln(K/x_{t,T}^{df|i}), w = w_t^{df}(T)\right) \tag{108}$$

where total variance is

$$w_t^{df}(T) = \int_t^T \left|\Sigma_{u,T}^{d|d} - \Sigma_{u,T}^{f|f} + \Sigma_{u,T}^{d\to f} + \sigma_u^{df}\right|^2 du \tag{109}$$

regardless of the collateral c.

6.2.2. *Caplets.*

Corrolary 2. Caplets prices are given by the BS formula

$$\mathrm{Caplet}_t^{e|i,\theta}(T,K) = Z_{t,T+\theta}^{e|i}\left(1+\theta L_{t,T;\theta}^{e|i}\right)\mathrm{BS}\left(y = \ln\left(\frac{1+\theta K}{1+\theta L_{t,T;\theta}^{e|i}}\right), w = w_t^{e,\theta}(T,K)\right). \tag{110}$$

where total variance is [FIXME: do not use the LGM parametrisation yet]

$$w_t^{e,\theta}(T,K) = \sum_{a=1}^{N_e}\sum_{b=1}^{N_e}\int_t^T \rho_u^{(e,a),(e,b)}\sigma_u^{e|e,a}\sigma_u^{e|e,b}G_{u,T;\theta}^{e|e,a}G_{u,T;\theta}^{e|e,b}\,du \tag{111}$$

[27] Looks like a repeat of a previous remark. Avoid duplication.

[28] Here the BS function is the function of the log-moneyness forward y and total variance w defined by

$$\mathrm{BS}(y,w) = \mathcal{N}(d_+(y,w)) - e^y\mathcal{N}(d_-(y,w)) \tag{106}$$

$$d_\pm(y,w) = -\frac{\ln y}{\sqrt{w}} \pm \frac{1}{2}\sqrt{w} \tag{107}$$

Proof.

$$\begin{aligned} & Z^{e|i}_{t,T+\theta}\mathbb{E}^{e|i,T+\theta}_t\left[\max\left(L^e_{T,T;\theta}-K,0\right)\right] \\ = \; & Z^{e|i}_{t,T+\theta}\mathbb{E}^{e|i,T+\theta}_t\left[\max\left((1+\theta L^e_{T,T;\theta})-(1+\theta K),0\right)\right] \\ = \; & Z^{e|i}_{t,T+\theta}\left(1+\theta L^{e|i}_{t,T;\theta}\right)\mathrm{BS}\left(y=\ln\left(\tfrac{1+\theta K}{1+\theta L^{e|i}_{t,T;\theta}}\right),w=w^{e,\theta}_t(T,K)\right). \end{aligned}$$

where total variance is [FIXME: do not use the LGM parametrisation yet]

$$\begin{aligned} & w^{e,\theta}_t(T,K) \\ = \; & \int_t^T \Sigma^{e|e,\top}_{u,T;\theta}\Sigma^{e|e}_{u,T;\theta}\,du \\ = \; & \int_t^T \left(\Sigma^{e|e}_{u,T+\theta}-\Sigma^{e|e}_{u,T}\right)^\top\left(\Sigma^{e|e}_{u,T+\theta}-\Sigma^{e|e}_{u,T}\right)\,du \\ = \; & \int_t^T \left(\Gamma_u\Pi^{e|e}\sigma^{e|e}_u H^{e|e,-1}_u(H^{e|e}_{T+\theta}-H^{e|e}_T)\right)^\top\left(\Gamma_u\Pi^{e|e}\sigma^{e|e}_u H^{e|e,-1}_u(H^{e|e}_{T+\theta}-H^{e|e}_T)\right)\,du \\ = \; & \int_t^T \left(\sigma^{e|e}_u H^{e|e,-1}_u(H^{e|e}_{T+\theta}-H^{e|e}_T)\right)^\top\Pi^{e|e,\top}\Gamma_u\Gamma_u\Pi^{e|e}\left(\sigma^{e|e}_u H^{e|e,-1}_u(H^{e|e}_{T+\theta}-H^{e|e}_T)\right)\,du \\ = \; & \sum_{a=1}^{N_e}\sum_{b=1}^{N_e}\int_t^T \rho^{(e,a),(e,b)}_u\sigma^{e|e,a}_u\sigma^{e|e,b}_u G^{e|e,a}_{u,T;\theta}G^{e|e,b}_{u,T;\theta}\,du \end{aligned}$$

□

6.2.3. *Dynamic of resetting forward Libors.*

Proposition 21. The resetting forward Libors $L^{f/d|i}_{t,T;\theta}$ is a shifted lognormal martingale under $\mathbb{Q}^{f|i,T+\theta}$:

$$\frac{d(1+\theta L^{f/d|i}_{t,T;\theta})}{1+\theta L^{f/d|i}_{t,T;\theta}} = \Sigma^{f|d}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t \tag{112}$$

so that

$$1+\theta L^{f/d|i}_{t,T;\theta} = (1+\theta R^{f|i}_{t,T;\theta})(1-\theta S^{d|i,\div}_{t,T;\theta})e^{-\int_t^T \Sigma^{d\to i}_{u,T;\theta}\cdot\Sigma^{df|i}_{u,T}\,du} \tag{113}$$

Proof. We use the dynamic of the funding spread to compute:

$$\begin{aligned} 1+\theta L^{f/d|i}_{t,T;\theta} &= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T;\theta}\right] \\ &= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[(1-\theta S^{d|i,\div}_{t,T;\theta})\mathcal{E}\left(-\int_t^T \Sigma^{d\to i}_{u,T;\theta}\cdot dW^{d|i,T}_u\right)\right] \\ &= (1+\theta R^{f|i}_{t,T;\theta})(1-\theta S^{d|i,\div}_{t,T;\theta})e^{-\int_t^T \Sigma^{d\to i}_{u,T;\theta}\cdot\Sigma^{df}_{u,T}\,du} \end{aligned}$$

We can now compute the volatility, knowing this is a $\mathbb{Q}^{f|i,T+\theta}$-martingale:

$$\begin{aligned} \frac{d(1+\theta L^{f/d|i}_{t,T;\theta})}{1+\theta L^{f/d|i}_{t,T;\theta}} &= \frac{d(1+\theta R^{f|i}_{t,T;\theta})}{1+\theta R^{f|i}_{t,T;\theta}}+\frac{d(1-\theta S^{d|i,\div}_{t,T;\theta})}{1-\theta S^{d|i,\div}_{t,T;\theta}}+O(dt) \\ &= \Sigma^{f|i}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t-\Sigma^{d\to i}_{t,T;\theta}\cdot dW^{d|i,T}_t+O(dt) \\ &= \left(\Sigma^{f|f}_{t,T;\theta}+\Sigma^{f\to i}_{t,T;\theta}-\Sigma^{d\to i}_{t,T;\theta}\right)\cdot dW^{f|i,T+\theta}_t \\ &= \Sigma^{f|d}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t \end{aligned}$$

□

Remark 23. Note that the volatility of the resetting forward Libor does not depend on the native rates volatility. This is quite surprising since the domestic Libor L^d appears explicitly in the definition.

The explanation is the following. Due to the resetting notional, the domestic discount rate volatility risk is compensated by the domestic component in the volatility of the FX-forward, or equivalently in the drift of the FX spot. The FX spot volatility also cancels out due to the fact that the resetting forward is paid in the foreign currency. The remaining volatility risks are the ones from foreign rates and collateral basis volatility[29].

6.2.4. *Dynamic of currency basis spreads.*

Proposition 22. The currency basis spread is the difference of two exponential $\mathbb{Q}^{f|c,T+\theta}$-martingales:

$$\begin{aligned} B^{f/d|c,r}_{t,T;\theta} &= L^{f/d|c,r}_{t,T;\theta} - L^{f|c}_{t,T;\theta} \\ &= \frac{1+\theta L^{f/d|c,r}_{t,T;\theta}}{\theta} - \frac{1+\theta L^{f|c}_{t,T;\theta}}{\theta} \end{aligned} \tag{114}$$

with dynamic:

$$dB^{f/d|c,r}_{t,T;\theta} = ??$$

Remark 24. Do we gain anything by deriving the full dynamic?

Even with deterministic spot covariance, the currency basis spread is the difference of two shifted lognormal processes which does not lead to a nice dynamic. In particular, computing the delivered volatility of a currency basis spread is equivalent to computing the implied volatility of a spread option in a two asset BS model. There is no exact formula and this will typically require 2D numerical integration. Instead, we will always approximate the dynamic of each Libor by a Gaussian one. These approximations are very accurate for low volatility regimes but they may fail for EM markets or stress scenarios.

Remark 25. Instead of the additive currency basis spread, it can be mathematically easier to consider the multiplicative basis spread defined by:

$$1-\theta B^{f/d|i,r,\div}_{t,T;\theta} = \frac{1+\theta L^{f|i}_{t,T;\theta}}{1+\theta(L^{f|i}_{t,T;\theta}+B^{f/d|i,r}_{t,T;\theta})} \tag{115}$$

$$= \frac{1+\theta L^{f|i}_{t,T;\theta}}{1+\theta L^{f/d|i,r}_{t,T;\theta}} \tag{116}$$

$$= \frac{1-\theta S^{f|i,\div}_{t,T;\theta}}{\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T,T;\theta}\right]} \tag{117}$$

If $i=d$, then $\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T,T;\theta}\right] = 1-\theta S^{d|d,\div}_{t,T;\theta}$ is constant and $1-\theta B^{f/d|d,r,\div}_{t,T;\theta}$ is an exponential martingale under $\mathbb{Q}^{f|d,T}$.

The additive currency basis spread can be recovered from the multiplicative one and $1+\theta L^{f|i}_{t,T;\theta}$.

Remark 26. In the DCB model, the currency basis spread was used to bootstrap the cross-currency discount curve $Z^{f|d}_{t,T}$.
How can we build this curve now?
Can we still bootstrap it?
The answer is yes, we can still bootstrap the cross-currency discount curve, but we have to bootstrap the collateral basis spot volatility at the same time[30]:

$$\theta B^{f/d|d,r}_{t,T;\theta} = (1+\theta R^{f|d}_{t,T;\theta})(1-\theta S^{d|d,\div}_{t,T;\theta}) - (1+\theta L^{f|f}_{t,T;\theta}) e^{-\int_t^T \Sigma^{f|f}_{u,T;\theta}\cdot\Sigma^{f\to d}_{u,T+\theta}\,du}. \tag{118}$$

[29] and the domestic funding basis which we assumed to be zero.

[30] The collateral basis volatility is integrated up to the payment date $T+\theta$, as in the collateral convexity of Libors. An earlier version had $\Sigma^{f\to d}_{u,T;\theta}$, which misses most of the adjustment: about 94% in the numerical example of the verification of Theorem 4.

so

$$1+\theta R^{f|d}_{t,T;\theta} = \frac{1+\theta R^{d|d}_{t,T;\theta}}{1+\theta L^{d|d}_{t,T;\theta}}\left\{(1+\theta L^{f|f}_{t,T;\theta})e^{-\int_t^T \Sigma^{f|f}_{u,T;\theta}\cdot\Sigma^{f\to d}_{u,T+\theta}\,du}+\theta B^{f/d|d,r}_{t,T;\theta}\right\} \tag{119}$$

If we assuming that we can observe the currency basis spread for all relevant expiries, either directly from market quotes or through an interpolation scheme, then the only unknown is the collateral basis spot vol which needs to be calibrated.

6.3. **Collateral convexity adjustments.** We derive closed form formulas for collateral convexity adjustments.

6.3.1. *Change of collateral formula.* The following is a key formula for our computations.

Lemma 2. Let (X, Y) be a Gaussian vector then

$$\frac{\mathbb{E}[e^X e^Y]}{\mathbb{E}[e^X]\mathbb{E}[e^Y]} = \exp\left(\mathrm{Cov}[X, Y]\right) \tag{120}$$

Proposition 23 (Collateral convexity)**.** Let $v_t^{e|i}$ be a lognormal process under $\mathbb{Q}^{e|i,T}$

$$\frac{dv_t^{e|i}}{v_t^{e|i}} = \sigma_{t,T}^v \cdot dW_t^{e|i,T} \tag{121}$$

then the collateral convexity adjustment is given by

$$\Xi_{t,T}^{e|i\to j}[v_T^e] = \frac{\mathbb{E}_t^{e|j,T}[v_T^e]}{\mathbb{E}_t^{e|i,T}\left[v_T^e\right]} = \exp\left\{-\int_t^T \sigma_{u,T}^v \cdot \Sigma_{u,T}^{i\to j}\, du\right\} \tag{122}$$

Proof. We apply the previous lemma with $X = \ln v_{T,T}^{e|i}$ and $Y = \ln \Xi_{T,T}^{e|i\to j}$. □

6.3.2. *Collateral convexity of FX forwards.* Applying the collateral convexity formula to FX forwards, we find.

Proposition 24 (Collateral convexity of FX forwards)**.**

$$\Xi_{t,T}^{d|i\to j}[x_T^{df}] = \frac{x_{t,T}^{df|j}}{x_{t,T}^{df|i}} = \exp\left\{-\int_t^T \Sigma_{u,T}^{df} \cdot \Sigma_{u,T}^{i\to j}\, du\right\} \tag{123}$$

Remark 27 (Generalised FX forward invariance)**.** The previous result provides a generalisation of FX forward invariance. Indeed, under zero collateral basis volatility, the formula reduces to $x_{t,T}^{df|j} = x_{t,T}^{df|i}$. Under non zero collateral volatility, it still provides a formula for off-market discount curves by setting:

$$Z_{t,T}^{d|f} = Z_{t,T}^{d|d} \frac{Z_{t,T}^{f|f}}{Z_{t,T}^{f|d}} \exp\left\{\int_t^T \left(\Sigma_{u,T}^{df} - \Sigma_{u,T}^{f|f} + \Sigma_{u,T}^{d\to f} + \sigma_u^{df}\right) \cdot \Sigma_{u,T}^{d\to f}\, du\right\} \tag{124}$$

The first part is the usual formula from deterministic basis model. The exponential term quantifies the convexity adjustment ignored by the FX forward invariant construction.

But one has to be careful though as the discount curve $Z_{t,T}^{f|d}$ that appears in this formula is not exactly the same as the curve $Z_{t,T}^{f|d,\mathrm{DCB}}$ built with the DCB model, because the former takes into account collateral convexity adjustment of Libors $L_{t,T;\theta}^{f|d}$ while the DCB model ignores it. [31]

6.3.3. *Collateral convexity of Libors.* Applying the collateral convexity formula to forward Libors, we find.

Proposition 25. Under the deterministic multiplicative funding spread assumption,

$$\Xi_{t,T+\theta}^{e|i\to j}\left[1 + \theta L_{t,T;\theta}^e\right] = \frac{1 + \theta L_{S,T;\theta}^{e|j}}{1 + \theta L_{t,T;\theta}^{e|i}} = \exp\left\{-\int_t^T \Sigma_{u,T;\theta}^{e|e} \cdot \Sigma_{u,T+\theta}^{i\to j}\, du\right\} \tag{125}$$

Remark 28 (Collateral convexity of pseudo-discount factors)**.** Let us assume the existence of projection pseudo-discount factors:

$$1 + \theta L_{t,T;\theta}^{e|c} = \frac{Z_{t,T}^{e,\theta|c}}{Z_{t,T+\theta}^{e,\theta|c}}. \tag{126}$$

[31] TODO: Q: Do the two convexity adjustments on $Z_{t,T}^{f|d}$ and $Z_{t,T}^{d|f}$ compensate each other or do they compound to make $Z_{t,T}^{f|d}$ diverge even more from $Z_{t,T}^{f|d,\mathrm{DCB}}$?

Can we decompose the convexity adjustment of forward Libors as a convexity adjustment of the pseudo-discount factors? Namely, can we find $\Xi Z_{t,T}^{e,\theta|e\to c}$ such that

$$\Xi Z_{t,T}^{e,\theta|e\to c} = \frac{Z_{t,T}^{e,\theta|c}}{Z_{t,T}^{e,\theta|e}} \tag{127}$$

and the above relation remains valid for all T?

In other words, can we find $\Xi Z_{t,T}^{e,\theta|e\to c}$ such that

$$\frac{\Xi Z_{t,T}^{e,\theta|e\to c}}{\Xi Z_{t,T+\theta}^{e,\theta|e\to c}} = \exp\left(-\int_t^T \Sigma_{u,T;\theta}^{e|e}\cdot\Sigma_{u,T+\theta}^{e\to c}\,du\right) \tag{128}$$

Note that AOA does not determine the drift of projection pseudo-discount factors uniquely as they are not uniquely determined by asset prices. So we need to add some constraint on them.

A natural extension of our funding spread hypothesis is to assume that the following ratio of *native* (pseudo)-discount factors is constant:

$$\frac{Z_{t,T}^{e,\theta|e}}{Z_{t,S}^{e|e}} = \frac{Z_{t_0,T}^{e,\theta|e}}{Z_{t_0,S}^{e|e}}. \tag{129}$$

This assumption is equivalent to imposing the following dynamic[32]:

$$\frac{dZ_{t,T}^{e,\theta|e}}{Z_{t,T}^{e,\theta|e}} = \frac{dZ_{t,T}^{e|e}}{Z_{t,T}^{e|e}} = r_t^{e|e}\,dt - \Sigma_{t,T}^{e|e}\cdot dW_t^{e,\mathrm{RN}} \tag{134}$$

However, this is not necessarily the most relevant or useful assumption.

TODO: Complete this investigation.

6.3.4. *Collateral convexity of compounded coupons.* We now compute the collateral convexity $K_t^C(a,b)$ of (73) for the three coupon factors (20) of a foreign index paid at b under domestic collateral. We write $\Sigma^c = \Sigma^{f\to d}$ and, for $u<b$, $\nu(u) = \min\left(\{t_k : k<n,\ t_k>u\}\cup\{b\}\right)$ for the first fixing after u (with $\nu(u)=a$ for $u<a$).

Proposition 26 (Collateral convexity of compounded coupons)**.** Let $C_{a,b}^f$ be a Libor, daily or continuous coupon factor (20) of the foreign currency, paid at b under domestic collateral.

(1) (HJM.) In any model of the section on arbitrage free models, for $t\le b$,

$$K_t^C(a,b) = \frac{\mathbb{E}_t^{f|f,b}\left[C_{a,b}^f\,\xi_{t,b}\right]}{\mathbb{E}_t^{f|f,b}\left[C_{a,b}^f\right]}, \qquad \xi_{t,b} = \frac{\xi_{b,b}^{f|f\to d}}{\xi_{t,b}^{f|f\to d}} = \exp\left(-\int_t^b \Sigma_{u,b}^c\cdot dW_u^{f|f,b} - \frac{1}{2}\int_t^b \left|\Sigma_{u,b}^c\right|^2 du\right), \tag{135}$$

with the density (74). The factors of $C_{a,b}^f$ fixed at t cancel, so only the fixings after t contribute.

(2) (Deterministic volatilities, in particular the LGCM.) If the volatilities are deterministic, $K_t^C(a,b)$ is deterministic and

$$\ln K_t^C(a,b) = -\int_t^b \varphi_u^C\cdot\Sigma_{u,b}^c\,du, \tag{136}$$

[32] We can check that this is compatible with the AOA condition on forward Libors, Namely,

$$\frac{d(1+\theta L_{t,T;\theta}^{e|e})}{(1+\theta L_{t,T;\theta}^{e|e})} = \frac{dZ_{t,T}^{e,\theta|e}}{Z_{t,T}^{e,\theta|e}} - \frac{dZ_{t,T+\theta}^{e,\theta|e}}{Z_{t,T+\theta}^{e,\theta|e}} - \left\langle \frac{dZ_{t,T+\theta}^{e,\theta|e}}{Z_{t,T+\theta}^{e,\theta|e}}, \frac{dZ_{t,T}^{e,\theta|e}}{Z_{t,T}^{e,\theta|e}} - \frac{dZ_{t,T+\theta}^{e,\theta|e}}{Z_{t,T+\theta}^{e,\theta|e}} \right\rangle \tag{130}$$

$$= \Sigma_{t,T;\theta}^{e|e}\cdot dW_t^{e,\mathrm{RN}} + \Sigma_{t,T+\theta}^{e|e}\cdot\Sigma_{t,T;\theta}^{e|e}\cdot dt \tag{131}$$

$$= \Sigma_{t,T;\theta}^{e|e}\cdot\left(dW_t^{e,\mathrm{RN}} + \Sigma_{t,T+\theta}^{e|e}dt\right) \tag{132}$$

$$= \Sigma_{t,T;\theta}^{e|e}\cdot dW_t^{e|e,T+\theta}. \tag{133}$$

so forward Libors are indeed $\mathbb{Q}^{e|e,T+\theta}$ as required by AOA.

where φ^C_u is the volatility loading of $\ln C^f_{a,b}$ on dW_u:

$$\begin{aligned}\varphi^{\mathrm{L}}_u &= \left(\Sigma^f_{u,b} - \Sigma^f_{u,a}\right)\mathbf{1}_{u<a},\\ \varphi^{\delta}_u &= \sum_{k:\, t_k>u}\left(\Sigma^f_{u,t_{k+1}} - \Sigma^f_{u,t_k}\right) = \Sigma^f_{u,b} - \Sigma^f_{u,\nu(u)},\\ \varphi^{0}_u &= \Sigma^f_{u,b} - \Sigma^f_{u,a}\mathbf{1}_{u<a}.\end{aligned}$$

Explicitly, with $a' = \max(t,a)$:

$$\ln K^{\mathrm{L}}_t(a,b) = -\int_t^a \left(\Sigma^f_{u,b} - \Sigma^f_{u,a}\right)\cdot \Sigma^c_{u,b}\, du \qquad (t \le a), \tag{137}$$

$$\ln K^{\delta}_t(a,b) = -\sum_{k:\, t_k>t}\int_t^{t_k}\left(\Sigma^f_{u,t_{k+1}} - \Sigma^f_{u,t_k}\right)\cdot \Sigma^c_{u,b}\, du, \tag{138}$$

$$\ln K^{0}_t(a,b) = -\mathbf{1}_{t<a}\int_t^a \left(\Sigma^f_{u,b} - \Sigma^f_{u,a}\right)\cdot \Sigma^c_{u,b}\, du \;-\; \int_{a'}^b \Sigma^f_{u,b}\cdot \Sigma^c_{u,b}\, du. \tag{139}$$

The Libor formula is the one-step case $n = 1$ of the daily one, and the earlier proposition on the collateral convexity of Libors. The daily formula is exact for the traded coupon, also inside a partially accrued period ($a < t < b$). The continuous formula is its limit as $\delta \to 0$ (Proposition 27).

Proof. 1. By (74) with $e = i = f$, $j = d$ and $T = b$, $\mathbb{E}^{f|d,b}_t[C] = \mathbb{E}^{f|f,b}_t[C\,\xi^{f|f\to d}_{b,b}]/\xi^{f|f\to d}_{t,b}$, and the denominator of K^C is $\mathbb{E}^{f|f,b}_t[C]$. The fixed part A_t of Lemma 1 is $\mathcal{F}_t$-measurable and positive, so it factors out of both expectations.

2. Write $W = W^{f|f,b}$. Under $\mathbb{Q}^{f|f,b}$ the instantaneous forwards have dynamic $dr^{f|f}_{s,u} = \mu_{s,u}ds + \sigma^f_{s,u}\cdot dW_s$ with a drift μ that is deterministic when the volatilities are (HJM drift condition and deterministic Girsanov kernels). For a fixing $t_k > t$, by stochastic Fubini,

$$-\ln Z^{f|f}_{t_k,t_{k+1}} = \int_{t_k}^{t_{k+1}} r^{f|f}_{t_k,u}\, du = \text{deterministic} + \int_t^{t_k}\left(\Sigma^f_{s,t_{k+1}} - \Sigma^f_{s,t_k}\right)\cdot dW_s,$$

since $\int_{t_k}^{t_{k+1}}\sigma^f_{s,u}du = \Sigma^f_{s,t_{k+1}} - \Sigma^f_{s,t_k}$. Summing over the fixings after t gives $\ln C^{\delta}_{a,b} = (\mathcal{F}_t\text{-measurable}) + \int_t^b \varphi^{\delta}_s\cdot dW_s$, and the sum telescopes to $\Sigma^f_{s,b} - \Sigma^f_{s,\nu(s)}$. For the continuous factor, $r^{f|f}_u = r^{f|f}_{u,u} = r^{f|f}_{t,u} + \int_t^u \mu_{s,u}ds + \int_t^u \sigma^f_{s,u}\cdot dW_s$ for $u \ge t$, and by stochastic Fubini

$$\int_{a'}^b\int_t^u \sigma^f_{s,u}\cdot dW_s\, du = \int_t^b\left(\int_{\max(s,a')}^b \sigma^f_{s,u}\, du\right)\cdot dW_s = \int_t^b\left(\Sigma^f_{s,b} - \Sigma^f_{s,\max(s,a')}\right)\cdot dW_s,$$

which is $\int_t^b \varphi^0_s\cdot dW_s$ since $\Sigma^f_{s,s} = 0$; the part $\int_a^t r^{f|f}_u du$ of a partially accrued period is $\mathcal{F}_t$-measurable. The Libor factor is the case $n = 1$. So $X = \ln C^f_{a,b}$ and $Y = \ln \xi_{t,b}$ are jointly Gaussian given $\mathcal{F}_t$ (deterministic integrands against the same Brownian motion), $\mathbb{E}^{f|f,b}_t[e^Y] = 1$, and by the lemma of the change of collateral formula

$$K^C_t(a,b) = \frac{\mathbb{E}_t[e^X e^Y]}{\mathbb{E}_t[e^X]\,\mathbb{E}_t[e^Y]} = \exp\left(\mathrm{Cov}_t[X,Y]\right) = \exp\left(-\int_t^b \varphi^C_s\cdot \Sigma^c_{s,b}\, ds\right).$$

The explicit forms follow from φ^C: for $u < a$ all three kernels equal $\Sigma^f_{u,b} - \Sigma^f_{u,a}$ (no fixing has occurred), and on $[a', b]$ the Libor kernel is 0, the daily kernel is $\Sigma^f_{u,b} - \Sigma^f_{u,\nu(u)}$ and the continuous one is $\Sigma^f_{u,b}$. Regrouping the daily integrand by fixing gives (138). □

Proposition 27 (In-arrears term and continuous compounding error)**.** Assume deterministic volatilities with $|\sigma^f_{s,u}| \le \bar\sigma^f$ and $|\sigma^c_{s,u}| \le \bar\sigma^c$ for $s \le u$ (Euclidean norms of the loadings on dW), and let $a' = \max(t,a)$.

(1) (In-arrears term.) $\ln K_t^0(a,b) - \ln K_t^{\mathrm{L}}(a,b) = -\int_{a'}^b \Sigma_{u,b}^f \cdot \Sigma_{u,b}^c \, du$, so

$$\left|\ln \frac{K_t^0(a,b)}{K_t^{\mathrm{L}}(a,b)}\right| \;\le\; \bar\sigma^f \bar\sigma^c \, \frac{(b-a')^3}{3} \;\le\; \bar\sigma^f \bar\sigma^c \, \frac{\theta^3}{3},$$

uniformly in the start date a. The same bound holds for $\ln(K_t^\delta / K_t^{\mathrm{L}})$.

(2) (Continuous vs daily, order one.) $\ln K_t^\delta(a,b) - \ln K_t^0(a,b) = \int_{a'}^b \Sigma_{u,\nu(u)}^f \cdot \Sigma_{u,b}^c \, du$, so, with $\delta = \max_k \delta_k$,

$$\left|\ln \frac{K_t^\delta(a,b)}{K_t^0(a,b)}\right| \;\le\; \bar\sigma^f \bar\sigma^c \, \frac{(b-a')^2}{2} \, \delta.$$

(3) (Leading term.) If moreover $|\sigma_{s,u}^f - \sigma_{s,s}^f| \le \Lambda(u-s)$, the grid is uniform with step δ (the last step shortened), and $g(u) = \sigma_{u,u}^f \cdot \Sigma_{u,b}^c$ is Lipschitz on $[a',b]$ outside finitely many points (e.g. piecewise constant parameters), then

$$\ln \frac{K_t^\delta(a,b)}{K_t^0(a,b)} \;=\; \frac{\delta}{2} \int_{a'}^b \sigma_{u,u}^f \cdot \Sigma_{u,b}^c \, du + O(\delta^2). \tag{140}$$

Proof. 1. Subtract (137) from (139). Then $|\Sigma_{u,b}^f| \le \int_u^b |\sigma_{u,v}^f| dv \le \bar\sigma^f (b-u)$, the same for Σ^c, and $|x \cdot y| \le |x||y|$, so the integrand is bounded by $\bar\sigma^f \bar\sigma^c (b-u)^2$. For the daily coupon the kernel on $[a',b]$ is $\Sigma_{u,b}^f - \Sigma_{u,\nu(u)}^f = \int_{\nu(u)}^b \sigma_{u,v}^f dv$, bounded by $\bar\sigma^f (b-u)$ too.

2. Subtract the kernels: $\varphi_u^0 - \varphi_u^\delta = \Sigma_{u,\nu(u)}^f \mathbf{1}_{u \ge a'}$ on $[t,b]$, and $|\Sigma_{u,\nu(u)}^f| \le \bar\sigma^f(\nu(u)-u) \le \bar\sigma^f \delta$, $|\Sigma_{u,b}^c| \le \bar\sigma^c (b-u)$.

3. $\Sigma_{u,\nu(u)}^f = (\nu(u)-u)\sigma_{u,u}^f + R_u$ with $|R_u| \le \Lambda(\nu(u)-u)^2/2 \le \Lambda\delta^2/2$, so the remainder contributes $O(\delta^2)$ after integration. On a full step $[t_k, t_k+\delta] \subset [a',b]$ where g is Lipschitz with constant Λ_g,

$$\left|\int_{t_k}^{t_k+\delta} (t_k+\delta-u) g(u)\, du - \frac{\delta}{2}\int_{t_k}^{t_k+\delta} g(u)\, du\right| \;=\; \left|\int_{t_k}^{t_k+\delta} \left(\tfrac{\delta}{2} - (u-t_k)\right)\left(g(u)-g(t_k)\right) du\right| \;\le\; \frac{\Lambda_g \delta^3}{4},$$

and there are at most θ/δ such steps, a total $O(\delta^2)$. The remaining steps (the one containing a' when t is inside the period, the shortened last step and the steps containing a break point of g) are finitely many, independent of δ, and each contributes at most $2 \sup|g| \, \delta^2$. □

Remark 29. (1) In item 3 the constant is a covariance integral: in the LGCM $\sigma_{u,u}^f$ has loadings $\sigma_u^{f,i}$ ($M_{u,u}^{f,i} = 1$), so the leading term is $\frac{\delta}{2}\sum_{i,j}\int_{a'}^b \rho_u^{(f,i),(c,j)} \sigma_u^{f,i} \sigma_u^{c,j} G_{u,b}^{c,j} \, du$. Its sign (one factor per block) is that of the (f,c) correlation: with $\rho^{f,c} < 0$, $K^0 > K^\delta$.

(2) The bounds are crude (they ignore the correlation). The measured sizes are in Remark 57: on the test model the in-arrears term is about 10^{-7} in $\ln K$, the Libor term 10^{-5} to $5 \cdot 10^{-4}$, and the continuous approximation is within 10^{-9} of the daily coupon.

(3) Item 1 bounds the in-arrears term uniformly in a; nothing is claimed about the Libor term (137), which need not grow with $a - t$ in general (mean reversion damps $\Sigma_{u,b}^f - \Sigma_{u,a}^f$, and the sign of $\rho^{f,c}$ can make the integrand change sign). On the test model it does grow (numerical observation, item 3 of Remark 57): $K^{\mathrm{L}} - 1 = 9.956 \cdot 10^{-6}$, $1.7963 \cdot 10^{-4}$, $5.4888 \cdot 10^{-4}$ at $a = 2, 10, 20$, so there the in-arrears term matters relatively more for short expiries.

6.3.5. *Collateral convexity of resetting forward Libors.*

Proposition 28. The convexity adjustment on resetting forward Libors $L_{t,T;\theta}^{f/d|i}$ is

$$\Xi_{t,T+\theta}^{f|i\to j}\left[1 + \theta L_{t,T;\theta}^{f/d|i}\right] \;=\; \exp\left\{-\int_t^S \Sigma_{u,T;\theta}^{f|d} \cdot \Sigma_{u,T+\theta}^{i\to j} \, du\right\} \quad ??? \tag{141}$$

6.3.6. *Collateral convexity of currency basis spreads.*

7. The Quasi-Gaussian Collateral Model

7.0. **Introduction.** The state variable in the generic HJM model is the whole forward curve which means that the dynamic is not a finite dimensional process in general[33]. The Cheyette separability assumption is well known to ensure a low dimensional Markovian dynamic in the classical framework.

We show that this is still valid in the collateralised multi-curve setting. The resulting models, which we call the Quasi-Gaussian Collateral Models, are the natural generalisation of the Cheyette model to stochastic collateral/currency basis.

7.1. Model specification.

7.1.1. *Separable volatility condition.*

Definition 4. An instantaneous forward rate volatility is *separable* if there exist

- a stochastic process $g : \Omega \times \mathbb{R}_+ \to M_d(\mathbb{R})$
- a deterministic function $h : \mathbb{R}_+ \to \mathbb{R}^d$

such that

$$\sigma_{t,T}(\omega) \quad = \quad g(\omega, t)h(T). \tag{142}$$

Hypothesis (Separable volatility)**.** The volatility of the following

- The instantaneous forward discount rates under native collateral
- The instantaneous forward collateral basis spreads.

are separable.

7.1.2. *HJM dynamic.*

Let us recall the dynamic of the generic HJM model under the domestic risk neutral measure:

$$dr^{d|d}_{t,T} = \left(\sigma^{d|d,\top}_{t,T}\Sigma^{d|d}_{t,T}\right)dt + \sigma^{d|d,\top}_{t,T}dW^{d,\mathrm{RN}}_t \tag{143}$$

$$dr^{f|f}_{t,T} = \left(\sigma^{f|f,\top}_{t,T}\Sigma^{f|f}_{t,T}\right)dt + \sigma^{f|f,\top}_{t,T}\left(dW^{d,\mathrm{RN}}_t - \sigma^{df}_t\,dt\right) \tag{144}$$

$$dq^{f|f\to d}_{t,T} = \left(\sigma^{f\to d,\top}_{t,T}\Sigma^{f\to d}_{t,T} + \sigma^{f\to d,\top}_{t,T}\Sigma^{f|f}_{t,T} + \sigma^{f|f,\top}_{t,T}\Sigma^{f\to d}_{t,T}\right)dt + \sigma^{f\to d,\top}_{t,T}\left(dW^{d,\mathrm{RN}}_t - \sigma^{df}_t\,dt\right) \tag{145}$$

[33] Here is a simple situation showing that the process $(r(t))$ is usually not Markov.
Assume that $\sigma(s,t) = t - s$ for every $s \leqslant t$ and every ν, then elementary computations yield

$$r(t) = F(0,t) + \tfrac{1}{8}t^4 + X_t$$

with

$$X_t = \int_0^t W_s \mathrm{d}s$$

hence the process $(r(t))$ is Markov if and only if the process (X_t) is. But the process (X_t) is not Markov (only the bivariate process (W_t, X_t) is).

The drift of the collateral basis is the difference $\sigma_{t,T}^{f|d,\top}\Sigma_{t,T}^{f|d} - \sigma_{t,T}^{f|f,\top}\Sigma_{t,T}^{f|f}$ of the HJM drifts of $r_{t,T}^{f|d}$ and $r_{t,T}^{f|f}$ (both under $\mathbb{Q}^{f,\mathrm{RN}}$), with $\sigma^{f|d} = \sigma^{f|f} + \sigma^{f\to d}$. In particular the term $\sigma_{t,T}^{f\to d,\top}\Sigma_{t,T}^{f\to d}$ appears once[34]. So that,

$$\frac{dZ_{t,T}^{d|d}}{Z_{t,T}^{d|d}} = r_t^{d|d}dt - \Sigma_{t,T}^{d|d,\top}dW_t^{d,\mathrm{RN}} \tag{146}$$

$$\frac{dZ_{t,T}^{f|f}}{Z_{t,T}^{f|f}} = r_t^{f|f}dt - \Sigma_{t,T}^{f|f,\top}\left(dW_t^{d,\mathrm{RN}} - \sigma_t^{df}dt\right) \tag{147}$$

$$\frac{dZ_{t,T}^{f|d}}{Z_{t,T}^{f|f}} = \left(r_t^{f|f} + q_t^{f\to d}\right)dt - \left(\Sigma_{t,T}^{f|f} + \Sigma_{t,T}^{f\to d}\right)^\top\left(dW_t^{d,\mathrm{RN}} - \sigma_t^{df}dt\right) \tag{148}$$

$$\frac{dx_t^{df}}{x_t^{df}} = \left(r_t^{d|d} - r_t^{f|f} - q_t^{f\to d}\right)dt + \sigma_t^{df,\top}dW_t^{d,\mathrm{RN}} \tag{149}$$

7.1.3. *Notations.* The model is stationary iff the volatilities are a function of time to expiry $\tau = T - t$ rather than both t and T. This suggests changing the parametrisation[35].

For each symbol a, let us write

$$H_t^a = \mathrm{diag}(h_t^a) \tag{150}$$

$$\sigma_{z,s}^a = g_s^a H_s^a \tag{151}$$

$$M_{t,T}^a = H_T^a H_t^{a,-1}\mathbf{1} = H_t^{a,-1}h_T^a \tag{152}$$

$$G_{t,T}^a = \int_t^T M_{t,u}^a\,du = H_t^{a,-1}\int_t^T h_u^a\,du \tag{153}$$

Let us define the mean reversion speeds as the coefficients of the diagonal matrix:

$$\beta_t^a = -H_t^{a,-1}\frac{d}{dt}H_t^a \tag{154}$$

One can represent the matrices more explicitly:

$$H_t^a = \mathrm{diag}\left(e^{-\int_0^t \beta_u^{a,1}\,du}, \ldots, e^{-\int_0^t \beta_u^{a,N_a}\,du}\right) \tag{155}$$

$$M_{t,T}^a = \left[e^{-\int_t^T \beta_u^{a,1}\,du}, \ldots, e^{-\int_t^T \beta_u^{a,N_a}\,du}\right]^\top \tag{156}$$

$$G_{t,T}^a = \left[\int_t^T e^{-\int_t^s \beta_u^{a,1}\,du}\,ds, \ldots, \int_t^T e^{-\int_t^s \beta_u^{a,N_a}\,du}\,ds\right]^\top \tag{157}$$

Remark 30. Moreover, one can reformulate in terms of 1D OU processes by setting:

$$\sigma_s^a = \mathrm{diag}\left(\sigma_{z,s}^{a,\top}\sigma_{z,s}^a\right)^{1/2} \tag{158}$$

$$\Omega_t = \sigma_s^{-1}\left(\sigma_{z,s}^\top\sigma_{z,s}\right)\sigma_s^{-1} \tag{159}$$

$$\Gamma_t^\top\Gamma_t = \Omega_t \tag{160}$$

[34] An earlier version of this formula had $\sigma_{t,T}^{f|d,\top}\Sigma_{t,T}^{f\to d}$ as last term, which counts it twice.

[35] This is also sensible for numerical reasons since, under standard parametrisation, the coefficients of g and h tend to have exponential growth.

The factor loading matrix $\Gamma_t \in M_{d,k}(\mathbb{R})$ is a square root of the instantaneous correlation matrix Ω_t whose bloc decomposition looks like:

$$\Omega_t \;=\; \Gamma_t^\top \Gamma \;=\; \begin{bmatrix} \rho_t^{d,d} & \rho_t^{d,f} & \rho_t^{d,c} & \rho_t^{d,x} \\ \rho_t^{f,d} & \rho_t^{f,f} & \rho_t^{f,c} & \rho_t^{f,x} \\ \rho_t^{c,d} & \rho_t^{c,f} & \rho_t^{c,c} & \rho_t^{c,x} \\ \rho_t^{x,d} & \rho_t^{x,f} & \rho_t^{x,c} & \rho_t^{x,x} \end{bmatrix} \tag{161}$$

This leads to

$$\sigma_{t,T}^{a,\top} dW_t \;=\; H_T^a H_t^{a,-1} \sigma_t^a \Pi_a^\top \underbrace{\Gamma_t^\top dW_t}_{d\widetilde{W}_t} \tag{162}$$

$$\Sigma_{t,T}^{a,\top} dW_t \;=\; G_{t,T}^a \sigma_t^a \Pi_a^\top \, d\widetilde{W}_t \tag{163}$$

7.1.4. *Dynamic.* We use the shorthands d, f, c and x for the blocks $d|d$, $f|f$, $f \to d$ and df, and write $W_t = W_t^{d,\mathrm{RN}}$. For each rates block $a \in \{d, f, c\}$, the volatility of instantaneous forwards is $\sigma_{t,T}^a = \sigma_{z,t}^a M_{t,T}^a$ with $\sigma_{z,t}^a = g_t^a H_t^a \in M_{k,N_a}(\mathbb{R})$, so that $\Sigma_{t,T}^a = \sigma_{z,t}^a G_{t,T}^a$. We also write

$$\mathcal{M}_{u,t}^a \;=\; \mathrm{diag}\left(M_{u,t}^a\right) \;=\; H_t^a H_u^{a,-1}, \qquad u \le t, \tag{164}$$

for the diagonal transport matrix. It satisfies $\mathcal{M}_{u,t}^a \mathcal{M}_{t,T}^a = \mathcal{M}_{u,T}^a$, $\partial_t \mathcal{M}_{u,t}^a = -\beta_t^a \mathcal{M}_{u,t}^a$ and $\partial_t G_{u,t}^a = \mathcal{M}_{u,t}^a \mathbf{1}$, and the forward volatilities split as $M_{u,T}^a = \mathcal{M}_{u,t}^a M_{t,T}^a$ and $G_{u,T}^a = G_{u,t}^a + \mathcal{M}_{u,t}^a G_{t,T}^a$.

Theorem 4 (Quasi-Gaussian Collateral Model)**.** Let us consider an arbitrage free model with instantaneous forward rates satisfying the separability condition. Let us define the vector-valued state variables

$$z_t^d \;=\; \int_0^t \mathcal{M}_{u,t}^d \sigma_{z,u}^{d,\top} \left(\sigma_{z,u}^d G_{u,t}^d \, du + dW_u \right) \tag{165}$$

$$z_t^f \;=\; \int_0^t \mathcal{M}_{u,t}^f \sigma_{z,u}^{f,\top} \left(\sigma_{z,u}^f G_{u,t}^f \, du - \sigma_u^{df} \, du + dW_u \right) \tag{166}$$

$$z_t^c \;=\; \int_0^t \mathcal{M}_{u,t}^c \sigma_{z,u}^{c,\top} \left(\sigma_{z,u}^c G_{u,t}^c \, du + \sigma_{z,u}^f G_{u,t}^f \, du - \sigma_u^{df} \, du + dW_u \right) \tag{167}$$

$$z_t^x \;=\; \ln\left(x_t^{df} / x_{0,t}^{df|d} \right) \tag{168}$$

together with the matrix-valued processes ($(a,b) \in \{(d,d),(f,f),(c,c),(c,f)\}$, $y_t^{f,c} = y_t^{c,f,\top}$)

$$y_t^{a,b} \;=\; \int_0^t \mathcal{M}_{u,t}^a \sigma_{z,u}^{a,\top} \sigma_{z,u}^b \mathcal{M}_{u,t}^b \, du \tag{169}$$

and the vector-valued process

$$v_t^{f,c} \;=\; \int_0^t \mathcal{M}_{u,t}^f \sigma_{z,u}^{f,\top} \sigma_{z,u}^c G_{u,t}^c \, du. \tag{170}$$

Then, the model is Markovian in these state variables with dynamic under $\mathbb{Q}^{d,\mathrm{RN}}$

$$dz_t^d \;=\; \left(y_t^{d,d} \mathbf{1}_d - \beta_t^d z_t^d \right) dt + \sigma_{z,t}^{d,\top} dW_t \tag{171}$$

$$dz_t^f \;=\; \left(y_t^{f,f} \mathbf{1}_f - \sigma_{z,t}^{f,\top} \sigma_t^{df} - \beta_t^f z_t^f \right) dt + \sigma_{z,t}^{f,\top} dW_t \tag{172}$$

$$dz_t^c \;=\; \left(y_t^{c,c} \mathbf{1}_c + y_t^{c,f} \mathbf{1}_f - \sigma_{z,t}^{c,\top} \sigma_t^{df} - \beta_t^c z_t^c \right) dt + \sigma_{z,t}^{c,\top} dW_t \tag{173}$$

$$dz_t^x \;=\; \left(\mathbf{1}_d^\top z_t^d - \mathbf{1}_f^\top z_t^f - \mathbf{1}_c^\top z_t^c - \mathbf{1}_f^\top v_t^{f,c} - \frac{1}{2} |\sigma_t^{df}|^2 \right) dt + \sigma_t^{df,\top} dW_t \tag{174}$$

$$dy_t^{a,b} \;=\; \left(\sigma_{z,t}^{a,\top} \sigma_{z,t}^b - \beta_t^a y_t^{a,b} - y_t^{a,b} \beta_t^b \right) dt \tag{175}$$

$$dv_t^{f,c} \;=\; \left(y_t^{f,c} \mathbf{1}_c - \beta_t^f v_t^{f,c} \right) dt \tag{176}$$

and the following reconstruction formulas:

(1) Instantaneous forward rates:

$$r^{d|d}_{t,T} = r^{d|d}_{0,T} + M^{d,\top}_{t,T}\left(z^d_t + y^{d,d}_t G^d_{t,T}\right) \tag{177}$$

$$r^{f|f}_{t,T} = r^{f|f}_{0,T} + M^{f,\top}_{t,T}\left(z^f_t + y^{f,f}_t G^f_{t,T}\right) \tag{178}$$

$$q^{f|f\to d}_{t,T} = q^{f|f\to d}_{0,T} + M^{c,\top}_{t,T}\left(z^c_t + y^{c,c}_t G^c_{t,T} + y^{c,f}_t G^f_{t,T}\right) + M^{f,\top}_{t,T}\left(v^{f,c}_t + y^{f,c}_t G^c_{t,T}\right) \tag{179}$$

(2) Zero coupon bonds and collateral basis curve $\Xi \mathrm{Z}^{f|f\to d}_{t,T} = Z^{f|d}_{t,T}/Z^{f|f}_{t,T}$:

$$Z^{d|d}_{t,T} = Z^{d|d}_{0,t,T}\exp\left(-G^{d,\top}_{t,T} z^d_t - \frac{1}{2} G^{d,\top}_{t,T} y^{d,d}_t G^d_{t,T}\right) \tag{180}$$

$$Z^{f|f}_{t,T} = Z^{f|f}_{0,t,T}\exp\left(-G^{f,\top}_{t,T} z^f_t - \frac{1}{2} G^{f,\top}_{t,T} y^{f,f}_t G^f_{t,T}\right) \tag{181}$$

$$\Xi \mathrm{Z}^{f|f\to d}_{t,T} = \Xi \mathrm{Z}^{f|f\to d}_{0,t,T}\exp\left(-G^{c,\top}_{t,T} z^c_t - G^{f,\top}_{t,T} v^{f,c}_t - \frac{1}{2} G^{c,\top}_{t,T} y^{c,c}_t G^c_{t,T} - G^{c,\top}_{t,T} y^{c,f}_t G^f_{t,T}\right) \tag{182}$$

(3) The FX spot: $x^{df}_t = x^{df|d}_{0,t} e^{z^x_t}$.

Remark 31. The state variables have a very simple interpretation:

$$\begin{aligned}
\mathbf{1}^\top_d z^d_t &= r^{d|d}_t - r^{d|d}_{0,t} \\
\mathbf{1}^\top_f z^f_t &= r^{f|f}_t - r^{f|f}_{0,t} \\
\mathbf{1}^\top_c z^c_t + \mathbf{1}^\top_f v^{f,c}_t &= q^{f\to d}_t - q^{f|f\to d}_{0,t} \\
z^x_t &= \ln\left(x^{df}_t / x^{df|d}_{0,t}\right).
\end{aligned}$$

So they are essentially the short rates and FX log moneyness forward spot with their initial conditions factored out.

The additional state variables y_t look like[36]

$$y^{(a,i),(b,j)}_t = \int_0^t \rho^{(a,i),(b,j)}_s \sigma^{a,i}_s \sigma^{b,j}_s e^{-\int_s^t \beta^{a,i}_u + \beta^{b,j}_u \, du} \, ds \tag{183}$$

They correspond to the evolution of covariance in the model and it is necessary to keep track of them in order to keep the dynamic Markovian. However, if the model parameters are deterministic, then so are the y_t's and $v^{f,c}_t$ and they can be pre-computed instead of diffused (see next section).

Remark 32 (Quanto drifts)**.** The quanto drift adjustments $-\sigma^{f,\top}_{z,t}\sigma^{df}_t$ and $-\sigma^{c,\top}_{z,t}\sigma^{df}_t$ are instantaneous: they do not require additional state variables. Only the cross-block covariance $y^{c,f}_t$ between collateral basis and foreign rates has to be tracked, together with $v^{f,c}_t$. The covariances $y^{d,f}_t$, $y^{d,c}_t$ and the FX covariances are not needed.

Remark 33 (The collateral basis is not driven by its own Cheyette state)**.** The answer to the question raised in the proof below, namely whether the collateral basis is driven by an OU process in its own factors only, is negative: the term $\sigma^{f,\top}_{t,T}\Sigma^c_{t,T}$ in the drift of $q^{f|f\to d}_{t,T}$ has the maturity dependence $M^f_{t,T}$ of the *foreign* factors. This is what the extra finite-variation state $v^{f,c}_t$ accounts for.

However, the stacked state

$$\hat{z}_t = \begin{bmatrix} z^f_t + v^{f,c}_t \\ z^c_t \end{bmatrix}, \quad \hat{G}_{t,T} = \begin{bmatrix} G^f_{t,T} \\ G^c_{t,T} \end{bmatrix}, \quad \hat{y}_t = \begin{bmatrix} y^{f,f}_t & y^{f,c}_t \\ y^{c,f}_t & y^{c,c}_t \end{bmatrix}, \tag{184}$$

[36] Assuming the volatilities are $\sigma_{t,T} = \sigma_t e^{-\int_t^T \beta_v \, dv}$.

is exactly a Cheyette state for the cross-currency discount rate $r^{f|d}_{t,T} = r^{f|f}_{t,T} + q^{f|f\to d}_{t,T}$: it has dynamic $d\hat{z}_t = (\hat{y}_t \mathbf{1} - \hat{\sigma}^\top_{z,t}\sigma^{df}_t - \hat{\beta}_t \hat{z}_t)\,dt + \hat{\sigma}^\top_{z,t} dW_t$ with $\hat{\sigma}_z = [\sigma^f_z | \sigma^c_z]$ and

$$Z^{f|d}_{t,T} = Z^{f|d}_{0,t,T} \exp\left(-\hat{G}^\top_{t,T}\hat{z}_t - \frac{1}{2}\hat{G}^\top_{t,T}\hat{y}_t\hat{G}_{t,T}\right). \tag{185}$$

So the two choices of state considered in the proof below (z^c_t driving $\Xi \mathrm{Z}^{f|f\to d}_{t,T}$, or $\hat{z}_t$ driving $Z^{f|d}_{t,T}$) differ only by the shift $v^{f,c}_t$. In the Quasi-Gaussian model $v^{f,c}_t$ is stochastic and must be carried as an extra state in either case, since $Z^{f|f}_{t,T}$ needs z^f_t and $Z^{f|d}_{t,T}$ needs $z^f_t + v^{f,c}_t$. In the LGCM it is deterministic and the two choices carry the same information.

Remark 34. Let us comment our parametrisation choices.

We chose to keep the quanto-drift adjustments inside the dynamic of the state variables. This ensures the definition of $z^{f|f}_t$ does not depend on the choice of the domestic currency d. But it implies that its dynamic under the domestic risk neutral measure does.

On the other hand, we chose to consider $z^{f|f\to d}_t$ as driving the collateral basis $\Xi \mathrm{Z}^{f|f\to d}_{t,T}$ rather than $\Xi \mathrm{Z}^{d|d\to f}_{t,T}$ or some other observable. This choice is motivated by the calibration of the model to the cross-currency market. Indeed, the most liquid instruments are collateralised in USD so it is natural to chose it as the domestic currency d of the model. But this implies that the first non-native discount curve to calibrate is $Z^{f|d}_{t,T}$. Factoring out the native discount curve $Z^{f|f}_{t,T}$ which is known from the native foreign market, we are led to consider $\Xi \mathrm{Z}^{f|f\to d}_{t,T}$ as containing mostly new information from the cross-currency market.

7.2. **Proof of the theorem.**

7.2.1. *Domestic rates.* Let us inject the separability condition in the dynamic of the instantaneous forward rate.

$$\begin{aligned}
r^{d|d}_{t,T} - r^{d|d}_{0,T} &= \int_0^t \sigma^{d|d,\top}_{u,T}\Sigma^{d|d}_{u,T}\,du + \int_0^t \sigma^{d|d,\top}_{u,T} dW^{d,\mathrm{RN}}_u \\
&= \int_0^t h^{d|d,\top}_T g^{d|d,\top}_u \int_u^T g^{d|d}_u h_v\,dv\,du + \int_0^t h^{d|d,\top}_T g^{d|d,\top}_u dW^{d,\mathrm{RN}}_u \\
&= r^{d|d}_{0,T} + h^{d|d,\top}_T \int_0^t g^{d|d,\top}_u g^{d|d}_u \int_u^T h_v\,dv\,du + h^{d|d,\top}_T \int_0^t g^{d|d,\top}_u dW^{d,\mathrm{RN}}_u
\end{aligned}$$

By writing $h^{d|d}_t = H^{d|d}_t \mathbf{1}_d$ and $g^{d|d}_t = \sigma^{d|d}_{z,t} H^{d|d,-1}_t$, this can be decomposed into

$$\begin{aligned}
r^{d|d}_{t,T} - r^{d|d}_{0,T} &= \mathbf{1}^\top_d H^{d|d}_T \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} \sigma^{d|d}_{z,u} H^{d|d,-1}_u \int_u^T h^{d|d}_v\,dv\,du \\
&\quad + \mathbf{1}^\top_d H^{d|d}_T \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} dW^{d,\mathrm{RN}}_u \\
&= \mathbf{1}^\top_d H^{d|d}_T \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} \sigma^{d|d}_{z,u} H^{d|d,-1}_u \int_u^T h^{d|d}_v\,dv\,du \\
&\quad + \mathbf{1}^\top_d H^{d|d}_T \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} dW^{d,\mathrm{RN}}_u
\end{aligned}$$

When $T = t$, we find the expression for the short rate:

$$\begin{aligned}
r^{d|d}_t - r^{d|d}_{0,t} &= \mathbf{1}^\top_d H^{d|d}_t \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} \sigma^{d|d}_{z,u} H^{d|d,-1}_u \int_u^t h^{d|d}_v\,dv\,du \\
&\quad + \mathbf{1}^\top_d H^{d|d}_t \int_0^t H^{d|d,-1}_u \sigma^{d|d,\top}_{z,u} dW^{d,\mathrm{RN}}_u \\
&= \mathbf{1}^\top_d z^{d|d}_t
\end{aligned}$$

Now coming back to the forward rate, we isolate the dependence in T and introduce $z_t^{d|d}$:

$$\begin{aligned}
& r_{t,T}^{d|d} - r_{0,T}^{d|d} \\
= \;& \mathbf{1}_d^\top H_T^{d|d} H_t^{d|d,-1} H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} \sigma_{z,u}^{d|d} H_u^{d|d,-1} \left(\int_u^t h_v^{d|d}\, dv + \int_t^T h_v^{d|d}\, dv \right) du \\
& + \; \mathbf{1}_d^\top H_T^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} dW_u^{d,\mathrm{RN}} \\
= \;& \mathbf{1}_d^\top H_T^{d|d} H_t^{d|d,-1} \left(H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} \sigma_{z,u}^{d|d} H_u^{d|d,-1} \int_u^t h_v^{d|d}\, dv\, du \right) \\
& + \; \mathbf{1}_d^\top H_T^{d|d} H_t^{d|d,-1} \left(H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} \sigma_{z,u}^{d|d} H_u^{d|d,-1}\, du\, H_t^{d|d} \right) \left(H_t^{d|d,-1} \int_t^T h_v^{d|d}\, dv \right) \\
& + \; \mathbf{1}_d^\top H_T^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} dW_u^{d,\mathrm{RN}} \\
= \;& M_{t,T}^{d|d,\top} \left(H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} \sigma_{z,u}^{d|d} H_u^{d|d,-1} \int_u^t h_v^{d|d}\, dv\, du + H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} dW_u^{d,\mathrm{RN}} \right) \\
& + \; M_{t,T}^{d|d,\top} \left(H_t^{d|d} \int_0^t H_u^{d|d,-1} \sigma_{z,u}^{d|d,\top} \sigma_{z,u}^{d|d} H_u^{d|d,-1}\, du\, H_t^{d|d} \right) \left(H_t^{d|d,-1} \int_t^T h_v^{d|d}\, dv \right) \\
= \;& M_{t,T}^{d|d,\top} \left(z_t^{d|d} + y_t^{d|d} G_{t,T}^{d|d} \right)
\end{aligned}$$

7.2.2. *Foreign rates.* Under $\mathbb{Q}^{d,\mathrm{RN}}$, the foreign forward rate has the extra quanto drift $-\sigma_{u,T}^{f,\top} \sigma_u^{df}$. With $\sigma_{u,T}^{f,\top} = M_{t,T}^{f,\top} \mathcal{M}_{u,t}^f \sigma_{z,u}^{f,\top}$, the same decomposition as for domestic rates gives

$$\begin{aligned}
r_{t,T}^{f|f} - r_{0,T}^{f|f} &= \int_0^t \sigma_{u,T}^{f,\top} \left(\Sigma_{u,T}^f - \sigma_u^{df} \right) du + \int_0^t \sigma_{u,T}^{f,\top} dW_u \\
&= M_{t,T}^{f,\top} \int_0^t \mathcal{M}_{u,t}^f \sigma_{z,u}^{f,\top} \left(\sigma_{z,u}^f \left(G_{u,t}^f + \mathcal{M}_{u,t}^f G_{t,T}^f \right) du - \sigma_u^{df}\, du + dW_u \right) \\
&= M_{t,T}^{f,\top} \left(z_t^f + y_t^{f,f} G_{t,T}^f \right).
\end{aligned}$$

The quanto term only contributes to z_t^f and does not depend on T beyond the factor $M_{t,T}^f$, which is why it does not require an additional state variable.

7.2.3. *Collateral basis.* The drift of $q_{t,T}^{f|f\to d}$ under $\mathbb{Q}^{d,\mathrm{RN}}$ is $\sigma_{u,T}^{c,\top}(\Sigma_{u,T}^c + \Sigma_{u,T}^f - \sigma_u^{df}) + \sigma_{u,T}^{f,\top} \Sigma_{u,T}^c$. The first part has maturity dependence $M_{t,T}^c$ and is handled exactly like the foreign rate:

$$\begin{aligned}
& \int_0^t \sigma_{u,T}^{c,\top} \left(\Sigma_{u,T}^c + \Sigma_{u,T}^f - \sigma_u^{df} \right) du + \int_0^t \sigma_{u,T}^{c,\top} dW_u \\
= \;& M_{t,T}^{c,\top} \int_0^t \mathcal{M}_{u,t}^c \sigma_{z,u}^{c,\top} \left(\sigma_{z,u}^c (G_{u,t}^c + \mathcal{M}_{u,t}^c G_{t,T}^c) du + \sigma_{z,u}^f (G_{u,t}^f + \mathcal{M}_{u,t}^f G_{t,T}^f) du - \sigma_u^{df}\, du + dW_u \right) \\
= \;& M_{t,T}^{c,\top} \left(z_t^c + y_t^{c,c} G_{t,T}^c + y_t^{c,f} G_{t,T}^f \right).
\end{aligned}$$

The second part has maturity dependence $M_{t,T}^f$:

$$\int_0^t \sigma_{u,T}^{f,\top} \Sigma_{u,T}^c\, du \;=\; M_{t,T}^{f,\top} \int_0^t \mathcal{M}_{u,t}^f \sigma_{z,u}^{f,\top} \sigma_{z,u}^c \left(G_{u,t}^c + \mathcal{M}_{u,t}^c G_{t,T}^c \right) du \;=\; M_{t,T}^{f,\top} \left(v_t^{f,c} + y_t^{f,c} G_{t,T}^c \right).$$

Summing the two gives the reconstruction formula for $q_{t,T}^{f|f\to d}$.

Remark 35. An alternative is to assume that $r^{f|d}_{t,T}$ rather than $q^{f|f\to d}_{t,T}$ has separable volatility. Since $\sigma^{f|d}_{t,T} = \sigma^{f}_{t,T} + \sigma^{c}_{t,T} = [\sigma^{f}_{z,t}|\sigma^{c}_{z,t}]\,[M^{f}_{t,T}; M^{c}_{t,T}]$, the two assumptions are the same. The corresponding Cheyette state for $r^{f|d}_{t,T}$ is the stacked state $\hat{z}_t$ of Remark 33.

7.2.4. *FX spot.* Since $\ln x^{df|d}_{0,t} = \ln x^{df}_0 + \int_0^t \left(r^{d|d}_{0,u} - r^{f|f}_{0,u} - q^{f|f\to d}_{0,u}\right) du$ and the FX spot has drift $r^{d|d}_t - r^{f|f}_t - q^{f\to d}_t$, we get

$$dz^x_t \;=\; \left((r^{d|d}_t - r^{d|d}_{0,t}) - (r^{f|f}_t - r^{f|f}_{0,t}) - (q^{f\to d}_t - q^{f|f\to d}_{0,t}) - \frac{1}{2}|\sigma^{df}_t|^2\right) dt + \sigma^{df,\top}_t dW_t$$

and we conclude with the reconstruction formulas for the short rates ($T = t$).

7.2.5. *Zero coupon bonds.* Since $\partial_T G_{t,T} = M_{t,T}$, for a symmetric matrix y we have $\int_t^T M^\top_{t,u} y G_{t,u}\, du = \frac{1}{2} G^\top_{t,T} y G_{t,T}$, and

$$\int_t^T \left(M^{c,\top}_{t,u} y^{c,f}_t G^f_{t,u} + M^{f,\top}_{t,u} y^{f,c}_t G^c_{t,u}\right) du \;=\; G^{c,\top}_{t,T} y^{c,f}_t G^f_{t,T}.$$

Integrating the instantaneous forward rates from t to T gives the ZCB formulas, with $Z_{0,t,T} = Z_{0,T}/Z_{0,t}$. Multiplying the formulas for $Z^{f|f}_{t,T}$ and $\Xi \mathrm{Z}^{f|f\to d}_{t,T}$ gives (185).

7.2.6. *Dynamic of the state variables.* The dynamic follows by differentiating the integral definitions with respect to t, using $\partial_t \mathcal{M}_{u,t} = -\beta_t \mathcal{M}_{u,t}$, $\partial_t G_{u,t} = \mathcal{M}_{u,t}\mathbf{1}$ and $G_{t,t} = 0$. For example,

$$\frac{d}{dt}\int_0^t \mathcal{M}^c_{u,t}\sigma^{c,\top}_{z,u}\sigma^f_{z,u} G^f_{u,t}\, du \;=\; -\beta^c_t \int_0^t \mathcal{M}^c_{u,t}\sigma^{c,\top}_{z,u}\sigma^f_{z,u} G^f_{u,t}\, du + \int_0^t \mathcal{M}^c_{u,t}\sigma^{c,\top}_{z,u}\sigma^f_{z,u}\mathcal{M}^f_{u,t}\, du\, \mathbf{1}_f,$$

which produces the term $y^{c,f}_t \mathbf{1}_f$ in the drift of z^c_t. This completes the proof of Theorem 4.

Remark 36 (Numerical verification)**.** The theorem was checked in the deterministic-volatility case (where the state is Gaussian) with two domestic factors, one foreign factor, one collateral basis factor and the FX factor. The parameters were piecewise constant with zero, 10^{-7} and negative mean reversions on some intervals, under a full 5×5 correlation matrix. The mean and covariance of $(z_t, \int_0^t (r^{d|d}_u - r^{d|d}_{0,u})\, du, \int_0^t (q^{f\to d}_u - q^{f|f\to d}_{0,u})\, du)$ were obtained by integrating the moment ODEs of the dynamic above. The three no-arbitrage identities

$$\mathbb{E}^{d,\mathrm{RN}}\left[e^{-\int_0^t r^{d|d}_u du} Z^{d|d}_{t,T}\right] = Z^{d|d}_{0,T}, \quad \mathbb{E}^{d,\mathrm{RN}}\left[e^{-\int_0^t r^{d|d}_u du} x^{df}_t Z^{f|d}_{t,T}\right] = x^{df}_0 Z^{f|d}_{0,T},$$
$$\mathbb{E}^{d,\mathrm{RN}}\left[e^{-\int_0^t (r^{d|d}_u - q^{f\to d}_u) du} x^{df}_t Z^{f|f}_{t,T}\right] = x^{df}_0 Z^{f|f}_{0,T},$$

then hold to machine precision (relative error below 10^{-15}) for $t \in \{0.8, 2.5, 7\}$ and $T - t \in \{0.25, 3, 15\}$. Dropping the quanto drift of z^f, the term $y^{c,f}\mathbf{1}_f$ or the term $\mathbf{1}_f^\top v^{f,c}$ breaks the second identity by $6.5 \cdot 10^{-2}$, $2.0 \cdot 10^{-3}$ and $8.3 \cdot 10^{-4}$ respectively.

7.3. **Reconstruction formulas for market observables.** All market observables of subsection 1.2 follow from Theorem 4. The reconstruction formulas for ZCBs give, for any collateral $i \in \{d, f\}$:

- FX forwards: $x^{df|i}_{t,T} = x^{df}_t \, Z^{f|i}_{t,T}/Z^{d|i}_{t,T}$. In particular

$$\ln \frac{x^{df|d}_{t,T}}{x^{df|d}_{0,T}} = z^x_t - \hat{G}^\top_{t,T}\hat{z}_t - \frac{1}{2}\hat{G}^\top_{t,T}\hat{y}_t\hat{G}_{t,T} + G^{d,\top}_{t,T}z^d_t + \frac{1}{2}G^{d,\top}_{t,T}y^{d,d}_t G^d_{t,T}. \tag{186}$$

- Forward discount rates: $1 + \theta R^{e|i}_{t,T;\theta} = Z^{e|i}_{t,T}/Z^{e|i}_{t,T+\theta}$.
- Native forward Libors, under the constant funding spread hypothesis: $1+\theta L^{e|e}_{t,T;\theta} = (1-\theta S^{e|e,\div}_{0,T;\theta})\, Z^{e|e}_{t,T}/Z^{e|e}_{t,T+\theta}$.
- Resetting forward Libors under domestic collateral: $1 + \theta L^{f/d|d}_{t,T;\theta} = (1 - \theta S^{d|d,\div}_{0,T;\theta})\, Z^{f|d}_{t,T}/Z^{f|d}_{t,T+\theta}$.
- Non-native forward Libors $L^{f|d}_{t,T;\theta} = \mathbb{E}^{f|d,T+\theta}_t[L^f_{T;\theta}]$ are not functions of the ZCBs. In the Quasi-Gaussian model the collateral convexity adjustment depends on the future path of the volatilities and has to be computed numerically. In the LGCM it is deterministic and given by the collateral convexity formula of the deterministic volatility section:

$$1 + \theta L^{f|d}_{t,T;\theta} = \left(1 + \theta L^{f|f}_{t,T;\theta}\right) \exp\left\{-\int_t^T \Sigma^{f|f}_{u,T;\theta} \cdot \Sigma^{f\to d}_{u,T+\theta}\, du\right\}. \tag{187}$$

  For a compounded overnight coupon, with the convention (21), $1 + \theta L^{f|d}_{t,a;\theta} = K^C_t(a,b)\, Z^{f|f}_{t,a}/Z^{f|f}_{t,b}$ for $t \le a$ (Lemma 1), where $K^C_t(a,b)$ is deterministic in the LGCM and given by Proposition 26: the exponential in (187) is the case K^{L}, and the daily and continuous coupons replace it by (138) and (139).
- Swap rates, currency basis spreads and annuities are then ratios of sums of the above.

7.4. **Gaussian par spread approximations.** The Gaussian approximations for swap rates and currency basis spreads are derived in the LGCM in subsection 8.3. They rely on freezing the gradient of the par spread with respect to the state variables. The same technique applies in the Quasi-Gaussian model, where the frozen gradient gives the volatility loading of the swap rate on $\hat{\sigma}_{z,t}$. Since y_t is then stochastic, the resulting swap rate dynamic is a stochastic volatility diffusion rather than a Gaussian one, cf. [1, §13].

8. The Linear Gaussian Collateral Model: Specification

8.0. **Introduction.** We now combine our two assumptions:

(1) spot volatilities are separable.
(2) spot volatilities are deterministic.

The resulting model, which we call the Linear Gaussian Collateral Model, is the natural generalisation of the classic Black-Scholes and Hull-White models to stochastic collateral/currency basis.

We get the best of both worlds:

(1) The model is Markovian. Moreover, compared to the Cheyette model, the variables y_t are deterministic so the dimensionality of the Markovian diffusion is greatly reduced[37].
(2) The model is Gaussian. It can be simulated exactly with arbitrary time steps and convexity adjustments are known in closed form.

Obviously, the main limitation is that the model only generates flat smiles[38].

8.1. Model specification.

8.1.1. *Notations.* For each symbol a, let us write

$$H_t^a = \mathrm{diag}(h_t^a) \tag{188}$$

$$\sigma_{z,s}^a = g_s^a H_s^a \tag{189}$$

$$\sigma_s^a = \mathrm{diag}\left(\sigma_{z,s}^{a,\top}\sigma_{z,s}^a\right)^{1/2} \tag{190}$$

$$M_{t,T}^a = H_t^{a,-1} H_T^a \mathbf{1} = H_t^{a,-1} h_T^a \tag{191}$$

$$G_{t,T}^a = \int_t^T M_{t,u}^a\, du = H_t^{a,-1}\int_t^T h_u^a\, du \tag{192}$$

$$\Omega_t = \sigma_t^{a,-1}\left(\sigma_{z,t}^{a,\top}\sigma_{z,t}^a\right)\sigma_t^{a,-1} \tag{193}$$

$$\Gamma_t^\top \Gamma_t = \Omega_t \tag{194}$$

Let us define the mean reversion speeds as the coefficients of the diagonal matrix:

$$\beta_t^a = -H_t^{a,-1}\frac{d}{dt}H_t^a \tag{195}$$

One can represent the matrices more explicitly:

$$H_t^a = \mathrm{diag}\left(e^{-\int_0^t \beta_u^{a,1}\,du},\dots,e^{-\int_0^t \beta_u^{a,N_a}\,du}\right) \tag{196}$$

$$\sigma_t^a = \mathrm{diag}\left(\sigma_t^{a,1},\dots,\sigma_t^{a,N_a}\right) \tag{197}$$

$$M_{t,T}^a = \left[e^{-\int_t^T \beta_u^{a,1}\,du},\dots,e^{-\int_t^T \beta_u^{a,N_a}\,du}\right]^\top \tag{198}$$

$$G_{t,T}^a = \left[\int_t^T e^{-\int_t^s \beta_u^{a,1}\,du}\,ds,\dots,\int_t^T e^{-\int_t^s \beta_u^{a,N_a}\,du}\,ds\right]^\top \tag{199}$$

8.1.2. *Model dynamic.* We parametrise the model by a set of correlated one dimensional OU state variables plus the FX state variable.

The model is driven by a k-dimensional Brownian motion $W_t^{\mathbb{P}}$ under the historical measure, and we write $W_t = W_t^{d,\mathrm{RN}}$ for the corresponding Brownian motion under the domestic risk-neutral measure. Let us write

[37] From $\frac{M(M+1)}{2} + N_x$ to $M + N_x$ where $M = N_d + N_f + N_c$.
[38] It is actually trivial to generate skew by replacing lognormal dynamics by shifted ones.

$\sigma_{z,t} = [\sigma_{z,t}^{d|d}|\sigma_{z,t}^{f|f}|\sigma_{z,t}^{f\to d}|\sigma_t^{df}] \in M_{k,N}(\mathbb{R})$, $N = N_d + N_f + N_c + 1$, for the matrix-valued volatility of the state variables. The vols $\sigma_{z,t}^a$ are deterministic. Let us set

$$\sigma_t = \operatorname{diag}\left(\sigma_{z,t}^\top \sigma_{z,t}\right)^{1/2} = \operatorname{diag}\left(\sigma_t^{d,1}, \dots, \sigma_t^{d,N_d}, \sigma_t^{f,1}, \dots, \sigma_t^{c,N_c}, \sigma_t^x\right) \tag{200}$$

$$\Omega_t = \sigma_t^{-1}\left(\sigma_{z,t}^\top \sigma_{z,t}\right)\sigma_t^{-1}. \tag{201}$$

Here $\sigma_t^x = |\sigma_t^{df}|$ is the FX spot volatility.

We define a correlated N-dimensional Brownian motion $\widetilde{W}$:

$$d\widetilde{W}_t = \Gamma_t^\top dW_t \tag{202}$$

where the factor loading matrix Γ_t is such that $\Gamma_t^\top \Gamma_t = \Omega_t$. Its components are denoted by $\widetilde{W}_t^{(a,i)}$, $a \in \{d, f, c\}$, $i = 1, \dots, N_a$, and $\widetilde{W}_t^x$, and $d\langle \widetilde{W}^{(a,i)}, \widetilde{W}^{(b,j)}\rangle_t = \rho_t^{(a,i),(b,j)} dt$. In this notation, the forward rate volatilities of Theorem 4 read

$$\phi_{t,T}^\top dW_t = \sum_{i=1}^{N_a} \sigma_t^{a,i} M_{t,T}^{a,i} d\widetilde{W}_t^{(a,i)}, \qquad \Sigma_{t,T}^{a,\top} dW_t = \sum_{i=1}^{N_a} \sigma_t^{a,i} G_{t,T}^{a,i} d\widetilde{W}_t^{(a,i)}, \qquad \sigma_t^{df,\top} dW_t = \sigma_t^x d\widetilde{W}_t^x. \tag{203}$$

We also write $\Pi_a : \mathbb{R}^N \to \mathbb{R}^{N_a}$ for the projection onto the bloc a. And for $M \in M_N(\mathbb{R})$, we write $M^{a,b} = \Pi_a M \Pi_b^\top$ for its (a, b) bloc. For example, the correlation matrix Ω_t admits the following bloc decomposition[39]:

$$\Omega_t = \Gamma_t^\top \Gamma_t = \begin{bmatrix} \Omega_t^{d,d} & \Omega_t^{d,f} & \Omega_t^{d,c} & \Omega_t^{d,x} \\ \Omega_t^{f,d} & \Omega_t^{f,f} & \Omega_t^{f,c} & \Omega_t^{f,x} \\ \Omega_t^{c,d} & \Omega_t^{c,f} & \Omega_t^{c,c} & \Omega_t^{c,x} \\ \Omega_t^{x,d} & \Omega_t^{x,f} & \Omega_t^{x,c} & \Omega_t^{x,x} \end{bmatrix} \tag{204}$$

where each bloc looks like:

$$\Omega_t^{a,b} = \begin{bmatrix} \rho_t^{(a,1),(b,1)} & \rho_t^{(a,1),(b,2)} & \cdots & \rho_t^{(a,1),(b,N_b)} \\ \rho_t^{(a,2),(b,1)} & \rho_t^{(a,2),(b,2)} & & \rho_t^{(a,2),(b,N_b)} \\ \vdots & & \ddots & \vdots \\ \rho_t^{(a,N_a),(b,1)} & \cdots & \cdots & \rho_t^{(a,N_a),(b,N_b)} \end{bmatrix} \tag{205}$$

Note that we don't require the matrix Ω_t to be definite: we can have $\Gamma_t \in M_{k,N}(\mathbb{R})$ with $k < N$.

Definition 5. The *Linear Gaussian Collateral Model* (LGCM) is defined by

(1) The following deterministic parameters:
   (a) IRD: spot volatilities $(\sigma_t^{d,j})$ and mean-reversions $(\beta_t^{d,j})$ for $j = 1, \dots, N_d$.
   (b) IRF: spot volatilities $(\sigma_t^{f,j})$ and mean-reversions $(\beta_t^{f,j})$ for $j = 1, \dots, N_f$.
   (c) CB: spot volatilities $(\sigma_t^{c,j})$ and mean-reversions $(\beta_t^{c,j})$ for $j = 1, \dots, N_c$.
   (d) FX: spot volatility σ_t^x.
   (e) Correlations: instantaneous correlations between Gaussian drivers $(\rho_t^{(a,j_a),(b,j_b)})$.

(2) The deterministic functions, for $a, b \in \{d, f, c\}$,

$$y_t^{(a,i),(b,j)} = \int_0^t \rho_u^{(a,i),(b,j)} \sigma_u^{a,i} \sigma_u^{b,j} M_{u,t}^{a,i} M_{u,t}^{b,j} \, du \tag{206}$$

$$v_t^{f,i} = \sum_{j=1}^{N_c} \int_0^t \rho_u^{(f,i),(c,j)} \sigma_u^{f,i} \sigma_u^{c,j} M_{u,t}^{f,i} G_{u,t}^{c,j} \, du \tag{207}$$

(the deterministic versions of $y_t^{a,b}$ and $v_t^{f,c}$ in Theorem 4).

[39] We use c as short for $f \to d$ and x as short for df.

(3) The state process $z_t = [z_t^d | z_t^f | z_t^c | z_t^x]^\top$, $z_0 = 0$, with the following dynamic under $\mathbb{Q}^{d,\mathrm{RN}}$:

$$dz_t^{d,i} = \left(\sum_{j=1}^{N_d} y_t^{(d,i),(d,j)} - \beta_t^{d,i} z_t^{d,i} \right) dt + \sigma_t^{d,i} \, d\widetilde{W}_t^{(d,i)} \tag{208}$$

$$dz_t^{f,i} = \left(\sum_{j=1}^{N_f} y_t^{(f,i),(f,j)} - \rho_t^{(f,i),x} \sigma_t^{f,i} \sigma_t^x - \beta_t^{f,i} z_t^{f,i} \right) dt + \sigma_t^{f,i} \, d\widetilde{W}_t^{(f,i)} \tag{209}$$

$$dz_t^{c,i} = \left(\sum_{j=1}^{N_c} y_t^{(c,i),(c,j)} + \sum_{j=1}^{N_f} y_t^{(c,i),(f,j)} - \rho_t^{(c,i),x} \sigma_t^{c,i} \sigma_t^x - \beta_t^{c,i} z_t^{c,i} \right) dt + \sigma_t^{c,i} \, d\widetilde{W}_t^{(c,i)} \tag{210}$$

$$dz_t^x = \left(\sum_{i=1}^{N_d} z_t^{d,i} - \sum_{i=1}^{N_f} \left(z_t^{f,i} + v_t^{f,i} \right) - \sum_{i=1}^{N_c} z_t^{c,i} - \frac{1}{2} (\sigma_t^x)^2 \right) dt + \sigma_t^x \, d\widetilde{W}_t^x. \tag{211}$$

(4) Reconstruction formulas, with $\hat{z}_t = [z_t^f + v_t^{f,c}; z_t^c]$, $\hat{G}_{t,T} = [G_{t,T}^f; G_{t,T}^c]$ and $\hat{y}_t$ the (f,c) bloc of y_t:

$$Z_{t,T}^{d|d} = Z_{0,t,T}^{d|d} \exp\left(-G_{t,T}^{d,\top} z_t^d - \frac{1}{2} G_{t,T}^{d,\top} y_t^{d,d} G_{t,T}^d \right) \tag{212}$$

$$Z_{t,T}^{f|f} = Z_{0,t,T}^{f|f} \exp\left(-G_{t,T}^{f,\top} z_t^f - \frac{1}{2} G_{t,T}^{f,\top} y_t^{f,f} G_{t,T}^f \right) \tag{213}$$

$$Z_{t,T}^{f|d} = Z_{0,t,T}^{f|d} \exp\left(-\hat{G}_{t,T}^\top \hat{z}_t - \frac{1}{2} \hat{G}_{t,T}^\top \hat{y}_t \hat{G}_{t,T} \right) \tag{214}$$

$$x_t^{df} = x_{0,t}^{df|d} \, e^{z_t^x} = x_0^{df} \frac{Z_{0,t}^{f|d}}{Z_{0,t}^{d|d}} \, e^{z_t^x}. \tag{215}$$

Remark 37. By Theorem 4, the LGCM is arbitrage free and reproduces the initial curves $Z_{0,\cdot}^{d|d}$, $Z_{0,\cdot}^{f|f}$, $Z_{0,\cdot}^{f|d}$ and the FX spot x_0^{df} for any choice of the parameters. The other discount curves follow from the reconstruction formulas: $Z_{t,T}^{d|f}$ from the change of collateral formula, and $\Xi \mathrm{Z}_{t,T}^{f|f\to d} = Z_{t,T}^{f|d} / Z_{t,T}^{f|f}$. Compared to the Quasi-Gaussian model, the only stochastic state variables are the $N-1$ OU factors $z_t^{a,i}$ and the log FX moneyness z_t^x. The y_t and $v_t^{f,c}$ are precomputed by the covariance integral algorithm of subsection 9.1.

Remark 38 (Stationarity)**.** Remember that, while we specify the dynamic under the domestic risk-neutral measure, we assumed that this measure is equivalent to the historical one. In particular, we expect delivered covariances in a well specified and calibrated model to be close to historically estimated ones.

Since statistical estimation relies on the stationarity of the stochastic process, our model should be stationary. This requires coefficients such mean reversion speeds to be constant. However, for numerical reasons, one often have to rely on term structure of parameters to match market prices. For this reason, we will keep deriving the formulas in the general setting.

8.1.3. *Dynamic of the the collateral basis.* We assume a Linear Gaussian dynamic for the instantaneous forward collateral basis $q_{t,T}^{f|f\to d}$, i.e. separable and deterministic volatility

$$\sigma_{t,T}^{f\to d,\top} dW_t = \sum_{i=1}^{N_c} \sigma_t^{c,i} M_{t,T}^{c,i} \, d\widetilde{W}_t^{(c,i)}. \tag{216}$$

Equivalently, the cross-currency discount rates $r_{t,T}^{f|d}$ follow a Linear Gaussian dynamic with $N_f + N_c$ factors, the foreign native ones and the collateral basis ones (see the remark on the cross-currency discount rate in the proof of Theorem 4).

Proposition 29. In the LGCM, the forward collateral basis has dynamic under $\mathbb{Q}^{d,\mathrm{RN}}$

$$\begin{aligned} dq_{t,T}^{f|f\to d} &= \sum_{i=1}^{N_c}\sigma_t^{c,i}M_{t,T}^{c,i}\left(\sum_{j=1}^{N_c}\rho_t^{(c,i),(c,j)}\sigma_t^{c,j}G_{t,T}^{c,j}+\sum_{j=1}^{N_f}\rho_t^{(c,i),(f,j)}\sigma_t^{f,j}G_{t,T}^{f,j}-\rho_t^{(c,i),x}\sigma_t^{x}\right)dt \\ &\quad +\sum_{i=1}^{N_f}\sum_{j=1}^{N_c}\rho_t^{(f,i),(c,j)}\sigma_t^{f,i}\sigma_t^{c,j}M_{t,T}^{f,i}G_{t,T}^{c,j}\,dt+\sum_{i=1}^{N_c}\sigma_t^{c,i}M_{t,T}^{c,i}\,d\widetilde{W}_t^{(c,i)} \end{aligned} \tag{217}$$

and is reconstructed from the state variables by

$$\begin{aligned} q_{t,T}^{f|f\to d} &= q_{0,T}^{f|f\to d}+\sum_{i=1}^{N_c}M_{t,T}^{c,i}\left(z_t^{c,i}+\sum_{j=1}^{N_c}y_t^{(c,i),(c,j)}G_{t,T}^{c,j}+\sum_{j=1}^{N_f}y_t^{(c,i),(f,j)}G_{t,T}^{f,j}\right) \\ &\quad +\sum_{i=1}^{N_f}M_{t,T}^{f,i}\left(v_t^{f,i}+\sum_{j=1}^{N_c}y_t^{(f,i),(c,j)}G_{t,T}^{c,j}\right) \end{aligned} \tag{218}$$

$$\Xi \mathrm{Z}_{t,T}^{f|f\to d} = \Xi \mathrm{Z}_{0,t,T}^{f|f\to d}\exp\left(-G_{t,T}^{c,\top}z_t^{c}-G_{t,T}^{f,\top}v_t^{f,c}-\frac{1}{2}G_{t,T}^{c,\top}y_t^{c,c}G_{t,T}^{c}-G_{t,T}^{c,\top}y_t^{c,f}G_{t,T}^{f}\right). \tag{219}$$

In particular, the short collateral basis is $q_t^{f\to d}=q_{0,t}^{f|f\to d}+\sum_i z_t^{c,i}+\sum_i v_t^{f,i}$.

Proof. The dynamic is the HJM dynamic of the collateral basis written with the correlated drivers. The reconstruction formulas are those of Theorem 4. □

Remark 39. The collateral basis has three sources of drift: its own convexity (Σ^c), the convexity with foreign rates (Σ^f, through both $\sigma^c\cdot\Sigma^f$ and $\sigma^f\cdot\Sigma^c$) and the quanto adjustment (σ^x). The second one is what makes the deterministic shift $v_t^{f,c}$ appear. It vanishes when foreign rates and collateral basis are uncorrelated, and then $q^{f|f\to d}$ is driven by the z^c factors only.

8.1.4. *Dynamic of the FX spot.*

Proposition 30. In the LGCM, the FX spot has dynamic under $\mathbb{Q}^{d,\mathrm{RN}}$

$$\frac{dx_t^{df}}{x_t^{df}} = \left(r_t^{d|d}-r_t^{f|f}-q_t^{f\to d}\right)dt+\sigma_t^{x}\,d\widetilde{W}_t^{x}, \tag{220}$$

where the short rates are $r_t^{d|d}=r_{0,t}^{d|d}+\sum_i z_t^{d,i}$, $r_t^{f|f}=r_{0,t}^{f|f}+\sum_i z_t^{f,i}$ and $q_t^{f\to d}$ is given above. The FX forward under collateral $i\in\{d,f\}$ is reconstructed by $x_{t,T}^{df|i}=x_t^{df}Z_{t,T}^{f|i}/Z_{t,T}^{d|i}$, and for $i=d$

$$\ln\frac{x_{t,T}^{df|d}}{x_{0,T}^{df|d}} = z_t^{x}-\hat{G}_{t,T}^{\top}\hat{z}_t-\frac{1}{2}\hat{G}_{t,T}^{\top}\hat{y}_t\hat{G}_{t,T}+G_{t,T}^{d,\top}z_t^{d}+\frac{1}{2}G_{t,T}^{d,\top}y_t^{d,d}G_{t,T}^{d}. \tag{221}$$

It is a lognormal $\mathbb{Q}^{d|d,T}$-martingale with volatility

$$\Sigma_{t,T}^{df,\top}dW_t = \sum_{i=1}^{N_d}\sigma_t^{d,i}G_{t,T}^{d,i}d\widetilde{W}_t^{(d,i)}-\sum_{i=1}^{N_f}\sigma_t^{f,i}G_{t,T}^{f,i}d\widetilde{W}_t^{(f,i)}-\sum_{i=1}^{N_c}\sigma_t^{c,i}G_{t,T}^{c,i}d\widetilde{W}_t^{(c,i)}+\sigma_t^{x}\,d\widetilde{W}_t^{x}, \tag{222}$$

so that the total implied variance of an FX option of expiry T is $w_0^{df}(T)=\int_0^T|\Sigma_{u,T}^{df}|^2du$, a sum of covariance integrals.

Proof. The FX dynamic is that of the HJM section. The FX forward follows from the reconstruction formulas and $x_t^{df}=x_{0,t}^{df|d}e^{z_t^x}$, using $x_{0,T}^{df|d}=x_{0,t}^{df|d}Z_{0,t,T}^{f|d}/Z_{0,t,T}^{d|d}$. Its volatility is given by the proposition on FX forwards in the HJM section. □

Remark 40. The FX spot volatility σ_t^x is the only FX parameter. The FX forward volatility also contains the rates and collateral basis volatilities through $G_{t,T}^{a,i}$, which grow with the time to expiry. For long expiries, they can account for a large part of the FX implied volatility, which is why bootstrapping σ_t^x to FX implied volatilities can fail (variance squeeze, see the remark on the FX options market in the calibration section).

8.1.5. *Correlation parametrisation.* The correlation matrix Ω_t has $N(N-1)/2$ entries, which is too many to be calibrated or estimated freely. We organise them by the blocs of Ω_t and by the quantities they drive in the model:

Bloc	Main impact
$\Omega^{d,d}$, $\Omega^{f,f}$	native swaption smiles: decorrelation of swap rates, calibrated in the single currency LGMs
$\Omega^{c,c}$	decorrelation of currency basis spreads of different tenors
$\Omega^{f,c}$	coupon collateral convexity K_j (Libor or compounded), drift of $q^{f\|f\to d}$, $v_t^{f,c}$
$\Omega^{d,x}$, $\Omega^{f,x}$, $\Omega^{c,x}$	quanto drifts, FX forward collateral convexity, FX option variance
$\Omega^{d,f}$, $\Omega^{d,c}$	joint distribution of domestic rates with foreign rates and basis (e.g. CVA)

The table follows from the formulas above. For example, $\Omega^{f,c}$ enters the Libor collateral convexity $\exp\{-\int \Sigma_{u,T;\theta}^{f|f} \cdot \Sigma_{u,T+\theta}^{f\to d} du\}$ and its compounded analogues (Proposition 26), and $\Omega^{c,x}$ enters the FX forward collateral convexity $\exp\{-\int \Sigma_{u,T}^{df} \cdot \Sigma_{u,T}^{i\to j} du\}$.

A natural parametrisation is the following:

(1) The diagonal blocs $\Omega^{d,d}$ and $\Omega^{f,f}$ are inherited from the single currency LGMs, which are calibrated independently to their native markets.
(2) The diagonal bloc $\Omega^{c,c}$ and the off diagonal blocs are marked, or estimated from historical time series of the corresponding market quotes (see the section on historical estimation). They are typically parametrised by a few numbers, e.g. one correlation per pair of blocs applied to the first factors.
(3) The resulting matrix is not necessarily positive semi-definite. It has to be projected onto the set of correlation matrices, for example by the nearest correlation matrix in Frobenius norm, or by flooring the negative eigenvalues of Ω_t at 0 and renormalising the diagonal. Alternatively, one parametrises the factor loading matrix Γ_t directly (e.g. by angles on the unit sphere for each column), which guarantees $\Omega_t = \Gamma_t^\top \Gamma_t$ is a correlation matrix and allows a rank $k < N$.

The projection modifies the native blocs in general. If the native calibrations must be preserved exactly, the projection should keep $\Omega^{d,d}$ and $\Omega^{f,f}$ fixed and only adjust the other blocs.

8.1.6. *Joint dynamic.* The LGCM is a linear Gaussian model. We write $X_t = [z_t^d | z_t^f | z_t^c | z_t^x]^\top$.

Proposition 31 (Joint law of the state)**.** Under $\mathbb{Q}^{d,\mathrm{RN}}$, X_t solves the linear SDE

$$dX_t = (A_t X_t + b_t)\, dt + \sigma_t\, d\widetilde{W}_t \tag{223}$$

where σ_t is the diagonal matrix of spot volatilities,

$$A_t = \begin{bmatrix} -\beta_t^d & 0 & 0 & 0 \\ 0 & -\beta_t^f & 0 & 0 \\ 0 & 0 & -\beta_t^c & 0 \\ \mathbf{1}_d^\top & -\mathbf{1}_f^\top & -\mathbf{1}_c^\top & 0 \end{bmatrix}, \qquad b_t = \begin{bmatrix} y_t^{d,d}\mathbf{1}_d \\ y_t^{f,f}\mathbf{1}_f - \sigma_t^f \Omega_t^{f,x} \sigma_t^x \\ y_t^{c,c}\mathbf{1}_c + y_t^{c,f}\mathbf{1}_f - \sigma_t^c \Omega_t^{c,x} \sigma_t^x \\ -\mathbf{1}_f^\top v_t^{f,c} - \frac{1}{2}(\sigma_t^x)^2 \end{bmatrix}.$$

X_t is Gaussian with mean m_t and covariance matrix C_t given by covariance integrals (with the notations of subsection 9.1):

$$\begin{aligned} m_t^{a,i} &= \sum_j J_{0,t}[M_{\cdot,t}^{a,i}, G_{\cdot,t}^{a,j}] + \mathbf{1}_{a=c} \sum_j J_{0,t}[M_{\cdot,t}^{c,i}, G_{\cdot,t}^{f,j}] - \mathbf{1}_{a\in\{f,c\}} J_{0,t}[M_{\cdot,t}^{a,i}, 1^x] \\ m_t^x &= -\frac{1}{2} J_{0,t}[\Sigma_{\cdot,t}^{df}, \Sigma_{\cdot,t}^{df}] + J_{0,t}[\Sigma_{\cdot,t}^{df}, \Sigma_{\cdot,t}^{d}] \\ C_t^{(a,i),(b,j)} &= y_t^{(a,i),(b,j)}, \qquad C_t^{(a,i),x} = J_{0,t}[M_{\cdot,t}^{a,i}, \Sigma_{\cdot,t}^{df}], \qquad C_t^{x,x} = J_{0,t}[\Sigma_{\cdot,t}^{df}, \Sigma_{\cdot,t}^{df}] = w_0^{df}(t). \end{aligned}$$

Proof. The SDE is 5 in matrix form. The law is the case $s = 0$ of Proposition 46, where $z_0 = 0$ and $y_0 = 0$. □

Remark 41. (1) The covariance of the rates factors is y_t itself: y_t is the covariance matrix of the OU factors, which is why it is called the covariance process in the Quasi-Gaussian model.

(2) Equivalently, $z_t^{a,i} = m_t^{a,i} + \xi_t^{a,i}$ where $\xi_t^{a,i} = \int_0^t \sigma_u^{a,i} M_{u,t}^{a,i} d\widetilde{W}_u^{(a,i)}$ are centred correlated one dimensional OU processes, $d\xi_t^{a,i} = -\beta_t^{a,i}\xi_t^{a,i}dt + \sigma_t^{a,i} d\widetilde{W}_t^{(a,i)}$. All the drifts are in the deterministic functions m_t.

(3) The formulas were checked against the moment ODEs of the linear SDE, with the model of the verification of Theorem 4, at $t = 0.8$ and $t = 7$: the means and covariances agree to $3 \cdot 10^{-16}$ (absolute), for means of order 10^{-2} and an FX variance of order 10^{-1}.

(4) The transitions between two dates and the drifts under the forward and jumping numéraire measures are given in subsection 9.4.

8.2. **Reconstruction of market observables.** We now derive the dynamic of market observables in terms of correlated Brownian motions.

8.2.1. *Forward discount rates.*

Proposition 32. The dynamic of forward discount rates is[40]:

$$\begin{aligned}\frac{d(1+\theta R^{e|i}_{t,T;\theta})}{1+\theta R^{e|i}_{t,T;\theta}} &= \sum_{k=1}^{N_e}\Sigma^{e|i}_{t,T;\theta}dW^{e|i,T+\theta,k}_t\\ &= \sum_{k=1}^{N_e}G^{e|e,k}_{t,T;\theta}\sigma^{e|e,k}_t d\widetilde{W}^{e|i,T+\theta,(e,k)}_t + \sum_{k=1}^{N_{e\to i}}G^{e\to i,k}_{t,T;\theta}\sigma^{e\to i,k}_t d\widetilde{W}^{e|i,T+\theta,(e\to i,k)}_t \end{aligned} \tag{224}$$

8.2.2. *Forward Libors.* [FIXME: Make the parametrisation explicit.]

Proposition 33. The dynamic of forward Libors is

$$\begin{aligned}\frac{d(1+\theta L^{e|i}_{t,T;\theta})}{1+\theta L^{e|i}_{t,T;\theta}} &= \Sigma^{e|e}_{t,T;\theta}dW^{e|i,T+\theta}_t\\ &= \sum_{k=1}^{N_e}G^{e|e,k}_{t,T;\theta}\sigma^{e|e,k}_t d\widetilde{W}^{e|i,T+\theta,(e,k)}_t \end{aligned} \tag{225}$$

8.2.3. *FX forwards.*

Proposition 34. The FX forward has dynamic

$$\begin{aligned}\frac{dx^{df|i}_{t,T}}{x^{df|i}_{t,T}} &= \left(\Sigma^{d|d}_{t,T}-\Sigma^{f|f}_{t,T}-\Sigma^{f\to d}_{t,T}+\sigma^{df}_t\right)\cdot dW^{d|i,T}_t\\ &= \sum_{k_d=1}^{N_d}G^{d|d,k_d}_{t,T}\sigma^{d|d,k_d}_t\, d\widetilde{W}^{d|i,T,(d,k_d)}_t - \sum_{k_f=1}^{N_f}G^{f|f,k_f}_{t,T}\sigma^{f|f,k_f}_t\, d\widetilde{W}^{d|i,T,(f,k_f)}_t\\ &\quad - \sum_{k_c=1}^{N_c}G^{f\to d,k_c}_{t,T}\sigma^{f\to d,k_c}_t\, d\widetilde{W}^{d|i,T,(f\to,k_c)}_t + \sum_{k_x=1}^{N_x}\sigma^{df,k_x}_t\, d\widetilde{W}^{d|i,T,(df,k_x)}_t. \end{aligned} \tag{226}$$

8.2.4. *Funding spreads.*

Proposition 35. The funding spreads have dynamic

$$\begin{aligned}\frac{d(1-\theta S^{e|i,\div}_{t,T;\theta})}{1-\theta S^{e|i}_{t,T;\theta}} &= -\Sigma^{e\to i}_{t,T;\theta}\cdot dW^{e|i,T}_t\\ &= -\sum_{k=1}^{N_e}G^{e\to i,k}_{t,T;\theta}\sigma^{e\to i,k}_t d\widetilde{W}^{e|i,T,(e\to i,k)}_t. \end{aligned} \tag{227}$$

[40]Here $\Sigma^a_{t,T;\theta}=\Sigma^a_{t,T+\theta}-\Sigma^a_{t,T}$ has components $\sigma^{a,k}_t G^{a,k}_{t,T;\theta}$ with

$$G^{a,k}_{t,T;\theta} \;=\; G^{a,k}_{t,T+\theta}-G^{a,k}_{t,T} \;=\; \int_T^{T+\theta} e^{-\int_t^u \beta^{a,k}_v\,dv}\,du \;=\; M^{a,k}_{t,T}\,G^{a,k}_{T,T+\theta}.$$

So $G^{a,k}_{t,T;\theta}$ is *not* equal to $G^{a,k}_{T,T+\theta}$: the missing amortising factor is $M^{a,k}_{t,T}=e^{-\int_t^T\beta^{a,k}_v dv}$. An earlier computation found a different expression because it used $\int_t^T e^{-\int_u^T\beta_v dv}du$, which is $\int_t^T M_{u,T}\,du$ rather than $G_{t,T}=\int_t^T M_{t,u}\,du$; the two agree only for constant mean reversion.

8.2.5. *Resetting Forward Libors.*

Proposition 36. The resetting forward Libor has dynamic

$$\frac{d(1+\theta L^{f/d|i,r}_{t,T;\theta})}{1+\theta L^{f/d|i,r}_{t,T;\theta}} = \Sigma^{f|d}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t \tag{228}$$

8.2.6. *Currency Basis Spreads.*

Proposition 37. The dynamic of the currency basis spread can be approximated by

$$B^{f/d|c,r}_{t,T;\theta} \approx B^{f/d|c,r}_{t_0,T;\theta} + \frac{1+\theta L^{f/d|c,r}_{t_0,T;\theta}}{\theta}\int_{t_0}^t \Sigma^{f\to d}_{u,T;\theta}\cdot dW^{f|c,T+\theta}_u + B^{f/d|c,r}_{t_0,T;\theta}\int_{t_0}^t \Sigma^{f|f}_{u,T;\theta}\cdot dW^{f|c,T+\theta}_u \tag{229}$$

Remark 42. Note that

$$L^{f/d|c,r}_{t_0,T;\theta} = L^{f|d}_{t_0,T;\theta} + B^{f/d|c,r}_{t_0,T;\theta} \tag{230}$$

so that

$$L^{f/d|c,r}_{t_0,T;\theta} = B^{f/d|c,r}_{t_0,T;\theta} + \frac{1}{\theta}\left((1+\theta L^{f|f}_{t_0,T;\theta})e^{-\int_{t_0}^T \Sigma^{f|f}_{u,T;\theta}\cdot\Sigma^{f\to d}_{u,T+\theta}\,du} - 1\right) \tag{231}$$

so the volatility of the currency basis spread can be explicitly computed from the foreign Libor, the currency basis spread itself and model parameters.

Remark 43. The first volatility term $\frac{1+\theta L^{f/d|c,r}_{t_0,T;\theta}}{\theta}\Sigma^{f\to d}_{u,T;\theta}$ corresponds to the collateral basis volatility. We expect it to be the the most significant one since the coefficient is of the order $1/\theta = 4$.

However, in periods of stress such as the Argentinian financial crisis, the currency basis spread will increase and the volatility of foreign rates will rise. As a result, the second term will make a material contribution to the realised volatility of the currency basis spread[41].

[41]Note however that if foreign rates volatility rises too high, the normal approximation for Libors may no longer be accurate.

8.3. **Gaussian par spread approximations.**

8.3.1. *Dynamic of swap rates.*

Proposition 38. The annuity has dynamic:

$$dA^{x|c}_{t,T;M} = \sum_{j=1}^{\ell M} \theta_i Z^{x|c}_{t,T_j} \left(r^{x|c}_t \, dt - \Sigma^{x|c,\top}_{t,T_j} dW^{x,\mathrm{RN}}_t \right)$$

Proposition 39. The floating leg has dynamic:

$$\begin{aligned} d\mathrm{Fl}^{x|c}_{t,T;M} &= \sum_{j=1}^{\ell M} \theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta} \left(r^{x|c}_t \, dt - \Sigma^{x|c,\top}_{t,T_j} dW^{x,\mathrm{RN}}_t \right) \\ &\quad + \sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j} (1 + \theta_j L^{x|c}_{t,T_{j-1};\theta}) \Sigma^{x|x,\top}_{t,T_{j-1};\theta} dW^{x|c,T_j}_t + O(dt) \end{aligned}$$

Proof.

$$\begin{aligned} \mathrm{Fl}^{x|c}_{t,T;M} &= \sum_{j=1}^{\ell M} \theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta} \\ d\mathrm{Fl}^{x|c}_{t,T;M} &= \sum_{j=1}^{\ell M} \theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta} \frac{dZ^{x|c}_{t,T_j}}{Z^{x|c}_{t,T_j}} + \sum_{j=1}^{\ell M} \theta_j Z^{x|c}_{t,T_j} dL^{x|c}_{t,T_{j-1};\theta} + O(dt) \\ &= \sum_{j=1}^{\ell M} \theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta} \left(r^{x|c}_t \, dt - \Sigma^{x|c,\top}_{t,T_j} dW^{x,\mathrm{RN}}_t \right) \\ &\quad + \sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j} (1 + \theta_j L^{x|c}_{t,T_{j-1};\theta}) \Sigma^{x|x,\top}_{t,T_{j-1};\theta} dW^{x|c,T_j}_t + O(dt) \end{aligned}$$

□

Proposition 40. The forward swap rate has dynamic:

$$dY^{x|c}_{t,T;M} = \frac{1}{A^{x|c}_{t,T;M}} \sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j} \left\{ (1 + \theta_j L^{x|c}_{t,T_{j-1};\theta}) \Sigma^{x|x}_{t,T_{j-1};\theta} + \theta_j (Y^{x|c}_{t,T;M} - L^{x|c}_{t,T_{j-1};\theta}) \Sigma^{x|c}_{t,T_j} \right\}^\top dW^{x|c,T;M}_t$$

Proof.

$$
\begin{aligned}
dY^{x|c}_{t,T;M} &= \frac{d\mathrm{Fl}^{x|c}_{t,T;M}}{A^{x|c}_{t,T;M}} - Y^{x|c}_{t,T;M}\frac{dA^{x|c}_{t,T;M}}{A^{x|c}_{t,T;M}} + O(dt) \\
&= -\frac{1}{A^{x|c}_{t,T;M}}\sum_{j=1}^{\ell M}\theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta}\Sigma^{x|c,\top}_{t,T_j} dW^{x,\mathrm{RN}}_t \\
&\quad + \frac{1}{A^{x|c}_{t,T;M}}\sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j}(1+\theta_j L^{x|c}_{t,T_{j-1};\theta})\Sigma^{x|x,\top}_{t,T_{j-1};\theta} dW^{x|c,T_j}_t \\
&\quad + Y^{x|c}_{t,T;M}\sum_{j=1}^{\ell M}\theta_j Z^{x|c}_{t,T_j}\Sigma^{x|c,\top}_{t,T_j} dW^{x,\mathrm{RN}}_t + O(dt) \\
&= -\frac{1}{A^{x|c}_{t,T;M}}\sum_{j=1}^{\ell M}\theta_j Z^{x|c}_{t,T_j} L^{x|c}_{t,T_{j-1};\theta}\Sigma^{x|c,\top}_{t,T_j} dW^{x|c,T;M}_t \\
&\quad + \frac{1}{A^{x|c}_{t,T;M}}\sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j}(1+\theta_j L^{x|c}_{t,T_{j-1};\theta})\Sigma^{x|x,\top}_{t,T_{j-1};\theta} dW^{x|c,T;M}_t \\
&\quad + Y^{x|c}_{t,T;M}\sum_{j=1}^{\ell M}\theta_j Z^{x|c}_{t,T_j}\Sigma^{x|c,\top}_{t,T_j} dW^{x|c,T;M}_t \\
&= \frac{1}{A^{x|c}_{t,T;M}}\sum_{j=1}^{\ell M} Z^{x|c}_{t,T_j}\left\{\theta_j Y^{x|c}_{t,T;M}\Sigma^{x|c}_{t,T_j} + (1+\theta_j L^{x|c}_{t,T_{j-1};\theta})\Sigma^{x|x}_{t,T_{j-1};\theta} - L^{x|c}_{t,T_{j-1};\theta}\Sigma^{x|c}_{t,T_j}\right\}^{\top} dW^{x|c,T;M}_t
\end{aligned}
$$

□

Remark 44. The component with collateral basis volatility is quite small since it depends on $Y^{x|c}_{t,T;M} - L^{x|c}_{t,T_{j-1};\theta}$ which is zero if the index curve is flat.

8.3.2. *Gaussian approximation of forward swap rates and currency basis spreads.* We now specialise the weight freezing technique to the LGCM. We consider a schedule $T_0 < T_1 < \ldots < T_n$ with accrual fractions θ_j, in the foreign currency f under domestic collateral d. This is the case where collateral basis volatility matters. The native case is recovered by switching off the collateral basis block (see Remark 47). We write

$$P_{t,j} = Z^{f|d}_{t,T_j}, \quad A_t = \sum_{j=1}^{n} \theta_j P_{t,j}, \quad L_{t,j} = L^{f|d}_{t,T_{j-1};\theta_j}, \quad L^r_{t,j} = L^{f/d|d,r}_{t,T_{j-1};\theta_j}, \quad B_{t,j} = L^r_{t,j} - L_{t,j},$$

for the discount factors, the annuity, the foreign and resetting forward Libors and the swaplet currency basis spreads. The forward swap rate and the forward resetting currency basis spread are

$$Y_t = Y^{f|d}_{t,T_0;M} = \frac{1}{A_t}\sum_{j=1}^{n} \theta_j P_{t,j} L_{t,j}, \qquad B_t = B^{f/d|d,r}_{t,T_0;M} = \frac{1}{A_t}\sum_{j=1}^{n} \theta_j P_{t,j} B_{t,j}. \tag{232}$$

Both are martingales under the annuity measure $\mathbb{Q}^{f|d,A}$. We use the stacked state $\hat{z}_t$, $\hat{G}_{t,T}$, $\hat{y}_t$ of Remark 33. Its volatility is $\hat{\sigma}_{z,t} = [\sigma^f_{z,t}|\sigma^c_{z,t}]$ and its transport matrix is $\hat{\mathcal{M}}_{u,t} = \mathrm{diag}(\mathcal{M}^f_{u,t}, \mathcal{M}^c_{u,t})$.

Lemma 3. In the LGCM, under the T-forward measure $\mathbb{Q}^{f|d,T}$, $\hat{z}_T$ is a centred Gaussian vector with covariance $\hat{y}_T$.

Lemma 4. In the LGCM, for $t \le T_0$, Y_t and B_t are explicit functions of $\hat{z}_t$:

$$\begin{aligned}
P_{t,j} &= Z^{f|d}_{0,t,T_j} \exp\left(-\hat{G}^\top_{t,T_j}\hat{z}_t - \frac{1}{2}\hat{G}^\top_{t,T_j}\hat{y}_t\hat{G}_{t,T_j}\right) \\
1+\theta_j L_{t,j} &= (1-\theta_j S^{f|f,\div}_{0,T_{j-1};\theta_j})\,\frac{Z^{f|f}_{t,T_{j-1}}}{Z^{f|f}_{t,T_j}}\,K_{t,j} \\
1+\theta_j L^r_{t,j} &= (1-\theta_j S^{d|d,\div}_{0,T_{j-1};\theta_j})\,\frac{P_{t,j-1}}{P_{t,j}}
\end{aligned}$$

where $Z^{f|f}_{t,T}$ is given by Theorem 4 with $z^f_t = \hat{z}^f_t - v^{f,c}_t$ and $v^{f,c}_t$ deterministic, and $K_{t,j} = K^C_t(T_{j-1}, T_j)$ is the deterministic collateral convexity of the foreign coupon factor (Proposition 26): $K_{t,j} = e^{-\int_t^{T_{j-1}} \Sigma^{f|f}_{u,T_{j-1};\theta_j}\cdot\Sigma^{f\to d}_{u,T_j}\,du}$ for a Libor coupon, (138) or (139) for a compounded one. For a compounded foreign (resp. domestic) coupon, $1-\theta_j S^{f|f,\div} = 1$ (resp. $1-\theta_j S^{d|d,\div} = 1$).

Proposition 41 (Gaussian approximation of forward swap rates)**.** Let us define the deterministic loadings, computed from forward values seen from $t_0 = 0$ ($P_j = Z^{f|d}_{0,T_0,T_j}$, $L_j = L^{f|d}_{0,T_{j-1};\theta_j}$, $Y = Y^{f|d}_{0,T_0;M}$, $A = \sum_j \theta_j P_j$):

$$q^f_Y = \frac{1}{A}\sum_{j=1}^{n} P_j\left((1+\theta_j Y)G^f_{T_0,T_j} - (1+\theta_j L_j)G^f_{T_0,T_{j-1}}\right) \tag{233}$$

$$q^c_Y = \frac{1}{A}\sum_{j=1}^{n} \theta_j P_j\,(Y - L_j)\,G^c_{T_0,T_j} \tag{234}$$

and $\hat{q}_Y = [q^f_Y; q^c_Y]$. Then, freezing the gradient of Y_t with respect to the state at its forward value, the swap rate has the Gaussian dynamic under $\mathbb{Q}^{f|d,A}$

$$dY_t \approx \hat{q}^\top_Y \hat{\mathcal{M}}_{t,T_0}\hat{\sigma}^\top_{z,t}\,dW^{f|d,A}_t \tag{235}$$

so that Y_{T_0} is approximately normal with mean Y_0 and variance

$$\mathbb{V}^{f|d,A}[Y_{T_0}] \approx \hat{q}^\top_Y \hat{y}_{T_0}\,\hat{q}_Y = \sum_{(a,i),(b,j)} q^{(a,i)}_Y q^{(b,j)}_Y \int_0^{T_0} \rho^{(a,i),(b,j)}_u \sigma^{a,i}_u \sigma^{b,j}_u M^{a,i}_{u,T_0} M^{b,j}_{u,T_0}\,du. \tag{236}$$

Proposition 42 (Gaussian approximation of currency basis spreads)**.** With the same notations and $B_j = B^{f/d|d,r}_{0,T_{j-1};\theta_j}$, $L^r_j = L_j + B_j$, $B = B^{f/d|d,r}_{0,T_0;M}$, let

$$q^f_B = \frac{1}{A}\sum_{j=1}^{n} \theta_j P_j \left(B\, G^f_{T_0,T_j} - B_j\, G^f_{T_0,T_{j-1}} \right) \tag{237}$$

$$q^c_B = \frac{1}{A}\sum_{j=1}^{n} P_j \left((1+\theta_j L_j + \theta_j B) G^c_{T_0,T_j} - (1+\theta_j L^r_j) G^c_{T_0,T_{j-1}} \right). \tag{238}$$

Then B_{T_0} is approximately normal under $\mathbb{Q}^{f|d,A}$ with mean B_0 and variance $\hat{q}_B^\top \hat{y}_{T_0}\, \hat{q}_B$.

The proofs are in Appendix D.

Remark 45 (Compounded overnight coupons)**.** Proposition 41 and Proposition 42 hold as stated when the foreign coupons are compounded overnight (daily or continuous), with the convention (21) for $L_{t,j}$:

(1) The loadings are unchanged in form. In Lemma 4 the coupon type only enters through the deterministic factor $K_{t,j}$ (and $\kappa^f = 1$), so $\nabla_{\hat{z}} \ln(1+\theta_j L_{t,j}) = [G^f_{t,T_j} - G^f_{t,T_{j-1}}; 0]$ for every coupon type, and the proofs of Appendix D go through verbatim. The coupon type changes the loadings only through the forward values $L_j = (\kappa^f K_{0,j} F_{j-1}/F_j - 1)/\theta_j$ ($F_j = Z^{f|f}_{0,T_0,T_j}$), B_j, Y and B that enter the formulas, with $K_{0,j} = K^C_0(T_{j-1}, T_j)$ seen from 0.

(2) The exact distribution of Y_{T_0} and B_{T_0} (2D quadrature) uses the state functions at T_0, hence $K_{T_0,j} = K^C_{T_0}(T_{j-1}, T_j)$ seen from T_0. Since T_0 is the start of the first period, every period starts at or after T_0 and no coupon is partially accrued at T_0: (138) and (139) apply with $t = T_0 \le T_{j-1}$.

(3) So the quadrature stays exact and the Gaussian approximation stays asymptotic in the volatility, as for Libor coupons. On the calibrated RFR example markets of the calibration section (daily SONIA and TONA coupons), the Gaussian CB volatilities differ from the quadrature by at most $7.1 \cdot 10^{-5}$ (GBPUSD) and $1.4 \cdot 10^{-4}$ (USDJPY) in relative terms, against $5.7 \cdot 10^{-5}$ for the EURIBOR coupons of the EURUSD market (measured, Remark 74). The same remark applies to non-native swap rates $Y^{f|d}$ with compounded coupons.

Remark 46 (Consistency checks)**.** (1) For a single swaplet ($n = 1$), $G^a_{T_0,T_0} = 0$ and the propositions give the loadings $q^c_B = \frac{1+\theta L^r_1}{\theta} G^c_{T_0,T_1}$ and $q^f_B = B\, G^f_{T_0,T_1}$. This is exactly the approximate dynamic of the swaplet currency basis spread in the section on reconstruction of market observables, since $\Sigma^a_{u,T_0;\theta} = \sigma^a_{z,u} \mathcal{M}^a_{u,T_0} G^a_{T_0,T_1}$.

(2) The collateral basis loading of the swap rate is proportional to $Y - L_j$. As already observed, it vanishes when the index curve is flat. In the numerical example below it contributes $3 \cdot 10^{-4}$ of the variance of Y.

(3) Symmetrically, the foreign rates loading of the currency basis spread is proportional to the level of the basis. The collateral basis loading is of order $(1+\theta L)/\theta$ times the collateral basis volatility of the swaplets.

Remark 47 (Native swaps)**.** For a swap in currency e under native collateral e, the collateral basis block disappears, $P_j = Z^{e|e}_{0,T_0,T_j}$, $1 + \theta_j L_j = (1 - \theta_j S^{e|e,\div}) P_{j-1}/P_j$, and (236) reduces to the classic LGM formula $q^{e,\top}_Y y^{e,e}_{T_0} q^e_Y$ with

$$q^e_Y = \frac{1}{A}\sum_{j=1}^{n} P_j \left((1+\theta_j Y) G^e_{T_0,T_j} - (1+\theta_j L_j) G^e_{T_0,T_{j-1}} \right),$$

cf. [1, §13]. The variance is a quadratic form in the deterministic covariance matrix y_{T_0} of the state, so it is computed by the covariance integral algorithm of subsection 9.1 at no extra cost.

Remark 48 (Accuracy)**.** The approximation is asymptotic in the volatility: it neglects the variation of the gradient $\nabla_{\hat{z}} Y$ along the path. We checked it against the exact distribution of Y_{T_0} and B_{T_0}, obtained from Lemma 3 and Lemma 4 by 2D Gauss-Hermite quadrature (160^2 nodes). The setup was a 5Y×10Y quarterly

swap with one foreign factor ($\sigma^f \approx 1.2\%$, $\beta^f \approx 3\%$) and one collateral basis factor ($\sigma^c \approx 0.4\%$, $\beta^c \approx 15\%$), $\rho^{f,c} = -0.4$, piecewise constant parameters and all volatilities scaled by λ. The exact pricer reproduces $\mathbb{E}^{f|d,A}[Y_{T_0}] = Y_0$ and $\mathbb{E}^{f|d,A}[B_{T_0}] = B_0$ to 10^{-16}. The relative errors of the approximate normal volatility $\sqrt{\mathbb{V}/T_0}$ are:

λ	normal vol of Y (bp)	rel. error	normal vol of B (bp)	rel. error
0.25	32.9	$2.0 \cdot 10^{-4}$	4.08	$-3.8 \cdot 10^{-5}$
0.5	65.7	$8.1 \cdot 10^{-4}$	8.16	$-1.5 \cdot 10^{-4}$
1	131.1	$3.3 \cdot 10^{-3}$	16.3	$-6.2 \cdot 10^{-4}$
2	259.0	$1.3 \cdot 10^{-2}$	32.5	$-2.6 \cdot 10^{-3}$
4	494.5	$5.1 \cdot 10^{-2}$	64.7	$-1.2 \cdot 10^{-2}$

The error is multiplied by 4 each time the volatilities are doubled, i.e. the relative error is $O(\lambda^2)$ (numerical observation). This is consistent with the first-order variation of the gradient averaging out in the variance. In this example the currency basis approximation is about five times more accurate than the swap rate one.

Remark 49 (Calibration)**.** Proposition 42 provides the model delivered volatility of currency basis spreads needed by the bootstrapping algorithm of the calibration section. It is explicit in the model parameters and only requires the covariance integrals of subsection 9.1, so it is cheap to evaluate inside the bootstrapping.

8.3.3. *Caplets on Libor-type and compounded coupons.* A Libor caplet fixes at the start a of its accrual period, a caplet on a compounded overnight coupon only at the end b: the variance of the latter keeps accumulating during the accrual period, with a weight that decreases as the coupon accrues. The LGCM prices both exactly (`market.caplet_total_variance`, `instruments.Caplet`).

Proposition 43 (Caplets on Libor-type and compounded coupons)**.** Let $e \in \{d, f\}$, C the coupon factor (20) of the index of currency e on $[a, b]$, $\theta = b - a$, and consider the caplet paying $(C - (1 + \theta K))^+$ at b under native collateral. With $\Sigma_{u,T} = \Sigma^{e|e}_{u,T}$ the bond volatility of the block, in the LGCM:

(1) under $\mathbb{Q}^{e|e,b}$, $\ln C$ is Gaussian, with mean such that $\mathbb{E}^{e|e,b}[C] = F := \kappa^e Z^{e|e}_{0,a}/Z^{e|e}_{0,b}$ ($\kappa^e = 1$ for compounded coupons) and variance

$$
\begin{aligned}
w^{\mathrm{L}} &= \int_0^a |\Sigma_{u,b} - \Sigma_{u,a}|^2 \, du \qquad \text{(Libor-type, fixed at } a\text{)}, \\
w^{\delta} &= w^{\mathrm{L}} + \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} |\Sigma_{u,b} - \Sigma_{u,t_{k+1}}|^2 \, du \qquad \text{(daily, grid } a = t_0 < \ldots < t_n = b\text{)}, \\
w^{0} &= w^{\mathrm{L}} + \int_a^b |\Sigma_{u,b}|^2 \, du \qquad \text{(continuous)};
\end{aligned}
\tag{239}
$$

(2) the caplet price is exactly $Z^{e|e}_{0,b}\,\mathrm{Black}\,(F, 1 + \theta K, w)$ with the variance of the coupon type;

(3) with constant volatility σ and no mean reversion (Ho–Lee), $w^{\mathrm{L}} = \sigma^2\theta^2 a$ and $w^0 = \sigma^2\theta^2(a + \theta/3)$: the compounded caplet has the variance of a Libor caplet with expiry $a + \theta/3$.

Proof. 1. By Theorem 4, $\ln Z^{e|e}_{t_k,t_{k+1}}$ is a deterministic function plus $-\int_0^{t_k} (\Sigma_{u,t_{k+1}} - \Sigma_{u,t_k}) \cdot dW_u$, with W a Brownian motion under $\mathbb{Q}^{e|e,b}$ (the measure changes have deterministic kernels in the LGCM, Proposition 47). So $\ln C = -\sum_k \ln Z^{e|e}_{t_k,t_{k+1}}$ is Gaussian, with stochastic part $\int_0^b \sum_k \mathbf{1}_{u<t_k} (\Sigma_{u,t_{k+1}} - \Sigma_{u,t_k}) \cdot dW_u$. For $u < a$ the sum telescopes to $\Sigma_{u,b} - \Sigma_{u,a}$, and for $u \in [t_j, t_{j+1})$ to $\Sigma_{u,b} - \Sigma_{u,t_{j+1}}$; the Itô isometry gives w^δ. The Libor factor $\kappa^e/Z^{e|e}_{a,b}$ is the one-step case ($n = 1$, the second term vanishes), and w^0 is the limit $\delta \to 0$, since $\Sigma_{u,t_{j+1}} \to \Sigma_{u,u} = 0$. The mean is Lemma 1. 2. C is lognormal under $\mathbb{Q}^{e|e,b}$ with mean F and log-variance w, so $\mathbb{E}^{e|e,b}[(C - (1 + \theta K))^+] = \mathrm{Black}(F, 1 + \theta K, w)$, and the price is $Z^{e|e}_{0,b}$ times it. 3. $\Sigma_{u,T} = \sigma(T - u)$ gives $\int_0^a \sigma^2\theta^2 du$ and $\int_a^b \sigma^2(b - u)^2 du = \sigma^2\theta^3/3$. □

Remark 50 (Swaptions are not affected)**.** (1) The in-accrual variance concerns options on a single coupon. A swaption on a native swap is not affected: at the exercise date T_0 the floating leg of the forward-starting swap is worth κ-weighted discount factors, $Z_{T_0,T_0} - Z_{T_0,T_n}$ for compounded coupons, by

Lemma 1. So the payoff only depends on the curve at T_0, whether the coupons fix in advance or in arrears, and the variance stops at T_0 (Proposition 51). Switching to compounded coupons only changes κ.

(2) A caplet on the current period ($a = 0$) has no time value if it is Libor-type ($w^{\mathrm{L}} = 0$), but a positive one if it is compounded.

(3) The ratio $w^{\delta}/w^{\mathrm{L}}$ is largest for short expiries and long accrual periods and tends to 1 as $a \to \infty$ (Ho–Lee: $1 + \theta/(3a)$). With piecewise constant volatilities it jumps when the accrual period crosses a knot, since the in-accrual variance then uses another volatility than the variance before a.

Remark 51 (Verification)**.** The tests are in `tests/test_caplets.py`; the figure and the table are generated by `python -m convexity.figures caplets`.

(1) The variance (239) agrees with the variance of $\ln C$ computed from the joint law of the state at the fixing dates (exact transitions, an independent code path) to 10^{-13} (relative), in both blocks, with monthly and weekly steps.

(2) The Ho–Lee limit holds to 10^{-12}; the one-step daily coupon is the Libor-type coupon; the daily variance converges to w^0 with an error proportional to the step (measured, not proved).

(3) Monte Carlo with the exact scheme and control variates of exactly known means ($\mathbb{E}[D_b C] = Z_{0,a}$, $\mathbb{E}[D_b] = Z_{0,b}$) reprices ATM compounded caplets within 4 standard errors, and rejects the same caplets priced with the Libor variance w^{L} (Table 1). Figure 1 shows the ratio of the compounded to the Libor caplet volatility.

block	period	$\sqrt{w_{\mathrm{RFR}}/w_{\mathrm{Libor}}}$	closed form (bp)	MC (bp)	s.e. (bp)	z	z with w_{Libor}
d	$[0.25, 0.5]$	1.1394	8.7561	8.7512	0.0069	-0.71	+154.6
d	$[2, 2.25]$	1.0181	17.5898	17.5838	0.0161	-0.37	+19.1
d	$[10, 10.25]$	1.0145	18.8810	18.8858	0.0191	+0.25	+14.4
f	$[1, 1.25]$	1.0174	12.9064	12.8799	0.0256	-1.03	+7.6
f	$[5, 6]$	1.0481	94.4479	94.2347	0.1972	-1.08	+20.9

TABLE 1. ATM caplets on the test model under native collateral, weekly compounding: volatility ratio, closed form of Proposition 43, Monte Carlo (2^{18} paths, seed 5) with its standard error and z-score, and the z-score of the price computed with the Libor variance w^{L}.

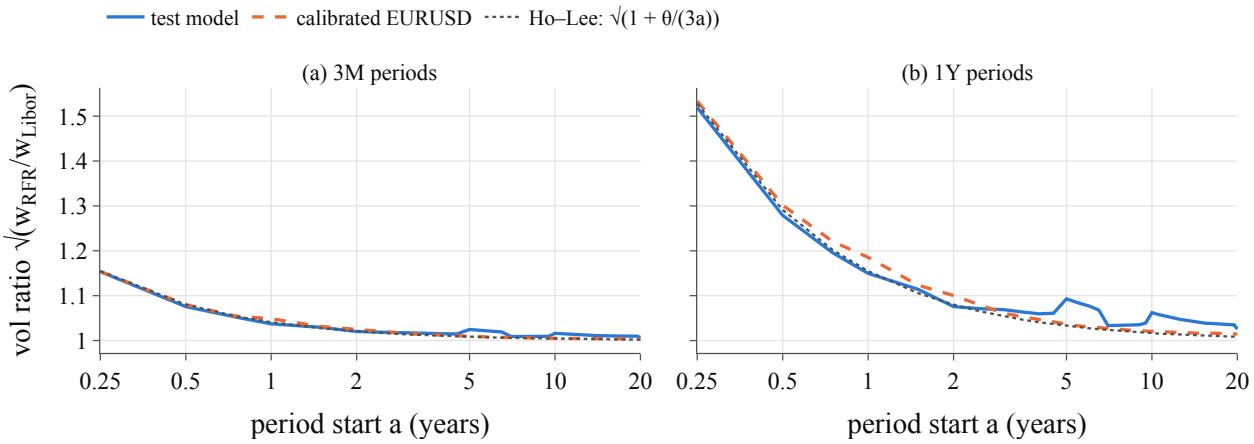


FIGURE 1. Ratio of the volatility of a daily compounded caplet to that of a Libor caplet on the same period, domestic block, against the start of the period: test model and LGCM calibrated to the EURUSD example, with the Ho–Lee reference $\sqrt{1 + \theta/(3a)}$.

8.3.4. *Correlations of FX and swap rates.* Correlations between FX and rates fixings drive quanto and hybrid FX/rates products, and correlations between swap rates of different tenors or fixing dates drive CMS spread and mid-curve products. In the LGCM they are outputs of the model parameters. This section gives a closed form approximation, an exact benchmark and a Monte Carlo estimator (`correlations.py`).

Definition 6 (Terminal correlations)**.** The observables are the log FX spot $\ln x_T^{df}$ and the native forward swap rates $Y_{T_0}^e = Y_{T_0,T_0;M}^{e|e}$, $e \in \{d, f\}$, fixing at T_0 with tenor M (the CMS fixing), each a function of the state at its date: $\ln x_T^{df} = \ln x_{0,T}^{df|d} + z_T^x$ and, by Theorem 4, $Y_{T_0}^e = \sum_m c_m Z_{T_0,t_m}^{e|e} / \sum_k \tau_k Z_{T_0,t_k}^{e|e}$ with $Z_{0,t}^{e|e} = Z_{0,T_0,t}^{e|e} \exp(-G_{T_0,t}^{e,\top} z_{T_0}^e - \frac{1}{2} G_{T_0,t}^{e,\top} y_{T_0}^{e,e} G_{T_0,t}^e)$. The terminal correlation of two observables X_1, X_2 with dates T_1, T_2 (equal for "same expiry", different for "intertemporal") under a measure $\mathbb{Q}$ is $\mathrm{Corr}^{\mathbb{Q}}(X_1, X_2)$. We use $\mathbb{Q}^{d,\mathrm{RN}}$ and the domestic forward measure $\mathbb{Q}^{d|d,T^*}$, $T^* \geq T_1, T_2$; a CMS coupon paid at T_p is valued under $\mathbb{Q}^{d|d,T_p}$.

Proposition 44 (Gaussian approximation of terminal correlations)**.** Let $s_t = (z_t^1, \ldots, z_t^N, z_t^x)$ be the rates factors and the log FX moneyness at t.

(1) For $T_1, T_2 > 0$ and $\underline{T} = T_1 \wedge T_2$, under $\mathbb{Q}^{d,\mathrm{RN}}$, under $\mathbb{Q}^{d|d,T^*}$ for any T^*, and under any measure whose density with respect to $\mathbb{Q}^{d,\mathrm{RN}}$ is the exponential of a stochastic integral with a deterministic kernel,

$$\begin{aligned} \mathrm{Cov}(z_{T_1}^i, z_{T_2}^j) &= \int_0^{\underline{T}} \rho_{ij} \sigma_u^i \sigma_u^j M_{u,T_1}^i M_{u,T_2}^j \, du, \\ \mathrm{Cov}(z_{T_1}^x, z_{T_2}^j) &= \int_0^{\underline{T}} \Sigma_{u,T_1}^{df} \cdot \sigma_u^j M_{u,T_2}^j e_j \, du, \qquad \mathrm{Cov}(z_{T_1}^x, z_{T_2}^x) = \int_0^{\underline{T}} \Sigma_{u,T_1}^{df} \cdot \Sigma_{u,T_2}^{df} \, du, \end{aligned}$$

with $\Sigma_{u,T}^{df} = \Sigma_{u,T}^{d|d} - \Sigma_{u,T}^{f|f} - \Sigma_{u,T}^{f\to d} + \sigma_u^{df}$ the FX forward volatility, $a \cdot b$ the product with the correlation matrix, and e_j the j-th driver. These are covariance integrals of subsection 9.1.

(2) Replace $Y_{T_0}^e$ by its frozen-weights linearisation $Y_0^e + q_Y^{e,\top} z_{T_0}^e$ (Remark 47) and keep $\ln x_T^{df}$, which is affine in s_T. With g_k the loadings of X_k on s_{T_k} and $C(T_1, T_2) = \mathrm{Cov}(s_{T_1}, s_{T_2})$ from item 1,

$$\mathrm{Corr}(X_1, X_2) \approx \frac{g_1^\top C(T_1, T_2)\, g_2}{\sqrt{g_1^\top C(T_1, T_1)\, g_1 \; g_2^\top C(T_2, T_2)\, g_2}}. \tag{240}$$

The right hand side does not depend on the measure, and it is exact when both observables are log FX spots.

Proof. 1. Under $\mathbb{Q}^{d,\mathrm{RN}}$ the drift of z^i is $-\beta_t^i z_t^i$ plus a deterministic function of t (y_t and $v_t^{f,c}$ are deterministic in the LGCM), so $z_T^i = m_T^i + \int_0^T M_{u,T}^i \sigma_u^i dW_u^i$ with m_T^i deterministic. By the FX forward dynamic of the reconstruction section, $x_{t,T}^{df|d}$ is a $\mathbb{Q}^{d|d,T}$-martingale with volatility $\Sigma_{t,T}^{df}$, and $W^{d|d,T} - W^{d,\mathrm{RN}}$ has a deterministic drift, so $z_T^x = \ln(x_{T,T}^{df|d} / x_{0,T}^{df|d}) = m_T^x + \int_0^T \Sigma_{u,T}^{df} \cdot dW_u$. The covariances follow from the Itô isometry on $[0, \underline{T}]$. A change of measure with a deterministic kernel only adds deterministic drifts, which do not change the covariances; the kernel of $\mathbb{Q}^{d|d,T^*}$ is $\Sigma_{\cdot,T^*}^{d|d}$ (Proposition 47). 2. The linearised observables are affine in the Gaussian vector (s_{T_1}, s_{T_2}); their covariance is $g_1^\top C(T_1, T_2) g_2$, by item 1 under any of these measures. □

Remark 52 (Exact benchmark and Monte Carlo estimator)**.** (1) *Exact.* Under $\mathbb{Q}^{d,\mathrm{RN}}$ or $\mathbb{Q}^{d|d,T^*}$, the state at the dates $T_1 \leq T_2$ is jointly Gaussian, with the law given by the exact transitions of Proposition 46 and Proposition 47: $X_{T_2} = F X_{T_1} + \mu + \varepsilon$ gives $\mathrm{Cov}(X_{T_1}, X_{T_2}) = \mathrm{Var}(X_{T_1}) F^\top$. The observables only depend on the components z^e of the swap blocks and z^x, so the moments $\mathbb{E}[X_k]$, $\mathrm{Var}(X_k)$, $\mathrm{Cov}(X_1, X_2)$ are computed by tensor Gauss–Hermite quadrature on the range of the covariance of these components (dimension at most 4 in the examples). This code path does not use the covariance integrals of item 1 of Proposition 44; the two agree to 10^{-16} (absolute) on the state covariances.

(2) *Monte Carlo.* The exact scheme of subsection 9.4 simulates the state under $\mathbb{Q}^{d,\mathrm{RN}}$ at the dates. For $\mathbb{Q}^{d|d,T^*}$ the paths are weighted by $D_{T_2} Z_{T_2,T^*}^{d|d}$ (its $\mathbb{Q}^{d,\mathrm{RN}}$ expectation is $Z_{0,T^*}^{d|d}$ by the tower property), and the correlation is the self-normalised weighted sample correlation. Its standard error is the delta

method one: with $\omega = w/\bar{w}$ and standardised $\tilde{x}$, $\tilde{y}$, the influence function is $\psi = \omega(\tilde{x}\tilde{y} - \frac{\rho}{2}(\tilde{x}^2 + \tilde{y}^2))$, averaged over antithetic pairs.

(3) *Control variate.* The linearised observables $L_k = g_k^\top s_{T_k}$, evaluated on the same paths, have the exactly known correlation (240), under either measure. The estimator $\hat{\rho}(X_1, X_2) - \hat{\rho}(L_1, L_2) + \rho^{\text{approx}}$, with influence function $\psi_X - \psi_L$, estimates the exact correlation with a much smaller standard error.

Remark 53 (Verification)**.** Table 2 checks the three estimators on the test model of Remark 57 (two domestic factors, one foreign, one collateral basis, full 5×5 correlation, synthetic initial curves; 10Y swaps with annual fixed and quarterly floating legs unless stated), and Figure 2 shows the convergence of the approximation (tests in `tests/test_correlations.py`; figures and tables by `python -m convexity.figures`).

(1) The quadrature is converged (12 and 24 nodes per dimension agree to 10^{-12}).

(2) The approximation error is $O(\lambda^2)$ under the scaling of all volatilities by λ (Figure 2 (a), ratio column of the table): a numerical observation. The correlation itself is invariant under λ at leading order. The dependence on the measure (panel (b)) is of the same order, except for $\ln x_5$ against Y_5^f, where it is below 10^{-8} for $\lambda \leq 1$.

(3) Monte Carlo (2^{17} paths, seed 1) agrees with the exact correlation within 4 standard errors, with and without control and under both measures. The control variate divides the standard error by about 10^3 under $\mathbb{Q}^{d,\mathrm{RN}}$, enough to reject the approximation (column "approx (z)"). Under $\mathbb{Q}^{d|d,15}$ the weights, whose coefficient of variation is 22% at 5Y, leave a controlled standard error of the order of the approximation error.

(4) The standard errors are calibrated: over 60 to 200 seeds the z-scores have a standard deviation between 0.92 and 1.09, with and without control and under both measures.

				MC plain		MC control			$Q^{d\|d,15}$
pair	exact	approx − exact	ratio $\lambda \times 2$	z	s.e.	z	s.e.	approx (z)	z
$\ln x_5$, Y_5^d	0.33952	+1.7e-04	4.00, 3.99	-0.41	3.5e-03	-0.59	3.1e-06	+57	+1.31
$\ln x_5$, Y_5^f	0.22120	+2.1e-05	3.99, 4.02	+0.47	3.7e-03	-1.48	7.1e-07	+31	+1.26
Y_5^d (2Y), Y_5^d	0.99171	-3.9e-05	3.99, 3.98	+2.15	6.3e-05	-1.75	6.6e-07	-57	-0.81
Y_5^d, Y_5^f	0.58812	+1.1e-04	4.00, 4.00	+1.29	2.5e-03	+2.41	3.8e-06	+28	-0.39
Y_2^d, Y_5^d	0.68002	+2.2e-04	4.00, 4.00	+0.86	2.1e-03	+0.78	4.4e-06	+49	-1.04
$\ln x_2$, Y_5^d	0.21403	+1.0e-04	4.00, 3.99	-0.95	3.7e-03	-0.49	3.1e-06	+33	+1.08

TABLE 2. Terminal correlations on the test model under $\mathbb{Q}^{d,\mathrm{RN}}$: exact value, error of the approximation (240) and its ratio when the volatilities are doubled ($\lambda = 0.5 \to 1 \to 2$), Monte Carlo z-scores against the exact value and standard errors (plain and with the control variate), the approximation in standard errors of the controlled Monte Carlo, and the controlled Monte Carlo under $\mathbb{Q}^{d|d,15}$ against the exact value under that measure.

Remark 54 (Correlation term structures)**.** Figure 3 compares the test model with the LGCM calibrated to the EURUSD example (one factor per block, Remark 74); Figure 4 shows correlations across tenors and fixing dates. Two qualitative features stand out.

(1) With one factor per block, two swap rates of the same currency and expiry are almost perfectly correlated whatever their tenors (Figure 4 (a)): decorrelation across tenors needs several factors per block, as in the test model. The correlations across currencies and with the FX spot are set by the marked correlations between the drivers and by the relative sizes of the volatilities, which change with the expiry (Figure 3).

(2) Intertemporal correlations (Figure 4 (b)) decay on both sides of the reference fixing date: the two fixings only share the variance accumulated up to the earlier date, damped by the mean reversion $M^i_{T_1,T_2}$ (item 1 of Proposition 44), while the later fixing adds the variance accumulated in between.

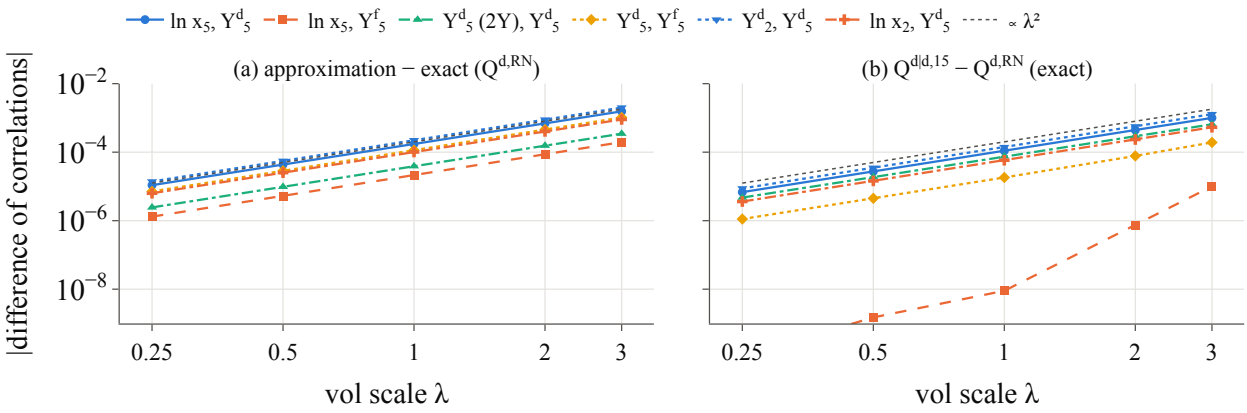


FIGURE 2. Absolute differences of correlations against the vol scale λ (test model): (a) approximation minus exact, (b) $\mathbb{Q}^{d|d,15}$ minus $\mathbb{Q}^{d,\mathrm{RN}}$, both exact. The dotted line has slope 2.

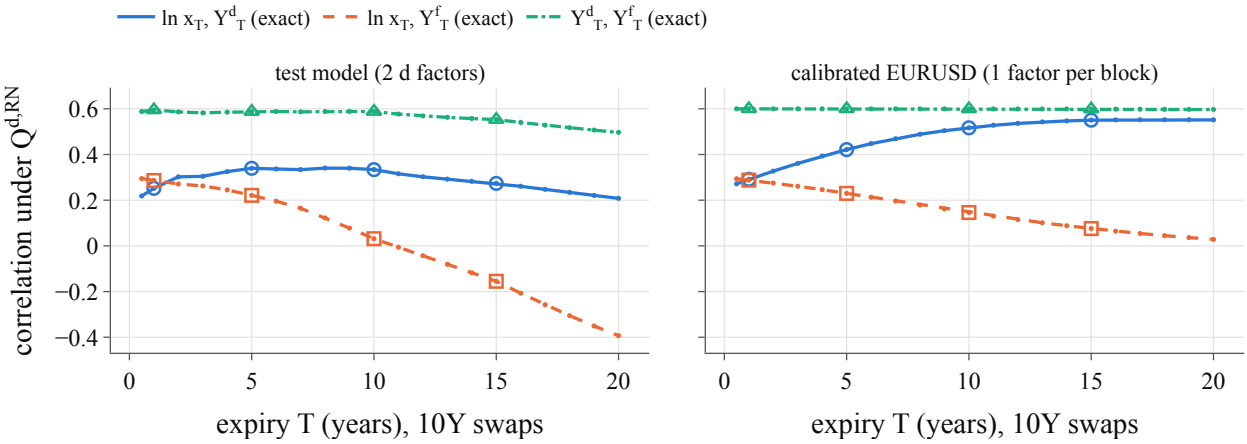


FIGURE 3. Terminal correlations under $\mathbb{Q}^{d,\mathrm{RN}}$ against the common expiry T (10Y swap rates): exact (lines), approximation (240) (small dots) and Monte Carlo with control variate, 2^{15} paths, ± 4 standard errors (open markers). Left: test model; right: LGCM calibrated to the EURUSD example.

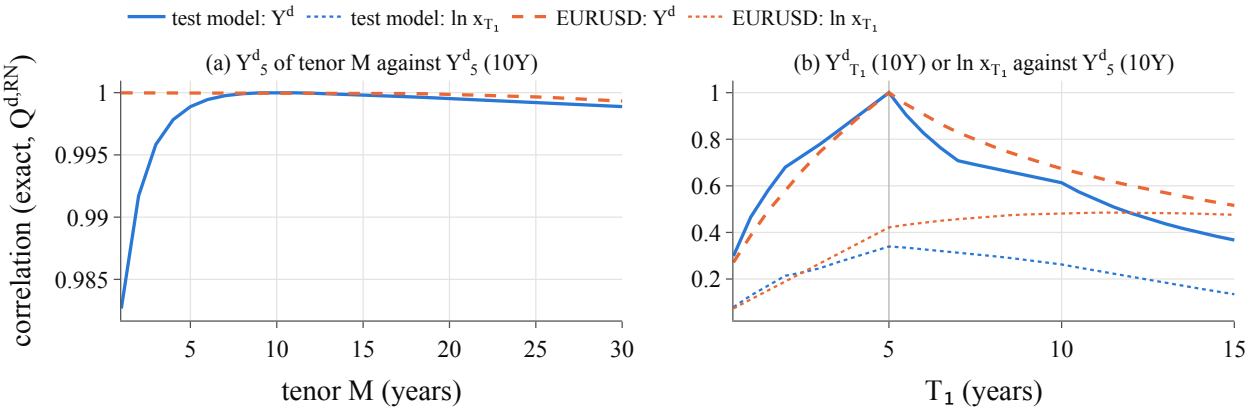


FIGURE 4. Exact correlations under $\mathbb{Q}^{d,\mathrm{RN}}$ with the $5\mathrm{Y}\times10\mathrm{Y}$ domestic swap rate: (a) the $5\mathrm{Y}$ swap rate of tenor M; (b) the $10\mathrm{Y}$ swap rate fixing at T_1 (intertemporal) and the log FX spot at T_1.

9. The Linear Gaussian Collateral Model: Implementation

9.1. **Covariance integrals.**

9.1.1. *Introduction.* The expectation and covariance matrix necessary for simulation, as well as all the convexity adjustments in the model, reduce to sums of integrals of the type:

$$I^{a,b}_{t,S;T_a,T_b} \;=\; \int_t^S \Sigma^a_{u,T_a}\cdot\Sigma^b_{u,T_b}\,du \;=\; \sum_{j_a=1}^{N_a}\sum_{j_b=1}^{N_b}\int_t^S \rho_u^{(a,j_a),(b,j_b)}\sigma_u^{a,j_a}\sigma_u^{b,j_b}G^{a,j_a}_{u,T_a}G^{b,j_b}_{u,T_b}\,du \tag{241}$$

where we set $G^x_{u,T}=1$ for the FX factor. The simulation also requires the same integrals with $G^{a,j}_{u,T}$ replaced by $M^{a,j}_{u,T}$. So we consider the elementary integrals

$$J_{t,S}\left[\ell^{a,i},\ell^{b,j}\right] \;=\; \int_t^S \rho_u^{(a,i),(b,j)}\sigma_u^{a,i}\sigma_u^{b,j}\,\ell_u^{a,i}\,\ell_u^{b,j}\,du \tag{242}$$

where each *loading* $\ell^{a,i}_u$ is one of $M^{a,i}_{u,T}$, $G^{a,i}_{u,T}$ or 1, for some maturity $T\geq S$. Sums of these give all the integrals we need. For example, $\Sigma^a_{u,T;\theta}$ has loadings $G^{a,i}_{u,T+\theta}-G^{a,i}_{u,T}$, and $y^{a,b}_t$ has entries $J_{0,t}[M^{a,i}_{\cdot,t},M^{b,j}_{\cdot,t}]$. For vector loadings $\Lambda_u=\sum_{(a,i)}c_{a,i}\,\sigma^{a,i}_u\ell^{a,i}_u\,\widetilde{e}_{(a,i)}$, we write $J_{t,S}[\Lambda,\Lambda']$ for the corresponding bilinear combination.

9.1.2. *Piecewise constant parameters.* We assume the spot vols, correlations and mean reversion parameters are piecewise constant on a grid (t_k). On a segment $[t_k,t_{k+1}]$ of length Δ_k, we write $\beta=\beta^{a,i}_k$ and $s=t_{k+1}-u\in[0,\Delta_k]$, and introduce

$$\varphi_\beta(s) \;=\; \int_0^s e^{-\beta v}\,dv \;=\; \frac{1-e^{-\beta s}}{\beta}\qquad(\varphi_0(s)=s). \tag{243}$$

Lemma 5. For $u\in[t_k,t_{k+1}]$ and $T\geq t_{k+1}$, every loading is affine in $\varphi_\beta(t_{k+1}-u)$:

$$\ell_u \;=\; p+q\,\varphi_\beta(t_{k+1}-u) \tag{244}$$

with

$$\begin{aligned}(p,q) &= \left(M_{t_{k+1},T}\,,\,-\beta M_{t_{k+1},T}\right)\quad\text{for }\ell=M_{\cdot,T},\\ (p,q) &= \left(G_{t_{k+1},T}\,,\,1-\beta G_{t_{k+1},T}\right)\quad\text{for }\ell=G_{\cdot,T},\\ (p,q) &= (1,0)\quad\text{for }\ell=1.\end{aligned}$$

Proof. We have $M_{u,T}=e^{-\beta s}M_{t_{k+1},T}$ and $G_{u,T}=\varphi_\beta(s)+e^{-\beta s}G_{t_{k+1},T}$, and $e^{-\beta s}=1-\beta\varphi_\beta(s)$. □

Proposition 45 (Covariance integrals)**.** With the notations of Lemma 5, for loadings $\ell^{a,i}$ and $\ell^{b,j}$ with coefficients (p_a,q_a), (p_b,q_b) and mean reversions $\alpha=\beta^{a,i}_k$, $\beta=\beta^{b,j}_k$ on the segment,

$$J_{t_k,t_{k+1}}\left[\ell^{a,i},\ell^{b,j}\right] \;=\; \rho_k\sigma^{a,i}_k\sigma^{b,j}_k\left(p_ap_b\Delta_k+p_aq_b\Phi_1(\beta)+q_ap_b\Phi_1(\alpha)+q_aq_b\Phi_2(\alpha,\beta)\right) \tag{245}$$

where

$$\Phi_1(\beta) \;=\; \int_0^\Delta \varphi_\beta(s)\,ds \;=\; \frac{\Delta-\varphi_\beta(\Delta)}{\beta}, \tag{246}$$

$$\Phi_2(\alpha,\beta) \;=\; \int_0^\Delta \varphi_\alpha(s)\varphi_\beta(s)\,ds \;=\; \frac{\Delta-\varphi_\alpha(\Delta)-\varphi_\beta(\Delta)+\varphi_{\alpha+\beta}(\Delta)}{\alpha\beta}. \tag{247}$$

Proof. Expand the product of the two affine loadings and use $\varphi_\alpha\varphi_\beta=(1-e^{-\alpha s}-e^{-\beta s}+e^{-(\alpha+\beta)s})/(\alpha\beta)$. □

Remark 55 (Stable evaluation)**.** The singularities of Φ_1 and Φ_2 at zero mean reversion are removable but the closed forms lose precision near them. Zero mean reversion is the Ho-Lee case and does occur in practice, for example for the FX factor or a level factor. Both functions are entire:

$$\Phi_1(\beta) \;=\; \sum_{n\geq0}\frac{(-\beta)^n\Delta^{n+2}}{(n+2)!},\qquad \Phi_2(\alpha,\beta) \;=\; \sum_{m,n\geq0}\frac{(-\alpha)^m(-\beta)^n\Delta^{m+n+3}}{(m+1)!\,(n+1)!\,(m+n+3)}.$$

We use these series when $|\beta|\Delta < 0.1$ (resp. $\max(|\alpha|,|\beta|)\Delta < 0.5$) and the closed forms, computed with `expm1`, when $\min(|\alpha|,|\beta|,|\alpha+\beta|)\Delta > 0.05$. In the remaining mixed case, where one rate is small and the other is not, we integrate $\varphi_\alpha\varphi_\beta$ by 64-point Gauss-Legendre quadrature. The integrand is entire, so this is exact to machine precision for the values of $|\beta|\Delta$ met in practice.

Algorithm 1 (Covariance integrals)**.** Given the parameter grid (t_k):

(1) Precompute, for each factor, $\int_0^{t_k}\beta_v dv$ on the grid. Then $M_{t,T} = e^{-\int_t^T \beta_v dv}$ for any t, T.
(2) Compute $G_{t_k,T}$ for all grid points $t_k \le T$ by the backward recursion

$$G_{t_k,T} \;=\; \varphi_{\beta_k}(\Delta_k) + e^{-\beta_k\Delta_k}\, G_{t_{k+1},T},$$

starting from the segment containing T. It avoids the representation $G_{t,T} = (\Lambda_T - \Lambda_t)/h_t$ with $\Lambda_T = G_{0,T}$, whose difference of two large numbers can lose precision when t is large or the mean reversion is negative.
(3) Split $[t,S]$ at the grid points. On each segment compute (p,q) for both loadings by Lemma 5 and the contribution by Proposition 45. Sum the contributions.

The cost is $O(K)$ per pair of factors, where K is the number of segments in $[t,S]$, i.e. $O(KN_aN_b)$ for $I^{a,b}_{t,S;T_a,T_b}$. All quantities are closed form.

Remark 56 (Numerical verification)**.** We compared Algorithm 1 to adaptive quadrature in which G itself is computed by quadrature of M. The test used 84 combinations of loadings $(M, G, 1)$, sub-intervals not aligned on the grid and maturities up to 12Y, with one factor having zero and 10^{-7} mean reversion and one factor negative mean reversion on some segments. The maximum relative error is $7\cdot 10^{-16}$. The kernels Φ_1, Φ_2 alone agree with quadrature to $4\cdot 10^{-16}$, including the cases $\alpha = 0$, $\alpha = 10^{-9}$, $\alpha+\beta = 0$ and $\beta\Delta = 10$.

9.2. **Convexity adjustments.** All collateral convexity and quanto adjustments of the deterministic volatility section are exponentials of covariance integrals, computed by Algorithm 1. For example:

$$\ln\frac{x^{df|j}_{t,T}}{x^{df|i}_{t,T}} \;=\; -\int_t^T \Sigma^{df}_{u,T}\cdot\Sigma^{i\to j}_{u,T}\,du,$$

$$\ln\frac{1+\theta L^{e|j}_{t,T;\theta}}{1+\theta L^{e|i}_{t,T;\theta}} \;=\; -\int_t^T \Sigma^{e|e}_{u,T;\theta}\cdot\Sigma^{i\to j}_{u,T+\theta}\,du,$$

where $\Sigma^{df}_{u,T}$ has loadings $G^d_{u,T}$, $-G^f_{u,T}$, $-G^c_{u,T}$ and 1, and $\Sigma^{e|e}_{u,T;\theta}$ has loadings $G^e_{u,T+\theta} - G^e_{u,T}$.

The coupon convexities of Proposition 26 are sums of covariance integrals too. The continuous one (139) needs three. The daily one (138) has one term per overnight fixing (92 for a quarterly coupon with $\delta = 1/365$: 91 full steps and a shortened last one), each with loadings $\Sigma^{f,i}_{u,t_{k+1}} - \Sigma^{f,i}_{u,t_k} = \sigma^{f,i}_u M^{f,i}_{u,t_k} G^{f,i}_{t_k,t_{k+1}}$. With $a' = \max(t,a)$ and the cocycle $M^{f,i}_{u,t_k} = M^{f,i}_{u,a'}M^{f,i}_{a',t_k}$, it is

$$-\ln K^\delta_t(a,b) \;=\; \sum_{i\in f,\,j\in c}\;\sum_{k:\,t_k>t} G^{f,i}_{t_k,t_{k+1}} M^{f,i}_{a',t_k}\left(J_{t,a'}[M^{f,i}_{\cdot,a'}, G^{c,j}_{\cdot,b}] + J_{a',t_k}[M^{f,i}_{\cdot,a'}, G^{c,j}_{\cdot,b}]\right),$$

one covariance integral over $[t,a']$ per pair of factors, plus the cumulative integrals J_{a',t_k} on the grid refined by the knots, accumulated in closed form with the loadings of Lemma 5 (`market.compounded_convexity`; the loading $M^{f,i}_{u,a'} = 1/M^{f,i}_{a',u}$ for $u > a'$ is still affine in φ_β). This takes 0.43ms for a daily quarterly coupon, against 48ms for the loop over fixings (`_compounded_convexity_scalar`, kept as the test oracle); the two agree to $|\Delta\ln K| \le 1.1\cdot 10^{-16}$.

Remark 57 (Verification of the coupon convexities)**.** The closed forms of Proposition 26 were checked on the test model (function `default_model` of `lgcm.py`: two domestic factors, one foreign, one collateral basis factor, $\rho^{f,c} = -0.4$, piecewise constant parameters) against an independent benchmark (function `coupon_expectation_exact` of `benchmarks.py`), which computes $\mathbb{E}^{f|d,b}_t[C]$ as an exact Gaussian expectation: the state is augmented with the integrated short rates and with $\ln C$, propagated with the exact transitions of Proposition 46 from fixing to fixing ($\ln C$ accumulates $-\ln Z^{f|f}_{t_k,t_{k+1}}$, an affine function of the

state, at each fixing), and the deflated payoff is the exponential of an affine function of a Gaussian vector. It does not use the covariance kernels of the closed forms.

(1) Over Libor, daily steps $1/52$ and $1/12$, $t \in \{0, 2, 5 + 5/52\}$ (the last one inside a period) and $a \in \{5, 10, 20\}$, $|\ln K_{\text{closed}} - \ln K_{\text{bench}}| \leq 4.4 \cdot 10^{-16}$, for $\ln K$ of order $5 \cdot 10^{-5}$. The benchmark under native collateral returns $Z^{f|f}_{t,a}/Z^{f|f}_{t,b}$ to $2.2 \cdot 10^{-16}$ (Lemma 1), and K does not depend on the state z_t (to 10^{-12}). The Libor formula equals the one-step daily formula to 10^{-14}, and $\ln(K^0/K^{\mathrm{L}})$ equals the in-arrears integral of Proposition 27 to 10^{-12} (relative).

(2) Monte Carlo (one quarterly period $[5, 5.25]$, daily step $1/52$, 2^{16} paths, seed 13): $z = -0.36$ for K^δ, and $K = 1$ is rejected at 33.4 standard errors (s.e. $1.47 \cdot 10^{-6}$ on K).

(3) Size of the in-arrears term (numerical observation, quarterly periods seen from $t = 0$): $K - 1$ is

a	Libor	daily $(1/365)$	continuous	$\ln(K^0/K^{\mathrm{L}})$
2	$9.956 \cdot 10^{-6}$	$1.0039 \cdot 10^{-5}$	$1.0040 \cdot 10^{-5}$	$8.4 \cdot 10^{-8}$
10	$1.7963 \cdot 10^{-4}$	$1.7975 \cdot 10^{-4}$	$1.7975 \cdot 10^{-4}$	$1.14 \cdot 10^{-7}$
20	$5.4888 \cdot 10^{-4}$	$5.4899 \cdot 10^{-4}$	$5.4899 \cdot 10^{-4}$	$1.11 \cdot 10^{-7}$

The in-arrears term is about 10^{-7} and roughly constant in a, as Proposition 27 predicts (bound $\bar\sigma^f \bar\sigma^c \theta^3/3 \approx 3.9 \cdot 10^{-7}$ with $\bar\sigma^f \approx 1.47\%$, $\bar\sigma^c \approx 0.51\%$, the largest spot volatilities), while the Libor term grows with a: the in-arrears term is 0.8% of the convexity at $a = 2$ and 0.06% at $a = 10$. A Monte Carlo cannot resolve it: in item 2 it is 0.07 standard errors (about 2^{28} paths would be needed for 4 standard errors). The exact benchmark of item 1 resolves it (10^{-7} against $4.4 \cdot 10^{-16}$).

(4) Continuous vs daily (numerical observation, $a = 10$): $|\ln(K^\delta/K^0)| = 1.380 \cdot 10^{-8}$, $6.771 \cdot 10^{-9}$, $3.343 \cdot 10^{-9}$, $1.661 \cdot 10^{-9}$ for $\delta = 1/26, 1/52, 1/104, 1/208$ (ratios 2.04, 2.03, 2.01) and $9.44 \cdot 10^{-10}$ for $\delta = 1/365$. The ratio of $\ln(K^\delta/K^0)$ to the leading term (140) is 1.0256, 1.0128, 1.0064, 1.0036, 1.0018 for $\delta = 1/52, 1/104, 1/208, 1/365, 1/730$ (the same at $a = 2$ and $a = 20$ to 10^{-3}): the first order term is confirmed and the remainder is $O(\delta^2)$ relative to δ, as proved. In these models the continuous coupon is within 10^{-9} of the daily one on $\ln K$.

The tests are in `tests/test_ois.py`.

9.3. **CurveEngine implementation.** In practice, all these computations can be abstracted in a helper class.

9.3.1. *Reconstruction formulas.*

9.4. **Monte Carlo simulation.** In the LGCM, $y_t^{a,b}$ and $v_t^{f,c}$ are deterministic. They are precomputed as $y_t^{(a,i),(b,j)} = J_{0,t}[M_{\cdot,t}^{a,i}, M_{\cdot,t}^{b,j}]$ and $v_t^{(f,i)} = \sum_j J_{0,t}[M_{\cdot,t}^{f,i}, G_{\cdot,t}^{c,j}]$. The state $z_t = [z_t^d | z_t^f | z_t^c | z_t^x]$ of Theorem 4 is Gaussian, and its transition between two dates $s < t$ can be sampled exactly, whatever the step size. For a vector loading Λ, we write $J[\Lambda, \Lambda'] = J_{s,t}[\Lambda, \Lambda']$ and use the loadings

$$\Sigma_{\cdot,t}^a : \left(G_{\cdot,t}^{a,j}\right)_j, \qquad \Sigma_{\cdot,t}^{df} = \Sigma_{\cdot,t}^d - \Sigma_{\cdot,t}^f - \Sigma_{\cdot,t}^c + \sigma^{df}, \qquad \hat{\Sigma}_{\cdot,t} = \Sigma_{\cdot,t}^f + \Sigma_{\cdot,t}^c = \Sigma_{\cdot,t}^{f|d}.$$

9.4.1. *Transition and covariance matrix.*

Proposition 46 (Exact transition under the risk-neutral measure)**.** Let $s < t$, $G^a = G_{s,t}^a$ and $\hat{G} = [G^f; G^c]$. Under $\mathbb{Q}^{d,\mathrm{RN}}$, conditionally on $\mathcal{F}_s$,

$$\begin{aligned}
z_t^a &= \mathcal{M}_{s,t}^a z_s^a + \mu_{s,t}^a + \varepsilon^a, \qquad a \in \{d, f, c\} \\
z_t^x &= z_s^x + G^{d,\top} z_s^d - \hat{G}^\top \hat{z}_s + \mu_{s,t}^x + \varepsilon^x \\
\int_s^t r_u^{d|d}\, du &= -\ln Z_{0,s,t}^{d|d} + G^{d,\top} z_s^d + \mu_{s,t}^r + \varepsilon^r \\
\int_s^t q_u^{f\to d}\, du &= -\ln \Xi \mathrm{Z}_{0,s,t}^{f|f\to d} + G^{c,\top} z_s^c + G^{f,\top} v_s^{f,c} + \mu_{s,t}^q + \varepsilon^q
\end{aligned}$$

with $\hat{z}_s = [z_s^f + v_s^{f,c}; z_s^c]$, deterministic means

$$\begin{aligned}
\mu_{s,t}^{d,i} &= \textstyle\sum_j J[M_{\cdot,t}^{d,i}, G_{\cdot,t}^{d,j}] + M_{s,t}^{d,i} \left(y_s^{d,d} G^d\right)_i \\
\mu_{s,t}^{f,i} &= \textstyle\sum_j J[M_{\cdot,t}^{f,i}, G_{\cdot,t}^{f,j}] + M_{s,t}^{f,i} \left(y_s^{f,f} G^f\right)_i - J[M_{\cdot,t}^{f,i}, 1^x] \\
\mu_{s,t}^{c,i} &= \textstyle\sum_j J[M_{\cdot,t}^{c,i}, G_{\cdot,t}^{c,j}] + \sum_j J[M_{\cdot,t}^{c,i}, G_{\cdot,t}^{f,j}] + M_{s,t}^{c,i} \left(y_s^{c,c} G^c + y_s^{c,f} G^f\right)_i - J[M_{\cdot,t}^{c,i}, 1^x] \\
\mu_{s,t}^{x} &= \frac{1}{2} G^{d,\top} y_s^{d,d} G^d - \frac{1}{2} \hat{G}^\top \hat{y}_s \hat{G} - \frac{1}{2} J[\Sigma_{\cdot,t}^{df}, \Sigma_{\cdot,t}^{df}] + J[\Sigma_{\cdot,t}^{df}, \Sigma_{\cdot,t}^{d}] \\
\mu_{s,t}^{r} &= \frac{1}{2} G^{d,\top} y_s^{d,d} G^d + \frac{1}{2} J[\Sigma_{\cdot,t}^{d}, \Sigma_{\cdot,t}^{d}] \\
\mu_{s,t}^{q} &= \frac{1}{2} G^{c,\top} y_s^{c,c} G^c + G^{c,\top} y_s^{c,f} G^f + \frac{1}{2} J[\Sigma_{\cdot,t}^{c}, \Sigma_{\cdot,t}^{c}] + J[\Sigma_{\cdot,t}^{f}, \Sigma_{\cdot,t}^{c}] - J[\Sigma_{\cdot,t}^{c}, \sigma^{df}]
\end{aligned}$$

(where 1^x denotes the unit loading of the FX factor) and a centred Gaussian noise

$$\varepsilon^{a,i} = \int_s^t \sigma_u^{a,i} M_{u,t}^{a,i}\, d\widetilde{W}_u^{(a,i)}, \quad \varepsilon^x = \int_s^t \Sigma_{u,t}^{df} \cdot dW_u, \quad \varepsilon^r = \int_s^t \Sigma_{u,t}^{d} \cdot dW_u, \quad \varepsilon^q = \int_s^t \Sigma_{u,t}^{c} \cdot dW_u,$$

independent of $\mathcal{F}_s$. Its covariance matrix is $C_{s,t} = (J[\Lambda_k, \Lambda_l])_{k,l}$, where Λ_k runs over the loadings of the noises: $M_{\cdot,t}^{a,i}$, $\Sigma_{\cdot,t}^{df}$, $\Sigma_{\cdot,t}^{d}$ and $\Sigma_{\cdot,t}^{c}$.

Proof. Rates factors. Integrating the dynamic of Theorem 4, $z_t^a = \mathcal{M}_{s,t}^a z_s^a + \int_s^t \mathcal{M}_{u,t}^a (\mathrm{drift}_u^a\, du + \sigma_{z,u}^{a,\top}\, dW_u)$. For the term $y_u^{a,b}\mathbf{1}$ of the drift, Fubini gives

$$\int_s^t \mathcal{M}_{u,t}^a y_u^{a,b} \mathbf{1}\, du = \int_s^t \mathcal{M}_{w,t}^a \sigma_{z,w}^{a,\top} \sigma_{z,w}^b G_{w,t}^b\, dw + \mathcal{M}_{s,t}^a y_s^{a,b} G_{s,t}^b,$$

and $\int_s^t \mathcal{M}_{u,t}^a \sigma_{z,u}^{a,\top} \sigma_u^{df}\, du$ gives the terms $J[M_{\cdot,t}^{a,i}, 1^x]$.

FX. The FX forward $x_{u,t}^{df|d}$ is a lognormal $\mathbb{Q}^{d|d,t}$-martingale with volatility $\Sigma_{u,t}^{df}$ and $x_{t,t}^{df|d} = x_t^{df}$. So $\ln x_t^{df} = \ln x_{s,t}^{df|d} - \frac{1}{2} J[\Sigma^{df}, \Sigma^{df}] + \int_s^t \Sigma_{u,t}^{df} \cdot dW_u^{d|d,t}$. We express $x_{s,t}^{df|d} = x_s^{df} Z_{s,t}^{f|d} / Z_{s,t}^{d|d}$ with the reconstruction formulas and use $dW_u^{d|d,t} = dW_u + \Sigma_{u,t}^d du$.

Numeraire. Integrating $d \ln Z_{u,t}^{d|d} = (r_u^{d|d} - \frac{1}{2}|\Sigma_{u,t}^d|^2) du - \Sigma_{u,t}^d \cdot dW_u$ from s to t gives $\int_s^t r_u^{d|d} du = -\ln Z_{s,t}^{d|d} + \frac{1}{2} J[\Sigma^d, \Sigma^d] + \varepsilon^r$, and we conclude with the reconstruction formula. The same argument applied to $Z_{u,t}^{f|d}$ and $Z_{u,t}^{f|f}$ under $\mathbb{Q}^{f,\mathrm{RN}}$, with $dW^{f,\mathrm{RN}} = dW - \sigma^{df} du$, gives $\int_s^t q_u^{f\to d}\, du$. □

Remark 58. The covariance matrix $C_{s,t}$ does not depend on the measure since all changes of measure we consider have deterministic Girsanov kernels. It is computed once per time step and its Cholesky (or eigenvalue) decomposition is stored. The state is then simulated as $X_t = F_{s,t}X_s + \mu_{s,t} + C_{s,t}^{1/2}\xi$ with $\xi \sim N(0, I)$. Only the components needed by the chosen numéraire are simulated: $\int r^{d|d}$ under $\mathbb{Q}^{d,\mathrm{RN}}$, nothing under the forward and jumping numéraire measures.

9.4.2. *Drifts under the risk-neutral, T-forward and jumping numéraire measures.*

Proposition 47 (Change of numéraire)**.** Let $\varepsilon^k = \int_s^t \Lambda_{k,u} \cdot dW_u$ be any of the noises of Proposition 46.

(1) Under the domestic T-forward measure $\mathbb{Q}^{d|d,T}$, $T \geq t$, the transition is the same with means

$$\mu_{s,t}^{k,T} \quad = \quad \mu_{s,t}^k - J[\Lambda_k, \Sigma_{\cdot,T}^d]$$

and the same covariance matrix. For example, for $T = t$ the domestic rates means reduce to $\mu_{s,t}^{d,t} = \mathcal{M}_{s,t}^d y_s^{d,d} G_{s,t}^d$, and the FX mean to $\frac{1}{2}G^{d,\top} y_s^{d,d} G^d - \frac{1}{2}\hat{G}^\top \hat{y}_s \hat{G} - \frac{1}{2}J[\Sigma^{df}, \Sigma^{df}]$.

(2) The jumping numéraire associated with a tenor structure $T_0 = 0 < T_1 < \dots$ is

$$N_t \quad = \quad Z_{t,T_{q(t)+1}}^{d|d} \prod_{k=0}^{q(t)} \frac{1}{Z_{T_k,T_{k+1}}^{d|d}}, \qquad q(t) = \max\{k : T_k \leq t\}. \tag{248}$$

It corresponds to rolling an investment in the zero coupon bond maturing at the next tenor date. On $[T_k, T_{k+1}]$ the associated measure $\mathbb{Q}^{d,J}$ coincides with the T_{k+1}-forward measure. So, when the simulation grid is contained in the tenor structure, the means of a step $[t_k, t_{k+1}]$ under $\mathbb{Q}^{d,J}$ are given by 1. with $T = t_{k+1}$.

Proof. 1. The Girsanov kernel is deterministic: $dW_u^{d|d,T} = dW_u + \Sigma_{u,T}^d du$, so $\varepsilon^k = \int_s^t \Lambda_{k,u} \cdot dW_u^{d|d,T} - J[\Lambda_k, \Sigma_{\cdot,T}^d]$. For the domestic factors with $T = t$, the term $\sum_j J[M_{\cdot,t}^{d,i}, G_{\cdot,t}^{d,j}]$ cancels with the shift. For the FX, the shift cancels the last term of μ^x.

2. N is continuous at the tenor dates. Between T_k and T_{k+1} it is proportional to $Z_{t,T_{k+1}}^{d|d}$, with an $\mathcal{F}_{T_k}$-measurable factor, so the density of the measure change is that of the T_{k+1}-forward measure on that interval. □

Remark 59. The jumping numéraire has several advantages:

- The numéraire update $N_{T_{k+1}} = N_{T_k}/Z_{T_k,T_{k+1}}^{d|d}$ only requires the reconstruction formula at T_k, whereas $\mathbb{Q}^{d,\mathrm{RN}}$ requires the simulation of $\int r_u^{d|d} du$.
- On each step, the means are the simplest ones: the domestic factors only carry $\mathcal{M}_{s,t}^d y_s^{d,d} G_{s,t}^d$, and the log FX spot is the log FX forward $\ln x_{s,t}^{df|d}$, lognormally corrected.

An earlier version of the formula for N_t used $Z_{t,T_{q(t)}}$, which is not defined since $T_{q(t)} \leq t$.

Remark 60 (Numerical verification)**.** We propagated the exact Gaussian law of the state through a quarterly grid up to 3Y with the transitions above and the model of the verification of Theorem 4. Under $\mathbb{Q}^{d,\mathrm{RN}}$, the mean and covariance agree with those obtained by integrating the moment ODEs of Theorem 4 to $4 \cdot 10^{-17}$ and $8 \cdot 10^{-17}$ (absolute). The martingale identities for $Z_{\cdot,T}^{d|d}$, $x^{df} Z_{\cdot,T}^{f|d}$ and $x^{df} Z_{\cdot,T}^{f|f}$ (under $\mathbb{Q}^{d,\mathrm{RN}}$), for $Z_{\cdot,T}^{d|d}/Z_{\cdot,T^*}^{d|d}$ and the FX forward (under $\mathbb{Q}^{d|d,T^*}$, $T^* = 20\mathrm{Y}$), and for $Z_{\cdot,T}^{d|d}/N$ and $x^{df} Z_{\cdot,T}^{f|d}/N$ (under $\mathbb{Q}^{d,J}$) all hold to machine precision. The measure shifts are not small: the means of the simulated variables at 3Y differ by up to $1.4 \cdot 10^{-4}$ between $\mathbb{Q}^{d,\mathrm{RN}}$ and $\mathbb{Q}^{d,J}$, and $1.4 \cdot 10^{-2}$ between $\mathbb{Q}^{d,\mathrm{RN}}$ and $\mathbb{Q}^{d|d,T^*}$.

Remark 61 (Initial curves)**.** The transitions only involve the initial curves through $Z_{0,s,t}$ at simulation dates and $Z_{0,t,T}$ at payment dates. The instantaneous forward curve $r_{0,\cdot}$ never needs to be evaluated pointwise. In particular, the discontinuities of $r_{0,\cdot}$ created by piecewise flat interpolation have no effect on the simulation (see subsection 10.7).

10. The Linear Gaussian Collateral Model: Calibration

10.0. **Introduction.**

10.0.1. *Single currency market.* The model incorporates a Linear Gaussian Model for each currency. This sub-model is meant to price instruments with native collateral.

In practice, this means that for each currency, we have an embedded single currency LGM which can be calibrated independently to its native IR market:

(1) Native discount $Z^{x|x}_{t,T}$ and index curves $L^{x|x}_{t,T;\theta}$ can be built by any standard method (independently of volatilities[42]).
(2) Volatilities $\sigma^{x|x}_{t,T}$ can be bootstrapped on the fly.
(3) Mean reversions can be marked/calibrated externally as usual to match other ATM vols or, in the multi-dimensional case, some delivered rates correlations.

10.0.2. *Cross-currency market.* The next step is to calibrate to the cross-currency market. We focus on the G10 currencies where liquid instruments are

(1) FX forwards (for short maturities)
(2) Resetting currency basis swaps
(3) FX options

We will assume that they are collateralised in currency d (USD in practice).

10.0.3. *Calibration targets and marked inputs.* The calibration described in this section, and implemented in `calibration.py` of the numerical companion, takes the following quotes:

(1) native swap rates in d and f: OIS swaps, whose floating leg pays compounded overnight (RFR) coupons, or Libor swaps with a constant funding spread;
(2) the FX spot x_0^{df} and d-collateralised FX forwards (outrights or points), in the direct quotation (x^{df}) or the inverted one ($1/x^{df}$);
(3) spot starting resetting currency basis (CB) swap spreads, d-collateralised, whose legs pay Libor or compounded overnight coupons;
(4) ATM normal volatilities of native swaptions in d and f;
(5) ATM Black volatilities of FX options;
(6) ATM normal volatilities of forward resetting CB spreads (implied, or historical estimates when there is no option market).

The following inputs are marked, not calibrated: the mean reversions $\beta^{a,i}$, the correlation matrix Ω, the native funding spreads κ^e and, within each block $a \in \{d, f, c\}$, fixed factor weights $w^{a,i}$. The spot volatilities are parametrised as

$$\sigma_t^{a,i} \quad = \quad \lambda_t^a \, w^{a,i}, \qquad a \in \{d, f, c\}, \tag{249}$$

where the scalar level λ_t^a and the FX spot volatility σ_t^x are piecewise constant on the expiries of the options that calibrate them, with flat extrapolation after the last expiry. The model knots are the union of all expiries. Calibrating one level per block with fixed weights is a design choice: it gives one unknown per option quote, and it keeps the shape of the factor structure that was marked. If a block has no option quotes (e.g. no CB volatilities), its level is marked.

The companion has three example markets (`market_data.example_quotes`), all with USD $= d$, SOFR collateral and SOFR domestic coupons: EURUSD (the original example, EUR $= f$ with ESTR collateral and EURIBOR 3M as the foreign index, i.e. Libor-type foreign coupons), GBPUSD (SONIA swaps and SONIA vs SOFR resetting CB swaps) and USDJPY (TONA swaps and TONA vs SOFR resetting CB swaps, FX quoted inverted). They are described in Remark 76. Their quotes are rounded and illustrative. They are not live market data. Unless stated otherwise, "the example market" below is EURUSD.

[42] Spot volatilities are included in Future convexity adjustments but they do not require a very precise calibration of the volatility so these convexity adjustments are typically marked externally if not hardcoded. Conversely, the values of the futures in the curve building have almost no impact on the volatilities calibrated to swaption vols.

10.1. **Calibration instruments.** We write $\kappa^e = 1 - \theta S^{e|e,\div} = 1 + \theta s^e$ for the constant multiplicative funding spread of the index of currency e with tenor θ (29), $\kappa^e = 1$ for an OIS index, whose coupons are compounded overnight (Remark 1). For a compounded coupon, L denotes the forward coupon rate (21). The spot date is 0.

Definition 7 (Calibration instruments)**.** (1) *Swaps.* A swap in currency e under native collateral on $[T_0, T_n]$ has floating dates $T_0 = t_0^f < \ldots < t_m^f = T_n$ every index tenor θ and fixed dates $t_1^x < \ldots < t_p^x = T_n$ with accruals τ_k (its own frequency). With $Z_t = Z_{0,t}^{e|e}$, its forward par rate is

$$Y = \frac{\mathrm{Fl}}{A}, \qquad \mathrm{Fl} = \sum_{i=1}^{m} \left(\kappa^e Z_{t_{i-1}^f} - Z_{t_i^f} \right) = \sum_{m'} c_{m'}^f Z_{t_{m'}^f}, \qquad A = \sum_{k=1}^{p} \tau_k Z_{t_k^x}, \tag{250}$$

since $1 + \theta L^{e|e}_{0,t^f_{i-1};\theta} = \kappa^e Z_{t^f_{i-1}} / Z_{t^f_i}$. The coefficients are $c^f = \kappa^e$ at T_0, $\kappa^e - 1$ at the intermediate floating dates and -1 at T_n. For an OIS (RFR) swap, $\kappa^e = 1$ holds exactly by Lemma 1, whatever the floating frequency and the compounding (daily or continuous): the floating leg is $Z_{T_0} - Z_{T_n}$. The GBP and JPY swaps of the examples are annual fixed against compounded SONIA or TONA.

(2) *Rate futures.* A future on the index of currency e over $[a, b]$ (e.g. 3M SOFR) is quoted as the price $100(1 - R^{\mathrm{fut}})$, with

$$R^{\mathrm{fut}} = L^{e|e}_{0,a;b-a} + \gamma_{a,b}, \qquad 1 + (b-a) L^{e|e}_{0,a;b-a} = \kappa^e \frac{Z^{e|e}_{0,a}}{Z^{e|e}_{0,b}}, \tag{251}$$

where the forward is model-free (Lemma 1 for compounded coupons) and $\gamma_{a,b} = \mathrm{Fut}^e_{0,a;b-a} - L^{e|e}_{0,a;b-a}$ is the futures convexity adjustment of (22). It depends on the model; as in market practice it is *marked* (an input of the quote, see the footnote of the introduction of this section), not computed from the LGCM. The example SOFR strip uses the Ho–Lee value $\gamma_{a,b} = \frac{1}{2}\sigma^2 ab$ with $\sigma = 95$bp (0 to 1.58bp over 0–2Y).

(3) *FX forwards.* d-collateralised, quoted as the outright $x_{0,T}^{df|d}$ or as points $(x_{0,T}^{df|d} - x_0^{df}) \cdot \pi$, with the pip factor $\pi = 10^4$. In the *inverted* quotation the market quotes the reciprocal pair (units of f per unit of d, e.g. USDJPY in JPY per USD when $d =$ USD and $f =$ JPY): the spot quote is $1/x_0^{df}$, the outright is $1/x_{0,T}^{df|d}$ and the points are $(1/x_{0,T}^{df|d} - 1/x_0^{df}) \cdot \pi$ ($\pi = 100$ for USDJPY). The model state is always $x = x^{df}$, so the model spot is the reciprocal of the quoted spot. This is exact: $1/x_{0,T}^{df|d} = x_0^{fd} Z_{0,T}^{d|d} / Z_{0,T}^{f|d}$ is the d-collateralised forward of the inverted pair.

(4) *Resetting CB swaps.* d-collateralised, on the schedule $T_j = T_0 + j\theta$, $j = 0, \ldots, n$, with θ the tenor of the foreign index. With $P_j = Z_{0,T_j}^{f|d}$ and $F_j = Z_{0,T_j}^{f|f}$, the par spread is

$$B = \frac{\sum_{j=1}^n \theta P_j B_j}{\sum_{j=1}^n \theta P_j}, \qquad \theta B_j = \kappa^d \frac{P_{j-1}}{P_j} - \kappa^f \frac{F_{j-1}}{F_j} K_j, \tag{252}$$

where $\theta B_j = (1 + \theta L^{f/d|d,r}_{0,T_{j-1};\theta}) - (1 + \theta L^{f|d}_{0,T_{j-1};\theta})$ is the swaplet spread of (27) and

$$K_j = \frac{1 + \theta L^{f|d}_{0,T_{j-1};\theta}}{1 + \theta L^{f|f}_{0,T_{j-1};\theta}} = K_0^C(T_{j-1}, T_j) \tag{253}$$

is the collateral convexity of the foreign coupon: the Libor collateral convexity for Libor coupons, the compounded one for RFR coupons. This is Proposition 12 at $t = 0$: the formula holds for both coupon types, the domestic (resetting) leg is model-free, and the coupon type only changes κ^d, κ^f and K_j.

(5) *Swaptions.* An ATM option on a native forward-starting swap with expiry T_0 (strike $K = Y$), quoted as the normal volatility σ_N defined from the payer price PV by $\mathrm{PV} = A_0 \, \sigma_N \sqrt{T_0/(2\pi)}$, $A_0 = \sum_k \tau_k Z^{e|e}_{0,t_k^x}$ (ATM Bachelier price under the annuity measure).

(6) *CB swaptions.* An ATM option on the forward resetting CB spread $B_{T_0} = B^{f/d|d,r}_{T_0,T_0;M}$, quoted as the normal volatility $\sigma_N = \mathbb{E}^{f|d,A}[(B_{T_0} - B_0)^+] \sqrt{2\pi/T_0}$ under the annuity measure.

(7) *FX options.* An ATM option on x_T^{df}, quoted as a Black volatility σ_{BS}. In the LGCM, $\ln x_T^{df}$ is Gaussian under $\mathbb{Q}^{d|d,T}$ with variance $w_0^{df}(T) = \int_0^T |\Sigma_{u,T}^{df}|^2 du$, which does not depend on the collateral, so the quote is $\sigma_{\mathrm{BS}}^2 T = w_0^{df}(T)$. In the inverted quotation the option is on $1/x_T^{df}$; its ATM Black volatility is the same (Proposition 48), so the quote is used as is.

Proposition 48 (FX options under the inverted quotation)**.** Let $x = x^{df}$ and $\tilde{x} = 1/x = x^{fd}$, and $F = x_{0,T}^{df|d}$. Consider a call on $\tilde{x}_T$ with strike $\tilde{K}$ paid in f and a put on x_T with strike $K = 1/\tilde{K}$ paid in d, both d-collateralised, and their Black volatilities, defined with the forwards $\tilde{F} = \mathbb{E}^{f|d,T}[\tilde{x}_T]$ and $F = \mathbb{E}^{d|d,T}[x_T]$.

(1) In any arbitrage free model, $\tilde{F} = 1/F$ and the two Black volatilities are equal. In particular the ATM Black volatility of $1/x$ (strike $1/F$) equals the ATM Black volatility of x (strike F), and the same holds for calls and puts by put-call parity.
(2) In the LGCM, both smiles are flat: $\sigma_{\mathrm{BS}}^2 T = w_0^{df}(T)$ at every strike.

Proof. 1. By the change of currency formula for FX forwards in the section on collateral convexity, $\mathbb{E}^{d|d,T}[x_T v_T^f] = F\,\mathbb{E}^{f|d,T}[v_T^f]$ for any payoff v_T^f in f. With $v^f = \tilde{x}_T$ this gives $\tilde{F} = 1/F$, and with $v^f = (\tilde{x}_T - \tilde{K})^+$,

$$\mathbb{E}^{f|d,T}\left[(\tilde{x}_T - \tilde{K})^+\right] \;=\; \frac{1}{F}\mathbb{E}^{d|d,T}\left[x_T\left(\tfrac{1}{x_T} - \tilde{K}\right)^+\right] \;=\; \frac{\tilde{K}}{F}\,\mathbb{E}^{d|d,T}\left[(K - x_T)^+\right].$$

Let σ be the Black volatility of the put, $\mathbb{E}^{d|d,T}[(K-x_T)^+] = KN(-d_2) - FN(-d_1)$ with $d_{1,2} = (\ln(F/K) \pm \frac{1}{2}\sigma^2 T)/(\sigma\sqrt{T})$. The right hand side is then $N(-d_2)/F - \tilde{K}N(-d_1)$. The Black call price of $\tilde{x}$ with forward $1/F$, strike $\tilde{K}$ and the same σ is $\frac{1}{F}N(\tilde{d}_1) - \tilde{K}N(\tilde{d}_2)$ with $\tilde{d}_{1,2} = (\ln(1/(F\tilde{K})) \pm \frac{1}{2}\sigma^2 T)/(\sigma\sqrt{T}) = -d_{2,1}$, the same number. Black prices are strictly increasing in the volatility, so σ is the Black volatility of the call on $\tilde{x}$. ATM, $\tilde{K} = 1/F$ corresponds to $K = F$.

2. $\ln x_T$ is Gaussian under $\mathbb{Q}^{d|d,T}$ with variance $w_0^{df}(T)$ (Proposition 53, item 3), so the put prices are Black prices with $\sigma^2 T = w_0^{df}(T)$ at every strike, and item 1 transfers this to $\tilde{x}$. □

Remark 62. Item 1 is the usual symmetry of FX volatilities: it uses only the FX forward identity, not the model. The implementation (`FXForward` with `inverted=True` and `MarketQuotes.fx_inverted` in `instruments.py`) converts the inverted spot and forwards to the model pair and leaves the FX volatility quotes unchanged. The test `test_fx_option_vol_is_invariant_under_inversion` checks the price identity of item 1 by adaptive quadrature of the lognormal density. In the Monte Carlo check the standard error of an inverted quote $g(F)$ is propagated by the delta method, $\pi s_F/F^2$ for points with s_F the standard error of the outright F (the second order bias, $O(s_F^2/F^3)$, is far below the standard error).

Remark 63 (Curve keys)**.** A discount curve $Z^{e|c}$ is identified by the pair (currency e, collateral overnight index c), e.g. $Z^{f|d}$ = (EUR, USD SOFR). The calibrated curves are $Z^{d|d}$, $Z^{f|f}$ and $Z^{f|d}$, the three initial curves of Theorem 4. The off-market curve $Z^{d|f}$ is a model output (see subsection 10.6): it is not bootstrapped from quotes.

10.2. **Discount curves.**

Proposition 49 (Model-free curves)**.** Assume constant funding spreads (29). Then, in any arbitrage free model:

(1) the par rates of native swaps are functions of the native discount curve $Z_{0,\cdot}^{e|e}$ only, given by (250);
(2) the cross-currency curve is determined by the d-collateralised FX forwards:

$$Z_{0,T}^{f|d} \;=\; \frac{x_{0,T}^{df|d}\, Z_{0,T}^{d|d}}{x_0^{df}}. \tag{254}$$

In particular, the native curves and $Z_{0,\cdot}^{f|d}$ up to the last FX forward pillar do not depend on the model. With piecewise flat forwards and one node per pillar, the native curves are obtained by a sequential bootstrap of the swap quotes.

Proof. 1. Each floating coupon is replicated by κ^e zero coupon bonds maturing at its start minus one at its end, by (29) and the definition of $L^{e|e}$ as an FRA par rate for Libor coupons, and by Lemma 1 (with $\kappa^e = 1$

and no hypothesis) for compounded ones; fixed coupons are zero coupon bonds. 2. This is (26) at $t = 0$ with $i = d$. □

Beyond the last FX forward, the resetting CB swaps determine $Z^{f|d}$, but (252) also involves the coupon collateral convexities $K_j = K_0^C(T_{j-1}, T_j)$. They depend on the model: $K_j = 1$ in the deterministic collateral basis (DCB) model, and in the LGCM they are given by Proposition 26 at $t = 0$: (138) for daily compounded coupons and, for Libor coupons, by (187),

$$K_j = \exp\left\{-\int_0^{T_{j-1}} \Sigma^{f|f}_{u,T_{j-1};\theta} \cdot \Sigma^{f\to d}_{u,T_j}\, du\right\}. \tag{255}$$

So the curve bootstrap only needs a *convexity provider* that returns K_j.

Algorithm 2 (Curve bootstrap with a convexity provider)**.** Input: linear quotes (swaps, FX forwards, resetting CB swaps) and a pricer that returns $K(T_{j-1}, T_j)$.

(1) Group the quotes by the curve they determine, and process the curves in dependency order: $Z^{d|d}$ and $Z^{f|f}$ first, then $Z^{f|d}$, whose FX forwards read $Z^{d|d}$ and whose CB swaps read $Z^{f|f}$.
(2) Native curves have piecewise flat instantaneous forwards. The curve $Z^{f|d}$ is the product of $Z^{f|f}$ and the collateral basis curve $\Xi Z^{f|f\to d}$, which has piecewise flat forwards (see the remark "Which curve to interpolate" in subsection 10.7).
(3) On each curve, sort the quotes by (priority, pillar), FX forwards before CB swaps. For each quote whose pillar lies beyond the last node, add a node at the pillar. The flat forward on the new segment is the only unknown: solve (model quote) = (market quote) by a bracketed 1D root search (Brent).
(4) A quote whose pillar is not beyond the last node of its curve is not used. It is reported as a check.
(5) Flat extrapolation after the last pillar.

The DCB market practice is the case $K_j = 1$ (`calibrate_curves(quotes)`). The LGCM uses the coupon convexity of the foreign index ((255) for Libor, (138) for daily compounding with the step of the index, $1/365$ by default) with the current model parameters.

Remark 64. (1) Flat forwards are an interpolation choice, not a consequence of the model. They make each bootstrap step a scalar equation, as in the DCB stripping of subsubsection 3.2.4, where the unknown is the average forward rate on the segment.
(2) Quotes inside an already built section (e.g. a CB swap shorter than the last FX forward) are checks, not targets. On the example market all 31 linear quotes set a node and there are no checks.
(3) In the example, $\rho^{f,c} < 0$ and $K_j > 1$ (Remark 75): the LGCM curve bootstrapped from the same quotes differs from the DCB curve after the last FX forward.
(4) Futures are bootstrapped like the other linear quotes, before them on the same curve (priority -1): each future sets the flat forward up to its end date. Swaps maturing inside the futures strip then become checks (see Remark 66).

10.2.1. *Global curve calibration.* The bootstrap ties the interpolation to the quotes: one node per quote, one scalar equation per node, and piecewise flat forwards, whose jumps at the pillars are visible in Figure 5. Current practice separates the two: the curve has its own parametrisation, piecewise flat where the market quotes forward rates directly (futures, FX forwards) and smooth where it quotes long par rates, and all its parameters are fitted together by non-linear least squares. Curves that depend on each other are calibrated as nested curve sets.

Definition 8 (Hybrid forward curve)**.** Let $0 = s_0 < s_1 < \ldots < s_m$ (flat nodes, $m \geq 1$) and $s_m = k_0 < k_1 < \ldots < k_n$ (spline knots, $n \geq 0$). The hybrid curve with parameters $x = (f_1, \ldots, f_m, a_1, \ldots, a_n)$ has the instantaneous forward

$$r_{0,t} = f_k \quad (s_{k-1} < t \leq s_k), \qquad r_{0,t} = I'(t) \quad (s_m < t \leq k_n), \qquad r_{0,t} = I'(k_n) \quad (t > k_n),$$

where I is the cubic spline on the knots $k_0, \ldots, k_n$ with $I(k_i) = \sum_{j\leq i} a_j(k_j - k_{j-1})$, $I'(k_0) = f_m$ and $I''(k_n) = 0$. So a_i is the average forward on the section $(k_{i-1}, k_i]$ and $\int_{s_m}^t r_{0,u}du = I(t)$ on the spline section; $-\ln Z_{0,t}$ is explicit everywhere. The forward is piecewise flat on $(0, s_m]$, continuous at s_m, piecewise quadratic and C^1 on $[s_m, k_n]$ and flat after k_n, with $r'_{0,k_n} = 0$, so it is C^1 at k_n as well. A cross-currency

curve is $Z^{f|d} = Z^{f|f}\,\Xi Z^{f|f\to d}$ with a hybrid collateral basis curve $\Xi Z^{f|f\to d}$ (the remark "Which curve to interpolate").

Proposition 50 (Smooth forward section)**.** Among the C^2 functions J on $[k_0, k_n]$ with $J(k_i) = I(k_i)$, $i = 0, \dots, n$, and $J'(k_0) = f_m$, the spline I of Definition 8 minimises $\int_{k_0}^{k_n} (J'')^2 = \int_{k_0}^{k_n} (r'_{0,t})^2 dt$: among the forward curves with the same section averages, continuous at the junction, it has the least squared slope.

Proof. Let $\phi = J - I$, so $\phi(k_i) = 0$ and $\phi'(k_0) = 0$. Then $\int (J'')^2 = \int (I'')^2 + 2\int I''\phi'' + \int (\phi'')^2$. On $[k_{i-1}, k_i]$, I''' is a constant c_i, and integrating by parts, $\int_{k_{i-1}}^{k_i} I''\phi'' = [I''\phi']_{k_{i-1}}^{k_i} - c_i[\phi]_{k_{i-1}}^{k_i} = [I''\phi']_{k_{i-1}}^{k_i}$. Summing over i (I'' is continuous), $\int I''\phi'' = I''(k_n)\phi'(k_n) - I''(k_0)\phi'(k_0) = 0$ by $I''(k_n) = 0$ and $\phi'(k_0) = 0$. Hence $\int (J'')^2 = \int (I'')^2 + \int (\phi'')^2 \geq \int (I'')^2$. □

Remark 65 (Why the spline is on the integrated forward)**.** Swap rates identify discount factors, i.e. integrals of the forward, not its point values. Table 3 compares a natural cubic spline of the forward through free knot values r_{0,k_i} (flat forward to 1Y), fitted exactly to the nine USD swaps of the example, with the hybrid curve fitted to the same swaps: the fit is exact in both cases, but the knot-value spline zig-zags (negative control in `tests/test_curve_fitting.py`). Parametrising by the section averages a_i keeps the parameters close to what the quotes measure, and Proposition 50 says what is smooth about the result.

knot (years)	1	2	3	5	7	10	15	20	30
spline through knot values (%)	4.21	3.11	3.82	2.98	4.58	2.88	5.73	1.33	9.88
hybrid curve (%)	4.21	3.29	3.47	3.56	3.83	4.04	4.07	3.70	3.10

TABLE 3. Instantaneous forward at the knots, USD swaps of the EURUSD example (DCB), both fits exact.

Algorithm 3 (Nested global curve calibration)**.** Input: linear quotes (swaps, futures, FX forwards, resetting CB swaps), a pricer that returns $K(T_{j-1}, T_j)$ (as in Algorithm 2), and for each target curve the nodes of Definition 8. By default, the flat nodes are the end dates of the futures and the FX forward maturities (or the first pillar if the curve has neither) and the spline knots are the remaining pillars.

(1) Group the target curves into *nested curve sets* by dependency level: level 0 are the curves whose quotes read no other target ($Z^{d|d}$, $Z^{f|f}$), level ℓ the curves whose quotes read curves of level $< \ell$ ($Z^{f|d}$, level 1).

(2) Level by level, with the curves of lower levels frozen, find the parameters x of all the curves of the level by Levenberg–Marquardt on

$$\min_x \sum_q w_q^2 \left(\frac{\text{model}_q(x) - \text{market}_q}{u_q} \right)^2 + \lambda\, 10^8 \sum_{\text{curves}} \int (r'_{0,t})^2\, dt,$$

with u_q = 1bp for rates and spreads, 0.01 for futures prices (1bp of rate) and 1 pip for FX forwards, weights w_q (1 by default) and a curvature penalty $\lambda \geq 0$ (0 by default). The penalty is a sum of squares in closed form: $r' = I''$ is linear on each section.

(3) Variant: a single global problem with all the target curves (`nested=False`).

With one quote per parameter and $\lambda = 0$ the problem is square and the fit is exact. With more quotes than parameters (e.g. 1Y and 2Y swaps inside a 2Y futures strip) it is a least squares compromise and the residuals are reported. With fewer, a positive λ is required. In the LGCM calibration, Algorithm 4 uses this algorithm instead of Algorithm 2 in steps 1 and 3 when `ModelSpec(curves=GlobalFit())`; the fixed point with λ^c is unchanged and warm starts each fit from the previous one.

Remark 66 (Global fit against the bootstrap)**.** Example markets, deterministic collateral basis (DCB) curves unless stated. The tests are in `tests/test_curve_fitting.py`; the figures and the table are generated by `python -m convexity.figures`.

(1) *Two independent algorithms, one curve.* With flat nodes at every pillar and no spline knots, the global fit returns the bootstrapped curves of Algorithm 2 (discount factors agree to 10^{-12} relative up to 40Y). Quotes generated from given hybrid curves are fitted back to the same parameters (10^{-11}), nested or in a single problem.
(2) *Shape.* Figure 5 compares the forwards. Both curves reprice every quote (Table 4); they agree where the quotes pin the discount factors and differ between the pillars and in the extrapolation, flat at r_{0,k_n} for the hybrid curve and at the last section average for the bootstrap.
(3) *Futures.* With the SOFR strip, the 1Y and 2Y swaps lie inside the futures section. The bootstrap reprices the futures and leaves the two swaps as checks; the least squares spreads the inconsistency over the futures and the swaps (Figure 6 (a)) and satisfies the first order conditions $J^\top r = 0$ (checked with a finite difference Jacobian).
(4) *LGCM.* With `GlobalFit()` the calibration reprices all 65 quotes of each market, in as many fixed point iterations as with the bootstrap (Table 4). The calibrated volatilities move by at most about 1% (the long λ^d, λ^f pieces, whose swaptions see the forwards between pillars). The collateral convexity of the 30Y $Z^{f|d}$ zero rate (LGCM minus DCB) is the same with both interpolations to a few hundredths of a bp: it is a property of the model, not of the interpolation (the remark "Interpolation and convexity adjustments decouple in the LGCM").
(5) *Locality.* The spline is not local (Figure 6 (b)): a bump of one long swap moves the bootstrapped forwards on its section and the next only, but the hybrid forwards on every section. Smoothness is paid for with less local hedge ratios.
(6) *Smoothing.* With more parameters than quotes the fit requires a penalty, and increasing it trades fit for smoothness monotonically.

	DCB curves			LGCM calibration			
market	max res. (bp)	evaluations	ms (B / G)	iterations	s (B / G)	max $\lvert\Delta\lambda\rvert$ (%)	30Y conv. (bp)
EURUSD	1.5e-12	11, 14	44 / 1065	5 / 5	5.8 / 10.4	1.22	3.75 / 3.76
GBPUSD	1.9e-12	11, 18	43 / 1566	5 / 5	6.2 / 11.7	1.22	2.94 / 2.97
USDJPY	2.7e-12	9, 13	42 / 1190	5 / 5	6.1 / 10.2	1.22	3.19 / 3.20

TABLE 4. Bootstrap (B) against global fit (G). DCB curves: largest residual of the global fit over all quotes (bp; pips for FX forwards), residual evaluations per nested level, wall time. LGCM calibration: fixed point iterations, wall time, largest relative change of $\lambda^d, \lambda^f, \lambda^c, \sigma^x$ on any piece, collateral convexity of the 30Y $Z^{f|d}$ zero rate (LGCM minus DCB on the same quotes).

10.3. **Calibration to single currency markets.**

10.3.1. *Calibration of spot-volatility to the cap market.* Not implemented: the companion calibrates the native blocks to swaptions only. Caplets on Libor-type and compounded coupons are priced exactly (Proposition 43) but are not yet calibration targets.

10.3.2. *Calibration of spot-volatility to the swaption market.* We consider a native block $a \in \{d, f\}$ and a swaption in currency a under native collateral. For a swap with cash flow dates t_m (the union of the floating and fixed dates), we write c_m for the coefficients of the payer swap at the ATM strike K: c_m^f of (250) on the floating dates and $-K\tau_k$ on the fixed dates, summed for equal dates. We also write $P_m = Z^{a|a}_{0,t_m}/Z^{a|a}_{0,T_0}$.

Proposition 51 (Exact one factor swaption)**.** Assume the block a has one factor, and write $z = z^a_{T_0}$, $y = y^{a,a}_{T_0}$, $G_m = G^a_{T_0,t_m}$. Then:

(1) Under $\mathbb{Q}^{a|a,T_0}$, $z \sim N(0, y)$ and $Z^{a|a}_{T_0,t} = P(t)\exp(-G^a_{T_0,t}z - \frac{1}{2}(G^a_{T_0,t})^2 y)$, with $P(t) = Z^{a|a}_{0,T_0,t}$.

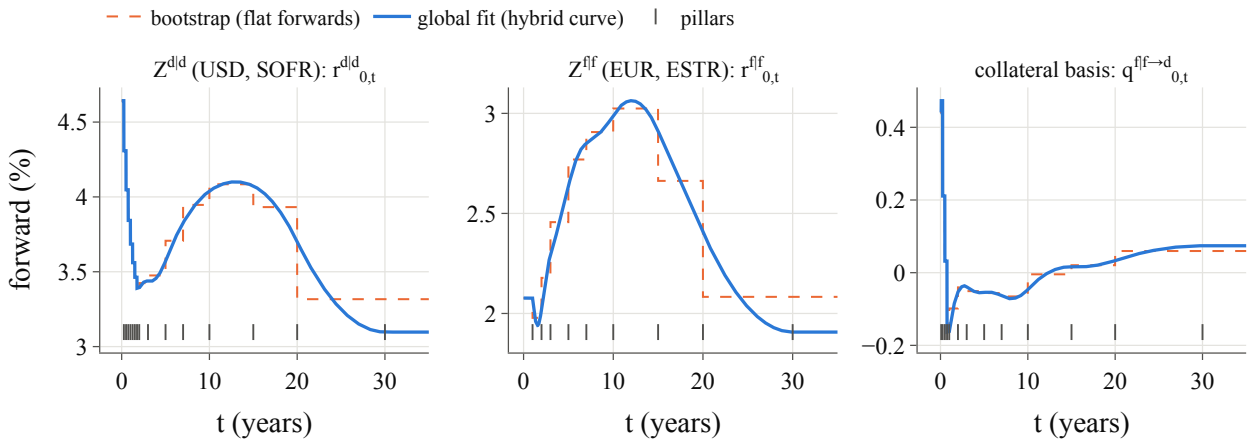


FIGURE 5. Instantaneous forwards of the EURUSD example with the SOFR futures (DCB): bootstrap with flat forwards (dashed) against the global fit of hybrid curves (solid); ticks mark the pillars. The collateral basis forward is $r^{f|d}_{0,t} - r^{f|f}_{0,t}$.

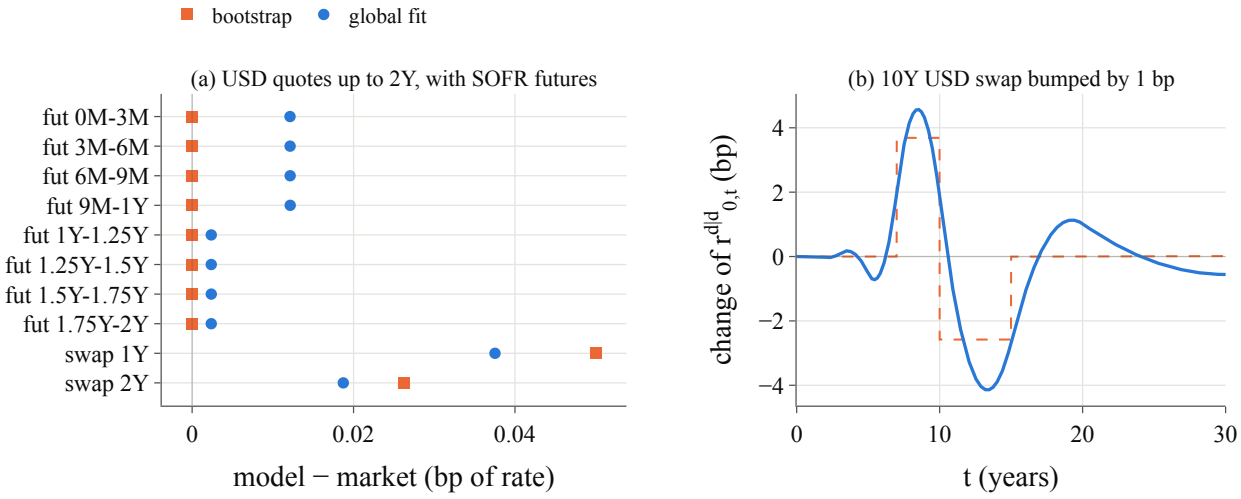


FIGURE 6. (a) Residuals of the USD quotes up to 2Y with the SOFR futures (bp of rate): the bootstrap fits the futures exactly, the least squares spreads the inconsistency. (b) Change of the USD forward when the 10Y SOFR swap is bumped by 1bp (EURUSD example without futures).

(2) The swap value at T_0 is $V(z) = \sum_m c_m P_m e^{-G_m z - \frac{1}{2}G_m^2 y}$. Assume there is z^* such that $V(z) > 0$ iff $z > z^*$ (so $V(z^*) = 0$). Then the payer swaption price is

$$\text{PV} \;=\; Z^{a|a}_{0,T_0}\,\mathbb{E}^{a|a,T_0}\left[V(z)^+\right] \;=\; Z^{a|a}_{0,T_0}\sum_m c_m P_m\,\mathcal{N}\left(-\frac{z^* + G_m y}{\sqrt{y}}\right), \tag{256}$$

and the quoted volatility is $\sigma_N = \text{PV}/(A_0\sqrt{T_0/(2\pi)})$.

(3) The assumption of 2. holds when the coefficients $c_m P_m$, ordered by increasing G_m, change sign exactly once, from positive to negative. This is the case for OIS swaps ($\kappa^a = 1$) with $K > -1/\tau_p$, τ_p the last fixed accrual: ordered by date, the coefficients are 1 at T_0 (where $G = 0$), then $-K\tau_k$ at the fixed dates before the end, all of the sign of $-K$, then $-1 - K\tau_p < 0$ at the end date. For $K \geq 0$ the only positive

coefficient is the first one; for $-1/\tau_p < K < 0$ every coefficient except the last is positive. In both cases there is exactly one sign change.

Proof. 1. The reconstruction formula of Definition 5 gives $Z^{a|a}_{T_0,t} = P(t)\exp(-G^a_{T_0,t}z - \frac{1}{2}(G^a_{T_0,t})^2 y)$ with y deterministic. The change of measure from $\mathbb{Q}^{d,\mathrm{RN}}$ to $\mathbb{Q}^{a|a,T_0}$ has a deterministic Girsanov kernel (Proposition 47 and its foreign analogue), so z is Gaussian, say $N(\mu, v)$. The martingale property $\mathbb{E}^{a|a,T_0}[Z^{a|a}_{T_0,t}] = P(t)$ gives $-G\mu + \frac{1}{2}G^2(v-y) = 0$ for all $G = G^a_{T_0,t}$, $t \geq T_0$, which cover an interval. Hence $\mu = 0$ and $v = y$ (compare Lemma 3).

2. The first equality is the forward measure pricing formula, and $V(z)^+ = V(z)\mathbf{1}_{z>z^*}$ by assumption. For each term (exponential tilt),

$$e^{-Gz-\frac{1}{2}G^2y}\,\frac{e^{-z^2/(2y)}}{\sqrt{2\pi y}} \;=\; \frac{e^{-(z+Gy)^2/(2y)}}{\sqrt{2\pi y}},$$

the density of $N(-Gy, y)$. So $\mathbb{E}[e^{-Gz-\frac{1}{2}G^2y}\mathbf{1}_{z>z^*}] = \mathbb{P}(N(-Gy,y) > z^*) = \mathcal{N}(-(z^*+Gy)/\sqrt{y})$.

3. $G^a_{T_0,t} = \int_{T_0}^t M^a_{T_0,u}du$ is increasing in t. Let G^* lie strictly between the largest G_m with a positive coefficient and the smallest G_m with a negative one. Then

$$e^{G^*z}V(z) \;=\; \sum_{c_m>0} c_m P_m e^{-\frac{1}{2}G_m^2 y}\, e^{(G^*-G_m)z} - \sum_{c_m<0} |c_m| P_m e^{-\frac{1}{2}G_m^2 y}\, e^{(G^*-G_m)z}$$

is strictly increasing (the first sum increases since $G_m < G^*$, the second decreases since $G_m > G^*$), and tends to $-\infty$ as $z \to -\infty$ and to $+\infty$ as $z \to +\infty$. It has a unique zero z^*, and V has the sign of $e^{G^*z}V$. For an OIS swap the floating coefficients c^f_m of (250) are 1 at T_0, 0 at the intermediate floating dates ($\kappa^a - 1 = 0$) and -1 at the end date, whatever the floating schedule. So the coefficients are 1 at T_0 and $-K\tau_k$, $-1-K\tau_p$ at the fixed dates, and the sign pattern of item 3 follows. □

Remark 67 (Libor swaps and what the code checks)**.** For a Libor swap ($\kappa^a > 1$) whose floating leg is more frequent than its fixed leg, the intermediate floating dates carry $\kappa^a - 1 > 0$, so the coefficients change sign many times (19 times for the EURIBOR 3M swaps of the example) and item 3 does not apply. The implementation (`pricers.native_swaption_exact_vol`) only checks $V(-12\sqrt{y}) < 0 < V(12\sqrt{y})$, finds z^* by Brent's method on this bracket and applies (256). It raises an error if there is no sign change on the bracket. It does not check that the root is unique. On the calibrated example market (16 swaptions 1Y to 20Y $\times$10Y in USD and EUR) we checked numerically that V changes sign exactly once on a grid of 20001 points of $[-12\sqrt{y}, 12\sqrt{y}]$, and that (256) agrees with an adaptive quadrature of $\mathbb{E}[V^+]$ to $2\cdot10^{-9}$ in relative terms for EUR 15Y$\times$10Y (the quadrature tolerance) and to $7\cdot10^{-14}$ for the others. This is a numerical observation for this market, not a proof for Libor swaps in general.

Proposition 52 (Gaussian swaption variance with separate fixed and floating schedules)**.** Let the block a have $N_a \geq 1$ factors and use the notations of (250) with $P_t = Z^{a|a}_{0,T_0,t}$, $A = \sum_k \tau_k P_{t^x_k}$ and $Y = \sum_m c^f_m P_{t^f_m}/A$. Define the loadings

$$q_i \;=\; \frac{1}{A}\left(-\sum_m c^f_m G^{a,i}_{T_0,t^f_m} P_{t^f_m} + Y\sum_k \tau_k G^{a,i}_{T_0,t^x_k} P_{t^x_k}\right), \qquad i = 1,\dots,N_a. \tag{257}$$

Freezing the gradient of the swap rate with respect to z^a at its forward value, as in Proposition 41, the swap rate at T_0 is approximately normal under the annuity measure with mean Y and variance $q^\top y^{a,a}_{T_0} q$. For equal fixed and floating schedules it is the formula of Remark 47.

Proof. By the reconstruction formula, $\partial_{z_i} Z^{a|a}_{T_0,t}|_{z=0} = -G^{a,i}_{T_0,t} Z^{a|a}_{T_0,t}$ at first order in the volatility. So $\partial_{z_i}\mathrm{Fl} = -\sum_m c^f_m G^{a,i}_{T_0,t^f_m} P_{t^f_m}$, $\partial_{z_i}A = -\sum_k \tau_k G^{a,i}_{T_0,t^x_k} P_{t^x_k}$ and $\partial_{z_i}Y = (\partial_{z_i}\mathrm{Fl} - Y\partial_{z_i}A)/A = q_i$. The covariance of $z^a_{T_0}$ is $y^{a,a}_{T_0}$ (Proposition 31). With equal schedules, $c^f_m = \kappa^a\mathbf{1}_{m=0} + (\kappa^a-1)\mathbf{1}_{0<m<n} - \mathbf{1}_{m=n}$ and $\tau = \theta$; regrouping the terms by period gives the loadings of Remark 47. □

Remark 68 (Accuracy of the Gaussian approximation)**.** Proposition 52 is an asymptotic approximation in the volatility. We measured the relative error of the approximate normal volatility against Proposition 51 on

a 5Y×10Y EURIBOR 3M swaption (test model of `tests/test_calibration.py`) with all volatilities scaled by $\lambda = 0.5, 1, 2$: $2.175 \cdot 10^{-4}$, $8.702 \cdot 10^{-4}$, $3.482 \cdot 10^{-3}$. The ratios are 4.0002 and 4.0010, so the error is $O(\lambda^2)$ in this example (numerical observation, not a proof). On the calibrated example market, the approximate volatility exceeds the exact one by $+2.9 \cdot 10^{-4}$ (EUR 1Y×10Y) to $+5.4 \cdot 10^{-3}$ (USD 20Y×10Y), i.e. up to 0.46bp. The equality with Remark 47 for equal schedules was checked to 10^{-12} against the independent code of `gaussian_approx.approx_native`, and the two factor variance against loadings and covariances computed by quadrature to 10^{-8}.

Remark 69 (Bootstrap of λ^a and initial guess)**.** Let $T_1 < \ldots < T_n$ be the expiries of the swaptions of block a (one swaption per expiry). The k-th swaption sets λ^a on $[T_{k-1}, T_k)$ ($T_0 = 0$), the last one on $[T_{n-1}, \infty)$. A swaption with expiry T_k only reads σ^a on $[0, T_k]$, so the bootstrap is triangular: the earlier swaptions are not moved by the later ones. On the segment, $y^{a,a}_{T_k}$ is the integral of $\sigma^{a,i}\sigma^{a,j}$ over $[0, T_k]$ and the loadings q do not depend on the volatilities, so the Gaussian variance is exactly

$$v(\lambda) \quad = \quad v_0 + \lambda^2 v_1, \qquad v_0 = v(0), \quad v_1 = v(1) - v(0), \tag{258}$$

and $\lambda = \sqrt{(\sigma_N^2 T_k - v_0)/v_1}$ matches the Gaussian approximation in closed form. With one factor, this is the initial guess of a Brent search on the exact volatility of Proposition 51. With several factors, it is the calibrated value: the Gaussian approximation is then the pricer, and the calibration diagnostics flag it. If $v_0 \geq \sigma_N^2 T_k$, the earlier segments already deliver the target variance (variance squeeze) and the calibration fails. Note that this test uses the Gaussian variance.

10.3.3. *Calibration of mean-reversion to ATM swaptions.* Not implemented: mean reversions are marked. The example uses $\beta^d = 3\%$, $\beta^f = 2\%$ and $\beta^c = 10\%$.

10.3.4. *Calibration of mean-reversion to swap rates correlations.* Not implemented: the correlations and the factor weights are marked.

10.4. **Calibration to the cross-currency market.**

10.4.1. *Necessity for joint calibration.* Let us consider things from the curve building point of view. On the cross-currency market, the most liquid long-dated linear instruments are resetting currency basis swaps. Say we calibrated the model up to tenor M, i.e. up to date $T_{\ell M}$ where $f = 4$ is the frequency of the swaps. We now consider the calibration to the tenor N. We have

(1) one new target: the currency basis spread $B^{x/u|u,r}_{t_0,T;N}$
(2) multiple new unknowns: for each $j = \ell M + 1, \dots, \ell N$
   (a) the discount factors $Z^{x|u}_{t_0,T_j}$
   (b) the foreign forward rates $L^{x|u}_{t_0,T_{j-1};\theta}$
   (c) the resetting forward rates $L^{x/u|u}_{t_0,T_{j-1};\theta}$

The problem is clearly under-determined.

However, the model provides convexity adjustments that relate the different forward rates to the ones from native single currency markets. But these convexity adjustments depend on the spot volatilities[43]. Obviously these should be calibrated to match implied volatilities of option prices (FX options and currency basis swaptions) or historical estimations of volatility.

If we assume that correlations are marked by traders, and MR are marked or calibrated externally, we are left with 3 types of unknown parameters:

(1) the cross-currency discount curve $Z^{f|d}_{t_0,T}$
(2) the collateral basis spot volatilities $\sigma^{f\to d}_{t,T}$
(3) the FX spot volatilities σ^{df}_t

And since they interact through convexity adjustments, they have to be jointly calibrated.

Moreover, we want the calibration to be local: for risk management purposes we do not want a 2Y FX option to be exposed to 10Y FX implied vols. This implies that we need to bootstrap the parameters.

10.4.2. *Bootstrapping.*

Introduction. Convexity adjustments induce a lot of interdependence between market quotes. For risk management, we still need to be able to localise the risk. This means that the value of a 1Y FX option for example, should not depend on any market quote with expiry/maturity beyond 1Y.

That is not the case in a global calibration.

The natural way to achieve that is to "bootstrap" all the curves and term structure of parameters.

This can be seen as breaking down the global calibration into steps by sorting calibration targets by expiry/maturity and calibrating model parameters iteratively.

Intuition. Let us assume we observe spot starting currency basis spreads for a given set of maturities M_i.

Let us assume we calibrated the model up to expiry M_i.

Then, we can compute the forward starting spread:

$$A^{f|d}_{0,T_0,T_{\ell N}} B^{f/d|d,r}_{0,T_0,T_{\ell N}} = A^{f|d}_{0,T_0,T_{\ell M}} B^{f/d|d,r}_{0,T_0,T_{\ell M}} + A^{f|d}_{0,T_{\ell M},T_{\ell N}} B^{f/d|d,r}_{0,T_{\ell M},T_{\ell N}} \tag{259}$$

$$B^{f/d|d,r}_{0,T_0,T_{\ell N}} \sum_{j=1}^{\ell N} \theta Z^{f|d}_{0,T_j} = A^{f|d}_{0,T_0,T_{\ell M}} B^{f/d|d,r}_{0,T_0,T_{\ell M}} + \left(\sum_{j=\ell M+1}^{\ell N} \theta Z^{f|d}_{0,T_j} \right) B^{f/d|d,r}_{0,T_{\ell M},T_{\ell N}} \tag{260}$$

$$A^{f|d}_{0,T_0,T_{\ell M}} \left(B^{f/d|d,r}_{0,T_0,T_{\ell N}} - B^{f/d|d,r}_{0,T_0,T_{\ell M}} \right) = \sum_{j=\ell M+1}^{\ell N} \theta Z^{f|d}_{0,T_j} \left(B^{f/d|d,r}_{0,T_{j-1},T_j} - B^{f/d|d,r}_{0,T_0,T_{\ell N}} \right) \tag{261}$$

Unknowns are

- Currency basis swaplet spreads $B^{f/d|d,r}_{0,T_j;\theta}$
- Discount rates $R^{f|d}_{0,T_j;\theta}$

[43] as well as the cross-asset correlations. For now, we assume that they are marked but the same approach could be extended to calibrate instantaneous correlations by adding some terminal correlations/covariances as targets.

The model formulas relate these to the model parameters of which

- Collateral basis spot vol $\sigma_t^{f\to d}$, $t \in [T_{\ell M}, T_{\ell N}[$,
- FX spot vol[44] σ_t^{df}, $t \in [T_{\ell M}, T_{\ell N}[$,

which we assume to be constant on the segment $[T_{\ell M}, T_{\ell N}[$, are unknown.

Let us choose a parametrisation for the discount curve $Z^{f|d}$ or at least the collateral spread $\Xi Z^{f|f\to d}$. For example, we can consider

- piecewise constant forwards
- quadratic forwards together with continuity and derivability constraint at $T_{\ell M}$

The point is that once the interpolation scheme has been chosen, the value of the discount curve on the segment is entirely specified by the average level of rates on the segment:

$$\overline{r}^{f|d}_{0,T_{\ell M},T_{\ell N}} = \frac{1}{T_{\ell N}-T_{\ell M}} \int_{T_{\ell M}}^{T_{\ell N}} r^{f|d}_{0,u}\, du \tag{262}$$

So we have reduced the determination of the discount factors $Z^{f|d}_{t_0,T_i}$ to that of a single parameter.

The problem is now well posed since we have 3 unknowns and 3 constraints as listed in Table 5.

Unknowns	Constraints
Average discount rate	Currency basis spread
ColBasis spot vol	Currency basis spread vol
FX spot vol	FX implied vol

TABLE 5. Bootstrapping

Note that, while mathematically complex, the correspondence between model parameters and calibration targets is very intuitive. The constraints are made explicit in Proposition 53 and the algorithm is Algorithm 4. The pillars of the three constraints need not coincide: in the implementation the collateral basis curve has one node per FX forward and CB swap maturity, λ^c one level per CB volatility expiry and σ^x one level per FX option expiry.

10.4.3. *Model usage and calibration targets.*

Remark 70 (Model usage and calibration targets)**.** The bootstrapping algorithm works as long as we have one target per expiry[45]. There is still a range of choices for the calibration targets that is up to discussion. The answer should be decided based on the model usage.

The model can be used in two distinct ways:

(1) Valuation and risk management. One should specify a set of hedging instruments and calibrate the model these. When an option market is available and these are included in hedging instruments, the internal parameters should be calibrated to options prices. When no option market is available as is the case for currency basis volatility risk, the covariance parameters should be calibrated to the historical covariances of the corresponding market quotes, i.e. currency basis spreads in this case. Ideally one would have only one set of parameters so that pricings are consistent across products and break-even levels are well defined.

(2) Benchmarking and reserve usage. One should mark covariance parameters that cannot be implied from market prices conservatively. In particular, these marks should depend on the structure of the portfolio.

Note that since CB delta is going to be hedged with spot-starting resetting cross-currency swaps, one can argue that we should calibrate to their vol. At the same time, it is trivial to synthesise a forward-starting swap.

[44] Actually it does not appear under constant funding spreads.

[45] With CB spread volatilities of tenor longer than the spacing of the expiries, the volatility of expiry T_0 reads σ^c beyond T_0 (through the coupon convexities $K_{T_0,j}$), so the expiry by expiry bootstrap is no longer triangular. The fixed point of Algorithm 4 resolves this dependency.

Remark 71 (FX options market)**.** The market ignores collateral convexity for FX options. It seems natural to assume FX forwards and volatilities correspond to USD collateralized products.

Note that in the Linear Gaussian model, the volatility of the FX process is invariant under collateral but the FX forward is not. This means that the calibration targets we should chose may not be the standard ATM options. We may want to calibrate to the volatilities corresponding to the adjusted FX forward. In the implementation, the ATM strike is the d-collateralised forward.

Assuming all other parameters are set and we chose target implied volatilities, the FX spot volatility $\sigma^{df}_\cdot$ can be bootstrapped in closed form to match ATM volatilities (Proposition 53). Calibration can only fail if we encounter a variance squeeze.

10.5. **Bootstrapping algorithm.**

10.5.1. *Calibration constraints.* We use the notations of Definition 7, and $\Sigma^c = \Sigma^{f\to d}$, so that $\Sigma^{d\to f} = -\Sigma^c$.

Proposition 53 (Calibration constraints)**.** (1) *CB spread.* Write $P_j = F_j\,\Xi\mathrm{Z}_j$ with $\Xi\mathrm{Z}_j = \Xi\mathrm{Z}^{f|f\to d}_{0,T_j}$. On a bootstrap segment, $\Xi\mathrm{Z}$ has a flat forward $\bar q$ after the last node, and the model CB spread is (252) with $K_j = K^C_0(T_{j-1},T_j)$ given by Proposition 26 ((255) for Libor coupons):

$$B(\bar q) \;=\; \frac{1}{\theta\sum_j P_j}\sum_{j=1}^{n} F_j\Xi\mathrm{Z}_j\left(\kappa^d\frac{F_{j-1}\Xi\mathrm{Z}_{j-1}}{F_j\Xi\mathrm{Z}_j} - \kappa^f\frac{F_{j-1}}{F_j}K_j\right). \tag{263}$$

It depends on the collateral basis volatility only through K_j, which reads σ^c and σ^f on $[0,T_{n-1}]$ for Libor coupons and up to the last overnight fixing before T_n for daily compounded ones (the in-arrears term of Proposition 27). The DCB model is the case $K_j = 1$.

(2) *CB spread volatility.* The model volatility of the forward CB spread of expiry T_0 is given by the exact distribution of B_{T_0} under $\mathbb{Q}^{f|d,A}$ (Lemma 3 and Lemma 4, 2D quadrature) and, approximately, by Proposition 42: $\sigma_N^2 T_0 \approx \hat q_B^\top \hat y_{T_0}\hat q_B$. The exact value uses the coupon convexities seen from T_0, $K_{T_0,j} = K^C_{T_0}(T_{j-1},T_j)$ (Remark 45), e.g. $K_{T_0,j} = \exp\{-\int_{T_0}^{T_{j-1}}\Sigma^{f|f}_{u,T_{j-1};\theta}\cdot\Sigma^c_{u,T_j}du\}$ for Libor coupons. So it reads σ^c on $[0,T_{n-1}]$ for Libor coupons (up to the last overnight fixing before T_n for daily compounded ones), beyond the expiry, and the curve $Z^{f|d}$ on $[T_0,T_n]$.

(3) *FX volatility.* The total variance of an FX option of expiry T is

$$w(T) \;=\; \sigma^2_{\mathrm{BS}}\,T \;=\; \int_0^T\left|\Sigma^{df}_{u,T}\right|^2 du, \qquad \Sigma^{df}_{u,T} \;=\; \Sigma^{d|d}_{u,T} - \Sigma^{f|f}_{u,T} - \Sigma^c_{u,T} + \sigma^{df}_u. \tag{264}$$

If $\sigma^x = s$ on the last segment $[t_0,T]$ and all other parameters are fixed, then $w(T) = as^2+bs+c$ with

$$\begin{aligned} a &= T - t_0 \;>\; 0,\\ b &= 2\sum_{(e,i)}\epsilon_e\int_{t_0}^T \rho^{(e,i),x}_u\sigma^{e,i}_u G^{e,i}_{u,T}\,du, \qquad \epsilon_d = 1,\ \epsilon_f = \epsilon_c = -1,\\ c &= \int_0^{t_0}\left|\Sigma^{df}_{u,T}\right|^2 du \\ &\quad + \int_{t_0}^T\left|\Sigma^{d|d}_{u,T} - \Sigma^{f|f}_{u,T} - \Sigma^c_{u,T}\right|^2 du, \end{aligned}$$

all covariance integrals (subsection 9.1). The FX volatility is matched by a non-negative s iff

$$\min_{s\ge 0}\left(as^2+bs+c\right) \;\le\; \sigma^2_{\mathrm{BS}}T,$$

i.e. iff $c \le \sigma^2_{\mathrm{BS}}T$, or $b<0$ and $c - b^2/(4a) \le \sigma^2_{\mathrm{BS}}T$. Otherwise there is a variance squeeze. The largest root is

$$s \;=\; \frac{-b+\sqrt{b^2-4a\left(c-\sigma^2_{\mathrm{BS}}T\right)}}{2a}. \tag{265}$$

Proof. 1. Substitute $P_j = F_j \Xi \mathrm{Z}_j$ in (252); K_j is Proposition 26 at $t = 0$ ((187) for Libor coupons). The daily kernel φ_u^δ vanishes after the last overnight fixing.

2. The CB spread is a function of $\hat{z}_{T_0}$ by Lemma 4, where the foreign coupon forward under domestic collateral at T_0 involves $K_{T_0,j}$ by Proposition 26 at $t = T_0$, and $\hat{z}_{T_0}$ is Gaussian by Lemma 3.

3. By the proposition on the FX spot dynamic in the LGCM specification,

$$\Sigma_{u,T}^{df} \cdot dW_u \quad = \quad \sum_{(e,i)} \epsilon_e \sigma_u^{e,i} G_{u,T}^{e,i} \, d\widetilde{W}_u^{(e,i)} + \sigma_u^x \, d\widetilde{W}_u^x .$$

On $[0, t_0)$ the integrand does not depend on s. On $[t_0, T]$, expand $|\Lambda_u + s\, e_x|^2 = |\Lambda_u|^2 + 2s\, \Lambda_u \cdot e_x + s^2$ with $\Lambda_u = \Sigma_{u,T}^{d|d} - \Sigma_{u,T}^{f|f} - \Sigma_{u,T}^{c}$ and $\rho^{x,x} = 1$. The values of σ^x after T do not enter. Since $a > 0$, the minimum of the quadratic over $s \geq 0$ is c if $b \geq 0$ and $c - b^2/(4a)$ if $b < 0$; a non-negative root exists iff this minimum is at most the target, and then the largest root (265) is non-negative. □

Remark 72. (1) In item 3, $\Sigma^{df} = \Sigma^{d|d} - \Sigma^{f|f} + \Sigma^{d\to f} + \sigma^{df}$, with $\Sigma^{d\to f} = -\Sigma^c$, as in the FX forward dynamic of the reconstruction section. We checked that the total variance of the implementation agrees with explicit loadings $(+G^d, -G^f, -G^c, 1)$ to rounding, and that on every FX segment of the calibrated example $w(2) - (4a + 2b + c) \leq 4 \cdot 10^{-15}$ (exact quadratic) and $a = T - t_0$ to 10^{-12}.

(2) The implementation computes a, b, c from $w(0)$, $w(1)$, $w(-1)$ and takes the largest root (265). When both roots are non-negative ($c > \sigma_{\mathrm{BS}}^2 T$ and $b < 0$) the choice of the largest one is a convention. On the example market $c < \sigma_{\mathrm{BS}}^2 T$ and $b < 0$ at every FX expiry, so the non-negative root is unique.

(3) The quantities of items 1 and 2 do not involve σ^x (K_j and the law of $\hat{z}$ under $\mathbb{Q}^{f|d,T_0}$ only involve the f and c blocks). So the FX volatilities can be calibrated last, without feedback.

10.5.2. *Bootstrapping algorithm.*

Algorithm 4 (LGCM calibration)**.** Input: the quotes of the introduction, the marked mean reversions, weights, correlations and funding spreads, and a marked level of λ^c if there are no CB volatility quotes. The model knots are the union of all option expiries and a horizon (60Y).

(1) *Native curves.* Bootstrap $Z^{d|d}$ and $Z^{f|f}$ from the swaps (Algorithm 2, model-free by Proposition 49).

(2) *Rates volatilities.* For $a \in \{d, f\}$, bootstrap λ^a expiry by expiry from the swaptions (Remark 69): Gaussian guess, then the exact price of Proposition 51 for one factor.

(3) *Fixed point on* $(\Xi \mathrm{Z}^{f|f\to d}, \lambda^c)$. Repeat:
   (a) bootstrap $Z^{f|d}$ from the FX forwards and the resetting CB swaps (Algorithm 2) with K_j from the current σ^c;
   (b) if CB volatilities are quoted, for each expiry T_k solve for λ^c on $[T_{k-1}, T_k)$ (the last one on $[T_{n-1}, \infty)$): Brent search on the Gaussian approximation of Proposition 42, then Brent search on the exact quadrature started there. Fail if the volatility at $\lambda^c = 0$ on the segment already reaches the target (variance squeeze). This test uses the Gaussian approximation, as for the swaptions (Remark 69);

   until the maximum change of the λ^c levels and of the forwards of $\Xi \mathrm{Z}^{f|f\to d}$ between two iterations is below 10^{-12} (at most 30 iterations, else fail). Without CB volatilities, one pass with the marked σ^c.

(4) *FX volatility.* For each FX expiry T_k, set σ^x on $[T_{k-1}, T_k)$ (the last one on $[T_{n-1}, \infty)$) by (265); fail on a variance squeeze.

Output: the model, the curves $Z^{d|d}$, $Z^{f|f}$, $Z^{f|d}$, and the off-market curve $Z^{d|f}$ (subsection 10.6).

Remark 73 (Why a fixed point, and why each expiry sets only its own segment)**.** (1) The curve $Z^{f|d}$ beyond the FX forwards depends on σ^c on $[0, 30\mathrm{Y}]$ through the K_j (Proposition 53, item 1). The CB volatility of expiry T_k depends on the curve (weights P_j, levels B_j) and, through $K_{T_k,j}$, on σ^c on $[T_k, T_k + \text{tenor} - \theta]$, i.e. on the next segment (item 2). Neither "curve first, then volatilities" nor "expiry by expiry" is triangular, hence the fixed point. The swaption and FX steps are triangular and need no iteration.

(2) A level set on $[T_{k-1}, \infty)$ instead of $[T_{k-1}, T_k)$ would overwrite the pieces read by the earlier CB volatilities. A draft implementation did this: the earlier CB volatilities were then missed by up to $3.8 \cdot 10^{-11}$. With one segment per expiry and the outer fixed point, they are repriced to $1.5 \cdot 10^{-17}$.

(3) On the example market the fixed point stops after 5 iterations, with changes ∞, $6.2 \cdot 10^{-4}$, $1.8 \cdot 10^{-7}$, $4.0 \cdot 10^{-11}$, $8.2 \cdot 10^{-15}$: successive ratios $2.8 \cdot 10^{-4}$, $2.3 \cdot 10^{-4}$, $2.1 \cdot 10^{-4}$, a contraction factor of about 2–$3 \cdot 10^{-4}$ per iteration. This is a numerical observation; we have no proof of contraction. It is consistent with the size of the coupling: $K_j - 1 \leq 2.24 \cdot 10^{-4}$ up to 30Y. With $\sigma^c = 0$ and no CB volatilities, one pass suffices and the LGCM curve equals the DCB curve to 10^{-14} (test `test_dcb_limit`).

(4) The whole calibration takes about 6s on the example market (65 quotes; 5.6 to 6.3s measured in different runs on this branch, 6.1 to 6.8s for GBPUSD and 6.0 to 6.6s for USDJPY with daily coupons, also 5 fixed point iterations). The calibrated levels are λ^d between 1.19% and 1.35%, λ^f between 0.89% and 1.05%, λ^c between 9.9bp and 18.6bp and σ^x between 7.0% and 8.3%. Round trip: quotes generated by the calibrated model, calibrated from another initial λ^c, give back the parameters to 10^{-8} (relative) and $Z^{f|d}$ to 10^{-12}.

Remark 74 (Verification)**.** All numbers are for the three example markets (Remark 76) and the default specification (one factor per block, $\beta^d = 3\%$, $\beta^f = 2\%$, $\beta^c = 10\%$, $\rho^{d,f} = 0.6$, $\rho^{f,c} = -0.4$, $\rho^{d,c} = -0.2$, $\rho^{d,x} = 0.25$, $\rho^{f,x} = 0.3$, $\rho^{c,x} = 0.35$), computed by `calibration_check.py`. The GBPUSD and USDJPY markets are calibrated with daily compounded foreign coupons ($\delta = 1/365$).

(1) *Analytic repricing* with the exact pricers (65 quotes per market). Maximum absolute errors, EURUSD / GBPUSD / USDJPY: swaps $2.3 \cdot 10^{-16}$ / $4.2 \cdot 10^{-17}$ / $4.2 \cdot 10^{-17}$, FX forwards $1.45 \cdot 10^{-12}$ / $1.9 \cdot 10^{-12}$ / $1.0 \cdot 10^{-12}$ pips, resetting CB swaps $5.0 \cdot 10^{-17}$ / $5.2 \cdot 10^{-17}$ / $1.0 \cdot 10^{-16}$, swaption normal volatilities $6.4 \cdot 10^{-17}$ / $1.9 \cdot 10^{-17}$ / $1.4 \cdot 10^{-17}$, FX Black volatilities $1.4 \cdot 10^{-17}$ / $2.8 \cdot 10^{-17}$ / $4.2 \cdot 10^{-17}$, CB normal volatilities $1.5 \cdot 10^{-17}$ / $2.3 \cdot 10^{-17}$ / $3.6 \cdot 10^{-17}$. The Gaussian approximations differ from the exact values by $+2.9 \cdot 10^{-4}$ to $+5.4 \cdot 10^{-3}$ (relative) for the EURUSD swaptions ($5.4 \cdot 10^{-3}$ at most in every market), and at most $5.7 \cdot 10^{-5}$ (15Y×1Y) / $7.1 \cdot 10^{-5}$ / $1.4 \cdot 10^{-4}$ for CB volatilities.

(2) *Monte Carlo repricing* of all 65 quotes with the exact LGCM scheme (quarterly grid refined by the payment and fixing dates), 2^{17} antithetic paths, with control variates whose means are exact no-arbitrage identities. For the compounded coupons of GBPUSD and USDJPY, the overnight fixings are streamed through the simulation and the coupon is valued on the paths with a *weekly* step $\delta = 1/52$, both on the paths and in the model (`mc_report(..., compounding_step=1/52)`: 1562 simulation dates instead of 122 for EURUSD). The closed form is exact for any step, and the daily to weekly change of the model quotes is at most 0.046 (GBPUSD) and 0.019 (USDJPY) standard errors, so the daily calibrated quotes remain the reference. With a *daily* grid on the paths (11040 dates, 183s), the 8 GBPUSD CB swaps have $|z| \leq 2.17$ (seed 7). Largest $|z|$ by type, seed 7:

	EURUSD	GBPUSD	USDJPY
swaps	2.15	1.74	1.76
FX forwards	1.62	1.82	1.81
resetting CB swaps	2.15 (20Y)	3.10 (30Y)	2.70 (30Y)
swaptions	2.22	1.86	1.89
FX options	1.84	1.73	1.83
CB volatilities	1.53	2.70	2.38
all quotes, seed 7 / 8 / 9	2.22 / 2.18 / 2.71	3.10 / 2.51 / 2.89	2.70 / 2.30 / 2.65

All quotes are within 4 standard errors. Median standard errors (seed 7, EURUSD / GBPUSD / USDJPY): 0.0039 / 0.0034 / 0.0071bp for CB spreads, 0.11 / 0.12 / 0.11bp for swaption volatilities, 0.024 / 0.017 / 0.029bp for CB volatilities. The CB spread standard errors grow with the maturity, from 0.00076bp (2Y) to 0.0155bp (30Y) for EURUSD. They were halved, compared with an earlier version of the check, by the control variate $C_3 = D_s^{f|f} \prod_{k<n-1} 1/Z_{t_k,t_{k+1}}^{f|f}$ (the native rolled account, whose mean $Z_{0,a}^{f|f}$ is the model-free value of Lemma 1); for compounded coupons the control $D_s^{f|d} \prod_{k<n-1} 1/Z_{t_k,t_{k+1}}^{f|d}$ (mean $Z_{0,a}^{f|d}$ by the same telescoping) was necessary: without it the standard error of a 5Y GBPUSD CB spread was 0.134bp instead of 0.0076bp (2^{15} paths). The option prices are compared as prices: the MC straddle is mapped to a volatility by the same linear map as the pricer, so no Gaussian assumption enters the MC side.

(3) *No bias across seeds.* The z of the CB swaps of one run are strongly correlated across maturities (same paths, shared periods), so a run with all z of the same sign is not by itself a sign of bias (e.g. seeds 8 and 9 of EURUSD, where 7 of the 8 CB swaps have $z < 0$). Over the 30 seeds 100 to 129 (EURUSD, 2^{17} paths), the mean z per maturity lies between -0.13 and $+0.26$ with a standard error of the mean of about 0.2, the mean over all CB swaps is $+0.01 \pm 0.13$, and the sample standard deviation of z per maturity is between 0.93 and 1.29 (about 1 ± 0.13 expected; 1.29 at 20Y). A larger sweep (EURUSD, 120 seeds 100 to 219, 2^{13} paths, CB swaps 10, 15, 20, 30Y) gives standard deviations of z of 0.94, 0.95, 0.97 and 1.04 (each ± 0.06), mean z between -0.04 and $+0.08$ (standard error of the mean about 0.09) and $\max|z| = 2.99$ over the 480 values. So the 1.29 at 20Y of the 30 seed sweep was sampling noise (the largest of 8 standard deviations, each with standard error about 0.13), and the standard error of the regression control variate (antithetic pairs averaged first, $p = 3$ controls, in-sample slope with a bias $O(p/N)$) is consistent with the observed spread: no batch-means correction is needed (numerical observation).

(4) *Power of the checks.* The CB swap check resolves the collateral convexity of the coupons: the DCB spread ($K_j = 1$) on the calibrated curve is $+35.7$ (2Y) to $+205.8$ (30Y) standard errors away from the MC value for EURUSD (seed 7), $+29.5$ to $+202.0$ for GBPUSD and $+12.5$ to $+156.6$ for USDJPY, and 37.9, 50.0, 65.8 standard errors at 5, 7, 10Y in the reduced EURUSD test run (2^{15} paths, seed 11). With 2^{18} paths (seed 3) the Gaussian swaption approximation is rejected at 7.8 and 7.0 standard errors for USD 15Y$\times$10Y and 20Y$\times$10Y, while the exact values are at $+0.35$ and $+0.94$. The Gaussian gap of the CB volatilities ($\leq 1.4 \cdot 10^{-4}$ relative) is below the MC resolution (about $2.5 \cdot 10^{-3}$ relative): only the quadrature distinguishes it.

(5) *Libor vs compounded coupons are not resolved by the MC.* On the RFR markets, the CB spread computed with the Libor convexity K_j^{L} in place of the compounded one differs from the model by $+0.00067$bp (2Y) to $+0.00099$bp (30Y) for GBPUSD and $+0.00059$ to $+0.00109$bp for USDJPY, i.e. between 0.05 and 0.9 standard errors at 2^{17} paths. The MC therefore checks the path valuation of the compounded coupons and rejects $K = 1$, but it cannot tell K^{δ} from K^{L}. That distinction rests on the exact benchmark of Remark 57 and on the deterministic test that the CB spread uses K^{δ} (`test_rfr_ccs_uses_the_compounded_convexity_not_libor`: on the test curves of a 20Y SONIA CB swap with a weekly step, the Libor-K spread differs by $3.7 \cdot 10^{-7}$, i.e. 0.004bp, and the compounded one is matched to 10^{-13} relative).

(6) *What is not tested.* The swap and FX forward MC checks are martingale tests of the deflators against the curves ($\mathbb{E}[D_t] = Z_{0,t}$, $\mathbb{E}[D_T x_T] = Z_{0,T} x_{0,T}^{df|d}$). They do not test the calibrated volatilities, and their closed forms are already repriced to 10^{-16} analytically. The CB volatility control variate assumes the closed form $K_{T_0,j}$; dropping $K_{T_0,j} - 1 \leq 7.2 \cdot 10^{-7}$ from it is not detected by the MC (it moves the spread by about 0.01bp, EURUSD).

The tests are in `tests/test_calibration.py` and `tests/test_ois.py`; the MC tests are marked `slow`.

Remark 75 (Collateral convexity in the calibrated curves)**.** We compare the LGCM calibration with the DCB curve bootstrap (Algorithm 2 with $K_j = 1$) on the same quotes, with the specification of Remark 74.

(1) The two $Z^{f|d}$ curves coincide up to the last FX forward (1Y), as they must (Proposition 49). After that, the LGCM zero rate of $Z^{f|d}$ is above the DCB one by $+0.17$bp at 5Y, $+0.63$bp at 10Y, $+2.05$bp at 20Y and $+3.75$bp at 30Y.

(2) The Libor collateral convexity of the last period, $K_j - 1$, is $4.4 \cdot 10^{-5}$ at 10Y and $2.24 \cdot 10^{-4}$ at 30Y. On the calibrated curve, it lowers the 30Y CB spread by 3.15bp compared with $K_j = 1$; the DCB curve absorbs this difference in its basis.

(3) The FX forward collateral convexity $x_{0,T}^{df|f} / x_{0,T}^{df|d} - 1 = \exp\{\int_0^T \Sigma_{u,T}^{df} \cdot \Sigma_{u,T}^{c} du\} - 1$ is $+3.7$bp at 5Y, $+15.3$bp at 10Y and $+119.6$bp at 30Y. The off-market curve $Z^{d|f}$ has a zero rate 1.53bp above the DCB off-market curve $Z^{d|d} Z^{f|f} / Z^{f|d}$ of (35) at 10Y.

(4) Sensitivity to $\rho^{f,c}$ (full recalibration, other inputs unchanged): 30Y shift of the $Z^{f|d}$ zero rate / 10Y FX forward collateral convexity of 5.63/20.95bp for $\rho^{f,c} = -0.6$, 3.75/15.30bp for -0.4, 0.00/3.98bp for 0 and -3.76/-7.36bp for $+0.4$. The curve shift vanishes at $\rho^{f,c} = 0$ because, with one factor

per block, K_j only involves the (f, c) covariance. The calibrated $\lambda^c(10\mathrm{Y})$ hardly moves (15.76 to 15.81bp). With the other default correlations, the correlation matrix is positive definite only for $\rho^{f,c}$ in about $[-0.7, 0.6]$.

The numbers of items 1 to 3 come from a script on the calibrated model; those of item 4 from the notebook `python/notebooks/lgcm_calibration.py`.

Remark 76 (Example markets)**.** The three example markets of `market_data.py` share the USD side: $d =$ USD, collateral and domestic coupons SOFR, SOFR swaps from 4.30% (1Y) to 3.80% (30Y), USD swaption volatilities 84 to 110bp. Each has 65 quotes: 18 swaps, 5 FX forwards (1M to 1Y, points), 8 resetting CB swaps (2Y to 30Y, quarterly), 16 swaptions (1Y to 20Y $\times$10Y), 10 FX options (1M to 15Y) and 8 CB volatilities (1Y to 20Y $\times$1Y). The quotes are rounded and illustrative, not live data, and they are not smoothed: the forward collateral basis q of the DCB curves (flat per bootstrap segment) jumps between the last FX forward and the first CB swaps. On the segments 0.75–1Y / 1–2Y / 2–3Y it is $+20.4$ / -9.6 / -4.0bp (EURUSD, a change of sign), -37.2 / -18.7 / -34.0bp (USDJPY) and -5.2 / -2.8 / -7.0bp (GBPUSD). The EURUSD sign flip predates the RFR markets (the EURUSD quotes are those of the original example). These jumps come from the illustrative inputs, not from the model, and the calibrations reprice them.

(1) EURUSD: EUR swaps 2.15% to 2.80% on EURIBOR 3M (funding spread 10bp, so Libor-type foreign coupons), spot 1.10 USD per EUR, CB spreads -8 to -14bp.
(2) GBPUSD: SONIA swaps (annual fixed against compounded SONIA) 4.05% (1Y) to 3.80% (30Y), spot 1.27 USD per GBP, points 3.1 (1M) to 37.0 (1Y), SONIA vs SOFR CB spreads -4 to -7bp, GBP swaption volatilities 80 to 104bp, FX volatilities 7.8% to 11.5%, CB volatilities 6.0 to 7.5bp.
(3) USDJPY: TONA swaps 0.55% (1Y) to 1.80% (30Y), spot quoted inverted, 150 JPY per USD (model spot $x_0^{df} = 1/150$ USD per JPY), points in $1/100$ JPY from -49 (1M) to -593 (1Y), TONA vs SOFR CB spreads -26 to -34bp, JPY swaption volatilities 45 to 66bp, FX volatilities 9.5% to 12.2%, CB volatilities 10 to 13bp.

The calibration (Algorithm 4) succeeds on the three markets without variance squeeze; every linear quote sets a node (31 per market, no checks). Calibrated levels (first piece, peak, last piece): $\lambda^f = 93$, 105, 89bp (EURUSD), 105, 119, 100bp (GBPUSD), 50, 84, 49bp (USDJPY); λ^c goes from 9.9 to 18.6bp (EURUSD), 6.6 to 12.9bp (GBPUSD) and 11.0 to 21.5bp (USDJPY) between the first and the last piece, mostly increasing with the expiry; $\sigma^x = 7.0$, 8.3, 7.2% (EURUSD), 7.8, 9.3, 8.1% (GBPUSD), 9.5, 10.5, 7.6% (USDJPY); λ^d is the same in the three markets (1.19% to 1.35%). The LGCM zero rate of $Z^{f|d}$ is above the DCB one by $+0.13$, $+0.50$, $+1.62$, $+2.94$bp at 5, 10, 20, 30Y for GBPUSD and $+0.13$, $+0.57$, $+1.86$, $+3.19$bp for USDJPY (EURUSD: Remark 75, item 1). The largest coupon convexity $K - 1$ of a quarterly period up to 30Y is $2.241 \cdot 10^{-4}$ (EURUSD, Libor), $1.7607 \cdot 10^{-4}$ (GBPUSD, daily SONIA) and $1.7314 \cdot 10^{-4}$ (USDJPY, daily TONA).

Remark 77 (Libor-type vs compounded convexity in the calibrated curves)**.** How much does the coupon type matter for the calibrated curves? We recalibrated the RFR markets with the Libor convexity K_j^{L} in place of the daily compounded one K_j^{δ} (foreign index replaced by a Libor-type index with zero funding spread and the same tenor, so the native swaps are unchanged by Lemma 1; every other input is the same). On the calibrated GBPUSD model, $K - 1$ of the period ending at T is $1.347954 \cdot 10^{-6}$ (Libor) against $1.367025 \cdot 10^{-6}$ (daily) at $T = 2$Y, $3.482846 \cdot 10^{-5}$ against $3.485167 \cdot 10^{-5}$ at 10Y and $1.760386 \cdot 10^{-4}$ against $1.760651 \cdot 10^{-4}$ at 30Y: $\ln(K^{\delta}/K^{\mathrm{L}}) = 1.9$, 2.3, $2.6 \cdot 10^{-8}$, roughly constant as Proposition 27 predicts, while $\ln(K^0/K^{\delta}) = 1.6$ to $2.2 \cdot 10^{-10}$. The zero rate of the calibrated $Z^{f|d}$ moves by (Libor K minus daily K)

	2Y	5Y	10Y	20Y	30Y
GBPUSD	-0.00067bp	-0.00086bp	-0.00091bp	-0.00097bp	-0.00100bp
USDJPY	-0.00060bp	-0.00097bp	-0.00109bp	-0.00101bp	-0.00096bp

and the calibrated volatilities by at most $7.4 \cdot 10^{-8}$ (GBPUSD) and $3.9 \cdot 10^{-8}$ (USDJPY) in relative terms. With the continuous convexity K^0 instead of K^{δ} the shift is $+0.000005$ to $+0.000009$bp. These are numerical observations on these markets: the in-arrears term of compounded coupons moves the curves by about 0.001bp, three orders of magnitude below the collateral convexity itself ($+2.9$ to $+3.2$bp at 30Y against the DCB curve, Remark 76), and far below the precision of the quotes. The Monte Carlo cannot resolve it (Remark 74, item

5); the exact benchmark can (Remark 57). So, for compounded coupons in these markets, the Libor-type convexity is an excellent approximation; the compounded one is used because it is exact and costs 0.43ms per period. Proposition 27 shows when this could change: the in-arrears term is bounded by $\bar{\sigma}^f \bar{\sigma}^c \theta^3/3$ and does not grow with the expiry, while the Libor term grows with the expiry in these examples (an observation, not a property of the model), so it matters relatively more for long accrual periods (annual coupons) and short expiries (0.8% of the convexity at 2Y on the test model of Remark 57).

10.6. **Off-market collateral.** Once the model is calibrated, off-market curves can be computed in closed form using the FX forward convexity adjustment:

$$Z^{d|f}_{0,T} \;=\; \frac{x_0^{df}\, Z^{f|f}_{0,T}}{x^{df|f}_{0,T}}, \qquad x^{df|f}_{0,T} \;=\; x^{df|d}_{0,T} \exp\left\{ \int_0^T \Sigma^{df}_{u,T} \cdot \Sigma^{c}_{u,T}\, du \right\}.$$

In the DCB model the exponential is 1 and $Z^{d|f}$ reduces to (35).

10.7. **Interpolation and consistency with the model dynamic.** The bootstrapping requires an interpolation scheme for the initial curves on each segment between calibration pillars. We discuss how this choice interacts with the model dynamic. Interpolation enters at two different levels: the initial curves at $t = 0$, and the curves at future dates $t > 0$.

10.7.1. *Initial curves.*

Remark 78 (Any interpolation is arbitrage free at $t = 0$)**.** In the HJM framework, the initial curves $Z^{d|d}_{0,\cdot}$, $Z^{f|f}_{0,\cdot}$ and $\Xi \mathrm{Z}^{f|f\to d}_{0,\cdot}$ are free initial conditions: the drifts are determined by the volatilities alone. So for any absolutely continuous interpolated initial curves (in particular piecewise flat forwards), Theorem 4 defines an arbitrage free model that reproduces them exactly. Contrary to Libor market models (see the remark on LMMs with stochastic currency basis), there is no stub interpolation to make consistent with the dynamic: discount factors at any date are given by the reconstruction formulas. Moreover, the exact simulation only uses the initial curves through discount factors at simulation and payment dates (Remark 61).

Remark 79 (Interpolation and convexity adjustments decouple in the LGCM)**.** In the LGCM, the convexity adjustments (e.g. $\exp\{-\int \Sigma^{f|f}_{u,T;\theta} \cdot \Sigma^{f\to d}_{u,T+\theta} du\}$) depend on the model parameters only, not on the initial curves. So, on a bootstrapping segment, the collateral basis spot volatility determines the convexity adjustments of all swaplets of the segment, and the interpolation scheme only determines how the average level of the collateral basis is distributed across these swaplets. The model delivered volatility of Proposition 42 depends on the initial curves only mildly, through the weights P_j and the levels L_j, B_j.

Remark 80 (Which curve to interpolate)**.** On a segment between two currency basis pillars, one can either interpolate the cross-currency curve $Z^{f|d}_{0,\cdot}$ or the collateral basis curve $\Xi \mathrm{Z}^{f|f\to d}_{0,\cdot} = Z^{f|d}_{0,\cdot}/Z^{f|f}_{0,\cdot}$, the native curve $Z^{f|f}_{0,\cdot}$ being known from the native market. Under the constant funding spread hypothesis and in the LGCM, the swaplet basis spreads are

$$\theta B_j \;=\; \kappa^d(1+\theta R^{f|d}_{0,T_{j-1};\theta}) - \kappa^f K_j(1+\theta R^{f|f}_{0,T_{j-1};\theta}) \;=\; (1+\theta R^{f|f}_{0,T_{j-1};\theta}) \left(\kappa^d \frac{\Xi \mathrm{Z}^{f|f\to d}_{0,T_{j-1}}}{\Xi \mathrm{Z}^{f|f\to d}_{0,T_j}} - \kappa^f K_j \right)$$

with $\kappa^e = 1 - \theta S^{e|e,\div}$ (1 for a compounded coupon) and K_j the (deterministic) coupon collateral convexity.

- With flat forwards for $r^{f|d}_{0,\cdot}$ on the segment, $R^{f|d}_{0,T_{j-1};\theta}$ is constant and B_j jumps one for one with the native forwards $R^{f|f}_{0,T_{j-1};\theta}$ at the native pillars. The implied collateral basis forward curve $q^{f|f\to d}_{0,\cdot} = r^{f|d}_{0,\cdot} - r^{f|f}_{0,\cdot}$ then carries all the term structure of the native curve with the opposite sign.
- With flat forwards for $q^{f|f\to d}_{0,\cdot}$ on the segment, B_j only varies through the factor $1+\theta R^{f|f}_{0,T_{j-1};\theta}$ and the convexity K_j.

Interpolating the collateral basis curve is therefore consistent with the choice of z^c as the state driving $\Xi \mathrm{Z}^{f|f\to d}$ (see the remark on our parametrisation choices in the Quasi-Gaussian section). It also avoids generating a term structure of the short collateral basis $q^{f\to d}_t = q^{f|f\to d}_{0,t} + \mathbf{1}^\top z^c_t + \mathbf{1}^\top v^{f,c}_t$, and hence of the FX drift, that only reflects the native curve pillars.

10.7.2. *Curves at future dates.*

Remark 81 (Calendar-pinned features)**.** By Theorem 4, for $t > 0$,

$$r^{d|d}_{t,T} - r^{d|d}_{0,T} \;=\; M^{d,\top}_{t,T}\left(z^d_t + y^{d,d}_t G^d_{t,T}\right)$$

and similarly for $r^{f|f}$ and $q^{f|f\to d}$. The right hand side is continuous in T, so every discontinuity or kink of the interpolated initial forward curve at a pillar date T_k is still there at the same *calendar* date T_k on every path. On the other hand, a curve re-bootstrapped at t with the same interpolation scheme has its pillars at $t + \tau_k$ for the benchmark tenors τ_k. In general, the two curves differ between the pillars, even if the re-bootstrapping uses the model values of the benchmark quotes.

The practical consequence is that future curves along a simulated path (for exposure, XVA or P&L explain) should be produced with the reconstruction formulas, not by re-stripping simulated quotes with the curve building code. The latter would price with curves that are not those of the model, which reintroduces exactly the inconsistency the model was designed to remove. For the same reason, the model's carry (roll-down) keeps the initial pillars at fixed calendar dates.

Whether a parametric interpolation family can be preserved by the dynamic is the consistency problem of Björk and Christensen [2]. A family $\mathcal{G}$ of forward curves $\tau \mapsto g(\tau)$, parametrised by time to maturity $\tau = T-t$, is *consistent* with the model if, whenever $\tau \mapsto r_{0,\tau}$ belongs to $\mathcal{G}$, $\tau \mapsto r_{t,t+\tau}$ belongs to $\mathcal{G}$ for all t and all paths.

Proposition 54 (Consistent families for the LGCM)**.** Assume the parameters of the LGCM are constant and all mean reversions are non zero.

(1) For a native rates block a with mean reversions $\beta^{a,1}, \dots, \beta^{a,N_a}$, the family

$$\mathcal{G}^a = \left\{ \tau \mapsto \zeta_0 + \sum_i \zeta_i e^{-\beta^{a,i}\tau} + \sum_{i \le j} \zeta_{ij} e^{-(\beta^{a,i}+\beta^{a,j})\tau} \right\}$$

is consistent with the dynamic of r^a. If some $\beta^{a,j}$ vanish, the functions $\tau e^{-\beta^{a,i}\tau}$ have to be added.

(2) If $\rho^{f,c} \neq 0$, a family of this exponential type is consistent with the dynamic of the collateral basis $q^{f|f\to d}$ only if it contains, in addition to the exponentials of the collateral basis block, the foreign rates exponentials $e^{-\beta^{f,i}\tau}$ and the mixed exponentials $e^{-(\beta^{c,i}+\beta^{f,j})\tau}$ (for generic parameters, i.e. unless their coefficients happen to cancel).

Proof. With constant parameters, $M^{a,i}_{t,t+\tau} = e^{-\beta^{a,i}\tau}$ and $G^{a,j}_{t,t+\tau} = (1 - e^{-\beta^{a,j}\tau})/\beta^{a,j}$ (or τ if $\beta^{a,j} = 0$). By Theorem 4, $r^a_{t,t+\tau} = r^a_{0,t+\tau} + \sum_i e^{-\beta^{a,i}\tau}(z^{a,i}_t + \sum_j y^{(a,i),(a,j)}_t G^{a,j}_{t,t+\tau})$. The second term lies in the span of $e^{-\beta^{a,i}\tau}$ and $e^{-(\beta^{a,i}+\beta^{a,j})\tau}$. The first term is in $\mathcal{G}^a$ if $r^a_{0,\cdot}$ is, since $e^{-\beta(t+\tau)} = e^{-\beta t}e^{-\beta\tau}$ and constants are shift invariant. For the collateral basis, the reconstruction formula of Theorem 4 contains $M^{f,\top}_{t,T}(v^{f,c}_t + y^{f,c}_t G^c_{t,T})$ and $M^{c,\top}_{t,T} y^{c,f}_t G^f_{t,T}$. When $\rho^{f,c} \neq 0$, $y^{c,f}_t$ and $v^{f,c}_t$ are non zero for $t > 0$, and these terms produce $e^{-\beta^{f,i}\tau}$ with coefficient $v^{f,c}_t + y^{f,c}_t(\beta^c)^{-1}\mathbf{1}$ and $e^{-(\beta^{c,i}+\beta^{f,j})\tau}$ with coefficient $-y^{(c,i),(f,j)}_t\left(1/\beta^{f,j} + 1/\beta^{c,i}\right)$. The exponential decompositions were checked symbolically (sympy). □

Remark 82.

(1) A consistent family has at most $1 + N_a + N_a(N_a+1)/2$ parameters, so it cannot reproduce an arbitrary set of market quotes. There is a genuine trade-off between exact fit of the initial curves (non-parametric interpolation, inconsistent families) and consistency (parametric curves). In the LGCM the exact fit is preferred, and the inconsistency is harmless as long as future curves are produced by the reconstruction formulas.

(2) The proposition is in line with [2], where the Nelson-Siegel family is shown to be inconsistent with the Hull-White model but can be augmented to achieve consistency. A one factor Hull-White model generates the exponential $e^{-2\beta\tau}$, which is not in the Nelson-Siegel family. See also [3]: no non-trivial HJM model is consistent with the Nelson-Siegel family.

(3) The second part of the proposition says that, if one uses a parametric family for the collateral basis curve, it must include the foreign rates exponentials. The collateral basis curve inherits part of the shape of the foreign curve through the convexity term $\sigma^f_{t,T} \cdot \Sigma^{f\to d}_{t,T}$ in its drift.

10.8. **Alternative calibrations.** For this purpose, we developed the Non Linear Curve Engine: a generic framework to jointly

(1) calibrate a stochastic spread model
(2) calibrate discount and index curves taking into account the convexity adjustment induced by the model.

Calibration targets for this joint calibration are

(1) FX forwards
(2) Cross-currency swaps (resetting and non-resetting)
(3) FX options

(4) Currency basis swaptions (since these are not liquid we cannot imply volatilities so we target historical estimates instead).

There are several ways to do the calibration.

10.8.1. *Global calibration.* We create a function with inputs

(1) curves parameters
(2) model parameters

and outputs

(1) model induced value of market quotes.

We run an optimiser to minimise the error between model induced quotes and market quotes.

10.8.2. *Nested calibration.* We decouple the calibration of model parameters and that of the curves. This leads to the following algorithm.

Until convergence is achieved

(1) Initialise the stochastic model:
  (a) Start with an initial guess for $\sigma_\cdot^{f\to d}$ and correlations.
  (b) Bootstrap the FX spot volatility $\sigma_\cdot^{df}$ to match FX options prices.
(2) Build the discount curve $Z_{t_0,\cdot}^{f|d}$ and index curve $L_{t_0,T;\theta}^{f|d}$.
  (a) Compute all the convexity adjustments.
  (b) Solve for the curve parameters to match currency basis spreads.
(3) Use the Gaussian approximation to compute the quadratic covariation of currency basis spreads

$$\int_{t_0}^{T_i\wedge T_j} d\left\langle B_{t_0,T_i;M_k}^{f/d|d,r}, B_{t_0,T_j;M_\ell}^{f/d|d,r}\right\rangle \quad = \quad \int_t^{T_i\wedge T_j} \dots\, dt \tag{266}$$

  as well as their covariance with other factors.
(4) Update $\sigma_\cdot^{f\to d}$ to match historical variances, update correlations to match historical covariances.

Note that the FX forward will move at each iteration. For simplicity, we could assume the FX vol quote do not.

The update (gradient descent) can be implemented in several ways: finite difference, adjoint differentiation etc...

10.9. **Multi-currency model.** [TODO: This seems really hard to do in closed form! Brute force is having a function that calibrates the ed market and the df market and computes the delivered vols on the cross-FX ef. Then run this function, seen as a function of correlations, though an optimiser to reduce the error between model delivered quotes and market observed vols.]

10.9.1. *Odd correlations.* So called odd-correlations such as EUR vs USDJPY correlations cannot be implied from market quotes because there are no liquid market instruments directly exposed to them. The lack of any natural economical interpretation increases the uncertainty associated with historical estimation. This makes this type of correlation unfit as model inputs. On the other hand, options on the FX cross EURUSD are liquid. One can then use a model to imply the odd-correlation EUR vs USDJPY from the natural ones: EUR vs EURUSD and EUR vs EURJPY.

In the BS-LGM model, this is very easy to do because the ATM vol of the cross is equivalent to a constraint on the FX-FX correlation.

In the LGCM model, we have many more unknown correlations. Can we still apply the same reasoning?

It seems like we need to bootstrap the FX-FX correlations and FX-CB correlations at the same time.

10.9.2. *Triangle relationship under stochastic collateral.* Let us write $\Sigma^{df}_{t,T}$ for the volatility of the FX-forward process:

$$\frac{dx^{df|c}_{t,T}}{x^{df|c}_{t,T}} = \Sigma^{df}_{t,T} \cdot dW^{d|c,T}_t \tag{267}$$

We saw that

$$\Sigma^{df}_{t,T} = \Sigma^{d|d}_{t,T} - \Sigma^{f|f}_{t,T} - \Sigma^{f\to d}_{t,T} + \sigma^{df}_t \tag{268}$$

does not depend on the collateral.

It follows that the total variance of an FX option with maturity T is:

$$\sigma^{df|c}_{\mathrm{BS},t}(T)^2(T-t) = \int_t^T \left|\Sigma^{d|d}_{u,T} - \Sigma^{f|f}_{u,T} + \Sigma^{d\to f}_{u,T} + \sigma^{df}_u\right|^2 du \tag{269}$$

regardless of the collateral[46].

Then, from $x^{ef}_{t,T} = x^{df}_{t,T}/x^{de}_{t,T}$, we find

$$\Sigma^{ef}_{t,T} = \Sigma^{df}_{t,T} - \Sigma^{de}_{t,T}. \tag{270}$$

It follows that implied vols of the cross-FX are given by:

$$\sigma^{ef}_{\mathrm{BS},t}(T)^2(T-t) = \int_t^T \left|\Sigma^{df}_{u,T}\right|^2 du + \int_t^T \left|\Sigma^{de}_{u,T}\right|^2 du - 2\int_t^T \Sigma^{df}_{u,T} \cdot \Sigma^{de}_{u,T}\, du \tag{271}$$

This can be viewed as a constraint on the correlation between the two FX rates.

When all factors are one-dimensional, assuming all other correlations are known, we can solve for the FX-FX spot correlation to match the delivered ATMF vols on the cross.

In the deterministic spread setting, one only has to mark IR-IR as well as FX-IRD and FX-IRF correlations. In this new context, one has to mark all collateral basis correlations as well.

This requires to define SDP matrices. Note that synchronicity of time series is crucial for correlation estimations.

10.9.3. *Implied correlations.* We can generalise the reasoning to imply so-called odd-correlations by considering the covariance between the cross-FX and other assets.

10.9.4. *Application to off-market discount curves.* What does the $Z^{\mathrm{EUR|JPY}}_{t,T}$ curve look like?

There is no market for cross-currency swaps collateralized in JPY. So we have to build it from the model calibrated to the EUR/JPY/USD market.

[46] But its moneyness does depend on the collateral.

11. BONDS, REPO AND REPO CONVEXITY

McCloud [7] explains the observed basis between repo discount factors, implied from repo rates, and bond discount factors, stripped from bond prices, as a convexity effect: the repo is valued with the derivative discount rate, while the bond delivered in the repo is priced with the bond discount rate. We show that this *repo convexity* is a forward collateral convexity adjustment in the sense of Proposition 23, with the bond curve playing the role of the native curve and the derivative curve that of the collateral. The LGCM gives it in closed form for any number of factors, and McCloud's Hull–White formulas are the one factor case.

11.1. Setting.

Hypothesis. Single currency, no default, no haircut. Under one risk-neutral measure, the *bond curve* and the *derivative curve* are

$$\bar{p}_t^T \;=\; \mathbb{E}_t\left[e^{-\int_t^T \bar{r}_u du}\right], \qquad p_t^T \;=\; \mathbb{E}_t\left[e^{-\int_t^T r_u du}\right], \qquad r_t \;=\; \bar{r}_t + b_t, \tag{272}$$

where $\bar{r}$ is the bond discount rate, r the derivative (OIS) discount rate and b the *discounting basis* (McCloud (11)). The volatilities are $d\bar{p}_t^T/\bar{p}_t^T = \bar{r}_t dt - \bar{\Sigma}_{t,T}\cdot dW_t$, $dp_t^T/p_t^T = r_t dt - \Sigma_{t,T}^p \cdot dW_t$, and $\Sigma_{t,T}^b := \Sigma_{t,T}^p - \bar{\Sigma}_{t,T}$. The forward measures of the two curves are $\bar{\mathbb{Q}}^T$ and $\mathbb{Q}^{p,T}$.

Remark 83 (Bonds as collateral)**.** This is the structure of native and non-native collateral of subsection 1.2: $\bar{p} = Z^{f|f}$, $p = Z^{f|d}$, $b = q^{f\to d}$ and $\Sigma^b = \Sigma^{f\to d}$, with the convention $\Sigma^{d|j} = \Sigma^{d|i} + \Sigma^{i\to j}$. As noted in the footnote on non-cash collateral in subsection 1.1, a rehypothecable security posted as collateral earns its repo rate: the bond curve is the curve of the bond as collateral. In the code, `repo.RepoModel` is an LGCM with the block f for the bond curve, the block c for the basis, no d factor and zero FX volatility.

Remark 84 (Liquidity basis)**.** McCloud splits the discount rate of the bond maturing at T as $\bar{r}_t^T = \bar{r}_t + \bar{z}_t^T$ (curve and liquidity contributions, (9)) and defines the liquidity adjustment $L_t^{eT} = \log \bar{\mathbb{E}}_t^{eT}[\exp\int_t^e \bar{s}_u^{eT} du]$, $\bar{s}^{eT} = \bar{z}^T - \bar{z}^e$, which is $\mu_S + \frac{1}{2}\sigma_S^2$ for a Gaussian $S = \int_t^e \bar{s}_u^{eT} du$ ((13), (16), (21)). It multiplies the repo discount factors by $e^{-L_t^{eT}}$. We set $\bar{z} = 0$ in the rest of this section.

11.2. Repo contracts.

Definition 9 (Repo)**.** A forward-starting repo agreed at t over $[s, e]$ on the bond maturing at T, $t \le s < e \le T$, exchanges at s one unit of cash for $1/\bar{p}_s^T$ bonds, which are returned at e against $1 + \delta f_{ts}^{eT}$. The *repo rate* f_{ts}^{eT} makes its value zero at t; the repo is discounted at r (McCloud (10)).

Remark 85 (Time and day count)**.** Model time is in ACT/365 years: the interval $e - s$ drives the convexity. The accrual fraction δ only converts the gross forward price into a quoted rate. It is a parameter of the contract: $\delta = (e-s)\cdot 365/360$ for ACT/360 (USD, EUR repo) and $\delta = e - s$ for ACT/365 (GBP). The convexity K^{repo} below does not depend on δ.

11.3. Repo rate as a collateral convexity adjustment.

Proposition 55 (Repo rate)**.**

$$1 + \delta f_{ts}^{eT} \;=\; \frac{\bar{p}_t^s}{\bar{p}_t^e} K_t^{\text{repo}}(s,e,T), \qquad K_t^{\text{repo}}(s,e,T) \;:=\; \frac{\mathbb{E}_t^{p,e}\left[\bar{p}_e^T/\bar{p}_s^T\right]}{\bar{\mathbb{E}}_t^e\left[\bar{p}_e^T/\bar{p}_s^T\right]} \;=\; \Xi_{t,e}^{\bar{p}\to p}\left[\bar{p}_e^T/\bar{p}_s^T\right]. \tag{273}$$

This is model-free.

Proof. The zero value of the repo at t is $\mathbb{E}_t[e^{-\int_t^e r_u du}(\bar{p}_e^T/\bar{p}_s^T - (1+\delta f))] = 0$, so $1 + \delta f = \mathbb{E}_t^{p,e}[\bar{p}_e^T/\bar{p}_s^T]$. Under $\bar{\mathbb{Q}}^e$, $\bar{p}^T/\bar{p}^e$ and $\bar{p}^s/\bar{p}^e$ are martingales, so $\bar{\mathbb{E}}_s^e[\bar{p}_e^T/\bar{p}_s^T] = 1/\bar{p}_s^e$ and $\bar{\mathbb{E}}_t^e[1/\bar{p}_s^e] = \bar{p}_t^s/\bar{p}_t^e$. Divide. □

11.4. Repo convexity in the LGCM.

Proposition 56 (LGCM repo convexity)**.** With deterministic volatilities,

$$\ln K_t^{\text{repo}}(s,e,T) \;=\; -\int_t^s \left(\bar{\Sigma}_{u,e} - \bar{\Sigma}_{u,s}\right)\cdot \Sigma_{u,e}^b\, du \;-\; \int_s^e \left(\bar{\Sigma}_{u,e} - \bar{\Sigma}_{u,T}\right)\cdot \Sigma_{u,e}^b\, du. \tag{274}$$

Proof. $v_u := \bar{\mathbb{E}}^e_u[\bar{p}^T_e/\bar{p}^T_s]$ is a $\bar{\mathbb{Q}}^e$-martingale with $v_u = \bar{p}^s_u/\bar{p}^e_u$ for $u \le s$ and $v_u = \bar{p}^T_u/(\bar{p}^e_u\,\bar{p}^T_s)$ for $s \le u \le e$. It is lognormal with volatility $\sigma^v_u = \bar{\Sigma}_{u,e} - \bar{\Sigma}_{u,s}$ on $[t,s]$ and $\bar{\Sigma}_{u,e} - \bar{\Sigma}_{u,T}$ on $[s,e]$. Apply Proposition 23 with $i \to j = \bar{p} \to p$ and maturity e. □

The first term is the Libor collateral convexity (187) of the forward rate on $[s,e]$; it is the whole convexity for a repo to maturity ($T = e$). The second term is new: during the repo window the bond still has $T - e$ to run, and its price at e is correlated with the basis.

Corrolary 3 (Maturity and forwardness)**.** Let $C^{eT}_{ts} = \ln K^{\text{repo}}_t(s,e,T)$, $M^{eT}_t = C^{eT}_{tt} = \int_t^e (\bar{\Sigma}_{u,T} - \bar{\Sigma}_{u,e}) \cdot \Sigma^b_{u,e}\,du$ and $F^{eT}_{ts} = C^{eT}_{ts} - (M^{eT}_t - M^{sT}_t)$ (McCloud (17)). Then $M^{ee}_t = 0$, $F^{eT}_{tt} = 0$ and, with the repo discount factor $\hat{p}^{eT}_t = \bar{p}^e_t e^{-M^{eT}_t}$,

$$1 + \delta f^{eT}_{ts} \;=\; \frac{\hat{p}^{sT}_t}{\hat{p}^{eT}_t} e^{F^{eT}_{ts}}. \tag{275}$$

Proof. The definitions and Proposition 55. □

11.5. **The Hull–White case.**

Proposition 57 (Hull–White repo convexity)**.** Let $d\bar{r}_t = \theta(\bar{r}^*_t - \bar{r}_t)dt + \sigma dx_t$ and $db_t = \kappa(b^*_t - b_t)dt + \varepsilon dy_t$ with $dx_t\,dy_t = \rho\,dt$ (McCloud (27)). With $E(\beta,\tau) = (1 - e^{-\beta\tau})/\beta$, $\bar{\Sigma}_{u,T} = \sigma E(\theta, T-u)$ and $\Sigma^b_{u,T} = \varepsilon E(\kappa, T-u)$. Then

$$\begin{aligned} C^{eT}_{ts} \;=\; \frac{\rho\sigma\varepsilon}{\theta\kappa}\Big(&(1 - e^{-\theta(T-e)}) \int_t^e e^{-\theta(e-u)}(1 - e^{-\kappa(e-u)})du \\ &-(1 - e^{-\theta(T-s)}) \int_t^s e^{-\theta(s-u)}(1 - e^{-\kappa(e-u)})du\Big), \end{aligned} \tag{276}$$

which is McCloud (30); integrating gives his (33) and (34).

Proof. $\bar{\Sigma}_{u,e} - \bar{\Sigma}_{u,s} = \sigma e^{-\theta(s-u)} E(\theta, e-s)$ and $\bar{\Sigma}_{u,T} - \bar{\Sigma}_{u,e} = \sigma e^{-\theta(e-u)} E(\theta, T-e)$. Substitute in Proposition 56 and split $\int_t^e = \int_t^s + \int_s^e$ in (30). □

Remark 86 (Check of McCloud's formulas)**.** McCloud's (30), (33) and (34) agree with Proposition 56, including the signs, with ρ the correlation of $\bar{r}$ and b: the measured agreement is 10^{-11} relative or better. The closed forms divide by $\theta\kappa(\theta+\kappa)$. The LGCM formula is stable at zero mean reversion, where

$$C^{eT}_{ts} \;=\; -\rho\sigma\varepsilon\left[\tfrac{1}{2}(e-s)\left((e-t)^2 - (e-s)^2\right) - \tfrac{1}{2}(T-e)(e-s)^2\right].$$

Both checks are in `tests/test_repo.py`.

11.6. **Repo discount curves.**

Proposition 58 (Hull–White repo curve)**.** In the Hull–White case of Proposition 57:

1. The infinite maturity repo curve $\hat{p}^e_t := \hat{p}^{e\infty}_t$ has instantaneous forwards $\hat{f}^e_t = \bar{f}^e_t + \frac{\rho\sigma\varepsilon}{\theta\kappa} e^{-\theta(e-t)}(1 - e^{-\kappa(e-t)})$ (McCloud (38)). For $e \ge E$, $\hat{f}^e_t - \bar{f}^e_t = (\hat{f}^E_t - \bar{f}^E_t)\, e^{-\theta(e-E)} \frac{1-e^{-\kappa(e-t)}}{1-e^{-\kappa(E-t)}}$ (McCloud (39)).
2. $\hat{p}^{eT}_t = (\bar{p}^e_t)^w (\hat{p}^e_t)^{1-w}$ with $w = e^{-\theta(T-e)}$ (McCloud (41)).
3. $F^{eT}_{ts} = (1 - e^{-(\theta+\kappa)(s-t)})\, F^{eT}_{-\infty,s}$, so $1 + \delta f^{eT}_{ts} = (1 + \delta f^{eT}_{ss})^{w'} (1 + \delta f^{eT}_{-\infty,s})^{1-w'}$ with $w' = e^{-(\theta+\kappa)(s-t)}$ at fixed $\hat{p}^{sT}_t/\hat{p}^{eT}_t$ (McCloud (43)).

Proof. By (34), $M^{eT}_t = (1 - e^{-\theta(T-e)}) M^{e\infty}_t$, which gives 2.; differentiate $M^{e\infty}_t$ in e for 1.; F^{eT}_{ts} in (34) has the factor $(1 - e^{-(\theta+\kappa)(s-t)})$, and its other factors depend on s, e, T only, which gives 3. □

Remark 87 (Beyond Hull–White)**.** The interpolations of Proposition 58 use a single mean reversion and do not hold with several factors or piecewise constant parameters. In a two-factor bond, one-factor basis piecewise constant model (`test_interpolation_fails_beyond_hull_white`), the mean reversion implied by $M^{eT}/M^{e\infty} = 1 - e^{-\theta(T-e)}$ is 0.0232 at $T - e = 2$ and 0.0296 at $T - e = 20$ (measured), and the ratio $F(\tau)/F(\infty)$ is 0.18, 0.31, 1.02, 1.53, 1.38 at $\tau = 0.5, 1, 3, 6, 10$: not monotone and above one. Use Proposition 56 directly.

11.7. **Coupon bonds.**

Definition 10 (Repo on a coupon bond)**.** A coupon bond pays c_k at T_k and is worth $\bar{P}_t = \sum_{T_k>t} c_k \bar{p}_t^{T_k}$ (ex-coupon at the payment dates). In the repo over $[s,e]$, one unit of cash buys $1/\bar{P}_s$ bonds at s; the bond matures at or after e; the cash flows paid in $(s,e]$, including a redemption at e, are passed through to the bond's economic owner at T_k (manufactured payments, discounted at r); a coupon at s belongs to the seller; at e the ex-coupon bonds are returned against $1+\delta f$.

Proposition 59 (Coupon repo rate)**.**

$$(277)\quad p_t^e\,(1+\delta f) \;=\; p_t^s\,\mathbb{E}_t^{p,s}[Y_s], \qquad Y_s \;=\; \frac{1}{\bar{P}_s}\left(\sum_{T_k>e} c_k\,p_s^e\,\frac{\bar{p}_s^{T_k}}{\bar{p}_s^e}\,K_s^{\mathrm{repo}}(s,e,T_k) + \sum_{s<T_k\le e} c_k\,p_s^{T_k}\right).$$

In the LGCM every term of Y_s is the exponential of an affine function of the Gaussian state z_s, and $\mathbb{E}^{p,s}[Y_s]$ is computed exactly by Gauss–Hermite quadrature. With a single cash flow it is Proposition 55. If $b=0$, then $Y_s=1$ and $1+\delta f = \bar{p}_t^s/\bar{p}_t^e$ for any coupons.

Proof. Condition on $\mathcal{F}_s$. By Proposition 55 at $t=s$, $\mathbb{E}_s^{p,e}[\bar{p}_e^{T_k}] = (\bar{p}_s^{T_k}/\bar{p}_s^e)K_s^{\mathrm{repo}}(s,e,T_k)$, and a coupon at $T_k\in(s,e]$ is worth $c_k p_s^{T_k}$ at s. If $b=0$ then $p=\bar{p}$, $K=1$ and the bracket is $\bar{P}_s$. □

Definition 11 (Frozen weights)**.** Write $Y_s = N(z)/D(z)$ with $N=\sum_k a_k e^{\alpha_k\cdot z}$, $D=\sum_j c_j e^{\beta_j\cdot z}$ and $z\sim\mathcal{N}(\mu,P)$ under $\mathbb{Q}^{p,s}$. Replace D by the lognormal with the same mean $\bar{D}=\sum_j c_j e^{\beta_j\cdot\mu+\frac12\beta_j^\top P\beta_j}$ and the loading $\bar{\beta}=\sum_j w_j\beta_j$, $w_j = c_j e^{\beta_j\cdot\mu+\frac12\beta_j^\top P\beta_j}/\bar{D}$. This gives, for any sum of exponentials $X=\sum_k x_k e^{\gamma_k\cdot z}$,

$$(278)\qquad \mathrm{FW}[X] \;:=\; \frac{1}{\bar{D}}\sum_k x_k \exp\left(\gamma_k\cdot\mu + \tfrac12\bar{\beta}^\top P\bar{\beta} + \tfrac12(\gamma_k-\bar{\beta})^\top P(\gamma_k-\bar{\beta})\right) \;\approx\; \mathbb{E}\left[X/D\right].$$

The frozen-weights approximation of the coupon repo rate uses $\mathbb{E}[D/D]=1$ as a control:

$$(279)\qquad \mathbb{E}^{p,s}[Y_s] \;\approx\; 1+\mathrm{FW}[N]-\mathrm{FW}[D].$$

Remark 88 (Accuracy of frozen weights)**.** The control makes the approximation exact when $N=D$, in particular for $b=0$ (Proposition 59), and for a single cash flow, where FW is exact. Without the control, $\mathrm{FW}[D]=\sum_j w_j e^{\bar{\beta}^\top P\bar{\beta}-\bar{\beta}^\top P\beta_j}\neq 1$, which breaks the $b=0$ identity; at the parameters of Remark 91 this artefact was 96% of the approximation error (6% coupon, window $[1.2,1.6]$). Measured with the control: the error of the repo rate scales as λ^4 when all volatilities are scaled by λ (errors $2.6\cdot10^{-5}$, $1.6\cdot10^{-6}$, $1.0\cdot10^{-7}$ for $\lambda=4,2,1$, ratios 16.2 and 16.0). This is an observation, not a proof. For a 10Y semi-annual bond with a coupon in the window $[1.2,1.6]$, the approximation misses at most 0.56% of the gap between the repo rate and the bond forward (coupons 0 to 8%).

Remark 89 (What the coupon adds)**.** For a coupon bond, the gap between the repo rate and the bond forward is not all convexity. A coupon in $(s,e]$ is passed through at T_k and discounted at r, while the bonds are priced at $\bar{r}$: with $b\neq0$ this moves the rate even at $\rho=0$. In the example above, at a 6% coupon, the gap is -0.124bp at $\rho=0$ and a further -0.057bp from the correlation; for a zero-coupon bond it is all correlation (-0.143bp). The notebook plots the two parts separately.

11.8. **Bond-collateralised derivatives.**

Proposition 60 (Overnight repo)**.** In the LGCM, as $h\to0$: $\ln K_t^{\mathrm{repo}}(t,t+h,T) = O(h^2)$, so the spot overnight repo rate on any bond of the curve converges to $\bar{r}_t$. For a forward start $s>t$,

$$(280)\qquad \frac{1}{h}\ln K_t^{\mathrm{repo}}(s,s+h,T) \;\to\; -\int_t^s \partial_s\bar{\Sigma}_{u,s}\cdot\Sigma^b_{u,s}\,du,$$

which is $-\rho\sigma\varepsilon\,(E(\theta,s-t)-E(\theta+\kappa,s-t))/\kappa$ in the Hull–White case: the forward overnight repo rate is not the bond forward rate.

Proof. For $s=t$ the first integral of Proposition 56 vanishes; in the second, $\bar{\Sigma}_{u,t+h}-\bar{\Sigma}_{u,T}$ is bounded and $\Sigma^b_{u,t+h}=O(h)$ on an interval of length h. For $s>t$, $\bar{\Sigma}_{u,s+h}-\bar{\Sigma}_{u,s} = h\,\partial_s\bar{\Sigma}_{u,s}+O(h^2)$ and the second integral is $O(h^2)$. □

Hypothesis (Security collateral)**.** Posting a fully rehypothecable security as collateral, with no haircut, is equivalent to posting cash collateral that earns the security's spot overnight repo rate: the effective discount rate of the collateralised trade is that repo rate. This is the extension to non-cash collateral suggested in the footnote of subsection 1.1 (see also Piterbarg [8]); it is an assumption on how the collateral is financed, not a consequence of the framework.

Proposition 61 (Discount curve of bond-collateralised derivatives)**.** Under the Hypothesis "Security collateral", a derivative collateralised by a bond of the curve (no haircut, $\bar{z} = 0$) has effective discount rate $\bar{r}$: its discount curve is the bond curve $\bar{p}$.

Proof. By Proposition 60, the spot overnight repo rate of the bond at u is $\bar{r}_u$. By the Hypothesis, $V_t = \mathbb{E}_t[e^{-\int_t^T \bar{r}_u du} V_T]$, and $V_T = 1$ gives $\bar{p}_t^T$. □

Remark 90 (Term repo curve or bond curve)**.** McCloud (§4) proposes the extrapolated term repo curve $\hat{p}^e$ for discounting bond-collateralised derivatives. That corresponds to a different financing assumption than the Hypothesis "Security collateral": the collateral financed with term repos locked at inception. Under rolling overnight financing, Proposition 61 gives $\bar{p}$, and the two differ by the maturity adjustment $M^{e\infty}$. Which one applies is a question about how the collateral is actually funded, not one the model decides. The size of the difference is shown in the notebook `repo_convexity.py` (in `python/notebooks`).

11.9. **Verification.**

Remark 91 (Verification)**.** The checks are in `python/convexity/tests/test_repo.py`. Hull–White parameters are $\theta = 5\%$, $\sigma = 1\%$, $\kappa = 40\%$, $\varepsilon = 0.8\%$, $\rho = -0.5$, with flat curves $\bar{r} = 3.5\%$, $b = -20$bp.

(1) Proposition 56 against nested adaptive quadrature in a two-factor bond, one-factor basis piecewise model with zero, tiny and negative mean reversions: $6 \cdot 10^{-15}$ relative. For $T = e$ it equals the Libor collateral convexity to 10^{-16} absolute (the round-off of `libor_convexity`; $1.5 \cdot 10^{-11}$ relative at $\ln K = 5 \cdot 10^{-6}$).

(2) Proposition 57: McCloud's closed forms to 10^{-11} relative, also past the last knot and for small and negative mean reversions; the polynomial limit at $\theta = \kappa = 0$ to 10^{-12}.

(3) Proposition 58: the interpolations (41), (43) and the extrapolation (39) hold to 10^{-12} in Hull–White; the forward spread (38) matches a central finite difference of $M_0^{e\infty}$ to 10^{-7}; the failure beyond Hull–White is quoted in the Remark above.

(4) Proposition 59: a one-cash-flow bond reproduces Proposition 55 to 10^{-13}; $b = 0$ gives the bond forward to 10^{-13}; Gauss–Hermite with 40 and 60 nodes agree to 10^{-14}.

(5) Monte Carlo (exact LGCM scheme, 2^{18} antithetic paths, seeds 1 to 4, volatilities scaled by 3, control variates $\mathbb{E}[D_{0,e}] = p_0^e$ and the bond-collateralised repo of value $\bar{p}_0^s$): the zero-coupon repo rate ($s = 1.3$, $e = 2.1$, $T = 7.5$, convexity -1.20bp) is at $z = -1.39, +0.57, +0.45, -0.23$; the bond forward ($K = 1$) is rejected at 7.8 to 9.8 standard errors. The coupon repo rate (10Y semi-annual 5%, two coupons in the window) is at $z = -0.88, +1.42, +0.72, -0.57$ against the quadrature; the frozen weights approximation (Definition 11) is also within 4 standard errors ($z = -1.41, +0.89, +0.19, -1.10$).

(6) Proposition 60: $\ln K_0(0, h, 10)$ scales as h^2; the forward overnight limit matches the Hull–White closed form (1.14bp at $s = 3$).

12. REMARKS

12.1. **Model usage.**

12.1.1. *Break even levels of covariances.* The model specifies break even levels of covariances.
This allows the trader to enter trades if the price offered.
Assess whether he is long or short correlations. etc...

12.1.2. *Benchmarking usage.* Because the model is consistent with hedging instruments, it can be used to assess the impact of stochastic currency basis on exotic products.

12.2. **Stochastic collateral and PnL leaks.** Q: How does the stochasticity of collateral impact the PnL of a trader hedging with a deterministic collateral model? Does he experience PnL leaks?
We have two impacts:

(1) Intrinsic value: collateral convexity impacts the forwards
(2) Time value: volatility of the collateral changes the volatility of market observables.

12.3. **Materiality.** Which trading desk would see the largest impacts?
A guess would be:

(1) DCU/Linear rates
(2) Counterparty credit risk
(3) Long dated FX

12.4. **Model assessment.**

12.4.1. *Pros and cons.* The model has the following:

(1) Pros:
  (a) Calibrates directly and consistently to liquid linear market instruments
  (b) The valuation of market instruments with native collateral is independent of other markets.
  (c) The valuation of liquid linear market instruments is independent of the diffusion parameters.
  (d) The model provides relevant intuition for the practitioner.
  (e) While designed primarily to match linear instruments, the model provides a fully arbitrage-free dynamic of that can be used to price exotics products.
(2) Cons:
  (a) Funding spreads are deterministic.
  (b) Lack of smile.
  (c) Calibration to historical data is somewhat uncertain.

12.4.2. *Generalisations.*
Quasi-Gaussian model. A natural extension of the LGCM model is to allow stochastic volatility.
Andreasen suggests the following parametrisation. Setting $g_i(\tau) = e^{-\kappa_i \tau}$ and

$$\eta_t = \begin{bmatrix} g_1(\theta_1) & \dots & g_n(\theta_1) \\ g_1(\theta_n) & \dots & g_n(\theta_n) \end{bmatrix}^{-1} \begin{bmatrix} \sigma_1(t,\theta_1) & \dots & \sigma_n(t,\theta_1) \\ \sigma_1(t,\theta_n) & \dots & \sigma_n(t,\theta_n) \end{bmatrix}$$

we have,

$$dr_{t,t+\theta} = \sum_i \sum_j e^{-\kappa_i \tau} \eta_t^{ij} dW_t^{T;j}$$

$$\begin{aligned} \sigma_j(t,\tau) &= \sqrt{z_t}\lambda(t,\tau)\left(b(t,\tau)r_t + (1-b(t,\tau))r_{0,t} - \zeta\right)\rho_j(\tau) \\ dz_t &= \theta(1-z_t)dt + \varepsilon(t)\sqrt{z_t}dZ_t \end{aligned}$$

Then,

- $\lambda(t,\tau_k)$ controls ATM vol for $r_{t,t+\tau_k}$
- $b(t,\tau_k)$ controls the skew
- $\varepsilon(t)$ controls the convexity of the smile

- $\rho(i)\rho^j(t)$ controls the correlation of forward rates

The dynamic of the swap rate

$$dY_t = \sum_i \partial_{x_i} Y \sum_j \eta_{ij} dW_t^{A,j}$$

is approximated by

$$\begin{aligned} dY_t &\approx \sqrt{z_t}\lambda(t)\left(b(t)Y_t + (1-b(t))Y_0 - \zeta\right) dW_t^A \\ dz_t &= \theta(1-z_t)dt + \varepsilon(t)\sqrt{z_t}dZ_t \end{aligned}$$

and one uses the parameter averaging method to get constant parameters followed by a Fourier transform.
LMM model with stochastic currency basis. Another alternative is to consider Libor Market Models with stochastic collateral.

The constant funding spread assumption and many of our computations can carry unchanged.

One interesting thing is that, as we saw in the calibration to currency basis spreads, we were lead to chose an interpolation scheme for the initial discount curve $Z_{t,T}^{f|d}$.

Similar problems arise in LMM models where one has to specify stub interpolation to create discount factors from the forward rates on a fixed grid. To avoid arbitrage, the interpolation of the dynamic should be consistent with the interpolation scheme for the curves. In the HJM-type models of these notes this issue does not arise, see subsection 10.7.

12.5. **Conclusion.** In these notes,

(1) We characterised absence of arbitrage for instantaneous forward rates models under perfect cash collateral.
(2) We proved that under the Cheyette separability condition, these models are Markovian.
(3) We derived explicit formulas for convexity adjustments.
(4) Finally, we introduced the Linear Gaussian Collateral Model and described efficient algorithms for its calibration and simulation.
(5) The convexity adjustments hold for Libor-type and compounded overnight coupons alike.
(6) The calibration builds the cross-currency discount curve jointly with the collateral basis volatility, the curves being either bootstrapped or fitted globally on hybrid (piecewise flat and smooth) curves; the collateral convexity moves the long end of the curve by a few basis points on the example markets.
(7) The Gaussian approximations of swap rates, currency basis spreads and terminal correlations are asymptotic in the volatility, with errors measured at $O(\lambda^2)$ under vol scaling; the same framework gives the repo convexity of bonds.
(8) Every closed form is checked against an exact benchmark (moment ODEs, quadrature, no-arbitrage identities) or Monte Carlo in the numerical companion. The open points are listed at the beginning of the notes.

Appendix

APPENDIX A. RISK NEUTRAL PRICING UNDER PERFECT COLLATERAL

A.1. **Risk neutral pricing under perfect collateral.** We assume perfect collateralization[47].

Theorem 5. Under the previous assumptions, for each currency d, there exist a unique measure $\mathbb{Q}^{d,\mathrm{RN}}$ equivalent to the historical measure $\mathbb{P}$ such that every domestic self-financed portfolio with domestic collateral rate r^c satisfies the martingale property

$$\mathbb{E}_t^{d,\mathrm{RN}}\left[e^{-\int_t^T r_u^c du} V_T^{d|c}\right] = V_t^{d|c}. \tag{281}$$

Proof. We follow Piterbarg's reasoning.

We consider a market with two collateralized assets S^1, S^2 with collateral rates r^1, r^2. An investment strategy is a portfolio:

$$\Pi_t = \Delta_t^1 S_t^1 + \Delta_t^2 S_t^2$$

At t, A enters the trade and at $t+dt$, he closes the position:

(1) at t,
  (a) A buys the assets from B paying him the amount $\Pi_t = \Delta_t^1 S_t^1 + \Delta_t^2 S_t^2$
  (b) B immediately posts the amount Π_t back to the collateral account (held by A or third party C).

(2) at $t+dt$,
  (a) A pays to B the interest rate on the collateral posted:
    - B receives: $\Delta_t^1 r_t^1 S_t^1 dt + \Delta_t^2 r_t^2 S_t^2 dt$.
  (b) A chooses to close the position:
    - The collateral is returned to B who receives: $\Delta_t^1 S_t^1 + \Delta_t^2 S_t^2$.
    - A sells the assets which are now valued at $\Delta_t^1 S_{t+dt}^1 + \Delta_t^2 S_{t+dt}^2$.

So A's P&L is

$$\begin{aligned}\mathrm{P\&L}_{t+dt} &= \left(\Delta_t^1 S_{t+dt}^1 + \Delta_t^2 S_{t+dt}^2\right) - \left(\Delta_t^1 r_t^1 S_t^1 dt\right) - \left(\Delta_t^2 r_t^2 S_t^2 dt\right) - \left(\Delta_t^1 S_t^1 + \Delta_t^2 S_t^2\right) \\ &= \Delta_t^1\left(dS_t^1 - r_t^1 S_t^1 dt\right) + \Delta_t^2\left(dS_t^2 - r_t^2 S_t^2 dt\right).\end{aligned}$$

Suppose our two assets have dynamic:

$$dS_t^i = \mu_t^i dt + \sigma_t^i dW_t^{\mathbb{P}}$$

driven by the same Brownian motion. Our P&L is

$$\mathrm{P\&L}_t = \left(\Delta_t^1(\mu_t^1 - r_t^1) + \Delta_t^2(\mu_t^1 - r_t^2)\right) dt + \left(\Delta_t^1\sigma_t^1 + \Delta_t^2\sigma_t^2\right)\cdot dW_t^{\mathbb{P}}$$

We can now choose non-zero weights that cancel the Brownian part:

$$\Delta_t^1\sigma_t^1 + \Delta_t^2\sigma_t^2 = 0$$

Namely[48]

$$\begin{aligned}\Delta_t^1 &= \sigma_t^2 \\ \Delta_t^2 &= -\sigma_t^1\end{aligned}$$

The P&L reduces to

$$\mathrm{P\&L} = \left(\Delta_t^1(\mu_t^1 - r_t^1) + \Delta_t^2(\mu_t^2 - r_t^2)\right) dt.$$

It is known at t so it has to be[49] 0, otherwise we would be able to make a risk-free profit without initial investment by buying or selling the strategy. This implies

$$\frac{r_t^1 - \mu_t^1}{\sigma_t^1} = \frac{r_t^2 - \mu_t^2}{\sigma_t^2}$$

[47] Hypothesis can be formulated explicitly.

[48] This can be generalised to the case of a multi-dimensional Brownian motion and an arbitrary number of assets S^i. One simply needs to make the following completeness assumption: the full volatility matrix Σ with rows σ^i is not invertible.

[49] Why? We should include our funding rate here!

Let us write λ_t for this quantity. We can define $\mathbb{Q}^{d,\mathrm{RN}}$ by setting:

$$\left.\frac{d\mathbb{Q}^{d,\mathrm{RN}}}{d\mathbb{P}}\right|_{\mathcal{F}_t} = \mathcal{E}\left(\int_0^\cdot \lambda_u\, dW_u^{\mathbb{P}}\right)$$

Then, under this measure,

$$dW_u^{d,\mathrm{RN}} = dW_u^{\mathbb{P}} - \lambda_t dt$$

is a Brownian motion and

$$dS_t^i = r_t^i dt + \sigma_t^i \cdot dW_t^{d,\mathrm{RN}}$$

so the discounted assets are $\mathbb{Q}^{d,\mathrm{RN}}$-martingales. □

Proposition 62. For any collateral rate c in currency f,

$$r^{d|c} = \mu^{df} + r^{f|c}.$$

Proof. Enter a collateralized trade at t. Close the position at $t+dt$. The value at t in currency d of this strategy is

$$\begin{aligned}
& \mathbb{E}_t^{d,\mathrm{RN}}\left[e^{-r_t^{d|c}dt}\left(V_{t+dt}^{d|c} - x_{t+dt}^{df}(1+r_t^{f|c})x_t^{fd}V_t^{d|c}dt\right)\right] \\
= & \mathbb{E}_t^{d,\mathrm{RN}}\left[e^{-r_t^{d|c}dt}\left(V_{t+dt}^{d|c} - (1+r_t^{f|c})\left(\mu_t^{df}dt + \sigma_t^{df}dW_t^{d,x}\right)V_t^{d|c}dt\right)\right] \\
= & e^{-r_t^{d|c}dt}\left(\mathbb{E}_t^{d,\mathrm{RN}}[dV_t^{d|c}] - (\mu_t^{df} + r_t^{f|c})V_t^{d|c}dt\right) \\
= & e^{-r_t^{d|c}dt}\left(r_t^{d|c} - \mu_t^{df} - r_t^{f|c}\right)V_t^{d|c}dt
\end{aligned}$$

Since we can enter a collateralized trade at no cost, no arbitrage implies this is 0 so $r_t^{d|c} = \mu_t^{df} + r_t^{f|c}$. □

Remark 92. Repo rates are equivalent to collateral.

Proposition 63.

$$\mathbb{E}_t^{d,\mathrm{RN}}\left[\frac{d\mathbb{Q}^{f,\mathrm{RN}}}{d\mathbb{Q}^{d,\mathrm{RN}}}\right] = e^{-\int_0^t \mu^{df}du}\frac{x_t^{df}}{x_0^{df}} = \mathcal{E}\left(\int_0^\cdot \sigma_v^x dW_v^x\right)_t \tag{282}$$

Proof. No arbitrage between foreign and domestic markets (carry trade). □

So the change from domestic to foreign measure is the usual one and the quanto adjustments are governed by the correlation with the FX driver.

Proposition 64 (Triangle relation for FX drift)**.**

$$\begin{aligned}
\mu^{df} + \mu^{fd} &= 0 \\
\mu^{df} + \mu^{fg} + \mu^{gd} &= 0.
\end{aligned}$$

Proof. Chain the change of measures. □

We can now extend our previous results to arbitrary collateral currency:

Proposition 65. For any collateral rate c in any currency,

$$r^{d|c} = \mu^{df} + r^{f|c}.$$

Proof. Let x be the currency of the collateral rate c, then

$$r^{d|c} = \mu^{dx} + r^{x|c} = \mu^{dx} + (r^{f|c} - \mu^{fx}) = \mu^{df} + r^{f|c}.$$

□

Proposition 66. For any currencies d, f and collaterals c_1, c_2,

$$r^{d|c_1} - r^{d|c_2} = r^{f|c_1} - r^{f|c_2}.$$

Proof.

$$r^{d|c_1} - r^{d|c_2} = (\mu^{df} + r^{f|c_1}) - (\mu^{df} + r^{f|c_2}) = r^{f|c_1} - r^{f|c_2}.$$

□

A.2. **HJM framework: proofs.**

A.2.1. *Dynamic of funding spreads.*

Lemma 6. Under our funding spread assumption, the multiplicative funding spread

$$1 - \theta S^{d|c,\div}_{t,T;\theta} = \frac{1 + \theta L^{d|c,\div}_{t,T;\theta}}{1 + \theta R^{d|c,\div}_{t,T;\theta}} \tag{283}$$

is an exponential martingale under $\mathbb{Q}^{d|c,T}$:

$$\frac{d(1 - \theta S^{d|c,\div}_{t,T;\theta})}{1 - \theta S^{d|c,\div}_{t,T;\theta}} = -\Sigma^{d\to c}_{t,T;\theta} \cdot dW^{d|c,T}_t \tag{284}$$

Proof. From Ito's lemma, we have:

$$\frac{d(X/Y)}{X/Y} = \frac{dX}{X} - \frac{dY}{Y} - \left\langle \frac{dY}{Y}, \frac{dX}{X} - \frac{dY}{Y} \right\rangle$$

So,

$$\begin{aligned}
&\frac{d(1 - \theta S^{d|c,\div}_{t,T;\theta})}{1 - \theta S^{d|c,\div}_{t,T;\theta}} \\
= \;& \frac{d(1 + \theta L^{d|c}_{t,T;\theta})}{1 + \theta L^{d|c}_{t,T;\theta}} - \frac{d(1 + \theta R^{d|c}_{t,T;\theta})}{1 + \theta R^{d|c}_{t,T;\theta}} - \left\langle \frac{d(1 + \theta R^{d|c}_{t,T;\theta})}{1 + \theta R^{d|c}_{t,T;\theta}}, \frac{d(1 + \theta L^{d|c}_{t,T;\theta})}{1 + \theta L^{d|c}_{t,T;\theta}} - \frac{d(1 + \theta R^{d|c,\div}_{t,T;\theta})}{1 + \theta R^{d|c}_{t,T;\theta}} \right\rangle \\
= \;& \Sigma^{d|d}_{t,T;\theta} dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} \cdot (\Sigma^{d|d}_{t,T;\theta} - \Sigma^{d|c}_{t,T;\theta}) dt \\
= \;& -\Sigma^{d\to c}_{t,T;\theta} dW^{d|c,T+\theta}_t + \Sigma^{d|c}_{t,T;\theta} \cdot \Sigma^{d\to c}_{t,T;\theta}\, dt \\
= \;& -\Sigma^{d\to c}_{t,T;\theta} \left(dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} dt \right) \\
= \;& -\Sigma^{d\to c}_{t,T;\theta} \cdot dW^{d|c,T}_t
\end{aligned}$$

□

A.2.2. *Resetting forward Libors.*

We first compute the adjustment due to the resetting notional.

Lemma 7. Under the constant funding spread assumption,

$$\mathbb{E}^{d|d,T+\theta}_t \left[x^{df}_T (1 + \theta L^d_{T;\theta}) \right] = x^{df|d}_{t,T} \left(1 + \theta L^{d|d}_{t,T;\theta} \right) \tag{285}$$

Proof. Using the fact that the funding spreads are deterministic:

$$\begin{aligned}
\mathbb{E}_t^{d|d,T}\left[x_S^{df}(1+\theta L_{T;\theta}^{d|d})\right] &= \mathbb{E}_t^{d|d,T}\left[x_S^{df}(1+\theta R_{S,T}^{d|d})\left(1-\theta S_{T;\theta}^{d|d,\div}\right)\right] \\
&= \mathbb{E}_t^{d|d,T}\left[x_S^{df}(1+\theta R_{T;\theta}^{d|d})\right]\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right) \\
&= \mathbb{E}_t^{d|d,T}\left[\frac{Z_{S,S}^{d|d}}{Z_{S,T}^{d|d}}x_S^{df}\right]\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right) \\
&= \frac{Z_{t,S}^{d|d}}{Z_{t,T}^{d|d}}\mathbb{E}_t^{d|d,S}\left[x_S^{df}\right]\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right) \\
&= \left(1+\theta R_{t,T;\theta}^{d|d}\right)x_{t,S}^{df|d}\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right) \\
&= x_{t,S}^{df|d}\left(1+\theta L_{t,T;\theta}^{d|d}\right)
\end{aligned}$$

□

Proposition 67.

$$1+\theta L_{t,T;\theta}^{f/d|d} = \left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right) \tag{286}$$

A.2.3. *Resetting cross-currency swaplets.* We now proceed to the proof of Theorem 3.

Theorem. Under deterministic multiplicative funding spread assumptions, the currency basis spread of a resetting cross-currency swaplet is:

$$B_{t,T;\theta}^{f/d|d,r} = \left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(S_{t,T;\theta}^{f|d,\div}-S_{t,T;\theta}^{d|d,\div}\right).$$

Proof. By definition, the currency basis satisfies

$$\begin{aligned}
B_{t,T;\theta}^{f/d|d,r} &= L_{t,T;\theta}^{f/d|d}-L_{t,T;\theta}^{f|d} \\
&= (1+\theta L_{t,T;\theta}^{f/d|d})-(1+\theta L_{t,T;\theta}^{f|d}) \\
&= \left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(1-\theta S_{t,T;\theta}^{d|d,\div}\right)-\left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(1-\theta S_{t,T;\theta}^{f|d,\div}\right) \\
&= \left(1+\theta R_{t,T;\theta}^{f|d}\right)\left(S_{t,T;\theta}^{f|d,\div}-S_{t,T;\theta}^{d|d,\div}\right)
\end{aligned}$$

□

A.3. **Deterministic collateral basis models: proofs.**

A.3.1. *Triangle relation for currency basis.*

A.4. **Deterministic volatility models: proofs.**

A.4.1. *Dynamic of market observables.*
Funding basis spreads.

Proposition (Funding basis spreads)**.** Under our funding spread assumption, the multiplicative funding spread

$$1-\theta S_{t,T;\theta}^{d|c,\div} = \frac{1+\theta L_{t,T;\theta}^{d|c,\div}}{1+\theta R_{t,T;\theta}^{d|c,\div}} \tag{287}$$

is a martingale[50] under $\mathbb{Q}^{d|c,T}$:

$$\frac{d(1-\theta S_{t,T;\theta}^{d|c,\div})}{1-\theta S_{t,T;\theta}^{d|c,\div}} = -\Sigma_{t,T;\theta}^{d\to c}\cdot dW_t^{d|c,T} \tag{288}$$

[50]Note that, if $f=d$, the vol is zero and we recover that the spread is constant.

Proof. From Ito's lemma, we have:

$$\frac{d(X/Y)}{X/Y} = \frac{dX}{X} - \frac{dY}{Y} - \left\langle \frac{dY}{Y}, \frac{dX}{X} - \frac{dY}{Y} \right\rangle$$

So,

$$\begin{aligned}
& \frac{d(1-\theta S^{d|c,\div}_{t,T;\theta})}{1-\theta S^{d|c,\div}_{t,T;\theta}} \\
= & \frac{d(1+\theta L^{d|c}_{t,T;\theta})}{1+\theta L^{d|c}_{t,T;\theta}} - \frac{d(1+\theta R^{d|c}_{t,T;\theta})}{1+\theta R^{d|c}_{t,T;\theta}} - \left\langle \frac{d(1+\theta R^{d|c}_{t,T;\theta})}{1+\theta R^{d|c}_{t,T;\theta}}, \frac{d(1+\theta L^{d|c}_{t,T;\theta})}{1+\theta L^{d|c}_{t,T;\theta}} - \frac{d(1+\theta R^{d|c,\div}_{t,T;\theta})}{1+\theta R^{d|c}_{t,T;\theta}} \right\rangle \\
= & \Sigma^{d|d}_{t,T;\theta} \cdot dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} \cdot dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} \cdot (\Sigma^{d|d}_{t,T;\theta} - \Sigma^{d|c}_{t,T;\theta}) dt \\
= & -\Sigma^{d\to c}_{t,T;\theta} \cdot dW^{d|c,T+\theta}_t + \Sigma^{d|c}_{t,T;\theta} \cdot \Sigma^{d\to c}_{t,T;\theta} dt \\
= & -\Sigma^{d\to c}_{t,T;\theta} \cdot \left(dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T;\theta} dt \right) \\
= & -\Sigma^{d\to c}_{t,T;\theta} \cdot \left(dW^{d|c,T+\theta}_t - \Sigma^{d|c}_{t,T+\theta} dt + \Sigma^{d|c}_{t,T} dt \right) \\
= & -\Sigma^{d\to c}_{t,T;\theta} \cdot \left(dW^{d,\mathrm{RN}}_t + \Sigma^{d|c}_{t,T} dt \right) \\
= & -\Sigma^{d\to c}_{t,T;\theta} \cdot dW^{d|c,T}_t
\end{aligned}$$

□

Resetting forward Libors.

Proposition (Resetting forward Libors)**.** The resetting forward Libors

$$1 + \theta L^{f/d|i}_{t,T;\theta} = \mathbb{E}^{f|i,T+\theta}_t \left[\frac{x^{fd}_{T+\theta}}{x^{fd}_T} (1 + \theta L^d_{T;\theta}) \right] \tag{289}$$

is an exponential martingale under $\mathbb{Q}^{f|i,T+\theta}$-martingale:

$$\frac{d(1+\theta L^{f/d|i}_{t,T;\theta})}{1+\theta L^{f/d|i}_{t,T;\theta}} = \Sigma^{f|d}_{t,T;\theta} \cdot dW^{f|i,T+\theta}_t \tag{290}$$

So that

$$1 + \theta L^{f/d|i}_{t,T;\theta} = (1 + \theta R^{f|i}_{t,T;\theta})(1 - \theta S^{d|i,\div}_{t,T;\theta}) e^{-\int_t^T \Sigma^{d\to i}_{u,T;\theta} \cdot \Sigma^{df|i}_{u,T} \, du}. \tag{291}$$

Proof. We can rewrite this in terms of in terms of domestic T-martingales:

$$\begin{aligned}
1+\theta L^{f/d|i}_{t,T;\theta} &= \mathbb{E}^{f|i,T+\theta}_t\left[\frac{x^{fd}_{T+\theta}}{x^{fd}_T}(1+\theta L^d_{T;\theta})\right]\\
&= x^{fd|i}_{t,T+\theta}\mathbb{E}^{d|i,T+\theta}_t\left[x^{df}_T(1+\theta L^d_{T;\theta})\right]\\
&= x^{fd|i}_{t,T+\theta}\mathbb{E}^{d|i,T}_t\left[\frac{Z^{d|i}_{T,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}x^{df}_T(1+\theta L^d_{T;\theta})\right]\\
&= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]\\
&= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]\\
&= \frac{x^{fd|i}_{t,T+\theta}}{Z^{d|i}_{t,T,T+\theta}}(1-\theta S^{d|i,\div}_{t,T;\theta})\frac{\mathbb{E}^{d|i,T}_t\left[x^{df}_T(1-\theta S^{d|i,\div}_{T;\theta})\right]}{x^{df|i}_{t,T}(1-\theta S^{d|i,\div}_{t,T;\theta})}\\
&= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T;\theta}\right]
\end{aligned}$$

Now using what we know about the funding spread:

$$\begin{aligned}
1+\theta L^{f/d|i}_{t,T;\theta} &= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[1-\theta S^{d|i,\div}_{T;\theta}\right]\\
&= (1+\theta R^{f|i}_{t,T;\theta})\mathbb{E}^{f|i,T}_t\left[(1-\theta S^{d|i,\div}_{t,T;\theta})\mathcal{E}\left(-\int_t^T \Sigma^{d\to i}_{u,T;\theta}\cdot dW^{f|i,T}_u\right)\right]\\
&= (1+\theta R^{f|i}_{t,T;\theta})(1-\theta S^{d|i,\div}_{t,T;\theta})e^{-\int_t^T \Sigma^{d\to i}_{u,T;\theta}\cdot\Sigma^{df|i}_{u,T}\,du}
\end{aligned}$$

We can now compute the volatility:

$$\begin{aligned}
\frac{d(1+\theta L^{f/d|i}_{t,T;\theta})}{1+\theta L^{f/d|i}_{t,T;\theta}} &= \frac{d(1+\theta R^{f|i}_{t,T;\theta})}{1+\theta R^{f|i}_{t,T;\theta}}+\frac{d(1-\theta S^{d|i,\div}_{t,T;\theta})}{1-\theta S^{d|i,\div}_{t,T;\theta}}+O(dt)\\
&= \Sigma^{f|i}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t-\Sigma^{d\to i}_{t,T;\theta}\cdot dW^{d|i,T}_t+O(dt)\\
&= \left(\Sigma^{f|f}_{t,T;\theta}+\Sigma^{f\to i}_{t,T;\theta}-\Sigma^{d\to i}_{t,T;\theta}\right)\cdot dW^{f|i,T+\theta}_t\\
&= \Sigma^{f|d}_{t,T;\theta}\cdot dW^{f|i,T+\theta}_t
\end{aligned}$$

□

Currency basis spreads. We consider the extension of the previous theorem to a non-native collateral.

Theorem. Let us assume that

(1) native multiplicative funding spreads are constant
(2) volatilities are deterministic.

Then, the currency basis spread of a resetting cross-currency swaplet is:

$$B^{f/d|c,r}_{t,T;\theta} = L^{f/d|c}_{t,T;\theta}-L^{f|c}_{t,T;\theta} \tag{292}$$

$$\theta B^{f/d|c,r}_{t,T;\theta} = (1+\theta R^{f|c}_{t,T;\theta})\left\{(1-\theta S^{d|c,\div}_{t,T;\theta})e^{-\int_t^T \Sigma^{df}_{u,T}\cdot\Sigma^{d\to c}_{u,T;\theta}\,du}-(1-\theta S^{f|c,\div}_{t,T;\theta})\right\} \tag{293}$$

A.4.2. *Comparison to the Determinisctic Collateral Basis Model.*

Remark 93 (Comparison to the Deterministic collateral basis model)**.** According to Equation 32, in the Deterministic collateral basis models, discount rates are given by:

$$R^{f|d,\mathrm{DCB}}_{t,T;\theta} := L^{f|f}_{t,T;\theta} + B^{f/d|d,r}_{t,T;\theta} + S^{d|d,-}_{t,T;\theta}$$

How does our new formula compare to this?

According to our derivation,

$$\begin{aligned}
\theta B^{f/d|d,r}_{t,T;\theta} &= (1+\theta R^{f|d}_{t,T;\theta})\frac{1+\theta L^{d|d}_{t,T;\theta}}{1+\theta R^{d|d}_{t,T;\theta}} - \left(1+\theta L^{f|d}_{t,T;\theta}\right) \\
&= (1+\theta R^{f|d}_{t,T;\theta})(1-\theta S^{d|d,\div}_{t,T;\theta}) - \left(1+\theta L^{f|d}_{t,T;\theta}\right) \\
B^{f/d|d,r}_{t,T;\theta} &= R^{f|d}_{t,T;\theta} - L^{f|d}_{t,T;\theta} - S^{d|d,\div}_{t,T;\theta} - \theta R^{f|d}_{t,T;\theta} S^{d|d,\div}_{t,T;\theta} \\
\theta B^{f/d|d,r}_{t,T;\theta} &= R^{f|d}_{t,T;\theta} - L^{f|f}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta} \\
&\quad - \left(L^{f|d}_{t,T;\theta} - L^{f|f}_{t,T;\theta}\right) + \left(S^{d|d,-}_{t,T;\theta} - S^{d|d,\div}_{t,T;\theta}\right) - \theta R^{f|d}_{t,T;\theta} S^{d|d,\div}_{t,T;\theta}
\end{aligned}$$

So we find that the correction to the deterministic approximation consists of

- a convexity adjustment for the Libor $L^{f|d}_{t,T;\theta} - L^{f|f}_{t,T;\theta}$
- a second (third?) order term depending on the level of the funding spread $S^{d|d,-}_{t,T;\theta} - S^{d|d,\div}_{t,T;\theta}$
- a second order term $\theta R^{f|d}_{t,T;\theta} S^{d|d,\div}_{t,T;\theta}$.

Neglecting, the higher order terms, we find:

$$R^{f|d}_{t,T;\theta} \approx L^{f|d}_{t,T;\theta} + B^{f/d|d,r}_{t,T;\theta} + S^{d|d,-}_{t,T;\theta} \tag{294}$$

Note that the convexity adjustment on Libors can be computed from the model parameters.

The Theorem 3 and the previous analysis considers the currency basis spread as an output of the model when it is in fact a market observable. It would be more telling to compare discount rates $R^{f|d}_{t,T;\theta}$ when models are calibrated to the same currency basis spreads.

Proposition 68 (Linear expansion)**.** Following Equation 32, let us define

$$R^{f|d,\mathrm{DCB}}_{t,T;\theta} = L^{f|f}_{t,T;\theta} + B^{f/d|d,r}_{t,T;\theta} + S^{d|d,-}_{t,T;\theta}$$

Then

$$R^{f|d}_{t,T;\theta} = R^{f|d,\mathrm{DCB}}_{t,T;\theta} + \dots$$

[TODO: Finish]

Proof.

$$
\begin{aligned}
& R^{f|d}_{t,T;\theta} - R^{f|d,\mathrm{DCB}}_{t,T;\theta} \\
= \; & \frac{1}{\theta} B^{f/d|d,r}_{t,T;\theta} \frac{1}{S^{f|d,\div}_{t,T;\theta} - S^{d|d,\div}_{t_0,T;\theta}} - L^{f|f}_{t,T;\theta} - B^{f/d|d,r}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta} \\
= \; & B^{f/d|d,r}_{t,T;\theta} \left(\frac{1}{(1+\theta S^{f|d,\div}_{t_0,T;\theta}) - (1+\theta S^{d|d,\div}_{t_0,T;\theta})} - 1 \right) - L^{f|f}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta} \\
= \; & B^{f/d|d,r}_{t,T;\theta} \left(\frac{(1+\theta R^{f|f}_{t_0,T;\theta})(1+\theta R^{d|d}_{t_0,T;\theta})}{(1+\theta L^{f|f}_{t_0,T;\theta})(1+\theta R^{d|d}_{t_0,T;\theta}) - (1+\theta R^{f|f}_{t_0,T;\theta})(1+\theta L^{d|d}_{t_0,T;\theta})} - 1 \right) - L^{f|f}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta} \\
= \; & B^{f/d|d,r}_{t,T;\theta} \frac{1+\theta(R^{f|f}_{t_0,T;\theta} + R^{d|d}_{t_0,T;\theta}) + \theta^2 R^{f|f}_{t_0,T;\theta} R^{d|d}_{t_0,T;\theta} - \theta L^{f|f}_{t_0,T;\theta} + \theta L^{d|d}_{t_0,T;\theta}}{\theta(L^{f|f}_{t_0,T;\theta} - L^{d|d}_{t_0,T;\theta})} - L^{f|f}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta} \\
= \; & B^{f/d|d,r}_{t,T;\theta} \frac{1+\theta(S^{f|f,-}_{t_0,T;\theta} + S^{d|d,-}_{t_0,T;\theta}) + \theta^2 R^{f|f}_{t_0,T;\theta} R^{d|d}_{t_0,T;\theta}}{\theta(L^{f|f}_{t_0,T;\theta} - L^{d|d}_{t_0,T;\theta})} - L^{f|f}_{t,T;\theta} - S^{d|d,-}_{t,T;\theta}
\end{aligned}
$$

□

Remark 94 (Structure of the swaplet spread)**.** Equation 67 is surprisingly simple

(1) The currency basis spread $B^{f/d|d,r}_{t,T;\theta}$ is a $\mathbb{Q}^{f|d,T+\theta}$-martingale as expected.
(2) Equation 68 shows that $Z^{f|d}_{t,T;\theta} B^{f/d|d,r}_{t,T;\theta}$ is exactly equal to the difference in funding spreads which corresponds to the financial intuition for the existence of currency basis.
(3) The relation between the currency basis $B^{f/d|d,r}_{t,T;\theta}$ and the discount rate $R^{f|d}_{t,T;\theta}$ *does* depend on the covariance matrix of the diffusion, through the foreign funding spread under domestic collateral. By its definition and the collateral convexity K of the foreign coupon, $1-\theta S^{f|d,\div}_{t,T;\theta} = \kappa^f K\,(1+\theta R^{f|f}_{t,T;\theta})/(1+\theta R^{f|d}_{t,T;\theta})$, so $S^{f|d,\div} \neq S^{f|f,\div}$ as soon as $K \neq 1$, and (67) becomes the swaplet spread $\theta B_j = \kappa^d P_{j-1}/P_j - \kappa^f K_j F_{j-1}/F_j$ of (252). (An earlier version of this item stated the opposite, implicitly setting $S^{f|d,\div} = S^{f|f,\div}$, i.e. $K = 1$.) This is why the discount curve $\{Z^{f|d}_{t,T}\}$ is bootstrapped from resetting cross-currency swaps with a convexity provider (Algorithm 2) and calibrated jointly with the collateral basis volatility.
(4) The currency basis spread does not depend on the FX spot or its volatility.
(5) The volatility of the currency basis spread does depend on the volatility of the foreign rate.
(6) Equation 67 compares nicely to DCB Equation 32 (see below).

Remark 95. Let us assume that both the DCB model and the stochastic collateral model are calibrated to the same instruments so that currency basis spreads match. What is the impact of stochastic collateral on discount factors?

For projection rate pseudo-discount factors:

$$
\begin{aligned}
\frac{Z^{f|d,\mathrm{proj,DCB}}_{t,S,T}}{Z^{f|d,\mathrm{proj}}_{t,S,T}} &= \frac{1+\theta L^{f|d}_{t,S,T}}{1+\theta L^{f|d,\mathrm{DCB}}_{t,S,T}} \\
&= \frac{1+\theta L^{f|d}_{t,S,T}}{1+\theta L^{f|f}_{t,S,T}} \\
&= \exp\left(-\int_t^S \Sigma^{f|f}_{u,S,T} \cdot \Sigma^{f\to d}_{u,T}\, du \right)
\end{aligned}
$$

Compounding the forward pseudo-discount factors, we find

$$\begin{aligned}
\frac{Z^{f|d,\mathrm{proj,DCB}}_{t,T}}{Z^{f|d,\mathrm{proj}}_{t,T}} &= \prod_i \frac{Z^{f|d,\mathrm{proj,DCB}}_{t,T_{i-1},T_i}}{Z^{f|d,\mathrm{proj}}_{t,T_{i-1},T_i}} \\
&= \prod_i \exp\left(-\int_t^{T_{i-1}} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_i\int_t^{T_{i-1}} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_i\sum_{k=1}^{i-1}\int_{T_{k-1}}^{T_k} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_{k=1}^{q(T)}\sum_{i=k+1}^{q(T)}\int_{T_{k-1}}^{T_k} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_{k=1}^{q(T)}\int_{T_{k-1}}^{T_k}\sum_{i=k+1}^{q(T)} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_{k=1}^{q(T)}\int_{T_{k-1}}^{T_k}\sum_{i=k+1}^{q(T)} \Sigma^{f|f}_{u,T_i}\cdot\Sigma^{f\to d}_{u,T_i} - \sum_{i=k+1}^{q(T)}\Sigma^{f|f}_{u,T_{i-1}}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_{k=1}^{q(T)}\int_{T_{k-1}}^{T_k}\sum_{i=k+1}^{q(T)} \Sigma^{f|f}_{u,T_i}\cdot\Sigma^{f\to d}_{u,T_i} - \sum_{i=k}^{q(T)-1}\Sigma^{f|f}_{u,T_i}\cdot\Sigma^{f\to d}_{u,T_{i+1}}\,du\right) \\
&= \exp\left(-\sum_{k=1}^{q(T)}\int_{T_{k-1}}^{T_k} \Sigma^{f|f}_{u,T}\cdot\Sigma^{f\to d}_{u,T} - \Sigma^{f|f}_{u,T_k}\cdot\Sigma^{f\to d}_{u,T_k} - \sum_{i=k+1}^{q(T)-1}\Sigma^{f|f}_{u,T_i}\cdot\Sigma^{f\to d}_{u,T_{i-1},T_i}\,du\right)
\end{aligned}$$

$$\begin{aligned}
\frac{Z^{f|d,\mathrm{proj,DCB}}_{t,T}}{Z^{f|d,\mathrm{proj}}_{t,T}} &= \prod_i \frac{Z^{f|d,\mathrm{proj,DCB}}_{t,T_{i-1},T_i}}{Z^{f|d,\mathrm{proj}}_{t,T_{i-1},T_i}} \\
&= \prod_i \exp\left(-\int_t^{T_{i-1}} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_i\int_t^{T_{i-1}} \Sigma^{f|f}_{u,T_{i-1},T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right) \\
&= \exp\left(-\sum_i\int_t^{T_{i-1}} \Sigma^{f|f}_{u,T_i}\cdot\Sigma^{f\to d}_{u,T_i}\,du - \int_t^{T_{i-1}}\Sigma^{f|f}_{u,T_{i-1}}\cdot\Sigma^{f\to d}_{u,T_i}\,du\right)
\end{aligned}$$

A.4.3. *Resetting cross-currency swaps.*

A.4.4. *Dynamic.* From the previous proposition, it is standard to derive the dynamic of the currency basis spread of a swap. It can be written

$$\frac{dB^{f/d|c,r}_{t,T;M}}{B^{f/d|c,r}_{t,T;M}} = \left(\sum_{i=1}^{\ell M} w^i_t \Sigma^{f|d}_{t,T_{i-1},T_i}\right) dW^{f|d,T_0,T_M}_t \tag{295}$$

where the stochastic weights are

$$w^j_t = \frac{B^{f/d|c,r}_{t,T_{j-1},T_j}}{B^{f/d|c,r}_{t,T;M}} \frac{\partial B}{\partial B_j} \tag{296}$$

We can then freeze those to their initial values.

Since the forward spreads $B^{f/d|c,r}_{t,T_{j-1},T_j}$ are not directly observables, it would be better to express the vol directly as a function of the currency basis spread of swaps: $B^{f/d|c,r}_{t,T_0,T_k}$. This is essentially equivalent to stripping the discount curve $Z^{f|d}_{t,T}$.

A.4.5. *Relation to swaplet spreads.* Let us define

$$\omega_j = S^{f|f,\div}_{t_0,T_{j-1},T_j} - S^{d|d,\div}_{t_0,T_{j-1},T_j} \tag{297}$$

This is a sort of "funding basis basis". We also introduce

$$\Omega_n = \prod_{j=1}^{n} \frac{1}{\omega_j} \tag{298}$$

so that

$$Z^{f|d}_{t,T_0,T_n} = \prod_{j=1}^{n} Z^{f|d}_{t,T_{j-1},T_j} = \prod_{j=1}^{n} \frac{1}{\omega_j} B^{f/d|d,r}_{t,T_{j-1},T_j} = \Omega_n \prod_{k=1}^{j} \frac{1}{B^{f/d|d,r}_{t,T_{j-1},T_j}}$$

Then the currency-basis spread of a swap reads

$$\begin{aligned}
B^{f/d|d,r}_{t,T;M} &= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} B^{f/d|d,r}_{t,T_{j-1},T_j}}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\
&= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} \omega_j \left(1 + \theta R^{f|d}_{t,T_{j-1},T_j}\right)}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\
&= \frac{\sum_{j=1}^{M} \omega_j \prod_{k=1}^{j-1} Z^{f|d}_{t,T_{k-1},T_k}}{\sum_{j=1}^{M} \prod_{k=1}^{j} Z^{f|d}_{t,T_{k-1},T_k}} \\
&= \frac{\sum_{j=1}^{M} \Omega_j \prod_{k=1}^{j-1} \frac{1}{B^{f/d|d,r}_{t,T_{k-1},T_k}}}{\sum_{j=1}^{M} \Omega_j \prod_{k=1}^{j} \frac{1}{B^{f/d|d,r}_{t,T_{k-1},T_k}}}.
\end{aligned}$$

A.4.6. *Relations between currency basis spreads.*

$$\left(\sum_{j=1}^{M+1} Z^{f|d}_{t,T_0,T_j}\right) B^{f/d|d,r}_{t,T,M+1} = \left(\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}\right) B^{f/d|d,r}_{t,T_0,T_M} + Z^{f|d}_{t,T,M+1} B^{f/d|d,r}_{t,T_M,T_{M+1}}$$

So,

$$B^{f/d|d,r}_{t,T_b,T_c} = \frac{A^{f|d}_{t,T_a,T_c} B^{f/d|d,r}_{t,T_a,T_c} - A^{f|d}_{t,T_a,T_b} B^{f/d|d,r}_{t,T_a,T_b}}{A^{f|d}_{t,T_a,T_c} - A^{f|d}_{t,T_a,T_b}}$$

As a result,

$$\begin{aligned} B^{f/d|d,r}_{t,T;M} &= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} B^{f/d|d,r}_{t,T_{j-1},T_j}}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\ &= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} \omega_j \left(1 + \theta Z^{f|d}_{t,T_{j-1},T_j}\right)}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\ &= \frac{\sum_{j=1}^{M} \omega_j \prod_{k=1}^{j-1} Z^{f|d}_{t,T_{k-1},T_k}}{\sum_{j=1}^{M} \prod_{k=1}^{j} Z^{f|d}_{t,T_{k-1},T_k}} \end{aligned}$$

A.4.7. *Dynamic of currency basis spreads.*

Proposition 69. The currency basis spread of a swap has dynamic

$$dB^{f/d|c,r}_{t,T;M} = \left(\sum_{i=1}^{\ell M} w^i_t \Sigma^{f|d}_{t,T_{i-1},T_i}\right) dW^{f|d,T_0,T_M}_t \tag{299}$$

where the stochastic weights w^j_t are defined by

$$w^j_t = B^{f/d|c,r}_{t,T_{j-1},T_j} \frac{\partial B}{\partial B_j} \tag{300}$$

$$= \dots \tag{301}$$

Proof. Let us define the "funding basis basis" as

$$\omega_j = S^{f|f,\div}_{t_0,T_{j-1},T_j} - S^{d|d,\div}_{t_0,T_{j-1},T_j} \tag{302}$$

$$\Omega_n = \prod_{j=1}^{n} \frac{1}{\omega_j} \tag{303}$$

Then,

$$\begin{aligned} Z^{f|d}_{t,T_0,T_n} &= \prod_{j=1}^{n} Z^{f|d}_{t,T_{j-1},T_j} \\ &= \prod_{j=1}^{n} \frac{1}{\omega_j} B^{f/d|d,r}_{t,T_{j-1},T_j} \\ &= \Omega_n \prod_{k=1}^{j} B^{f/d|d,r}_{t,T_{j-1},T_j} \end{aligned}$$

Then

$$\begin{aligned} B^{f/d|d,r}_{t,T;M} &= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} B^{f/d|d,r}_{t,T_{j-1},T_j}}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\ &= \frac{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j} \omega_j \left(1 + \theta Z^{f|d}_{t,T_{j-1},T_j}\right)}{\sum_{j=1}^{M} Z^{f|d}_{t,T_0,T_j}} \\ &= \frac{\sum_{j=1}^{M} \omega_j \prod_{k=1}^{j-1} Z^{f|d}_{t,T_{k-1},T_k}}{\sum_{j=1}^{M} \prod_{k=1}^{j} Z^{f|d}_{t,T_{k-1},T_k}} \end{aligned}$$

We have a function

$$B(z_1,\dots,z_M) \quad = \quad \frac{\sum_{j=1}^{M}\omega_j\prod_{k=1}^{j-1}z_k}{\sum_{j=1}^{M}\prod_{k=1}^{j}z_k}$$

Its derivatives are

$$z_i\partial_i B \quad = \quad \frac{\left(\sum_{j=i+1}^{M}\omega_j\prod_{k=1}^{j-1}z_k\right)\left(\sum_{j=1}^{M}\prod_{k=1}^{j}z_k\right)-\left(\sum_{j=1}^{M}\omega_j\prod_{k=1}^{j-1}z_k\right)\left(\sum_{j=i+1}^{M}\prod_{k=1}^{j}z_k\right)}{\left(\sum_{j=1}^{M}\prod_{k=1}^{j}z_k\right)^2} \tag{304}$$

□

Proposition 70 (Gaussian approximations)**.** The currency basis spread of a resetting swaplet are approximately exponential martingales:

$$\frac{dB_{t,T;M}^{f/d|c,r})}{B_{t,T;M}^{f/d|c,r}} \quad \approx \quad \widetilde{\Sigma}_{t,T;M}^{f/d|c,r}dW^{f|d,T_0,T_M}. \tag{305}$$

where

$$\widetilde{\Sigma}_{t,T;M}^{f/d|c,r} \quad = \quad \dots \tag{306}$$

Proof. We know that $B_{t,T;M}^{f/d|c,r}$ is a martingale under the measure corresponding to the annuity numéraire $A_{\cdot,T;M}^{f|d}$ so it is enough to compute its volatility. Moreover, we have the relation

$$B_{t,T;M}^{f/d|c,r} \quad = \quad \frac{\sum_{j=1}^{M}\theta_j^f Z_{t,T_j}^{f|c}B_{t,T_{j-1},T_j}^{f/d|c,r}}{\sum_{j=1}^{M}\theta_j^f Z_{t,T_j}^{f|c}}$$

so

$$\begin{aligned} dB_{t,T;M}^{f/d|c,r} \quad &= \quad \sum_j \frac{\partial Y}{\partial B_j}dB_{t,T_{j-1},T_j}^{f/d|c,r}+O(dt) \\ &= \end{aligned}$$

□

A.5. **[TODO: Remove this] Calibration to currency basis swaps.**

A.5.1. *Discount curve with domestic collateral.* According to Equation 294, the discount rate with domestic collateral is

$$\begin{aligned} R_{t,T;\theta}^{f|d} \quad &\approx \quad L_{t,T;\theta}^{f|d}+B_{r,T;\theta}^{f/d|d,r}+S_{t,T;\theta}^{d|d,-} \\ &= \quad \frac{1}{\theta}\left(\left(1+\theta L_{t,T;\theta}^{f|f}\right)e^{-\int_t^S \Sigma_{u,T;\theta}^{f|f}\cdot\Sigma_{u,T}^{f\to d}du}-1\right)+B_{r,T;\theta}^{f/d|d,r}+S_{t,T;\theta}^{d|d,-} \end{aligned}$$

It follows that, given the spot volatilities, the discount factors are:

$$\begin{aligned}
Z_{t,T}^{f|d} &= \prod_i \frac{1}{1+\theta R_{t,T_i;\theta}^{f|d}} \\
&= \prod_i \frac{1}{1+\theta R_{t,T_i;\theta}^{f|d,\mathrm{DCB}}} \frac{1+\theta R_{t,T_i;\theta}^{f|d,\mathrm{DCB}}}{1+\theta R_{t,T_i;\theta}^{f|d}} \\
&= Z_{t,T}^{f|d,\mathrm{DCB}} \prod_i \frac{1}{1+\frac{\theta(R_{t,T_i;\theta}^{f|d}-R_{t,T_i;\theta}^{f|d,\mathrm{DCB}})}{1+\theta R_{t,T_i;\theta}^{f|d,\mathrm{DCB}}}} \\
&= Z_{t,T}^{f|d,\mathrm{DCB}} \prod_i \left(1-\frac{\theta(R_{t,T_i;\theta}^{f|d}-R_{t,T_i;\theta}^{f|d,\mathrm{DCB}})}{1+\theta R_{t,T_i;\theta}^{f|d,\mathrm{DCB}}}\right) \\
&= Z_{t,T}^{f|d,\mathrm{DCB}} \prod_i \left(1-\frac{\theta(L_{t,T_i;\theta}^{f|d}-L_{t,T_i;\theta}^{f|f})}{1+\theta R_{t,T_i;\theta}^{f|d,\mathrm{DCB}}}\right)
\end{aligned}$$

Remark 96 (Bootstrapping of cross-currency curves)**.** The above data allows us to bootstrap curves and price any fixed or Libor cashflow under any collateral. The internal parameters are only required for off-market collateral.

The currency basis spread of a resetting cross-currency swap is:

$$B_{t,T_M,T_{M+1}}^{f/d|d,r} = \frac{A_{t,T;M}^{f|d}}{\theta Z_{t,T_{M+1}}^{f|d}}\left(B_{t,T,M+1}^{f/d|d,r}-B_{t,T;M}^{f/d|d,r}\right)+B_{t,T,M+1}^{f/d|d,r} \tag{307}$$

We have:

$$\begin{aligned}
A_{t,T,M+1}^{f|d}B_{t,T,M+1}^{f/d|d,r} &= A_{t,T;M}^{f|d}B_{t,T;M}^{f/d|d,r}+Z_{t,T_{M+1}}^{f|d}B_{t,T_M,T_{M+1}}^{f/d|d,r} \\
\left(A_{t,T_0,T_M}^{f|d}+\theta Z_{t,T_{M+1}}^{f|d}\right)B_{t,T,M+1}^{f/d|d,r} &= A_{t,T;M}^{f|d}B_{t,T;M}^{f/d|d,r}+\theta Z_{t,T_{M+1}}^{f|d}B_{t,T_M,T_{M+1}}^{f/d|d,r} \\
A_{t,T_0,T_M}^{f|d}\left(B_{t,T,M+1}^{f/d|d,r}-B_{t,T_0,T_M}^{f/d|d,r}\right) &= \theta Z_{t,T_{M+1}}^{f|d}\left(B_{t,T_M,T_{M+1}}^{f/d|d,r}-B_{t,T,M+1}^{f/d|d,r}\right) \\
\frac{A_{t,T_0,T_M}^{f|d}}{\theta Z_{t,T_M}^{f|d}}\left(B_{t,T,M+1}^{f/d|d,r}-B_{t,T_0,T_M}^{f/d|d,r}\right) &= Z_{t,T_M,T_{M+1}}^{f|d}\left(B_{t,T_M,T_{M+1}}^{f/d|d,r}-B_{t,T,M+1}^{f/d|d,r}\right) \\
&= S_{t,T_M,T_{M+1}}^{f|f,\div}-S_{t,T_M,T_{M+1}}^{d|d,\div}-Z_{t,T_M,T_{M+1}}^{f|d}B_{t,T,M+1}^{f/d|d,r}
\end{aligned}$$

$$Z_{t,T_M,T_{M+1}}^{f|d}B_{t,T,M+1}^{f/d|d,r} = S_{t,T_M,T_{M+1}}^{d|d,\div}-S_{t,T_M,T_{M+1}}^{f|f,\div}+\frac{A_{t,T_0,T_M}^{f|d}}{\theta Z_{t,T_M}^{f|d}}\left(B_{t,T,M+1}^{f/d|d,r}-B_{t,T_0,T_M}^{f/d|d,r}\right) \tag{308}$$

We can write that as a system of linear equations for the discount factors.

Interpolation methods can be included.

APPENDIX B. REVIEW OF LINEAR GAUSSIAN MODEL

B.1. **Reminder on LGM parametrisations: Single currency case.** This is detailed in [1, §10].

B.1.1. *Separable volatilities.*

Hypothesis (Separable volatility)**.** We assume:

$$\sigma_{t,T} = g(t)h(T) \tag{309}$$

where

(1) $h : \mathbb{R}_+ \to \mathbb{R}^d$ is a deterministic differentiable vector valued function.
(2) $g : \mathbb{R}_+ \to M_{k,d}(\mathbb{R})$ is a matrix-valued stochastic process satisfying

such that

(1) $\sigma_{t,T}^\top \sigma_{t,T}$ is invertible for all t, T.

B.1.2. *Parametrisation by a Gaussian martingale.*

Proposition 71. The model can be described by the martingale state variable:

$$Z_t = \int_0^t g(u)^\top dW_u \tag{310}$$

and the following reconstruction formulas:

(1) The forward rate is:

$$r_{t,T} = r_{0,t} + \Omega_{t,T} + h_t^\top Z_t \tag{311}$$

where

$$\Omega_{t,T} = h(T)^\top \int_0^t g(s)^\top g(s) \int_0^s h(u)\, du \tag{312}$$

(2) The ZCB is:

$$Z_{t,T} = Z_{0,t,T} \exp(??) \tag{313}$$

B.1.3. *Parametrisation by mean reverting state variables.*

Proposition 72. The model can be described by the variables

$$z_t = H_t \int_0^t g_s^\top g_s \int_s^t h_u \, du \, ds + H_t \int_0^t H_s^{-1} \sigma_{z,s}^\top dW_s \tag{314}$$

$$y_t = H_t \int_0^t g_s^\top g_s \, ds H_t \tag{315}$$

with dynamic

$$dz_t = (y_t \mathbf{1} - \beta_t z_t)\, dt + \sigma_{z,t}^\top dW_t \tag{316}$$

$$dy_t = H_t g_t^\top g_t H_t - \beta_t y_t - y_t \beta_t \tag{317}$$

where

$$H_t = \operatorname{diag}(h_t) \tag{318}$$

$$\sigma_{z,s} = g_s H_s \tag{319}$$

$$M_{t,T} = H_T H_t^{-1} \mathbf{1} \tag{320}$$

$$G_{t,T} = \int_t^T M_{t,u}\, du \tag{321}$$

and the following reconstruction formulas:

(1) The forward rate is:

$$r_{t,T} = r_{0,t} + M_{t,T}^\top \left(z_t + y_t \int_t^T M_{t,u}\, du \right) \tag{322}$$

(2) The ZCB is:

$$Z_{t,T} = Z_{0,t,T} \exp\left(-G_{t,T}^\top z_t - \frac{1}{2} G_{t,T}^\top y_t G_{t,T}\right) \tag{323}$$

B.1.4. *Parametrisation by correlated Brownian motions.* It is much more intuitive to have a finite number of state variables corresponding to each asset. We now explain how OU processes driven by correlated Brownian motions lead to a multi-dimensional OU process. Conversely, we investigate under which conditions a multi-dimensional OU process can de decomposed into one dimensional ones[51].

Proposition 73. The model can be described by the variables with dynamic

$$dx_t^i = \left(\sum_j y_t^{ij} - \beta_t^i x_t^i\right) dt + \sigma_t^i d\widetilde{W}_t^i \tag{324}$$

$$d\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \rho_t^{ij}\, dt \tag{325}$$

where

$$\sigma_t = \mathrm{diag}(\sigma_{z,t}^\top \sigma_{z,t})^{1/2} \tag{326}$$

$$\sigma_t^i = \|\sigma_x^{\cdot i}(t)\|_2 = \sqrt{\sum_j \sigma_z^{ji}(t)^2} \tag{327}$$

$$d\widetilde{W}_t^i = \frac{1}{\sigma_t^i} \sum_j \sigma_z^{ji}(t) dW_t^j \tag{328}$$

$$\rho^{ij}(t) = \frac{[\sigma_z^\top \sigma_z]_t^{ij}}{\sqrt{[\sigma_z^\top \sigma_z]_t^{ii} [\sigma_z^\top \sigma_z]_t^{jj}}} \tag{329}$$

Proof. It is trivial to check that the $\widetilde{W}_t^i$ defined by

$$\widetilde{\sigma}_x^i(t) = \|\sigma_x^{\cdot i}(t)\| = \sqrt{\sum_j \sigma_z^{ji}(t)^2} \tag{330}$$

$$d\widetilde{W}_t^i = \frac{1}{\widetilde{\sigma}_x^i(t)} \sum_j \sigma_z^{ji}(t) dW_t^j. \tag{331}$$

are Brownian motions. Moreover, the instantaneous covariances are given by :

$$d\langle x_t^i, x_t^j \rangle = d\left\langle \sum_k \sigma_z^{ki}(t) dW_t^k, \sum_\ell \sigma_z^{\ell j}(t) dW_t^\ell \right\rangle \tag{332}$$

$$= \sum_k \sigma_z^{ki}(t) \sigma_z^{kj}(t) dt \tag{333}$$

$$= [\sigma_z^\top \sigma_z]^{ij}(t) dt. \tag{334}$$

Hence the expression of the instantaneous correlations. □

Remark 97. In vectorial form, the dynamic is

$$dz_t = (y_t \mathbf{1} - \beta_t z_t)\, dt + \sigma_{z,t} \Gamma_t^\top d\widehat{W}_t^{\mathrm{RN}} \tag{335}$$

where the factor loading matrix Γ_t is a square root of the correlation matrix

$$\Gamma_t^\top \Gamma_t = (\rho_t^{ij}). \tag{336}$$

with the same reconstruction formulas:

[51] It seems that this breaks symmetry: this property is not invariant by change of reference currency.

(1) The forward rate is:

$$r_{t,T} = r_{0,t} + M_{t,T}^{\top}\left(z_t + y_t \int_t^T M_{t,u}\, du\right) \tag{337}$$

(2) The ZCB is:

$$Z_{t,T} = Z_{0,t,T} \exp\left(-G_{t,T}^{\top} z_t - \frac{1}{2} G_{t,T}^{\top} y_t G_{t,T}\right) \tag{338}$$

B.1.5. *Simulation.*

Proposition 74. Let z_t be a multi-dimensional OU process:

$$dz_t = (y_t \mathbf{1} - \beta_t z_t)\, dt + \sigma_{z,t}^{\top} d\widehat{W}_t \tag{339}$$

then

$$\mathbb{E}[z_{t+h}|z_t] = e^{-\int_t^{t+h} \beta_u\, du} z_t + \int_t^{t+h} e^{-\int_s^{t+h} \beta_u\, du} y_s \mathbf{1}\, ds \tag{340}$$

$$\mathbb{V}[z_{t+h}|z_t] = \int_t^{t+h} e^{-\int_s^{t+h} \beta_u\, du} \sigma_{z,s}^{\top} \sigma_{z,s} e^{-\int_s^{t+h} \beta_u\, du}\, ds \tag{341}$$

B.1.6. *Correlation structure of instantaneous forward rates.*

B.1.7. *Correlation structure of forward Libor rates.*

B.1.8. *Correlation structure of forward swap rates.*

B.2. **Reminder on LGM parametrisations: Multi-currency case.**

B.2.1. *FX dynamic.*

B.2.2. *Quanto-drift adjustment.*

B.2.3. *Full dynamic.*

B.3. **[TODO: Remove this]: Parametrisation.**

B.4. **Precomputation of model quantities.** Model prescribes quantities that can be precomputed:

(1) Covariance of increments of the state variable
(2) Drift adjustments of increments of the state variable
(3) Quanto adjustments
(4) Collateral convexity adjustments of market observables.

All of these are expressed as covariance integrals of the form:

$$I^{a,b}_{t,S;T_a,T_b} = \int_t^S \Sigma^a_{u,T_a} \cdot \Sigma^b_{u,T_b}\, du \tag{342}$$

The objective of this section is to describe the algorithm for their computation.

B.4.1. Remember that quantities of interest are

$$\sigma^{d|d}_{t,T} = \sigma^{d|d}_r(t) \begin{bmatrix} e^{\int_0^t \beta^1_v\, dv} \\ \vdots \\ e^{\int_0^t \beta^d_v\, dv} \end{bmatrix} \begin{bmatrix} e^{-\int_0^T \beta^1_v\, dv} \\ \vdots \\ e^{-\int_0^T \beta^d_v\, dv} \end{bmatrix} \tag{343}$$

$$\Sigma^{d|d}_{t,T} = \int_t^T \sigma^{d|d}_{t,u}\, du \tag{344}$$

$$= \sigma^{d|d}_r(t) \begin{bmatrix} e^{\int_0^t \beta^1_v\, dv} \\ \vdots \\ e^{\int_0^t \beta^d_v\, dv} \end{bmatrix} \begin{bmatrix} \int_t^T e^{-\int_0^u \beta^1_v\, dv}\, du \\ \vdots \\ \int_t^T e^{-\int_0^u \beta^d_v\, dv}\, du \end{bmatrix} \tag{345}$$

$$= g(t)G(t,T) \tag{346}$$

Proposition 75. Let us introduce:

$$h^i(T) = e^{-\int_0^T \beta^i_v\, dv} \tag{347}$$

$$\Lambda^i(T) = \int_0^T e^{-\int_0^u \beta^i_v\, dv}\, du. \tag{348}$$

The quantities $h^i(t_k)$ and $\Lambda^i(t_k)$ are precomputed and stored. Since we assume piecewise constant mean reversion, we compute these integrals:

$$-\ln h^i(t_{k+1}) = -\ln h^i(t_k) + \int_{t_k}^{t_{k+1}} \beta^i_v\, dv \tag{349}$$

$$= -\ln h^i(t_k) + \sum_{\tau_j \in [t_k, t_{k+1}[} \beta^i(\tau_j)(\min(\tau_{j+1}, t_{k+1}) - \tau_j). \tag{350}$$

We then exponentiate these to get the values of $h^i(t_k)$.

It could be easier to refine the term structures so that the mean reversion is piecewise constant on all time steps.

Because, β^i is piece-wise constant, its integral is piecewise linear so the exponential can be integrated.

$$\begin{aligned}
\Lambda^i(t_{k+1}) &= \int_0^{t_{k+1}} e^{-\int_0^u \beta_v^i\, dv}\, du \\
&= \int_0^{t_k} e^{-\int_0^u \beta_v^i\, dv}\, du + \int_{t_k}^{t_{k+1}} e^{-\int_0^u \beta_v^i\, dv}\, du \\
&= \Lambda^i(t_{k+1}) + e^{-\int_0^{t_k} \beta_u^i\, dv} \int_{t_k}^{t_{k+1}} e^{-\int_{t_k}^u \beta_v^i\, dv}\, du \\
&= \Lambda^i(t_{k+1}) + h^i(t_k) \int_{t_k}^{t_{k+1}} e^{-\beta_k^i (u-t_k)}\, du \\
&= \Lambda^i(t_{k+1}) + \begin{cases} h^i(t_k) \frac{1-e^{-\beta_k^i (t_{k+t}-t_k)}}{\beta_k^i} & \text{if } \beta_k^i \neq 0 \\ h^i(t_k)(t_{k+t}-t_k) & \text{else} \end{cases}
\end{aligned}$$

From these we can compute any requested $G^i(t_j, t_k)$ as

$$\begin{aligned}
G^i(t,T) &= \int_t^T e^{-\int_0^u \beta_v^i\, dv}\, du \\
&= \left(\Lambda^i(T) - \Lambda^i(t)\right)/h^i(t).
\end{aligned}$$

Then we need to compute the $y(t_k)$'s.

Proposition 76. We have

$$y(t) = \left(\Lambda^i(T) - \Lambda^i(t)\right)/h^i(t). \tag{351}$$

Proposition 77. We have

$$G^i(t,T) = \left(\Lambda^i(T) - \Lambda^i(t)\right)/h^i(t). \tag{352}$$

Proposition 78. The short rate is

$$r_t = r_{0,t} + 1^\top z_t. \tag{353}$$

Proposition 79 (Zero coupon reconstruction)**.** We have

$$Z_{t,T}^{d|d} = Z_{0,t,T}^{d|d} \exp\left(-G(t,T)^\top z_t - \frac{1}{2} G(t,T)^\top y(t) G(t,T)\right) \tag{354}$$

Proposition 80 (Caplet volatility)**.** Let us introduce:

$$w_{t,T;\theta}^{d|d} = \int_t^T \left|\Sigma_{u,T;\theta}^{d|d}\right|^2 du \tag{355}$$

Then

Proof.

$$\frac{d(1+\delta L_{t,T;\delta})}{1+\delta L_{t,T;\delta}} = \left(\Sigma_{t,T+\delta} - \Sigma_{t,T}\right)^\top dW^{d|d,T+\delta} \tag{356}$$

So:

$$(1+\delta L_{S,T;\delta} = (1+\delta L_{t,T;\delta}) \exp\left(\int_t^S \Sigma_{u,T;\delta} \cdot dW_u^{d|d,T+\delta} - \frac{1}{2}\int_t^S |\Sigma_{u,T;\delta}|^2\, du\right) \tag{357}$$

and

$$\mathbb{V}_t\left[\ln(1+\delta L_{T,T;\delta}\right] = \int_t^T \left|\Sigma_{u,T+\delta} - \Sigma_{u,T}\right|^2 du \tag{358}$$

□

Proposition 81 (Gaussian swap rate approximation)**.** Let us introduce:

$$\mathbb{V}_t^{d|d,T}[Y_{T,T;M}^{d|d}] = \tag{359}$$

B.5. **Calibration.**

Proposition 82 (Spot volatility bootstrapping)**.**

Proposition 83 (Mean reversion calibration)**.**

B.6. **Monte-Carlo simulation.**

B.6.1. *Monte-Carlo simulation of Ornstein-Uhlenbeck processes.*

$$\begin{aligned}
dX_t &= -\beta_t X_t\,dt + \sigma_t\,dW_t \\
d\left(e^{\int_0^t \beta_u\,du} X_t\right) &= \sigma_t e^{\int_0^t \beta_u\,du}\,dW_t \\
e^{\int_0^{t_{k+1}} \beta_u\,du} X_{t_{k+1}} &= e^{\int_0^{t_k} \beta_u\,du} X_{t_k} + \int_{t_k}^{t_{k+1}} \sigma_u e^{\int_0^u \beta_v\,dv}\,dW_u \\
X_{t_{k+1}} &= e^{-\int_{t_k}^{t_{k+1}} \beta_v\,dv} X_{t_k} + e^{-\int_0^{t_{k+1}} \beta_v\,dv} \int_{t_k}^{t_{k+1}} \sigma_u e^{\int_0^u \beta_v\,dv}\,dW_u \\
&= e^{-\int_{t_k}^{t_{k+1}} \beta_v\,dv} X_{t_k} + \int_{t_k}^{t_{k+1}} \sigma_u e^{-\int_u^{t_{k+1}} \beta_v\,dv}\,dW_u
\end{aligned}$$

Proposition 84. Conditionally on X_{t_k}, $X_{t_{k+1}}$ is normal with moments:

$$\mathbb{E}[X_{t_{k+1}}|X_{t_k}] = e^{-\int_{t_k}^{t_{k+1}} \beta_v\,dv} X_{t_k} \tag{360}$$

$$\mathbb{V}[X_{t_{k+1}}|X_{t_k}] = \int_{t_k}^{t_{k+1}} \sigma_u^2 e^{-2\int_u^{t_{k+1}} \beta_v\,dv}\,du \tag{361}$$

This allows for the exact simulation of the process using the scheme:

$$X_{t_{k+1}} = \alpha_{k+1} X_{t_k} + \alpha_{k+1}\Sigma_{k+1}\xi_{k+1}$$

where

$$\begin{aligned}
\alpha_{k+1} &= e^{-\int_{t_k}^{t_{k+1}} \beta_v\,dv} \\
\Sigma_{k+1} &= \sqrt{\int_{t_k}^{t_{k+1}} \sigma_u^2 e^{2\int_{t_k}^u \beta_v\,dv}\,du}
\end{aligned}$$

(so that $\alpha_{k+1}^2\Sigma_{k+1}^2$ is the conditional variance of the proposition) and (ξ_k) is a Gaussian white noise.

For correlated OU processes,

$$X_{t_{k+1}}^i = e^{-\int_{t_k}^{t_{k+1}} \beta_v^i\,dv} X_{t_k}^i + (L_{k+1}\xi_{k+1})^i$$

where ξ_{k+1} is a standard Gaussian vector,

$$\Gamma_{k+1}^{ij} = \int_{t_k}^{t_{k+1}} \rho_u^{ij}\sigma_u^i\sigma_u^j e^{-\int_u^{t_{k+1}} (\beta_v^i+\beta_v^j)\,dv}\,du$$

is the conditional covariance matrix, and L_{k+1} is a square root of Γ_{k+1}, meaning that $L_{k+1}L_{k+1}^\top = \Gamma_{k+1}$, as given by the Cholesky or the eigenvalue decomposition (the latter when Γ_{k+1} is singular).

B.7. **Precomputation of covariance matrix.**

B.8. **The Single Currency Linear Gaussian Model: Calibration.**

B.8.1. *Calibration of native rates markets.*

B.8.2. *Calibration of spot volatility.*

B.8.3. *Calibration of mean reversion.*

Appendix C. Linear Gaussian Collateral Model: proofs

APPENDIX D. GAUSSIAN APPROXIMATION OF PAR SWAP SPREADS

In this section, we prove the Gaussian approximations of forward swap rates and forward currency basis spreads stated in subsubsection 8.3.2.

D.0.1. *Proof of Lemma 3.* By Girsanov, $dW_t^{d,\mathrm{RN}} = dW_t^{f|d,T} + \left(\sigma_t^{df} - \Sigma_{t,T}^{f|d}\right) dt$ and $\Sigma_{t,T}^{f|d} = \hat{\sigma}_{z,t}\hat{G}_{t,T}$. So, by Remark 33,

$$d\hat{z}_t \;=\; \left(\hat{y}_t\mathbf{1} - \hat{\sigma}_{z,t}^\top\hat{\sigma}_{z,t}\hat{G}_{t,T} - \hat{\beta}_t\hat{z}_t\right) dt + \hat{\sigma}_{z,t}^\top dW_t^{f|d,T}.$$

The coefficients are deterministic in the LGCM, so $\hat{z}_T$ is Gaussian with covariance $\int_0^T \hat{\mathcal{M}}_{u,T}\hat{\sigma}_{z,u}^\top\hat{\sigma}_{z,u}\hat{\mathcal{M}}_{u,T}\,du = \hat{y}_T$ and mean

$$\int_0^T \hat{\mathcal{M}}_{u,T}\left(\hat{y}_u\mathbf{1} - \hat{\sigma}_{z,u}^\top\hat{\sigma}_{z,u}\hat{G}_{u,T}\right) du \;=\; 0,$$

since, by Fubini and $\hat{\mathcal{M}}_{u,T}\hat{\mathcal{M}}_{w,u} = \hat{\mathcal{M}}_{w,T}$,

$$\int_0^T \hat{\mathcal{M}}_{u,T}\hat{y}_u\mathbf{1}\,du \;=\; \int_0^T \hat{\mathcal{M}}_{w,T}\hat{\sigma}_{z,w}^\top\hat{\sigma}_{z,w}\int_w^T \hat{\mathcal{M}}_{w,u}\mathbf{1}\,du\,dw \;=\; \int_0^T \hat{\mathcal{M}}_{w,T}\hat{\sigma}_{z,w}^\top\hat{\sigma}_{z,w}\hat{G}_{w,T}\,dw.$$

D.0.2. *Proof of Lemma 4.* The formula for $P_{t,j}$ is (185). Under the constant funding spread hypothesis, $1 + \theta L_{t,T;\theta}^{f|f} = (1 - \theta S_{0,T;\theta}^{f|f,\div})Z_{t,T}^{f|f}/Z_{t,T+\theta}^{f|f}$, and the foreign Libor under domestic collateral follows from (187). For a compounded coupon, $1 + \theta L_{t,T;\theta}^{f|d} = K_t^C(T, T+\theta)Z_{t,T}^{f|f}/Z_{t,T+\theta}^{f|f}$ by Lemma 1 and Proposition 26, with K^C deterministic. The resetting forward Libor is given by the proposition on resetting forward Libors in the HJM section, or by (71) for any coupon type.

D.0.3. *Proof of Proposition 41.* Write $\ell_{t,j} = 1 + \theta_j L_{t,j}$, so that $Y_t = A_t^{-1}\sum_j P_{t,j}(\ell_{t,j} - 1)$. By Lemma 4, $Y_t = Y(t, \hat{z}_t)$ with

$$\nabla_{\hat{z}} \ln P_{t,j} \;=\; -\hat{G}_{t,T_j}, \qquad \nabla_{\hat{z}} \ln \ell_{t,j} \;=\; \begin{bmatrix} G_{t,T_j}^f - G_{t,T_{j-1}}^f \\ 0 \end{bmatrix}.$$

Since Y_t is a $\mathbb{Q}^{f|d,A}$-martingale, its dynamic is $dY_t = \nabla_{\hat{z}}Y(t,\hat{z}_t)^\top\hat{\sigma}_{z,t}^\top dW_t^{f|d,A}$, and a direct computation gives

$$\nabla_{\hat{z}}Y \;=\; \frac{1}{A_t}\sum_{j=1}^n P_{t,j}\begin{bmatrix} (1+\theta_j Y_t)G_{t,T_j}^f - \ell_{t,j}G_{t,T_{j-1}}^f \\ \theta_j(Y_t - L_{t,j})G_{t,T_j}^c \end{bmatrix}.$$

For $T \geq T_0$, $G_{t,T} = G_{t,T_0} + \mathcal{M}_{t,T_0}G_{T_0,T}$. The terms in G_{t,T_0} cancel since $\sum_j P_{t,j}\left((1+\theta_j Y_t) - \ell_{t,j}\right) = \sum_j \theta_j P_{t,j}(Y_t - L_{t,j}) = 0$, so

$$\nabla_{\hat{z}}Y(t,\hat{z}_t) \;=\; \hat{\mathcal{M}}_{t,T_0}\,\hat{q}_Y(t)$$

where $\hat{q}_Y(t)$ is given by the formulas of the proposition evaluated with the time-t values of the ratios $P_{t,j}/A_t$, $L_{t,j}$ and Y_t. The weight freezing approximation consists in replacing $\hat{q}_Y(t)$ by its value $\hat{q}_Y$ computed from forward values seen from 0. Since $\hat{q}_Y(t)$ only depends on ratios of discount factors, we can use the forward discount factors $Z_{0,T_0,T_j}^{f|d}$. Then $Y_{T_0} - Y_0 = \int_0^{T_0}\hat{q}_Y^\top\hat{\mathcal{M}}_{t,T_0}\hat{\sigma}_{z,t}^\top dW_t^{f|d,A}$ is centred Gaussian with variance $\int_0^{T_0}\hat{q}_Y^\top\hat{\mathcal{M}}_{t,T_0}\hat{\sigma}_{z,t}^\top\hat{\sigma}_{z,t}\hat{\mathcal{M}}_{t,T_0}\hat{q}_Y\,dt = \hat{q}_Y^\top\hat{y}_{T_0}\hat{q}_Y$.

D.0.4. *Proof of Proposition 42.* Write $\ell_{t,j}^r = 1 + \theta_j L_{t,j}^r$, so that $B_t = A_t^{-1}\sum_j P_{t,j}(\ell_{t,j}^r - \ell_{t,j})$. By Lemma 4, $P_{t,j}\ell_{t,j}^r = (1 - \theta_j S^{d|d,\div})P_{t,j-1}$ and $P_{t,j}\ell_{t,j}$ is proportional to $P_{t,j}Z_{t,T_{j-1}}^{f|f}/Z_{t,T_j}^{f|f}$, so

$$\nabla_{\hat{z}}\left(P_{t,j}\ell_{t,j}^r\right) = -P_{t,j}\ell_{t,j}^r\hat{G}_{t,T_{j-1}}, \qquad \nabla_{\hat{z}}\left(P_{t,j}\ell_{t,j}\right) = -P_{t,j}\ell_{t,j}\begin{bmatrix} G_{t,T_{j-1}}^f \\ G_{t,T_j}^c \end{bmatrix},$$

and $\nabla_{\hat{z}} A_t^{-1} = A_t^{-2} \sum_j \theta_j P_{t,j} \hat{G}_{t,T_j}$. Using $\ell^r_{t,j} - \ell_{t,j} = \theta_j B_{t,j}$, we get

$$\nabla_{\hat{z}} B \;=\; \frac{1}{A_t} \sum_{j=1}^{n} P_{t,j} \begin{bmatrix} \theta_j \left(B_t G^f_{t,T_j} - B_{t,j} G^f_{t,T_{j-1}} \right) \\ (\ell_{t,j} + \theta_j B_t) G^c_{t,T_j} - \ell^r_{t,j} G^c_{t,T_{j-1}} \end{bmatrix}.$$

The terms in G_{t,T_0} cancel in both blocks since $\sum_j \theta_j P_{t,j}(B_t - B_{t,j}) = 0$, and we conclude as for swap rates.

Remark 98. The weight freezing could be applied to other representations of the same par spread, for example the representation of B_t as a weighted sum of differences of funding spreads in the corollary on resetting cross-currency swaps. All representations have the same exact gradient at t, so they lead to the same frozen loadings when the gradient is frozen at the forward values. They differ only if one freezes the weights rather than the gradient.

APPENDIX E. NORMAL APPROXIMATIONS

E.1. **Shifted lognormal distributions.** In the Linear Gaussian Model, the simply compounded forward rates (Libors) follow shifted lognormal distributions. But, when calibrating the model to G10 market data, one finds that caplet smile is flat which means that the model distribution of Libors is very close to be Gaussian.

In this section, we derive Gaussian approximations of shifted lognormal distributions.

E.1.1. *Moment matching.*

Random variables. Let Z be a standard Gaussian variable and let X be a shifted lognormal variable with distribution:

$$1+\theta X = (1+\theta\mu_{\mathrm{LN}})e^{\sigma_{\mathrm{LN}}Z-\frac{1}{2}\sigma_{\mathrm{LN}}^2} \tag{362}$$

where Z is standard Gaussian variable.

We can compute the moments of X:

$$\mathbb{E}[X] = \mu_{\mathrm{LN}} \tag{363}$$

$$\mathbb{V}[X] = \left(\frac{1+\theta\mu_{\mathrm{LN}}}{\theta}\right)^2\left(e^{\sigma_{\mathrm{LN}}^2}-1\right) \tag{364}$$

It follows that X can be approximated by

$$Y = \mu_{\mathrm{N}}+\sigma_{\mathrm{N}}Z \tag{365}$$

where

$$\mu_{\mathrm{N}} = \mu_{\mathrm{LN}} \tag{366}$$

$$\sigma_{\mathrm{N}} = \frac{1+\theta\mu_{\mathrm{LN}}}{\theta}\sqrt{e^{\sigma_{\mathrm{LN}}^2}-1} \tag{367}$$

Martingales. The previous result easily extends to Gaussian martingales.

Proposition 85. Let X_t be a shifted exponential martingale:

$$\frac{d(1+\theta X_t)}{1+\theta X_t} = \sigma_{\mathrm{LN},t}dW_t \tag{368}$$

Then, when the lognormal volatility of marginal distribution is low, X_t is well approximated by a Brownian martingale:

$$dY_t = \sigma_{\mathrm{N},t}dW_t \tag{369}$$

where

$$Y_0 = X_0 \tag{370}$$

$$\sigma_{\mathrm{N},t} = \frac{1+\theta X_0}{\theta}\sigma_{\mathrm{LN},t} \tag{371}$$

Proof. We clearly match expectations so let us match total variance, that is that variance of each terminal distribution X_T:

$$\int_0^T \sigma_{\mathrm{N},t}^2\,dt = \left(\frac{1+\theta X_0}{\theta}\right)^2\left(e^{\int_0^T\sigma_{\mathrm{LN},t}^2\,dt}-1\right) \tag{372}$$

Taking the derivative wrt T, we find:

$$\sigma_{\mathrm{N},T}^2 = \left(\frac{1+\theta X_0}{\theta}\right)^2\sigma_{\mathrm{LN},T}^2\left(e^{\int_0^T\sigma_{\mathrm{LN},t}^2\,dt}-1\right). \tag{373}$$

And, at first order,

$$\sigma_{\mathrm{N},t} \approx \frac{1+\theta X_0}{\theta}\sigma_{\mathrm{LN},t}. \tag{374}$$

□

E.1.2. *Cumulative distribution.* Let Z be a standard Gaussian variable and $1+\theta X = (1+\theta\mu)e^{\sigma Z - \frac{1}{2}\sigma^2}$. The CDF of X is

$$\begin{aligned} F_X(x) &= \mathbb{P}\left(1+\theta X \leq 1+\theta x\right) \\ &= \mathbb{P}\left((1+\theta\mu)e^{\sigma Z-\frac{1}{2}\sigma^2} \leq 1+\theta x\right) \\ &= \mathbb{P}\left(e^{\sigma Z-\frac{1}{2}\sigma^2} \leq \frac{1+\theta x}{1+\theta\mu}\right) \\ &= \mathbb{P}\left(\sigma Z - \frac{1}{2}\sigma^2 \leq \ln\left(\frac{1+\theta x}{1+\theta\mu}\right)\right) \\ &= \mathbb{P}\left(Z \leq \ln\left(\frac{1+\theta x}{1+\theta\mu}\right) + \frac{1}{2}\sigma^2\right) \\ &= \Phi\left(\ln\left(\frac{1+\theta x}{1+\theta\mu}\right) + \frac{1}{2}\sigma^2\right) \end{aligned}$$

We can then expand[52] around $x=\mu$.

E.2. **Lognormal mixtures.**

E.2.1. *Additive mixtures of lognormal variables.* Additive mixtures appear in options on baskets in the BS model. One can moment match a shifted lognormal distribution to the first 3 moments of the basket. This can give very good results at very low cost.

E.2.2. *Difference of lognormal variables.* We consider

$$X = e^{\mu^1+\sigma^1 Z^1} - e^{\mu^2+\sigma^2 Z^2} \tag{375}$$

where (Z_1, Z_2) is Gaussian vector with unit variance but non-zero correlation.

For processes, the analog is:

$$X_t = e^{\mu_t^1+\int_0^t \sigma_u^1 dW_u^1} - e^{\mu_t^2+\int_0^t \sigma_u^2 dW_u^2} \tag{376}$$

where W^1, W^2 are correlated Brownian motions.

This is the case of the currency basis which is the difference of two lognormal processes. Note that this implies that the model assigns a non-zero probability to the basis changing sign. Moreover the marginal distributions have fatter tails than a Gaussian.

Nevertheless, we can approximate this process by a normal or shifted lognormal process by moment matching of the marginal distributions.

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[52] The lognormal random variable with parameters μ, σ is $e^{\mu+\sigma Z}$, where Z is standard normal, and so its CDF is $F(x) = \Phi\left(\frac{\ln(x)-\mu}{\sigma}\right)$ for $x>0$, where Φ is the standard normal CDF. When $x = e^\mu(1+t)$ is near e^μ,

$$\ln(x) = \mu + \ln(1+t) = \mu + t + O(t^2)$$

so that

$$F(x) \approx \Phi\left(\frac{t}{\sigma}\right)$$

In fact

$$F(x) - \Phi\left(\frac{t}{\sigma}\right) = \Phi\left(\frac{\ln(1+t)}{\sigma}\right) - \Phi\left(\frac{t}{\sigma}\right)$$

This will be a very good approximation for t near 0. For fixed t, it goes to 0 as $\sigma \to 0$ because both Φ's go to 0 if $t<0$ and both Φ's go to 1 if $t>0$. When σ is small there appear to be two local minima of $F(x) - \Phi(t/\sigma)$, at approximately $t = \pm\sqrt{2}\sigma$, where $F(x) - \Phi(t/\sigma) \approx -\sigma/(\sqrt{2\pi}e) \pm \sigma^2/(6\sqrt{\pi}e)$. Thus the maximum error in F is approximately

$$\frac{\sigma}{\sqrt{2\pi}e} + \frac{\sigma^2}{6\sqrt{\pi}e}$$

For example, for $\sigma = 1/2$, the actual maximum error is approximately 0.0817231495769266, and the above approximation to the maximum error is 0.08202940444. As σ gets smaller, the approximation should get better.

Moments. Write $A = e^{\mu^1+\sigma^1 Z^1}$, $B = e^{\mu^2+\sigma^2 Z^2}$, ρ the correlation of Z^1 and Z^2, and

$$M_{mn} := \mathbb{E}[A^m B^n] = \exp\left(m\mu^1 + n\mu^2 + \tfrac{1}{2}\left(m^2(\sigma^1)^2 + n^2(\sigma^2)^2 + 2mn\rho\sigma^1\sigma^2\right)\right).$$

Then $X = A - B$ has

$$\mathbb{E}[X] = M_{10} - M_{01} = e^{\mu^1+\frac{1}{2}(\sigma^1)^2} - e^{\mu^2+\frac{1}{2}(\sigma^2)^2}, \tag{377}$$

$$\begin{aligned}\mathbb{V}[X] &= M_{20} - 2M_{11} + M_{02} - \mathbb{E}[X]^2 \\ &= M_{10}^2\left(e^{(\sigma^1)^2} - 1\right) + M_{01}^2\left(e^{(\sigma^2)^2} - 1\right) - 2M_{10}M_{01}\left(e^{\rho\sigma^1\sigma^2} - 1\right),\end{aligned} \tag{378}$$

$$\mathbb{SK}[X] = \frac{\mathbb{E}[X^3] - 3\mathbb{E}[X]\,\mathbb{E}[X^2] + 2\mathbb{E}[X]^3}{\mathbb{V}[X]^{3/2}}, \qquad \mathbb{E}[X^3] = M_{30} - 3M_{21} + 3M_{12} - M_{03}, \tag{379}$$

with $\mathbb{E}[X^2] = M_{20} - 2M_{11} + M_{02}$. (Checked by Monte Carlo with $8 \cdot 10^6$ draws: mean, variance and skewness agree within the sampling error.)

Normal approximation. X can be approximated by $\mathcal{N}(\mu, \sigma^2)$ with

$$\mu = \mathbb{E}[X], \qquad \sigma = \sqrt{\mathbb{V}[X]}. \tag{380}$$

Shifted lognormal approximation. A shifted lognormal $Y = a + \epsilon\, e^{\mu+\sigma Z}$, $\epsilon = \pm 1$, has, with $w = e^{\sigma^2}$,

$$\mathbb{E}[Y] = a + \epsilon\, e^{\mu+\frac{1}{2}\sigma^2}, \tag{381}$$

$$\mathbb{V}[Y] = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right) = e^{2\mu}w(w-1), \tag{382}$$

$$\mathbb{SK}[Y] = \epsilon\,(w+2)\sqrt{w-1}. \tag{383}$$

Matching the mean, variance and skewness $\gamma = \mathbb{SK}[X]$ gives $\epsilon = \operatorname{sign}(\gamma)$ and the unique root $w > 1$ of $(w+2)^2(w-1) = \gamma^2$,

$$w = \beta^{1/3} + \beta^{-1/3} - 1, \qquad \beta = 1 + \tfrac{1}{2}\gamma^2 + |\gamma|\sqrt{1 + \tfrac{1}{4}\gamma^2},$$

then $\sigma^2 = \ln w$, $e^{\mu} = \sqrt{\mathbb{V}[X]/(w(w-1))}$ and $a = \mathbb{E}[X] - \epsilon\, e^{\mu}\sqrt{w}$. (Checked numerically: the root solves the cubic to 10^{-15} and the matched Y reproduces the three moments within the Monte Carlo error.)

E.3. **Second order mixtures.** In the case of an option on a non-resetting cross-currency swaplet on a non-USD pair, the underlying is a basket with 3 terms:

(1) $B^{d|c,nr}_{T,T;\theta}$: Normal
(2) $S\left(\frac{x_T^{df}}{x_t^{df}} - 1\right)$: Lognormal.
(3) $x_t^{df} B^{f|c,nr}_{T,T;\theta}$: product of Normal and Lognormal

Let us moment match this distribution to a shifted lognormal.

[TODO: Write down the computation]

APPENDIX F. ESTIMATION OF HISTORICAL VOLATILITIES AND CORRELATIONS

F.1. **Stationarity assumption.**

F.1.1. *Market quotes and stationarity.*

F.1.2. *Statistical evidence.*

F.2. **Model.**

F.2.1. *Statistical evidence.*

F.3. **Estimator.**

F.3.1. *Unexplained.*

Remark 99. There is a significant gap between implied and realised vols. Here are a list of possible explanations:

(1) The model assumes Gaussian returns while we know they are fat-tailed.
(2) The implied volatility quoted in the market account for transaction/liquidity costs that the model ignores.

Remark 100 (Rough volatility)**.** Empirical studies show that volatility is rough, i.e. similar to a fractional Brownian motion with Hurst exponent $H \approx 0.1$.

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